

| <b>Lesson No.</b> | <b>TOPICS</b>                                            | <b>Page No.</b> |
|-------------------|----------------------------------------------------------|-----------------|
| 1                 | Markets and prices                                       | 1               |
| 2                 | Markets and prices (continued)                           | 3               |
| 3                 | The basics of supply and demand                          | 5               |
| 4                 | The basics of supply and demand (continued)              | 9               |
| 5                 | Elasticities of supply and demand                        | 16              |
| 6                 | Consumer behavior                                        | 22              |
| 7                 | Consumer behavior (continued)                            | 27              |
| 8                 | Consumer behavior (continued)                            | 34              |
| 9                 | Marginal utility and consumer choice                     | 36              |
| 10                | Individual and market demand                             | 41              |
| 11                | Individual and market demand (continued)                 | 46              |
| 12                | Individual and market demand (continued)                 | 51              |
| 13                | Choice under uncertainty                                 | 56              |
| 14                | Choice under uncertainty (continued)                     | 60              |
| 15                | Choice under uncertainty (continued)                     | 64              |
| 16                | Production                                               | 68              |
| 17                | Production (continued)                                   | 73              |
| 18                | Theory of costs                                          | 78              |
| 19                | Theory of costs (continued)                              | 82              |
| 20                | Theory of costs (continued)                              | 85              |
| 21                | Theory of costs (continued)                              | 89              |
| 22                | Perfectly competitive markets                            | 93              |
| 23                | Perfectly competitive markets (continued)                | 97              |
| 24                | Equilibrium in perfectly competitive markets             | 101             |
| 25                | Equilibrium in perfectly competitive markets (continued) | 104             |
| 26                | Profit maximization and competitive supply               | 108             |
| 27                | The analysis of competitive markets                      | 112             |
| 28                | The analysis of competitive markets (continued)          | 115             |
| 29                | The analysis of competitive markets (continued)          | 119             |
| 30                | Market structure and competitive strategy                | 122             |
| 31                | Market structure and competitive strategy (continued)    | 126             |
| 32                | Market structure and competitive strategy (continued)    | 129             |
| 33                | Market structure and competitive strategy (continued)    | 133             |
| 34                | Pricing with market power                                | 137             |
| 35                | Pricing with market power (continued)                    | 141             |
| 36                | Pricing with market power (continued)                    | 144             |
| 37                | Pricing with market power (continued)                    | 148             |
| 38                | Pricing with market power (continued)                    | 151             |
| 39                | Monopolistic competition                                 | 156             |
| 40                | Oligopoly                                                | 158             |
| 41                | Competition versus collusion                             | 163             |
| 42                | Markets for factor inputs                                | 168             |
| 43                | Markets for factor inputs (continued)                    | 171             |
| 44                | Markets for factor inputs (continued)                    | 174             |
| 45                | Markets for factor inputs (continued)                    | 177             |

## LESSON 1

## MARKETS AND PRICES

**Economics** is the study of how societies use scarce resources to produce valuable commodities and distribute them among different people.

**MICROECONOMICS VS. MACROECONOMICS**

Microeconomics deals with behavior of individual units.

- When Consuming; How we choose what to buy
- When Producing; How we choose what to produce
- Markets: The interaction of consumers and producers.

Macroeconomics deals with analysis of aggregate issues:

- Economic growth
- Inflation
- Unemployment

Microeconomics is the foundation of macroeconomic analysis.

**THEMES OF MICROECONOMICS**

According to Mick Jagger & the Rolling Stones, “You can’t always get what you want”. Why Not?

- ♦ Limited Resources
- ♦ Unlimited Wants

Allocation of Scarce Resources and Trade-offs

- ♦ In a planned economy
- ♦ In a market economy

Microeconomics and Optimal Trade-offs

1. Consumer Theory
2. Workers
3. Theory of the Firm

Microeconomics and Prices

- The role of prices in a market economy
- How prices are determined

**THEORIES AND MODELS**

Microeconomic Analysis

Theories are used to explain observed phenomena in terms of a set of basic rules and assumptions. For example, the theory of the firm or the theory of consumer behavior.

Models:

A mathematical representation of a theory used to make a prediction.

Validating a Theory

The validity of a theory is determined by the quality of its prediction, given the assumptions.

Evolving the Theory

Testing and refining theories is central to the development of the science of economics.

**POSITIVE VERSUS NORMATIVE ECONOMICS**

Positive economics deals with the observations or predictions of the facts of economic life. For example:

What will be the impact of an increase in wages on the price of a product?

Normative Economics is the value judgments about how economics should operate, based on certain moral principles or preferences?" For example:

What wage rate should be paid to the auto workers to make them an active member of the society?

## WHAT IS A MARKET?

### ♦ Markets vs. Industries

Market is a geographically defined area where buyers and sellers interact to determine the price of a product or a set of products.

Industries are the supply side of the market.

### ♦ Defining the Market

The market parameters must be set before an analysis of the market can take place.

### ♦ Arbitrage

Buying a product at a low price in one location and selling at a high price in another.

### ♦ Competitive vs. Noncompetitive Markets

– In competitive Markets, because of the large number of buyers and sellers, no individual buyer or seller can influence the price.

♦ Example: Most agricultural markets

– Noncompetitive Markets are the markets where individual producers can influence the price.

♦ Example: OPEC

### ♦ Market Price

– Competitive markets establish one price.

– Noncompetitive markets may set many prices for the same product.

### ♦ Market Definition - The Extent of a Market

– Market Definition

Which buyers and sellers should be included in a given market?

– Market Extent

Defines the boundaries of the market

♦ Geographic

♦ Range of products

– Examples

– Geographic boundaries

♦ Gold: Lahore vs. Karachi

♦ Housing: Islamabad vs. Rawalpindi

– Range of Products

♦ Gasoline: regular, super, & diesel

♦ Cameras: Polaroid, point & shoot, digital

– Markets for Prescription Drugs

♦ Well-defined markets - therapeutic drugs

♦ Ambiguous markets – painkillers

**LESSON 2****MARKETS AND PRICES (Continued)****ECONOMICS; ANOTHER PERSPECTIVE**

Economics is the study of the choices made by people who are faced with scarcity.

Scarcity is a situation in which resources are limited but can be used in different ways; so one good or service must be sacrificed for another.

**SOCIETY'S CHOICES**

The decisions of producers, consumers and government determine how an economic system answers three fundamental questions:

1. What products do we produce?
2. How do we produce these products?
3. Who consumes the products?

**FACTORS OF PRODUCTION**

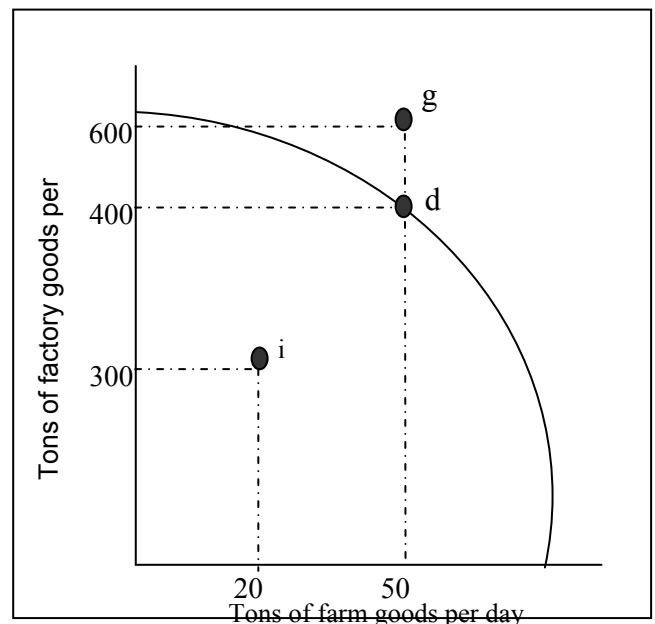
Factors of production are the resources that are used to produce goods and services:

1. Natural resources:  
The things created by acts of nature such as land, water, mineral, oil and gas deposits, renewable and nonrenewable resources.
2. Labor:  
The human effort, physical and mental, used by workers in the production of goods and services.
3. Physical capital.  
All the machines, buildings, equipment, roads and other objects made by human beings to produce goods and services.
4. Human capital:  
The knowledge and skills acquired by a worker through education and experience.
5. Entrepreneurship:  
The effort to coordinate the production and sale of goods and services.  
Entrepreneurs take risk and commit time and money to a business without any guarantee of profit.

**THE PRODUCTION POSSIBILITIES FRONTIER (PPF)**

The PPF curve shows the possible combinations of goods and services available to an economy, given that all productive resources are fully and efficiently employed.

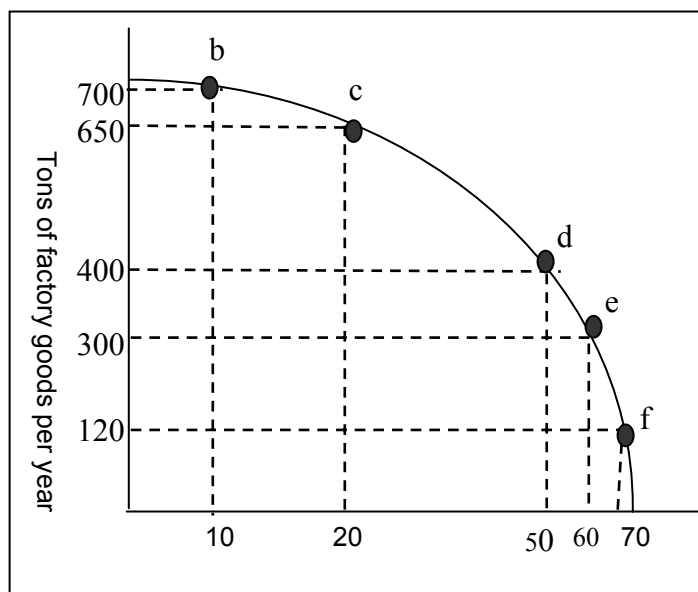
When the economy is at point *i*, resources are not fully employed and/or they are not used efficiently. Point *g* is desirable because it yields more of both goods, but not attainable given the amount of resources available. Point *d* is one of the possible combinations of goods produced when resources are fully and efficiently employed.



### SCARCITY AND THE PPF

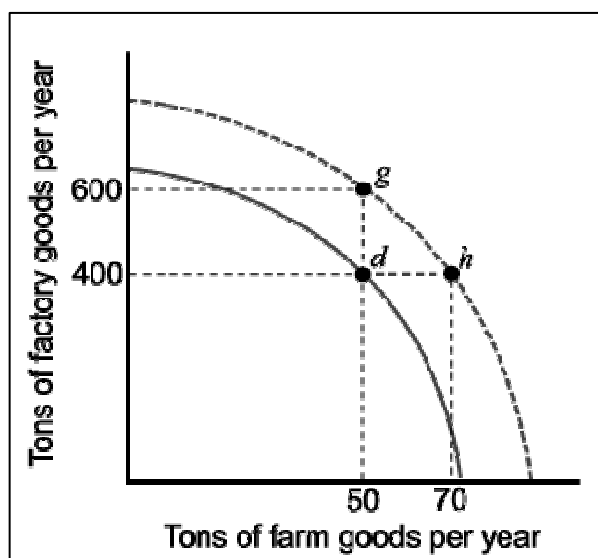
To increase the amount of farm goods by 10 tons, we must sacrifice 100 tons of factory goods.

The PPF curve is bowed out because resources are not perfectly adaptable to the production of the two goods. As we increase the production of one good, we sacrifice progressively more of the other.



### SHIFTING THE PPF CURVE

To increase the production of one good without decreasing the production of the other, the PPF curve must shift outward. The PPF curve shifts outward as a result of an increase in the economy's resources OR a technological innovation that increases the output obtained from a given amount of resources. From point *d*, an additional 200 tons of factory goods or 20 tons of farm goods are now possible (or any combination in between).



## LESSON 3

## THE BASICS OF SUPPLY AND DEMAND

## REAL VERSUS NOMINAL PRICES

Nominal price is the absolute or current dollar price of a good or service when it is sold. Real price is the price relative to an aggregate measure of prices or constant dollar price. The Consumer Price Index (CPI) is an aggregate measure. Real prices are emphasized to permit the analysis of relative prices.

## CALCULATING REAL PRICES

$$\text{Real Price} = \frac{\text{CPI}_{\text{base year}}}{\text{CPI}_{\text{current year}}} \times \text{Nominal Price}_{\text{current year}}$$

## CALCULATING THE REAL PRICE OF MILK

| Year | Nominal Price of Milk | CPI   | Real Price of Milk in 1970 dollars |
|------|-----------------------|-------|------------------------------------|
| 1970 | .40                   | 38.8  | .40=38.8/38.8x .40                 |
| 1980 | .65                   | 82.4  | .31=38.8/82.4x .65                 |
| 1999 | 1.05                  | 167.0 | .24=38.8/167.0x 1.05               |

## CALCULATING REAL PRICES: AN EXAMPLE - EGGS &amp; COLLEGE

$$\text{Real Price of Eggs}_{1998 \text{ (1970 = 100)}} = \frac{38.8_{1970}}{163} \times 1.04$$

$$\text{Real Price of College Education}_{1998 \text{ (1970 = 100)}} = \frac{38.8}{163.0} \times \$19,213 = \$4,573$$

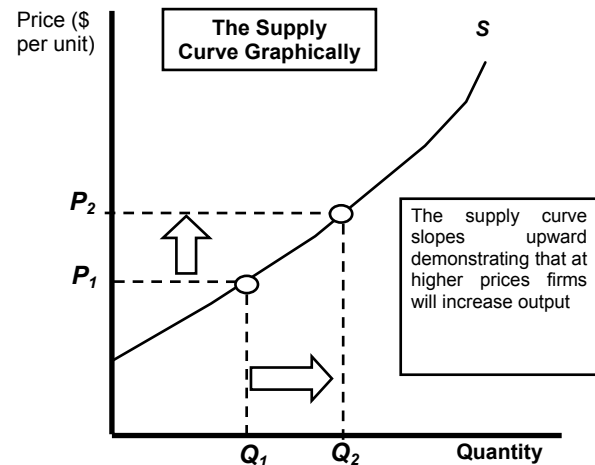
|                                   | 1970 | 1975 | 1980 | 1985  | 1990  | 1998  |
|-----------------------------------|------|------|------|-------|-------|-------|
| Consumer Price Index (1983 = 100) | 38.3 | 53.8 | 82.4 | 107.6 | 130.7 | 163.0 |
| <b>Nominal Prices (\$)</b>        |      |      |      |       |       |       |
| Grade A Large Eggs                | 0.61 | 0.77 | 0.84 | 0.80  | 0.98  | 1.04  |
| College Education                 | 2530 | 3403 | 4912 | 8156  | 12800 | 19213 |
| <b>Real Prices (\$1970)</b>       |      |      |      |       |       |       |
| Grade A Large Eggs                | 0.61 | 0.56 | 0.40 | 0.29  | 0.30  | 0.25  |
| College Education                 | 2530 | 2454 | 2313 | 2941  | 3800  | 4573  |

## SUPPLY AND DEMAND

### THE SUPPLY CURVE

The supply curve shows how much of a good producers are willing to sell at a given price, holding constant other factors that might affect quantity supplied. This price-quantity relationship can be shown by the equation:

$$Q_s = Q_s(P)$$



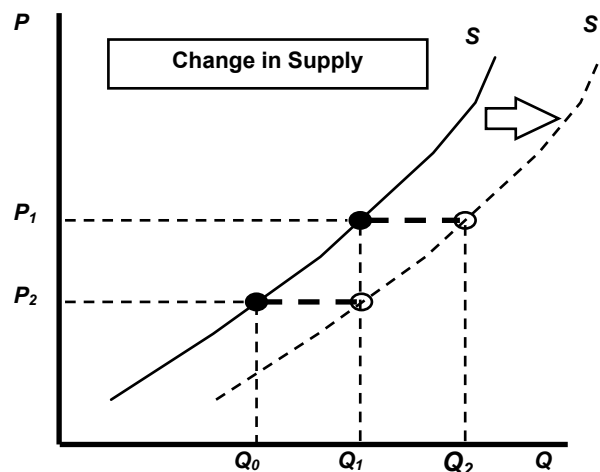
### NON-PRICE DETERMINING VARIABLES OF SUPPLY

Costs of Production

- Labor
- Capital
- Raw Materials

The cost of raw materials falls

- At  $P_1$ , produce  $Q_2$
- At  $P_2$ , produce  $Q_1$
- Supply curve shifts right to  $S'$
- More produced at any price on  $S'$  than on  $S$



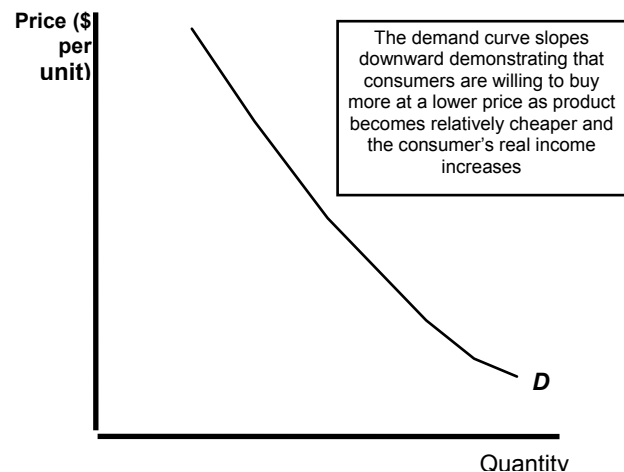
### Supply - A Review

- Supply is determined by non-price supply-determining variables as such as the cost of labor, capital, and raw materials.
- Changes in supply are shown by shifting the entire supply curve.
- Changes in quantity supplied are shown by movements along the supply curve and are caused by a change in the price of the product.

### THE DEMAND CURVE

The demand curve shows how much of a good consumers are willing to buy as the price per unit changes holding non-price factors constant. This price-quantity relationship can be shown by the equation:

$$Q_D = Q_D(P)$$

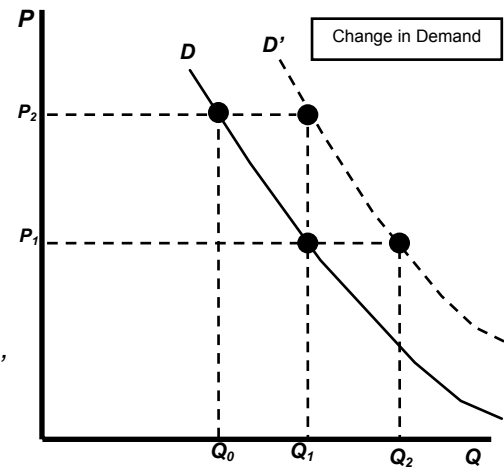


## NON-PRICE DETERMINING VARIABLES OF DEMAND

- Income
- Consumer Tastes
- Price of Related Goods
  - Substitutes
  - Complements

### Income Increases

- At  $P_1$ , produce  $Q_2$
- At  $P_2$ , produce  $Q_1$
- Demand Curve shifts right
- More purchased at any price on  $D'$  than on  $D$



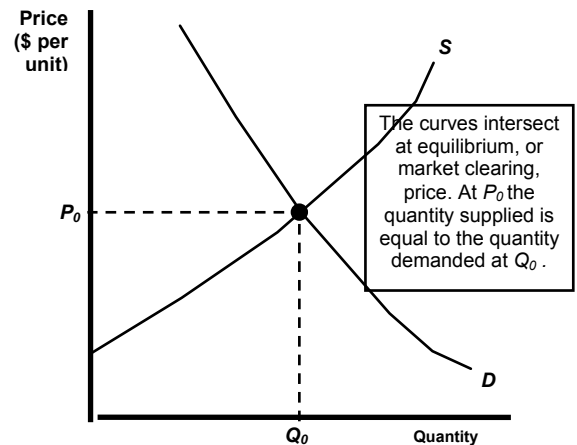
### Demand - A Review

- Demand is determined by non-price demand-determining variables, such as, income, price of related goods, and tastes.
- Changes in demand are shown by shifting the entire demand curve.
- Changes in quantity demanded are shown by movements along the demand curve.

## THE MARKET MECHANISM

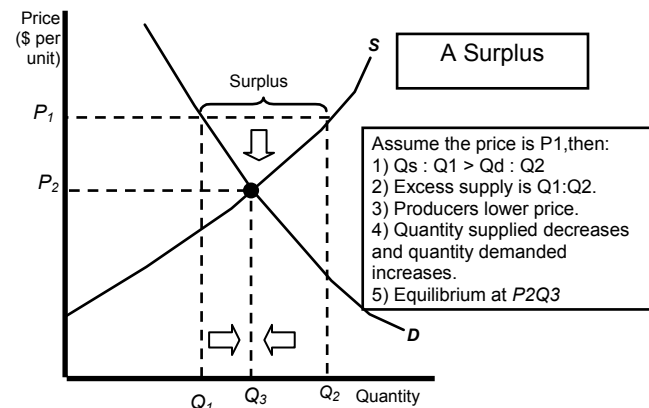
Characteristics of the equilibrium or market clearing price:

- $QD = QS$
- No shortage
- No excess supply
- No pressure on the price to change



The market price is above equilibrium

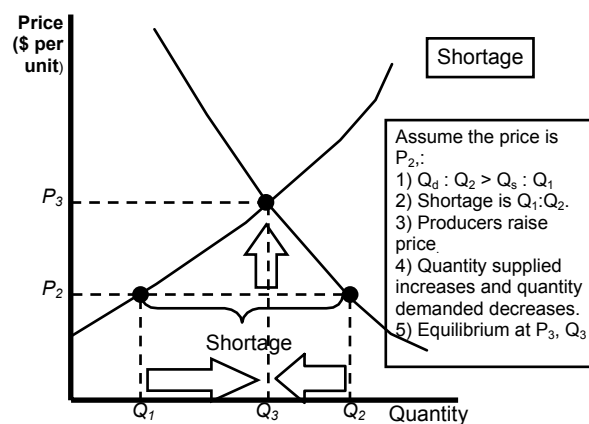
- There is excess supply
- Producers lower prices
- Quantity demanded increases and quantity supplied decreases
- The market continues to adjust until the equilibrium price is reached.





The market price is below equilibrium:

- There is a shortage
- Producers raise prices
- Quantity demanded decreases and quantity supplied increases
- The market continues to adjust until the new equilibrium price is reached.



### Market Mechanism Summary

- 1) Supply and demand interacts to determine the market-clearing price.
- 2) When not in equilibrium, the market will adjust to alleviate a shortage or surplus and return the market to equilibrium.
- 3) Markets must be competitive for the mechanism to be efficient.

## LESSON 4

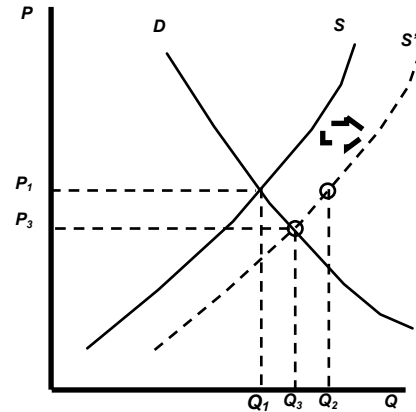
## THE BASICS OF SUPPLY AND DEMAND (Continued)

**CHANGES IN MARKET EQUILIBRIUM**

Equilibrium prices are determined by the relative level of supply and demand. Supply and demand are determined by particular values of supply and demand determining variables. Changes in any one or combination of these variables can cause a change in the equilibrium price and/or quantity. For example:

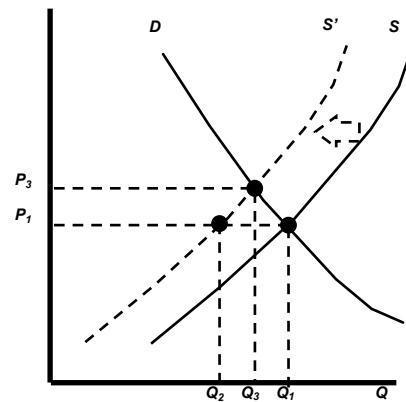
Raw material prices fall

- S shifts to  $S'$
- Surplus @  $P_1$  of  $Q_1, Q_2$
- Equilibrium @  $P_3, Q_3$



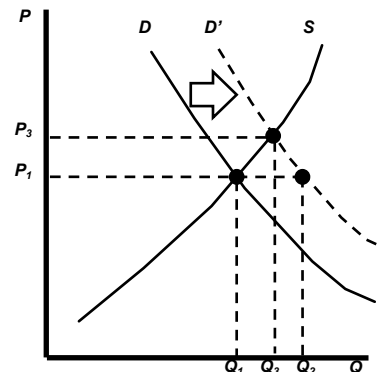
Raw material prices rise

- S shifts to  $S'$
- Shortage @  $P_1$  of  $Q_1, Q_2$
- Equilibrium @  $P_3, Q_3$



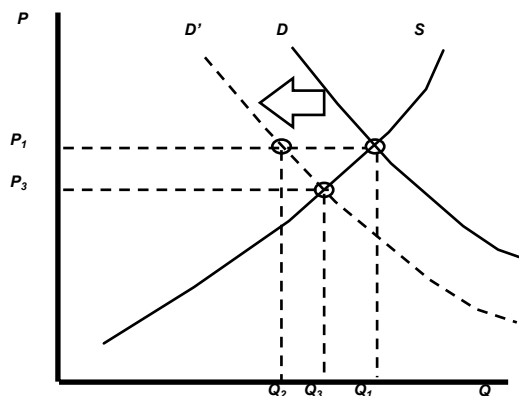
Income Increases

- Demand shifts to  $D'$  Shortage @  $P_1$  of  $Q_1, Q_2$
- Equilibrium @  $P_3, Q_3$



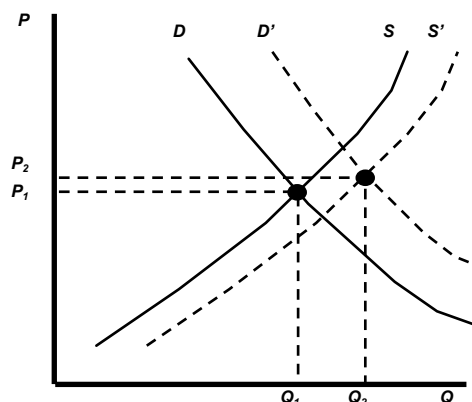
## Income Decreases

- Demand shifts to  $D'$
- Surplus @  $P_1$  of  $Q_1, Q_2$
- Equilibrium @  $P_3, Q_3$



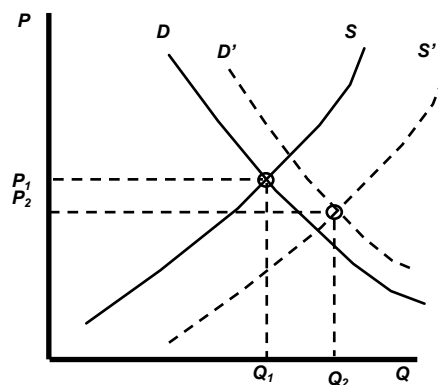
## Income Increases &amp; raw material prices fall

- The increase in  $D$  is greater than the increase in  $S$
- Equilibrium price and quantity increase to  $P_2, Q_2$



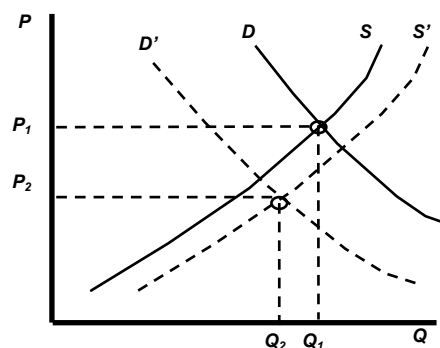
## Income Increases &amp; raw material prices fall

- The increase in  $D$  is less than the increase in  $S$
- Equilibrium price decrease to  $P_2$  and quantity increase to  $Q_2$

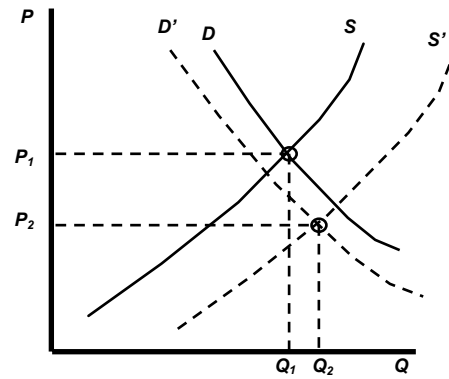


## Income Decreases &amp; raw material prices Fall

- The decrease in  $D$  is greater than the increase in  $S$
- Equilibrium price and quantity decrease to  $P_2, Q_2$



- ♦ Income decreases & raw material prices fall
  - The decrease in  $D$  is less than the increase in  $S$
  - Equilibrium price decrease to  $P_2$  and quantity increase to  $Q_2$



### SHIFTS IN SUPPLY AND DEMAND

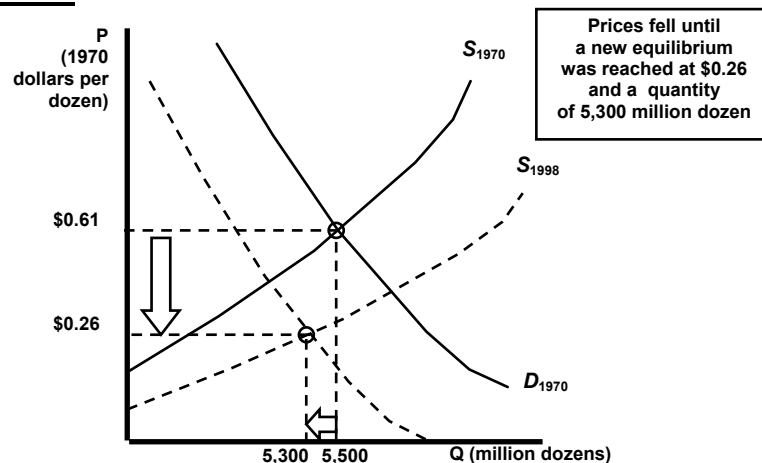
When supply and demand change simultaneously, the impact on the equilibrium price and quantity is determined by:

- 1) The relative size and direction of the change
- 2) The shape of the supply and demand curves

### THE PRICES OF EGGS & EDUCATION REVISITED

The real price of eggs fell 59% from 1970 to 1998. Supply increased due to the increased mechanization of poultry farming and the reduced cost of production. Demand decreased due to the increasing consumer concern over the health and cholesterol consequences of eating eggs.

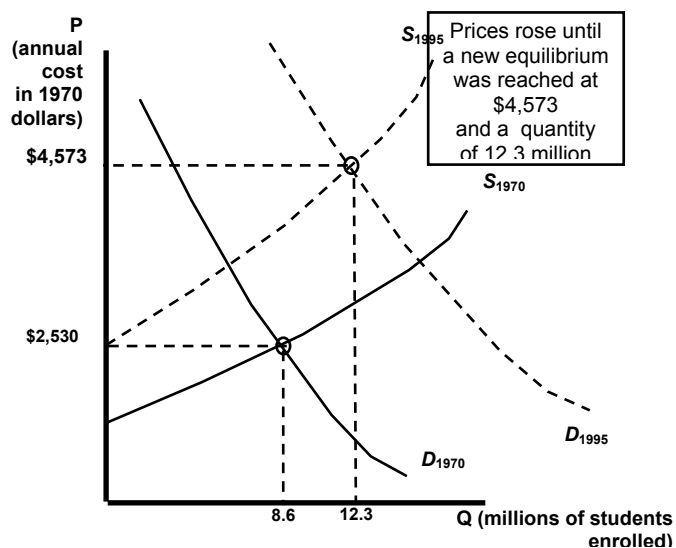
### MARKET FOR EGGS



### PRICE OF COLLEGE EDUCATION

- ♦ The real price of a college education rose 68 percent from 1970 to 1995.
- ♦ Supply decreased due to higher costs of equipping and maintaining modern classrooms, laboratories and libraries, and higher faculty salaries.
- ♦ Demand increased due a larger percentage of a larger number of high school graduates attending college.

## MARKET FOR COLLEGE EDUCATION

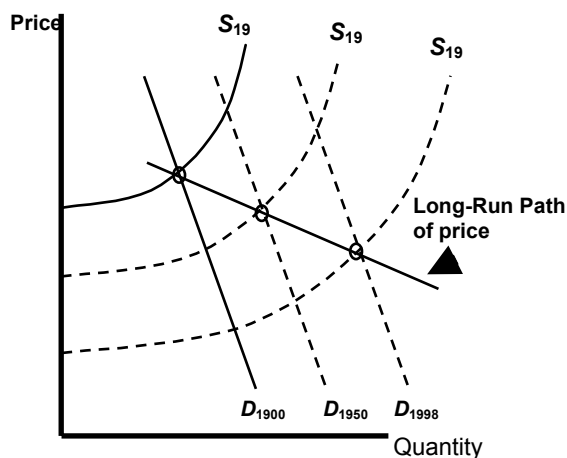


## THE LONG-RUN BEHAVIOR OF NATURAL RESOURCE PRICES

### OBSERVATIONS

- Consumption of copper has increased about a hundred fold from 1880 through 1998 indicating a large increase in demand.
- The real price for copper has remained relatively constant.

### CHANGES IN MARKET EQUILIBRIUM



### CONCLUSION

Decreases in the costs of production have increased the supply by more than enough to offset the increase in demand.

**FACTORS SHIFTING DEMAND CURVE**

| <b>Factors Changing Demand</b>            | <b>Effect on Demand</b> | <b>Direction of Shift in Demand Curve</b> | <b>Effect on Equilibrium Price</b> | <b>Effect on Equilibrium Quantity</b> |
|-------------------------------------------|-------------------------|-------------------------------------------|------------------------------------|---------------------------------------|
| Increase in income (normal good)          | Increase                | Rightward                                 | Increase                           | Increase                              |
| Decrease in income(normal good)           | Decrease                | Leftward                                  | Decrease                           | Decrease                              |
| Increase in income (inferior good)        | Decrease                | Leftward                                  | Decrease                           | Decrease                              |
| Decrease in income(inferior good)         | Increase                | Rightward                                 | Increase                           | Increase                              |
| Increase in price of Substitute           | Increase                | Rightward                                 | Increase                           | Increase                              |
| Decrease in price of substitute           | Decrease                | Leftward                                  | Decrease                           | Decrease                              |
| Increase in price of complement           | Decrease                | Leftward                                  | Decrease                           | Decrease                              |
| Decrease in price of complement           | Increase                | Rightward                                 | Increase                           | Increase                              |
| Increase in taste and preference for good | Increase                | Rightward                                 | Increase                           | Increase                              |
| Decrease in taste and preference for good | Decrease                | Leftward                                  | Decrease                           | Decrease                              |
| Increase in number of consumers           | Increase                | Rightward                                 | Increase                           | Increase                              |
| Decrease in number of consumers           | Decrease                | Leftward                                  | Decrease                           | Decrease                              |

**FACTORS SHIFTING SUPPLY CURVE**

| <b>Factors Changing Supply</b>  | <b>Effect on Supply</b> | <b>Direction of Shift in Supply Curve</b> | <b>Effect on Equilibrium Price</b> | <b>Effect on Equilibrium Quantity</b> |
|---------------------------------|-------------------------|-------------------------------------------|------------------------------------|---------------------------------------|
| Increase in resource price      | Decrease                | Leftward                                  | Increase                           | Decrease                              |
| Decrease in resource price      | Increase                | Rightward                                 | Decrease                           | Increase                              |
| Improved technology             | Increase                | Rightward                                 | Decrease                           | Increase                              |
| Decline in technology           | Decrease                | Leftward                                  | Increase                           | Decrease                              |
| Expect a price increase         | Decrease                | Leftward                                  | Increase                           | Decrease                              |
| Expect a price decrease         | Increase                | Rightward                                 | Decrease                           | Increase                              |
| Increase in number of suppliers | Increase                | Rightward                                 | Decrease                           | Increase                              |
| Decrease in number of suppliers | Decrease                | Leftward                                  | Increase                           | Decrease                              |

## ELASTICITIES OF SUPPLY AND DEMAND

Generally, elasticity is a measure of the sensitivity of one variable to another. It tells us the percentage change in one variable in response to a one percent change in another variable.

### PRICE ELASTICITY OF DEMAND

Price Elasticity of Demand measures the sensitivity of quantity demanded to price changes. It measures the percentage change in the quantity demanded for a good or services that results from a one percent change in the price of that good or service.

The price elasticity of demand is:

$$\frac{\text{Percentage change in Quantity Demanded}}{\text{Percentage change in Price}}$$

$$E_P = (\% \Delta Q) / (\% \Delta P)$$

The percentage change in a variable is the absolute change in the variable divided by the original level of the variable. So the price elasticity of demand is also:

$$E_P = \frac{\Delta Q / Q}{\Delta P / P} = \frac{P}{Q} \frac{\Delta Q}{\Delta P}$$

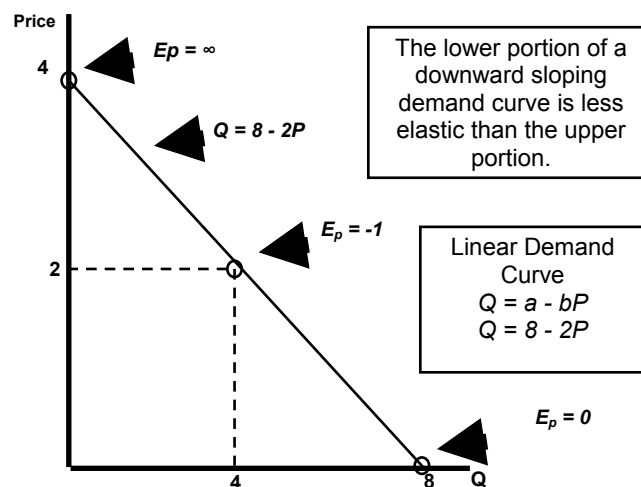
### INTERPRETING PRICE ELASTICITY OF DEMAND VALUES

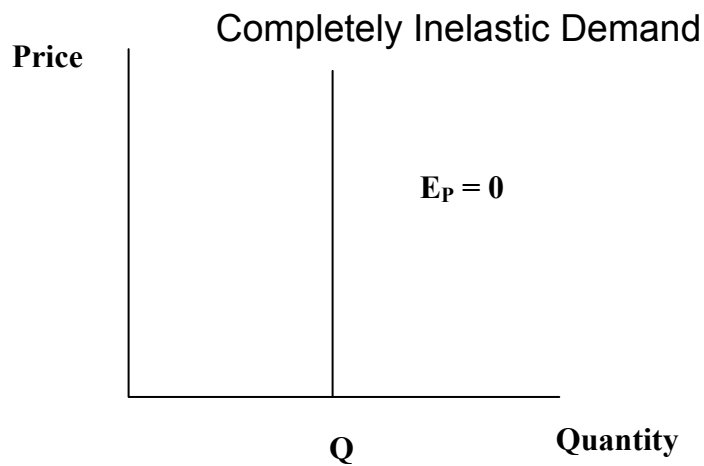
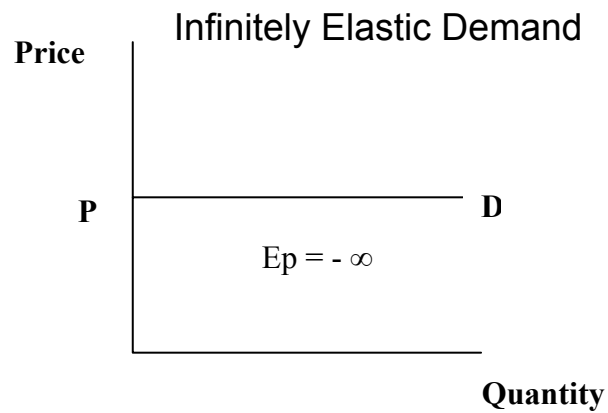
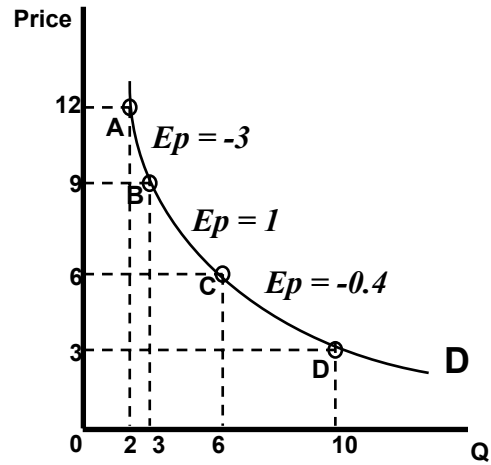
- 1) Because of the inverse relationship between P and Q;  $E_P$  is negative.
- 2) If  $|E_P| > 1$ , the percent change in quantity is greater than the percent change in price. We say the demand is price elastic.
- 3) If  $|E_P| < 1$ , the percent change in quantity is less than the percent change in price. We say the demand is price inelastic.

The primary determinant of price elasticity of demand is the availability of substitutes.

- Many substitutes demand is price elastic
- Few substitutes demand is price inelastic
- 

### PRICE ELASTICITIES OF DEMAND







**LESSON 5****ELASTICITIES OF SUPPLY AND DEMAND****INCOME ELASTICITY OF DEMAND**

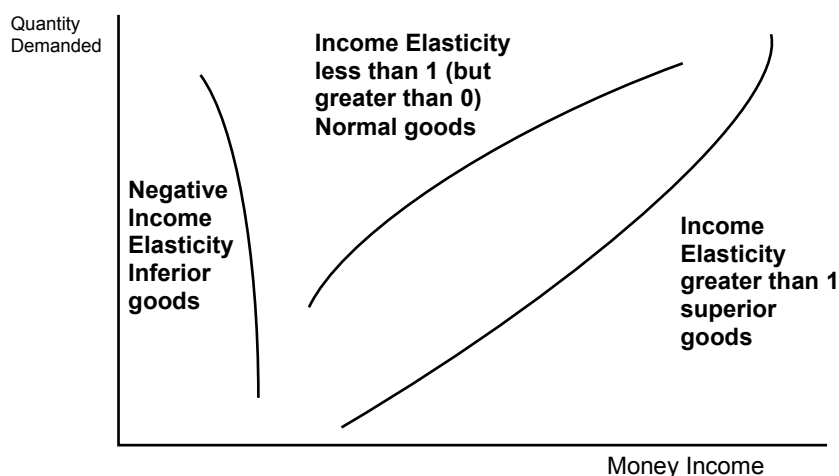
Income elasticity of demand measures the percentage change in quantity demanded resulting from a one percent change in income.

The income elasticity of demand is:

$$E_I = \frac{\Delta Q/Q}{\Delta I/I} = \frac{I}{Q} \frac{\Delta Q}{\Delta I}$$

Income Elasticity of Demand for:

- Normal goods
- Superior goods
- Inferior goods

**CROSS ELASTICITY OF DEMAND**

Cross elasticity of demand measures the percentage change in the quantity demanded of one good that results from a one percent change in the price of another good. For example consider the substitute goods, butter and margarine.

The cross elasticity of demand is:

$$E_{Q_b P_m} = \frac{\Delta Q_b / Q_b}{\Delta P_m / P_m} = \frac{P_m}{Q_b} \frac{\Delta Q_b}{\Delta P_m}$$

Cross elasticity for substitutes is positive and Cross elasticity for complements is negative.

**PRICE ELASTICITY OF SUPPLY**

Price Elasticity of supply measures the percentage change in quantity supplied resulting from a 1 percent change in price. This elasticity is usually positive because price and quantity supplied are directly related.

We can refer to elasticity of supply with respect to interest rates, wage rates, and the cost of raw materials.

| Price (\$) | Quantity Demanded | Quantity Supplied |
|------------|-------------------|-------------------|
| 60         | 22                | 14                |
| 80         | 20                | 16                |
| 100        | 18                | 18                |
| 120        | 16                | 20                |

Recall

$$E_P = \frac{\Delta Q/Q}{\Delta P/P} = \frac{P}{Q} \frac{\Delta Q}{\Delta P}$$

- ♦ Elasticity of demand when price is \$80 is  
 $E_p = 80/20 \times -2/20 = -0.40$
- ♦ Elasticity of demand when price is \$100 is  
 $E_p = 100/18 \times -2/20 = -0.56$
- ♦ Elasticity of supply when price is \$80 is  
 $E_p = 80/16 \times 2/20 = 0.50$
- ♦ Elasticity of supply when price is \$100 is  
 $E_p = 100/18 \times 2/20 = 0.56$

### THE MARKET FOR WHEAT

- 1981 Supply Curve for Wheat
  - $QS = 1,800 + 240P$
- 1981 Demand Curve for Wheat
  - $QD = 3,550 - 266P$
- Equilibrium:  $Q_S = Q_D$

$$1,800 + 240P = 3,550 - 266P$$

$$506P = 1,750$$

$$P = 3.46 / \text{bushel}$$

$$Q = 1,800 + (240)(3.46) = 2,630 \text{ million bushels}$$

$$E_p^D = \frac{P}{Q} \frac{\Delta Q_D}{\Delta P} = \frac{3.46}{2,630} (-266) = -.035 \text{ Inelastic}$$

$$E_p^S = \frac{P}{Q} \frac{\Delta Q_S}{\Delta P} = \frac{3.46}{2,630} (240) = .032 \text{ Inelastic}$$

- Assume the price of wheat is \$4.00/bushel

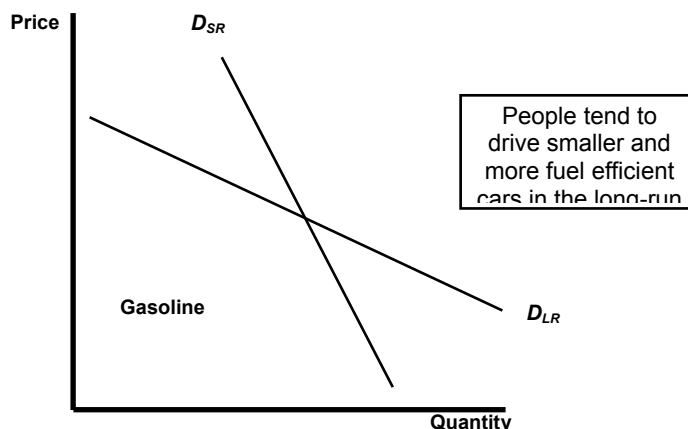
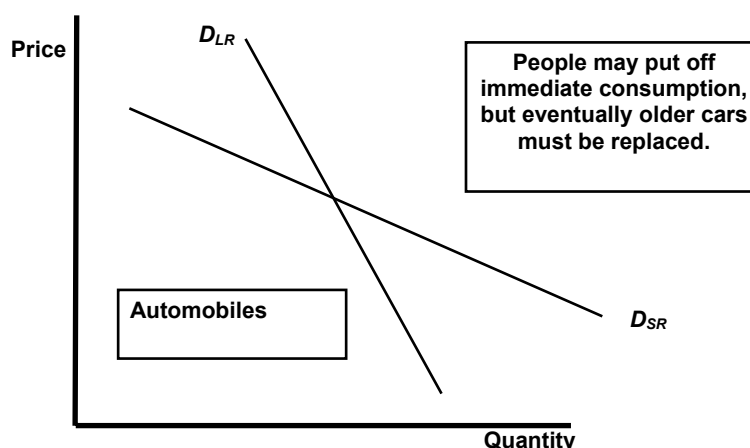
$$Q_D = 3,550 - (266)(4.00) = 2,486$$

$$E_p^D = \frac{4.00}{2,486} (-266) = -0.43$$

|      | Supply (Qs)   | Demand (Qd)   | Equilibrium Price<br>Qs = Qd                                                        |
|------|---------------|---------------|-------------------------------------------------------------------------------------|
| 1981 | $1800 + 240P$ | $3550 - 266P$ | $1800 + 240P = 3550 - 266P$<br>$506P = 1750$<br>$P_{1981} = \$3.46 / \text{bushel}$ |
| 1998 | $1944 + 207P$ | $3244 - 283P$ | $1944 + 207P = 3244 - 283P$<br>$P_{1998} = \$2.65 / \text{bushel}$                  |

**SHORT-RUN VERSUS LONG-RUN ELASTICITIES**

Price elasticity of demand varies with the amount of time consumers have to respond to a price. For most goods and services, short-run elasticity is less than long-run elasticity. (e.g. gasoline, Drs.). For other Goods (durables), short-run elasticity is greater than long-run elasticity (e.g. automobiles)

**GASOLINE: SHORT-RUN AND LONG-RUN DEMAND CURVES****AUTOMOBILES: SHORT-RUN AND LONG-RUN DEMAND CURVES**

Income elasticity also varies with the amount of time consumers have to respond to an income change. For most goods and services, income elasticity is greater in the long-run than in the short run. For example, higher incomes may be converted into bigger cars so the income elasticity of demand for gasoline increases with time.

For other Goods (durables), Income elasticity is less in the long-run than in the short-run. For example, originally, consumers will want to hold more cars. Later, purchases will only be to replace old cars.

Gasoline and Automobiles are complementary goods.

For gasoline, the long-run price and income elasticities are larger than the short-run elasticities. For automobiles, the long-run price and income elasticities are smaller than the short-run elasticities.

**THE DEMAND FOR GASOLINE**

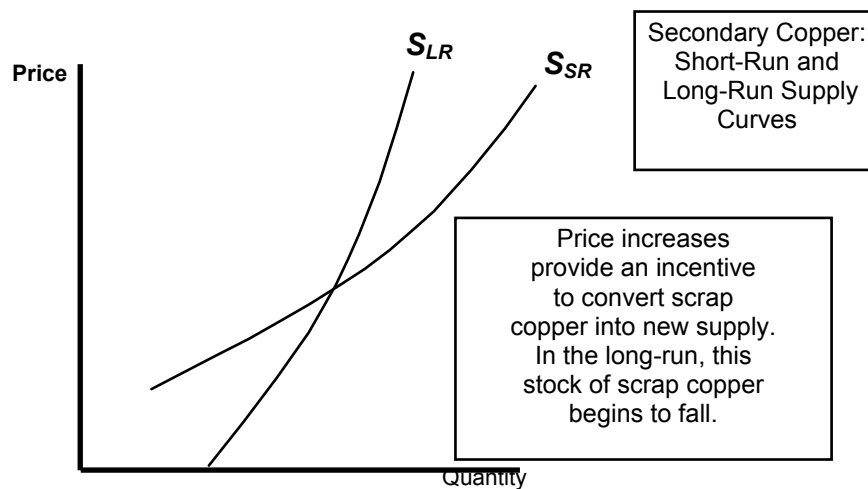
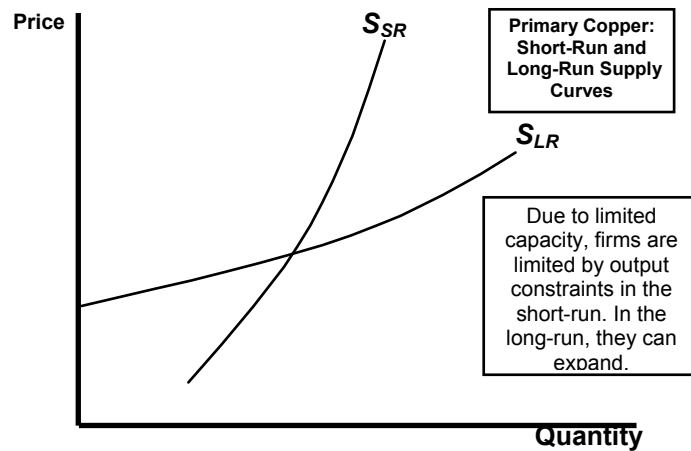
| Elasticity | Years Following price or income change |       |       |       |       |       |
|------------|----------------------------------------|-------|-------|-------|-------|-------|
|            | 1                                      | 2     | 3     | 4     | 5     | 6     |
| Price      | -0.11                                  | -0.22 | -0.32 | -0.49 | -0.82 | -1.17 |
| Income     | 0.07                                   | 0.13  | 0.20  | 0.32  | 0.54  | 0.78  |

**THE DEMAND FOR AUTOMOBILES**

| Elasticity | Years Following price or income change |       |       |       |       |       |
|------------|----------------------------------------|-------|-------|-------|-------|-------|
|            | 1                                      | 2     | 3     | 4     | 5     | 6     |
| Price      | -1.20                                  | -0.93 | -0.75 | -0.55 | -0.42 | -0.40 |
| Income     | 3.00                                   | 2.33  | 1.88  | 1.38  | 1.02  | 1.00  |

**SUPPLY**

For Most goods and services, long-run price elasticity of supply is greater than short-run price elasticity of supply. For other Goods (durables, recyclables), long-run price elasticity of supply is less than short-run price elasticity of supply



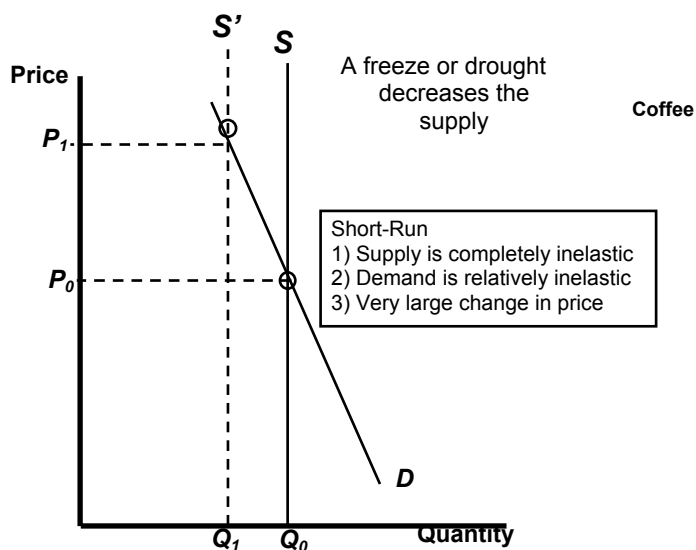
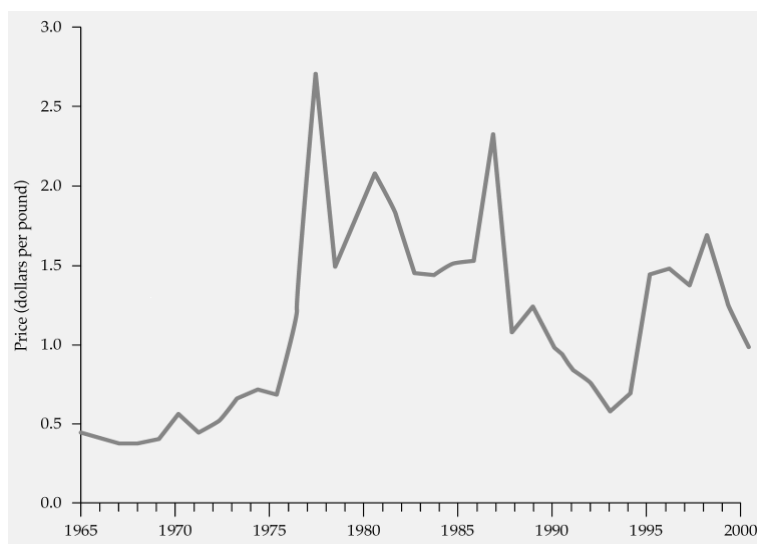
**SUPPLY OF COPPER**

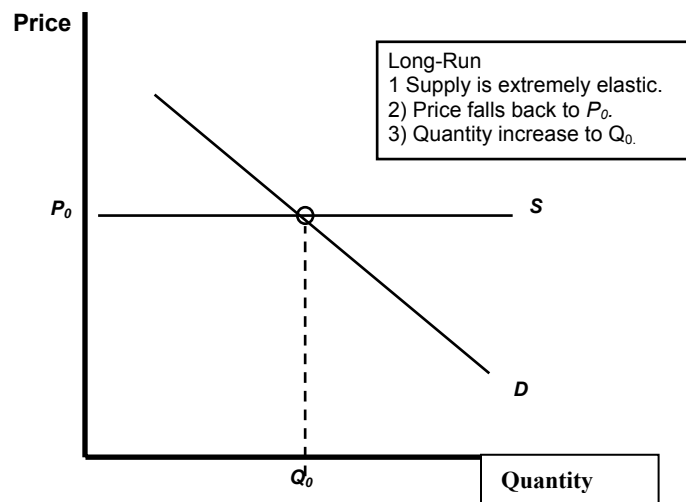
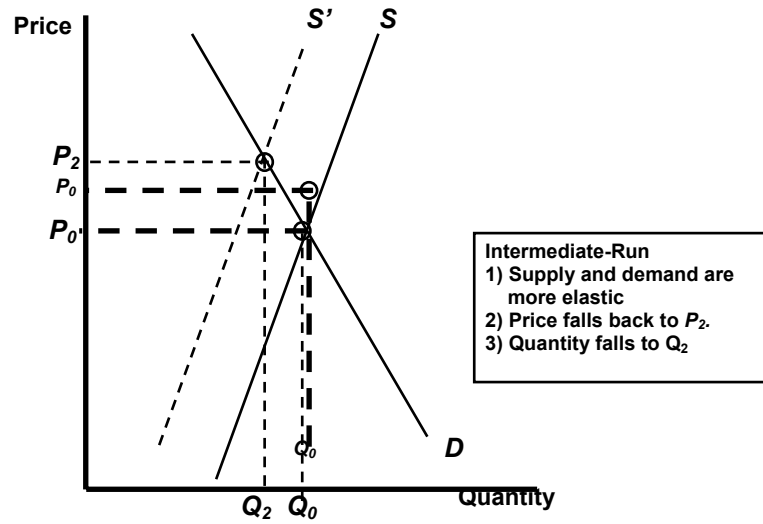
| Price Elasticity of: | Short Run | Long run |
|----------------------|-----------|----------|
| Primary Supply       | 0.20      | 1.60     |
| Secondary Supply     | 0.43      | 0.31     |
| Total Supply         | 0.25      | 1.50     |

**WEATHER IN BRAZIL AND THE PRICE OF COFFEE IN NEW YORK**

Elasticity explains why coffee prices are very volatile.

- Due to the differences in supply elasticity in the long-run and short run.





**LESSON 6****CONSUMER BEHAVIOR**

The explanation of how consumers allocate their resources (income) to the purchase of different goods and services to maximize their well being.

There are three steps involved in the study of consumer behavior.

1) We will study consumer preferences to describe how and why people prefer one good to another.

2) Then we will turn to budget constraints because people have limited incomes.

3) Finally, we will combine consumer preferences and budget constraints to determine consumer choices.

**WHAT COMBINATION OF GOODS WILL CONSUMERS BUY TO MAXIMIZE THEIR SATISFACTION?****CONSUMER PREFERENCES**

**A market basket** is a collection of one or more commodities. One market basket may be preferred over another market basket containing a different combination of goods.

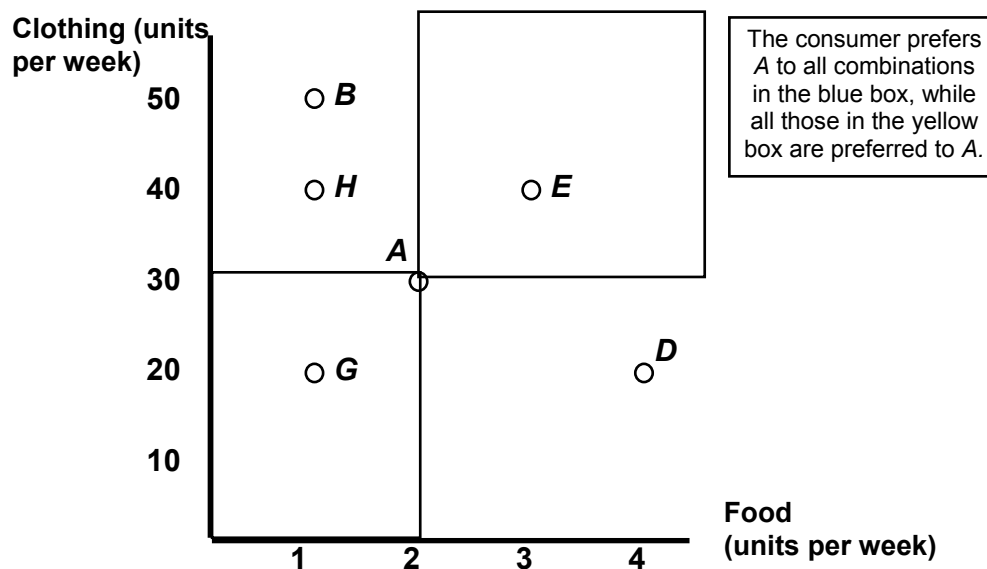
Three Basic Assumptions

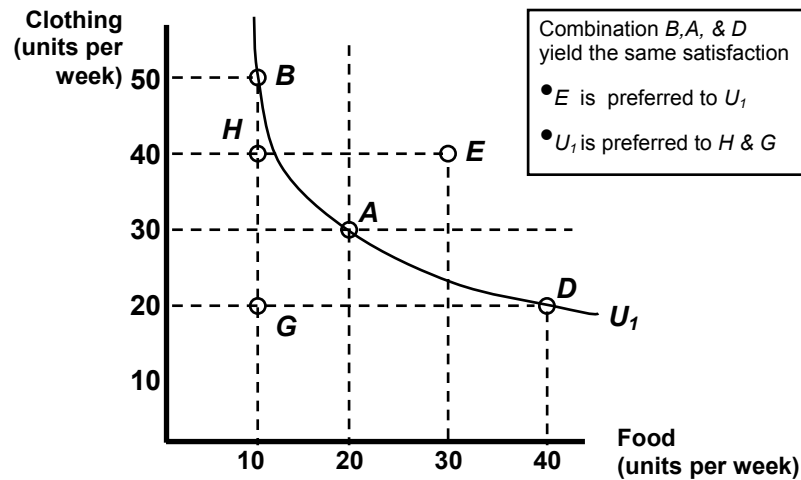
- 1) Preferences are complete.
- 2) Preferences are transitive.
- 3) Consumers always prefer more of any good to less.

| Market Basket | Units of Food | Units of Clothing |
|---------------|---------------|-------------------|
| A             | 20            | 30                |
| B             | 10            | 50                |
| D             | 40            | 20                |
| E             | 30            | 40                |
| G             | 10            | 20                |
| H             | 10            | 40                |

**INDIFFERENCE CURVES**

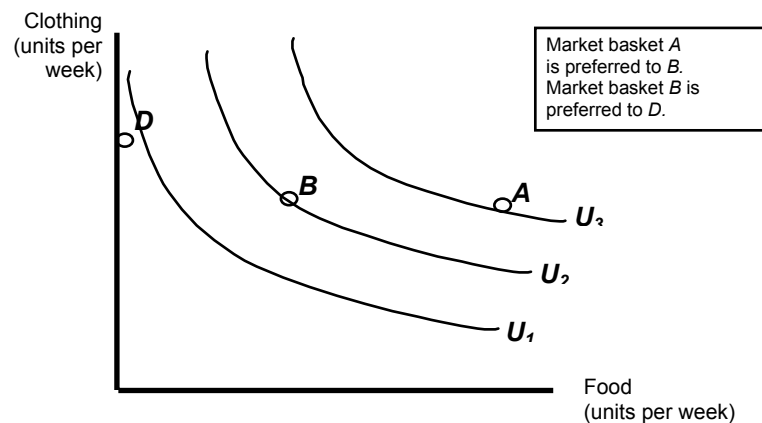
Indifference curves represent all combinations of market baskets that provide the same level of satisfaction to a person.



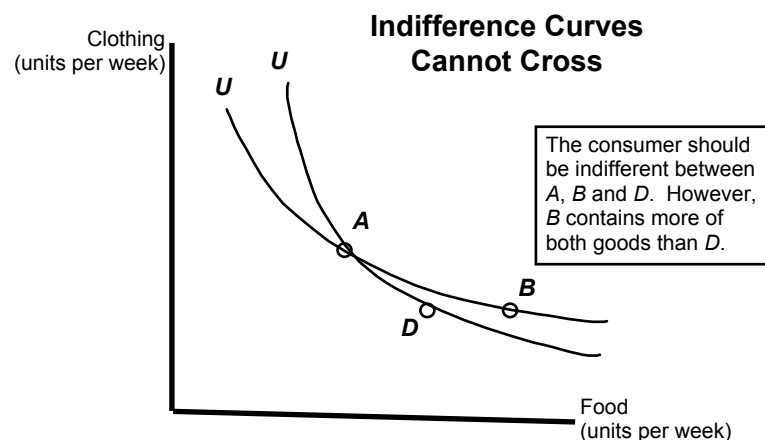


Indifference curves slope downward to the right. If it sloped upward it would violate the assumption that more of any commodity is preferred to less. Any market basket lying above and to the right of an indifference curve is preferred to any market basket that lies on the indifference curve.

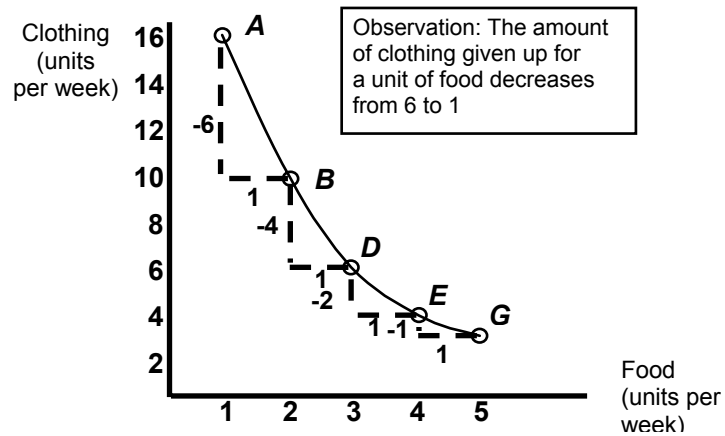
An **indifference map** is a set of indifference curves that describes a person's preferences for all combinations of two commodities. Each indifference curve in the map shows the market baskets among which the person is indifferent.



Finally, indifference curves cannot cross. This would violate the assumption that more is preferred to less.

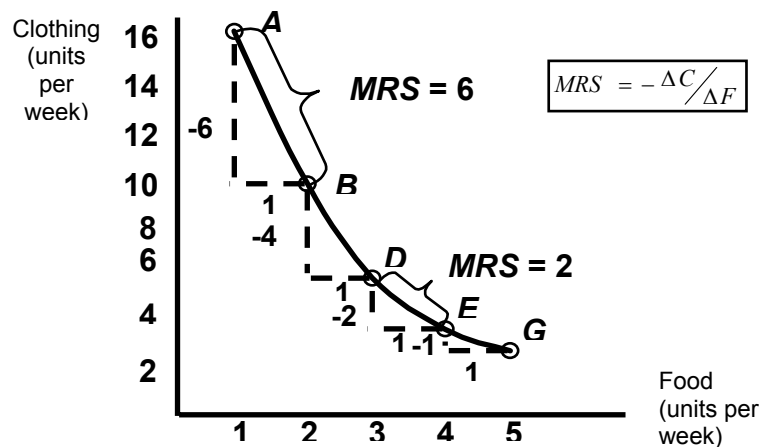






### MARGINAL RATE OF SUBSTITUTION

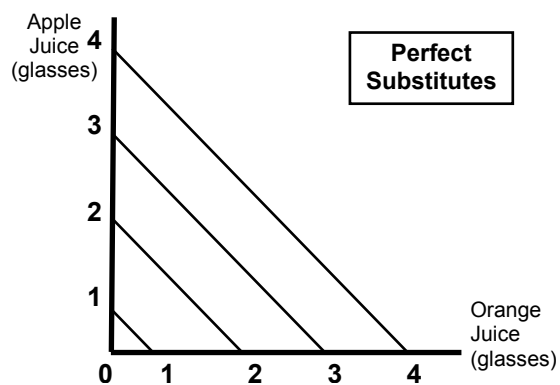
The marginal rate of substitution (MRS) quantifies the amount of one good a consumer will give up to obtain more of another good. It is measured by the slope of the indifference curve.



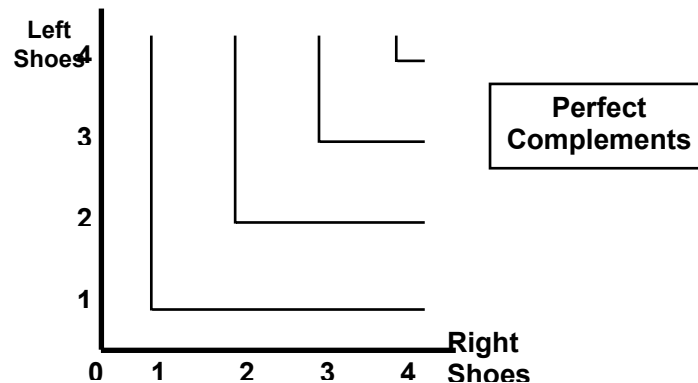
We will now add a fourth assumption regarding consumer preference: Along an indifference curve there is a diminishing marginal rate of substitution. Note the MRS for AB was 6, while that for DE was 2. Indifference curves are convex because as more of one good is consumed, a consumer would prefer to give up fewer units of a second good to get additional units of the first one. Consumers prefer a balanced market basket.

### PERFECT SUBSTITUTES AND PERFECT COMPLEMENTS

Two goods are perfect substitutes when the marginal rate of substitution of one good for the other is constant.



Two goods are complements when the indifference curves for the goods are shaped as right angles.

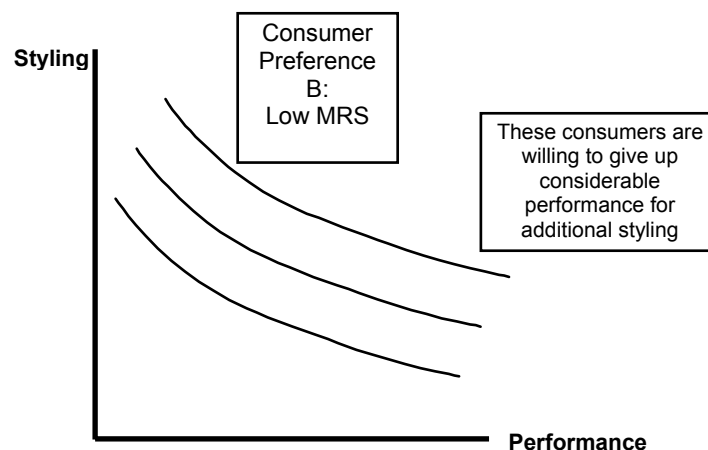
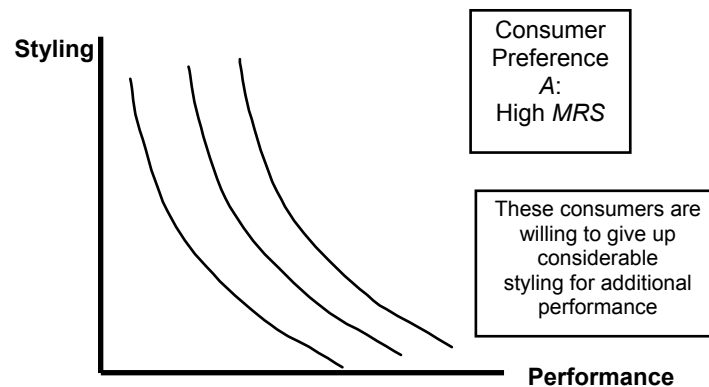


**BADS** are the things for which less is preferred to more. For example, air pollution

### DESIGNING NEW AUTOMOBILES

Automobile executives must regularly decide when to introduce new models and how much money to invest in restyling.

An analysis of consumer preferences would help to determine when and if car companies should change the styling of their cars.



**DESIGNING NEW AUTOMOBILES**

– What Do You Think?

**How can we determine the consumer's preference?**

A recent study of automobile demand in the USA shows that over the past two decades most consumers have preferred styling over performance.

Growth of Japanese Imports in 1970's and 1980's

- 15% of domestic cars underwent a style change each year
- This compares to 23% for imports

## LESSON 7

## CONSUMER BEHAVIOR (Continued)

## UTILITY

Utility is the numerical score representing the satisfaction that a consumer gets from a given market basket. If buying 3 copies of Microeconomics makes you happier than buying one shirt, then we say that the books give you more utility than the shirt.

## UTILITY FUNCTIONS

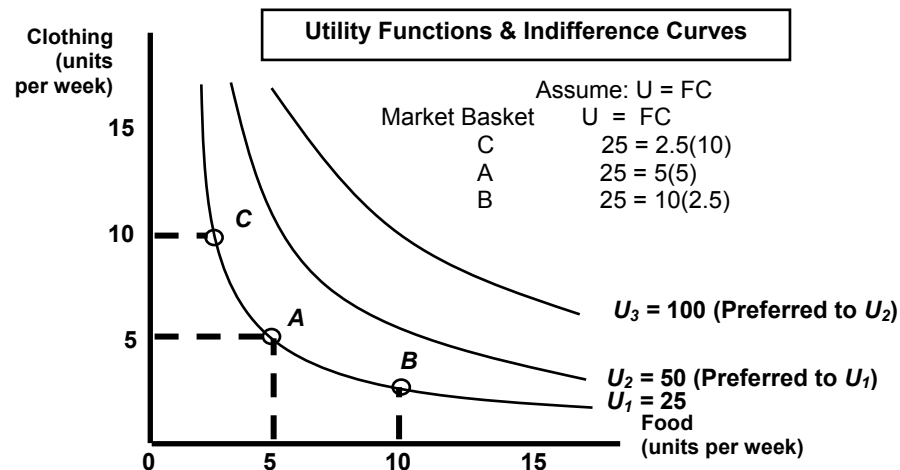
Assume: The utility function for food (F) and clothing (C)

$$U(F, C) = F + 2C$$

| Market Baskets: | F units | C units | $U(F, C) = F + 2C$ |
|-----------------|---------|---------|--------------------|
| A               | 8       | 3       | $8 + 2(3) = 14$    |
| B               | 6       | 4       | $6 + 2(4) = 14$    |
| C               | 4       | 4       | $4 + 2(4) = 12$    |

The consumer is indifferent to A & B

The consumer prefers A & B to C



## ORDINAL VERSUS CARDINAL UTILITY

Ordinal Utility Function places market baskets in the order of most preferred to least preferred, but it does not indicate how much one market basket is preferred to another. Cardinal Utility Function is a utility function describing the extent to which one market basket is preferred to another.

## ORDINAL VERSUS CARDINAL RANKINGS

The actual unit of measurement for utility is not important. Therefore, an ordinal ranking is sufficient to explain how most individual decisions are made.

## BUDGET CONSTRAINTS

Preferences do not explain all of consumer behavior. Budget constraints also limit an individual's ability to consume in light of the prices they must pay for various goods and services.

## THE BUDGET LINE

The budget line indicates all combinations of two commodities for which total money spent equals total income.

Let F equal the amount of food purchased, and C is the amount of clothing.

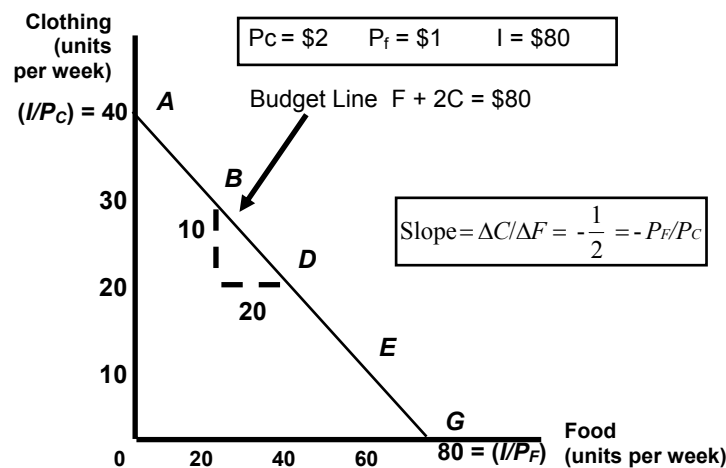
- ♦ Price of food =  $P_f$  and price of clothing =  $P_c$

Then  $P_f F$  is the amount of money spent on food, and  $P_c C$  is the amount of money spent on clothing.

The budget line then can be written:

$$P_F F + P_C C = I$$

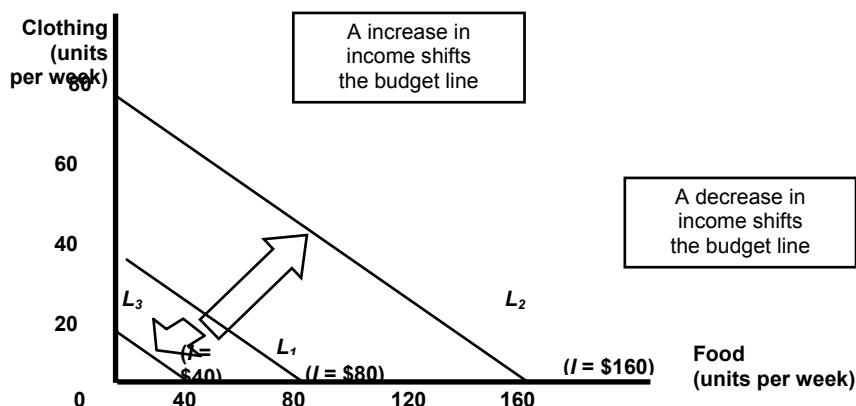
| Market Basket | Food (F)<br>$P_f = (\$1)$ | Clothing (C)<br>$P_c = (\$2)$ | Total Spending<br>$P_f F + P_c C = I$ |
|---------------|---------------------------|-------------------------------|---------------------------------------|
| A             | 0                         | 40                            | \$80                                  |
| B             | 20                        | 30                            | \$80                                  |
| D             | 40                        | 20                            | \$80                                  |
| E             | 60                        | 10                            | \$80                                  |
| G             | 80                        | 0                             | \$80                                  |



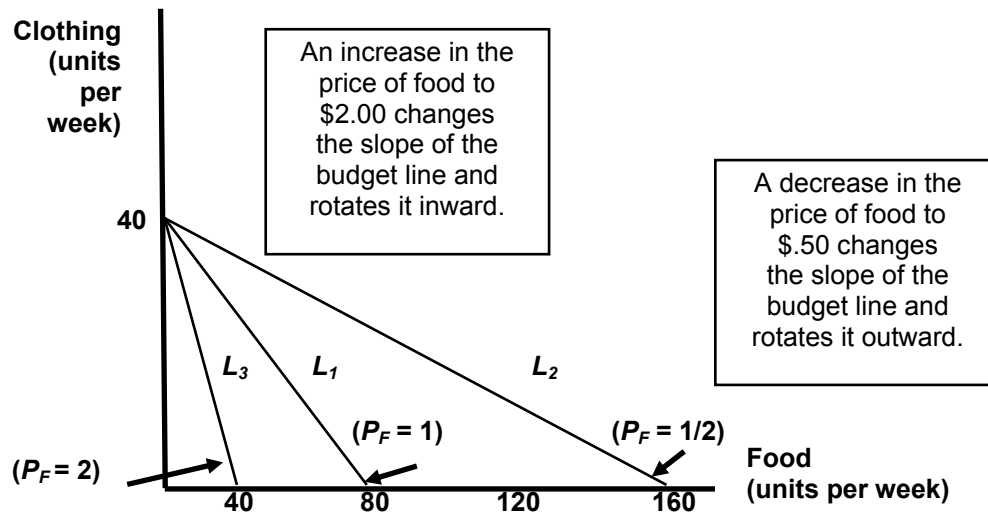
As consumption moves along a budget line from the intercept, the consumer spends less on one item and more on the other. The slope of the line measures the relative cost of food and clothing. The slope is the negative of the ratio of the prices of the two goods. The slope indicates the rate at which the two goods can be substituted without changing the amount of money spent. The vertical intercept ( $I/P_C$ ), illustrates the maximum amount of C that can be purchased with income I. The horizontal intercept ( $I/P_F$ ), illustrates the maximum amount of F that can be purchased with income I.

### THE EFFECTS OF CHANGES IN INCOME AND PRICES

An increase in income causes the budget line to shift outward, parallel to the original line (holding prices constant). A decrease in income causes the budget line to shift inward, parallel to the original line (holding prices constant).



If the price of one good increases, the budget line shifts inward, pivoting from the other good's intercept. If the price of one good decreases, the budget line shifts outward, pivoting from the other good's intercept.



If the two goods increase in price, but the *ratio* of the two prices is unchanged, the slope will not change. However, the budget line will shift inward to a point parallel to the original budget line.

If the two goods decrease in price, but the ratio of the two prices is unchanged, the slope will not change. However, the budget line will shift outward to a point parallel to the original budget line.

### CONSUMER CHOICE

Consumers choose a combination of goods that will maximize the satisfaction they can achieve, given the limited budget available to them.

The maximizing market basket must satisfy two conditions:

- 1) It must be located on the budget line.
- 2) Must give the consumer the most preferred combination of goods and services.

Recall, the slope of an indifference curve is:

$$MRS = -\frac{\Delta C}{\Delta F}$$

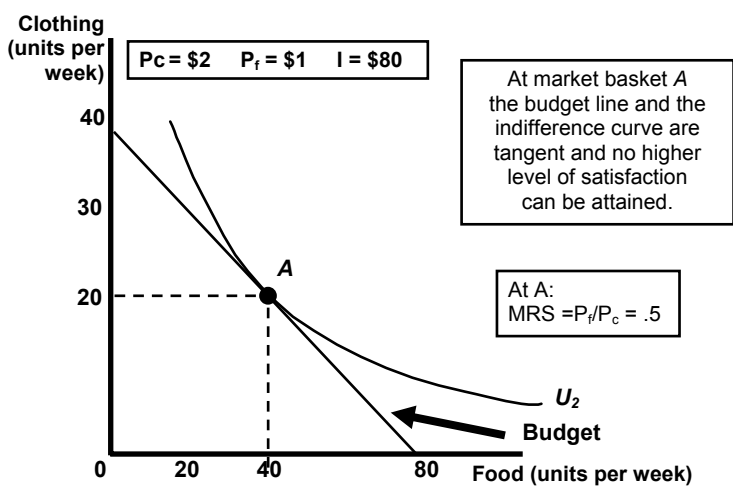
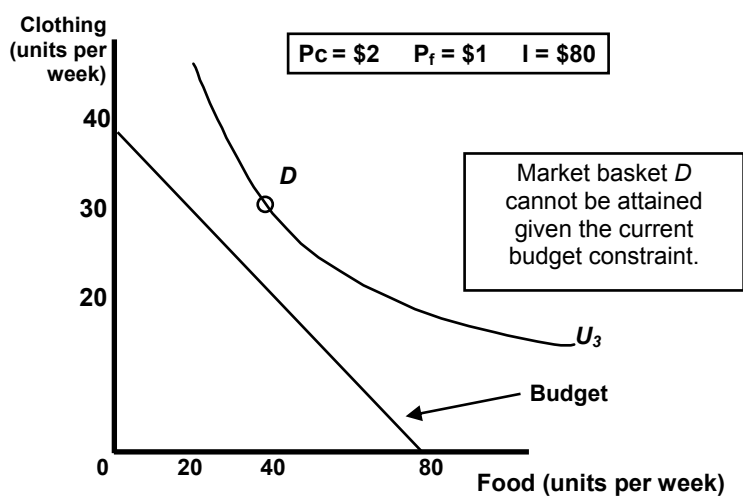
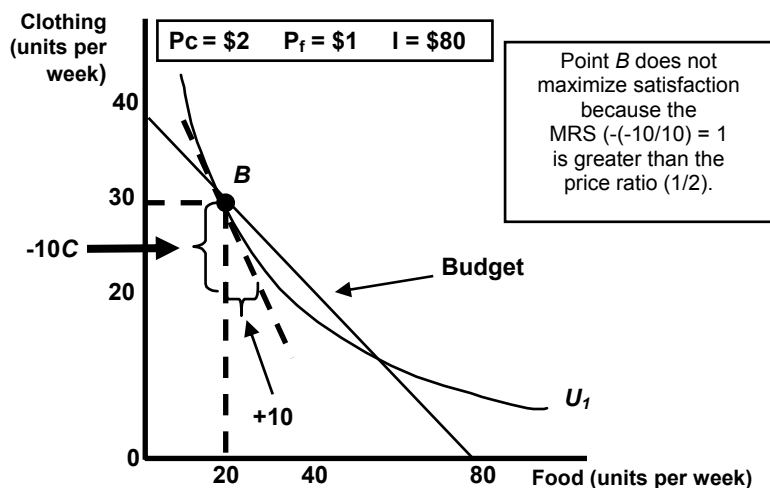
Further, the slope of the budget line is:

$$Slope = -\frac{P_F}{P_C}$$

Therefore, it can be said that satisfaction is maximized where:

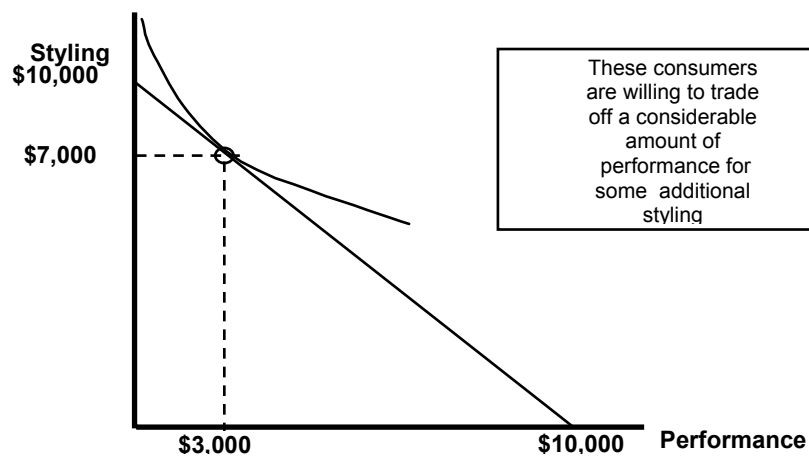
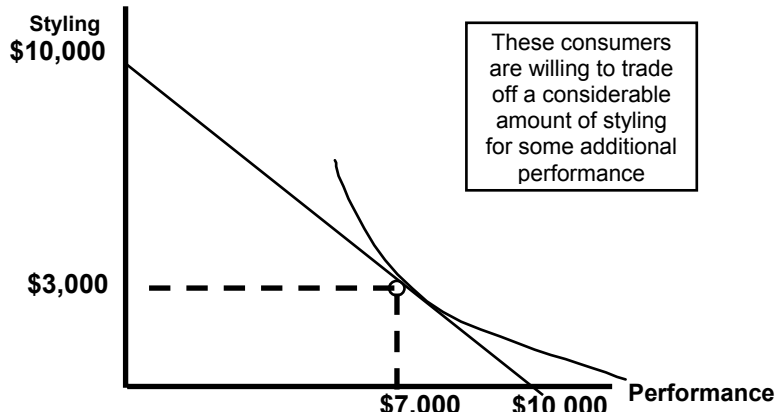
$$MRS = \frac{P_F}{P_C}$$

It can be said that satisfaction is maximized when marginal rate of substitution (of F and C) is equal to the ratio of the prices (of F and C).

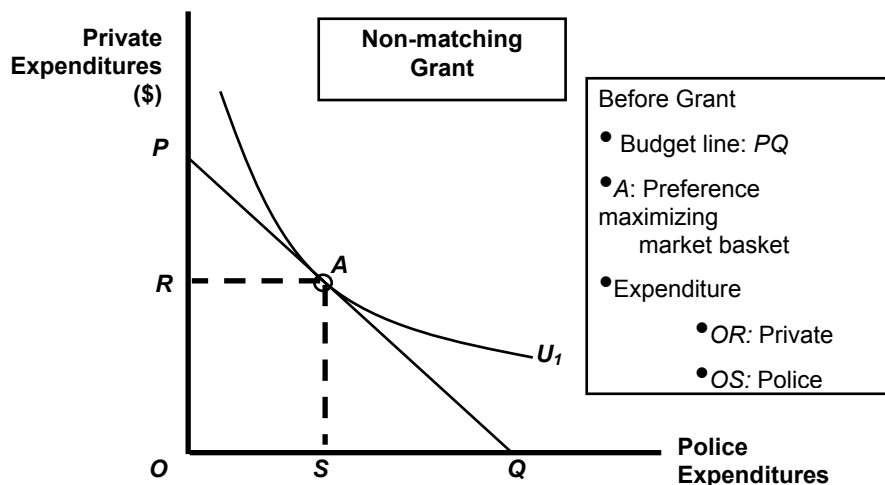


**DESIGNING NEW AUTOMOBILES (II)**

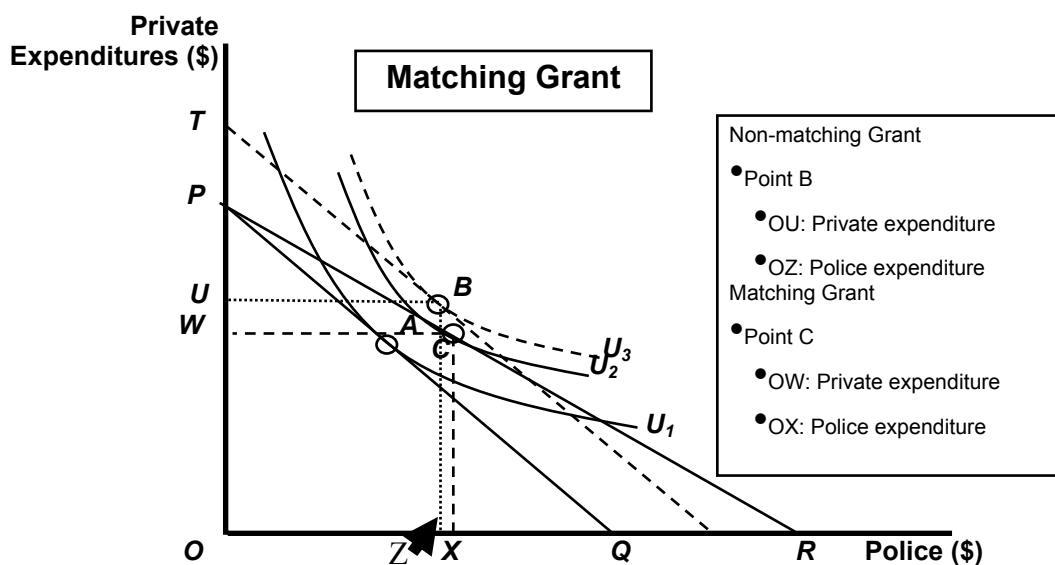
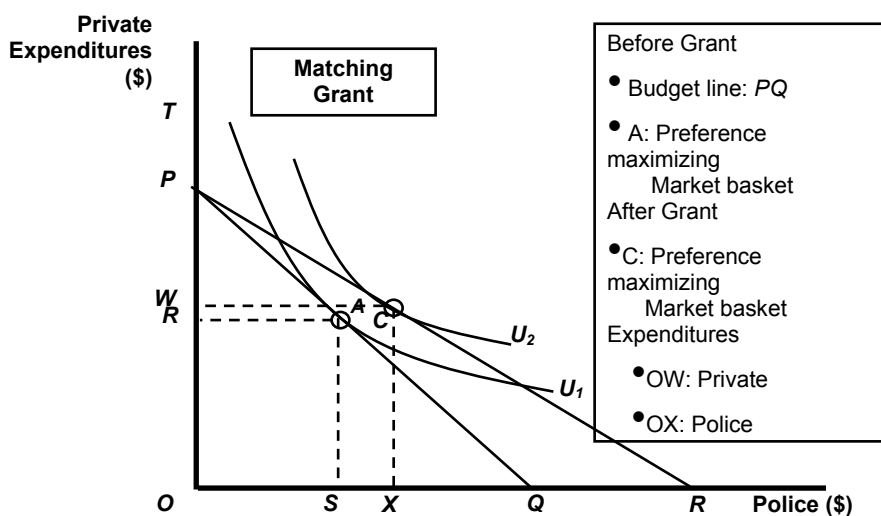
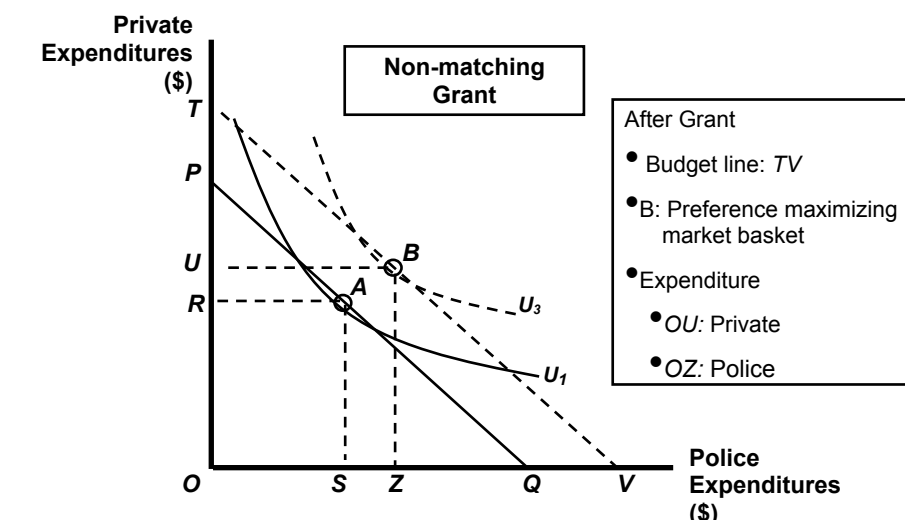
Consider two groups of consumers, each wishing to spend \$10,000 on the styling and performance of cars. Each group has different preferences. By finding the point of tangency between a group's indifference curve and the budget constraint auto companies can design a production and marketing plan.

**DECISION MAKING & PUBLIC POLICY**

Choosing between a non-matching and matching grant to fund police expenditures

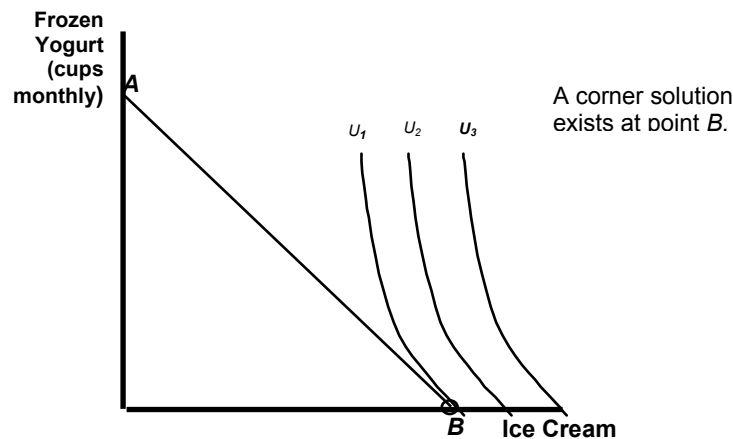






### CORNER SOLUTION

A corner solution exists if a consumer buys in extremes, and buys all of one category of good and none of another. This exists where the indifference curves are tangent to the horizontal and vertical axis. MRS is not equal to  $P_A/P_B$ .



### A CORNER SOLUTION

At point B, the MRS of ice cream for frozen yogurt is greater than the slope of the budget line. This suggests that if the consumer could give up more frozen yogurt for ice cream he would do so. However, there is no more frozen yogurt to give up! When a corner solution arises, *the consumer's MRS does not necessarily equal the price ratio.*

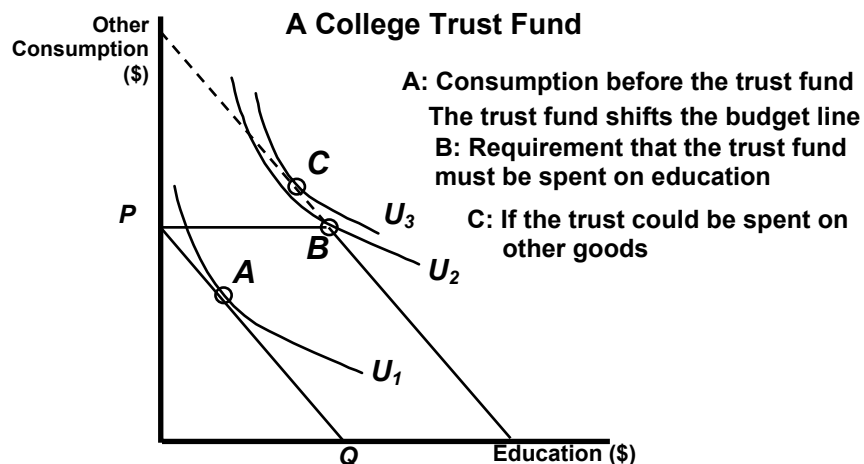
In this instance it can be said that:

$$MRS \geq P_{IceCream} / P_{Frozen\ Yogurt}$$

If the MRS is, in fact, significantly greater than the price ratio, then a small decrease in the price of frozen yogurt *will not* alter the consumer's market basket.

### A COLLEGE TRUST FUND

Suppose Jane Doe's parents set up a trust fund for her college education. Originally, the money must be used for education. If part of the money could be used for the purchase of other goods, her consumption preferences change.



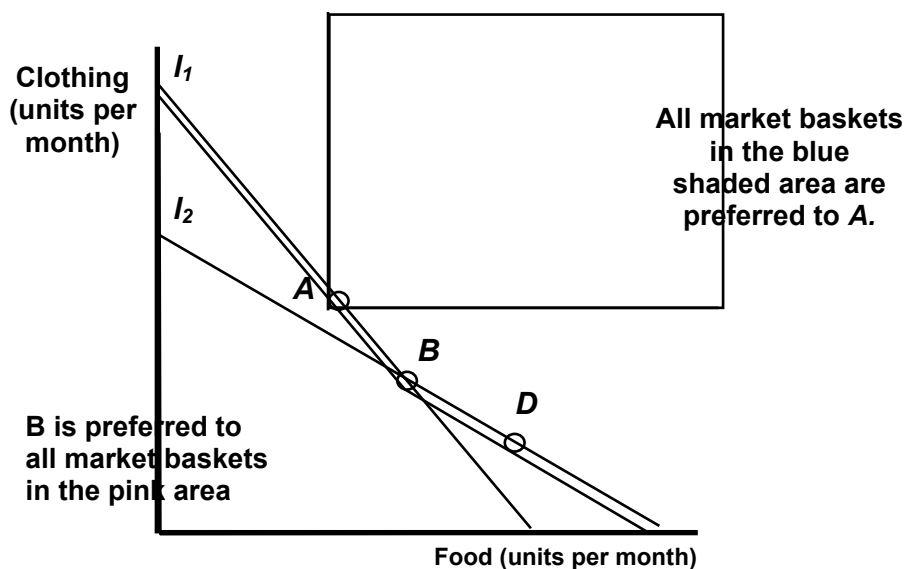
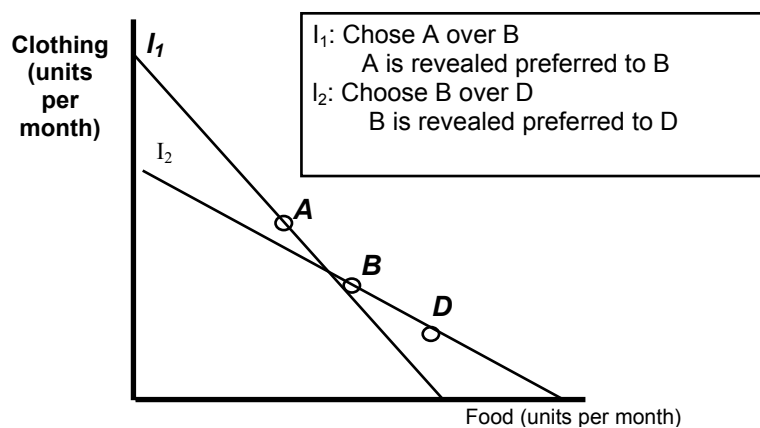
## LESSON 8

## CONSUMER BEHAVIOR (Continued)

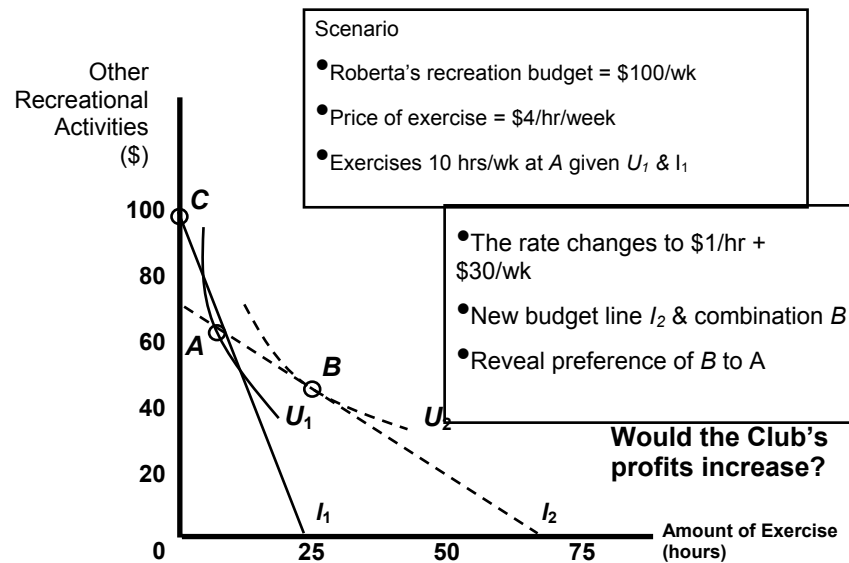
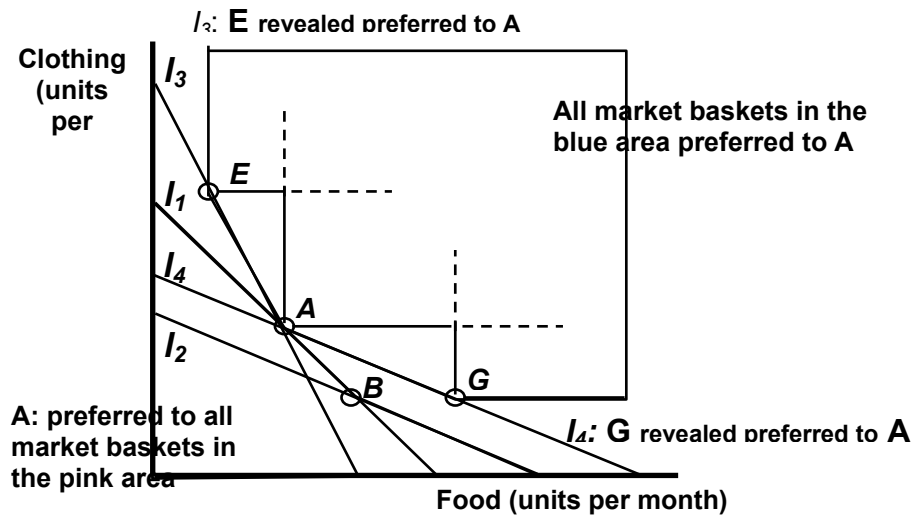
## REVEALED PREFERENCES

If we know the choices a consumer has made, we can determine what her preferences are if we have information about a sufficient number of choices that are made when prices and incomes vary.

## REVEALED PREFERENCES--TWO BUDGET LINES



## REVEALED PREFERENCES--FOUR BUDGET LINES



## LESSON 9

**MARGINAL UTILITY AND CONSUMER CHOICE****MARGINAL UTILITY**

Marginal utility measures the additional satisfaction obtained from consuming one additional unit of a good.

**MARGINAL UTILITY: AN EXAMPLE**

The marginal utility derived from increasing from 0 to 1 units of food might be 9, increasing from 1 to 2 might be 7, increasing from 2 to 3 might be 5

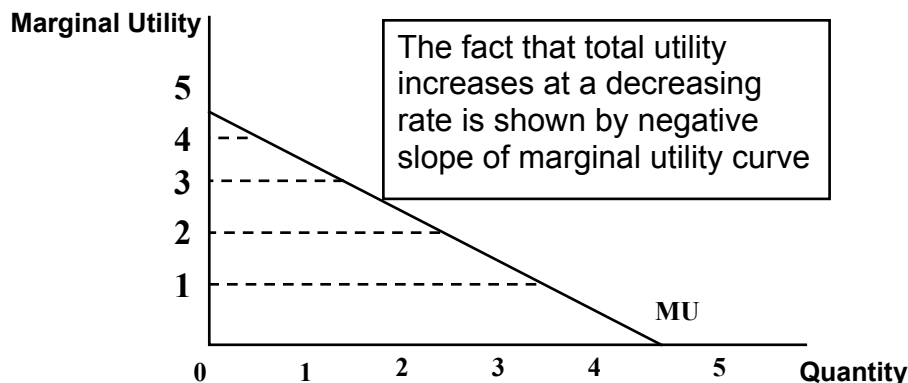
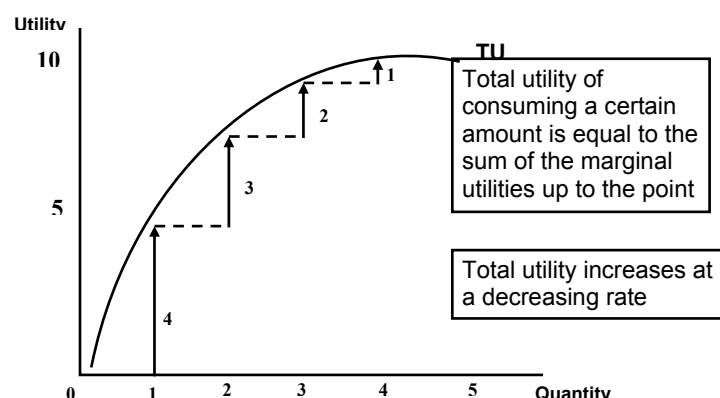
Observation: Marginal utility is diminishing

**DIMINISHING MARGINAL UTILITY**

The principle of diminishing marginal utility states that as more and more of a good is consumed, consuming additional amounts will yield smaller and smaller additions to utility.

**RELATIONSHIP OF TOTAL AND MARGINAL UTILITY****Diminishing Marginal Utility: An Example**

| Quantity of good consumed | Total utility | Marginal utility |
|---------------------------|---------------|------------------|
| 0                         | 0             |                  |
| 1                         | 4             | 4                |
| 2                         | 7             | 3                |
| 3                         | 9             | 2                |
| 4                         | 10            | 1                |
| 5                         | 10            | 0                |



## MARGINAL UTILITY AND CONSUMER CHOICE

### MARGINAL UTILITY AND THE INDIFFERENCE CURVE

If consumption moves along an indifference curve, the additional utility derived from an increase in the consumption of one good, food (F), must balance the loss of utility from the decrease in the consumption in the other good, clothing (C).

Formally:

$$\text{Rearranging:} \quad 0 = MU_F(\Delta F) + MU_C(\Delta C)$$

$$-(\Delta C / \Delta F) = MU_F / MU_C$$

Because:

$$-(\Delta C / \Delta F) = MRS \text{ of F for C}$$

$$MRS = MU_F / MU_C$$

When consumers maximize satisfaction the:

$$MRS = P_F / P_C$$

Since the MRS is also equal to the ratio of the marginal utilities of consuming F and C, it follows that:

$$MU_F / MU_C = P_F / P_C$$

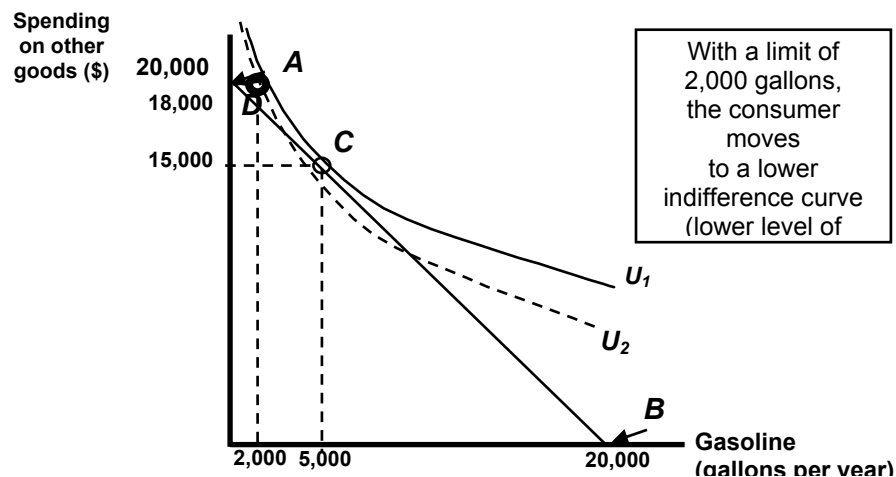
Which gives the equation for utility maximization?

$$MU_F / P_F = MU_C / P_C$$

Total utility is maximized when the budget is allocated so that the marginal utility per dollar of expenditure is the same for each good. This is referred to as the **equal marginal principle**.

### GASOLINE RATIONING

In 1974 and again in 1979, the government imposed price controls on gasoline. This resulted in shortages and gasoline was rationed. Non-price rationing is an alternative to market rationing. Under one form everyone has an equal chance to purchase a rationed good. Gasoline is rationed by long lines at the gas pumps. Rationing hurts some by limiting the amount of gasoline they can buy. This can be seen in the following model. It applies to a woman with an annual income of \$20,000.



### COST-OF-LIVING INDEXES

The CPI is calculated each year as the ratio of the cost of a typical bundle of consumer goods and services today in comparison to the cost during a base period.

Example

Two sisters, Raheela and Sarah, have identical preferences. Sarah began college in 1987 with a \$500 discretionary budget. In 1997, Raheela started college and her parents promised her a budget that was equivalent in purchasing power.

|                 |           |            |
|-----------------|-----------|------------|
| Price of books  | \$20/book | \$100/book |
| Number of books | 15        | 6          |
| Price of food   | \$2.00/lb | \$2.20/lb  |
| Pounds of food  | 100       | 300        |
| Expenditure     | \$500     | \$1,260    |

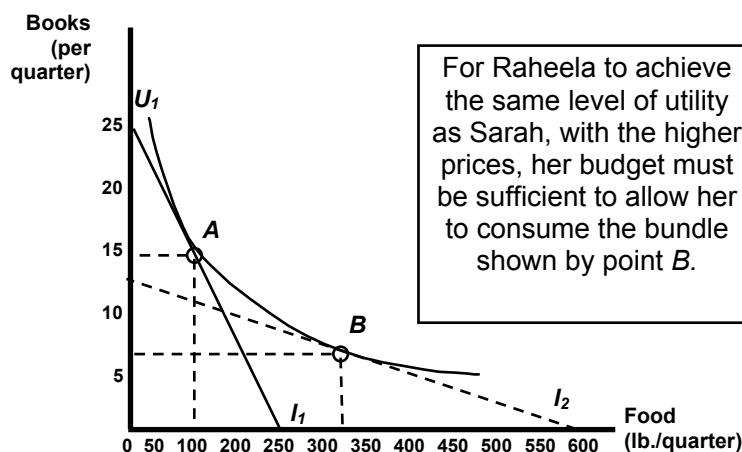
Sarah' Expenditure

$$\$500 = 100 \text{ lbs of food} \times \$2.00/\text{lb} + 15 \text{ books} \times \$20/\text{book}$$

Raheela' Expenditure for Equal Utility

$$\$1,260 = 300 \text{ lbs of food} \times \$2.20/\text{lb} + 6 \text{ books} \times \$100/\text{book}$$

The ideal cost-of-living adjustment for Raheela is \$760. The ideal cost-of-living index is  $\$1,260/\$500 = 2.52$  or 252. This implies a 152% increase in the cost of living.



The ideal cost of living index represents the cost of attaining a given level of utility at current (1997) prices relative to the cost of attaining the same utility at base (1987) prices. To do this on an economy-wide basis would entail large amounts of information.

### LASPEYRES PRICE INDEX

Price indexes, like the CPI, use a fixed consumption bundle in the base period called a Laspeyres price index. The Laspeyres index tells us the amount of money at current year prices that an individual requires to purchase the bundle of goods and services that was chosen in the base year divided by the cost of purchasing the same bundle at base year prices.

**CALCULATING RAHEELA'S LASPEYRES COST OF LIVING INDEX**

Setting the quantities of goods in 1997 equal to what were bought by her sister, but setting their prices at their 1997 levels result in an expenditure of

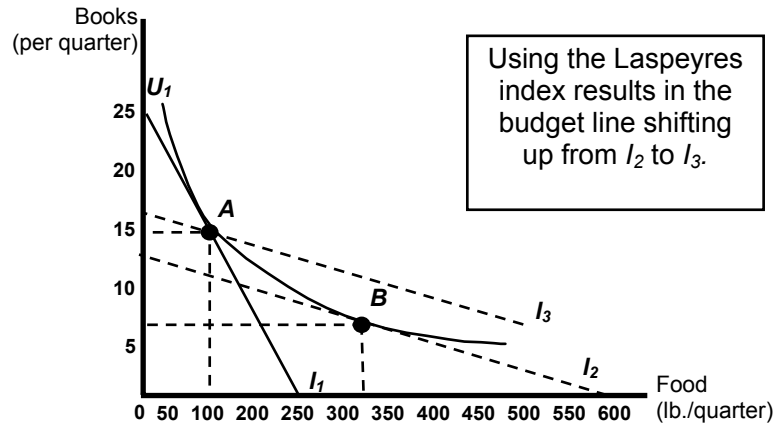
$$\$1,720 (100 \times 2.20 + 15 \times \$100)$$

Her cost of living adjustment would now be \$1,220.

The Laspeyres index is:

$$\$1,720/\$500 = 344.$$

This overstates the true cost-of-living increase.



What Do You Think? Does the Laspeyres index always overstate the true cost-of-living index? Yes! The Laspeyres index assumes that consumers do not alter their consumption patterns as prices change. By increasing purchases of those items that have become relatively cheaper, and decreasing purchases of the relatively more expensive items consumers can achieve the same level of utility without having to consume the same bundle of goods.

**THE PAASCHE INDEX**

The Paasche Index calculates the amount of money at current-year prices that an individual requires to purchase a current bundle of goods and services divided by the cost of purchasing the same bundle in the base year.

**COMPARING THE TWO INDEXES**

Suppose there are two goods: Food (F) and Clothing (C)

Let:

- $P_{Ft}$  &  $P_{Ct}$  be current year prices
- $P_{Fb}$  &  $P_{Cb}$  be base year prices
- $F_t$  &  $C_t$  be current year quantities
- $F_b$  &  $C_b$  be base year quantities

Both indexes involve ratios that involve today's current year prices,  $P_{Ft}$  and  $P_{Ct}$ . However, the Laspeyres index relies on base year consumption,  $F_b$  and  $C_b$ . Whereas, the Paasche index relies on today's current consumption,  $F_t$  and  $C_t$ . Then a comparison of the Laspeyres and Paasche indexes gives the following equations:

$$LI = \frac{P_{Ft} F_b + P_{Ct} C_b}{P_{Fb} F_b + P_{Cb} C_b}$$

$$PI = \frac{P_{Ft} F_t + P_{Ct} C_t}{P_{Fb} F_t + P_{Cb} C_t}$$



**Sarah (1990)**

Cost of base-year bundle at current prices equals

$$\$1,720 (100 \text{ lbs} \times \$2.20/\text{lb} + 15 \text{ books} \times \$100/\text{book})$$

Cost of same bundle at base year prices is

$$\$500 (100 \text{ lbs} \times \$2.00/\text{lb} + 15 \text{ books} \times \$20/\text{book})$$

The Laspeyres index is:

$$LI = \frac{\$1,720}{\$500} = 344$$

Cost of buying current year bundle at current year prices is

$$\$1,260 (300 \text{ lbs} \times \$2.20/\text{lb} + 6 \text{ books} \times \$100/\text{book})$$

Cost of the same bundle at base year prices is

$$\$720 (300 \text{ lbs} \times \$2/\text{lb} + 6 \text{ books} \times \$20/\text{book})$$

The Paasche index is:

$$PI = \frac{\$1,260}{\$720} = 175$$

The Paasche index will understate the cost of living because it assumes that the individual will buy the current year bundle in the base year.

## LESSON 10

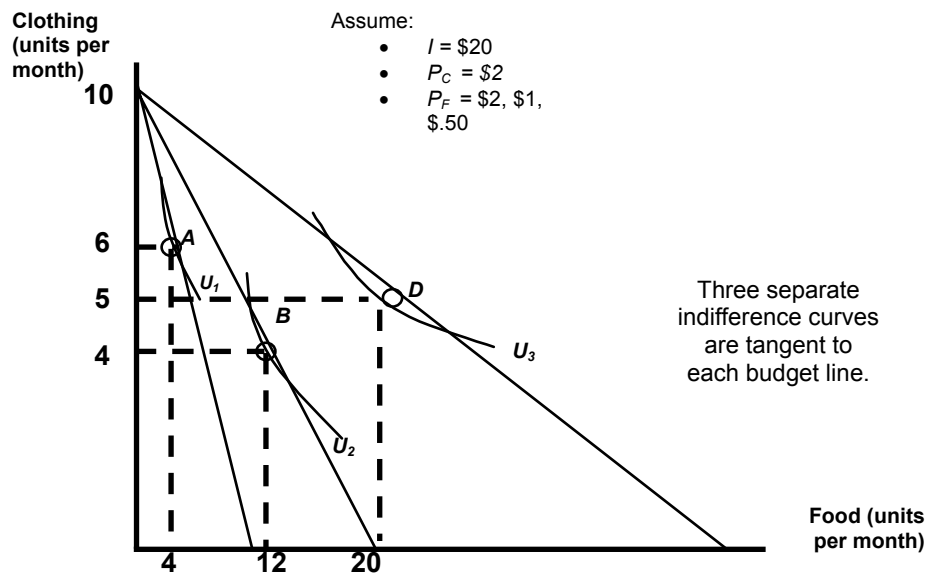
## INDIVIDUAL AND MARKET DEMAND

## INDIVIDUAL DEMAND

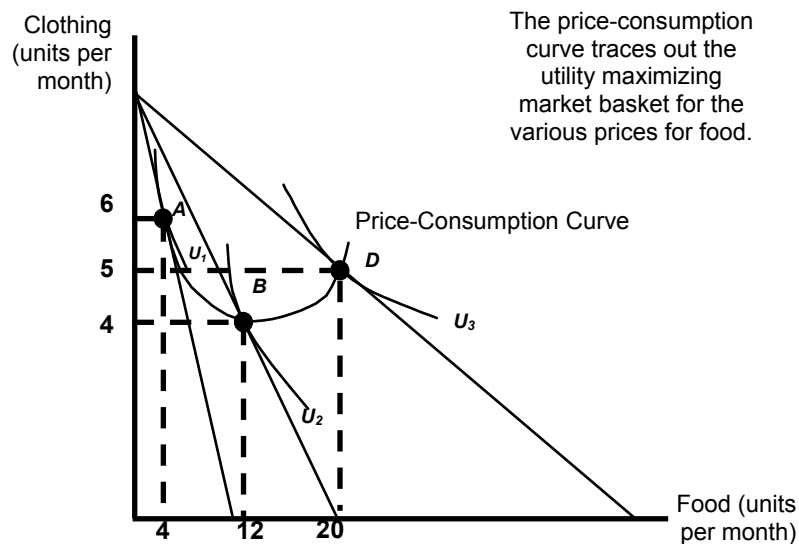
## PRICE CHANGES

Using the figures developed in the previous chapter, the impact of a change in the price of food can be illustrated using indifference curves.

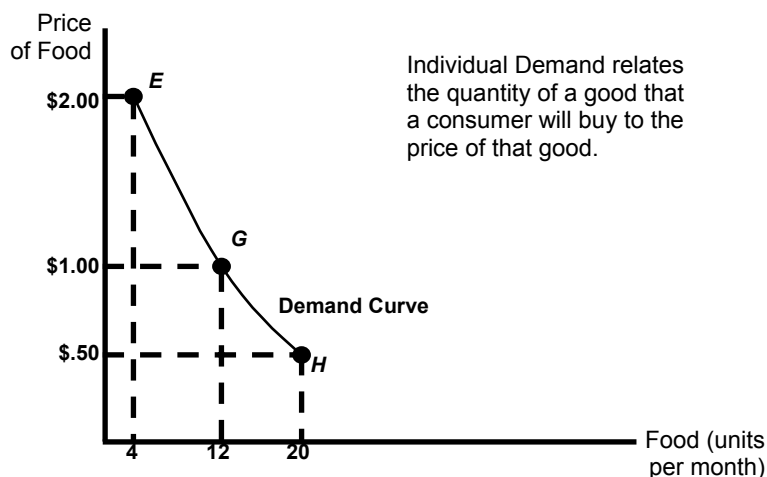
## Effect of a Price Change



## Effect of a Price Change



### Effect of a Price Change

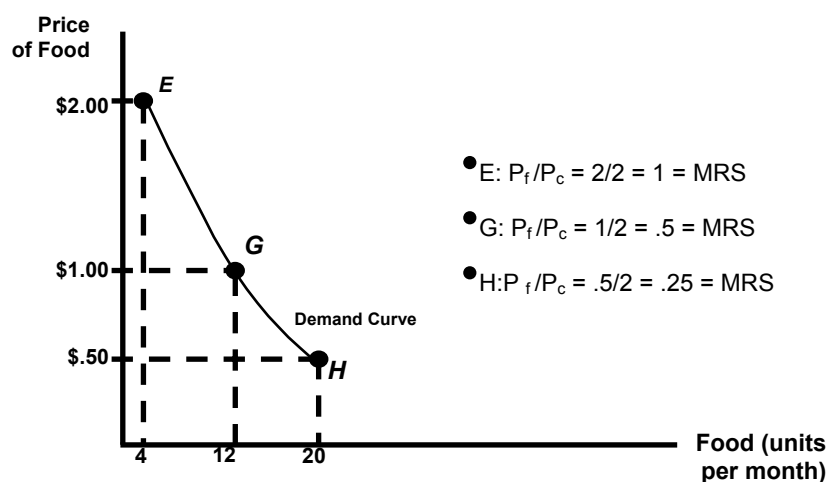


### TWO IMPORTANT PROPERTIES OF DEMAND CURVES

1. The level of utility that can be attained changes as we move along the curve.
2. At every point on the demand curve, the consumer is maximizing utility by satisfying the condition that the *MRS* of food for clothing equals the ratio of the prices of food and clothing.

### Effect of a Price Change

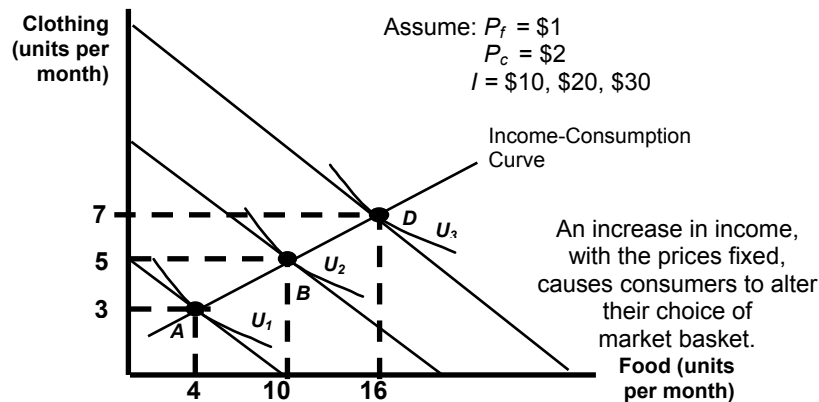
When the price falls:  $P_f/P_c$  & *MRS* also fall



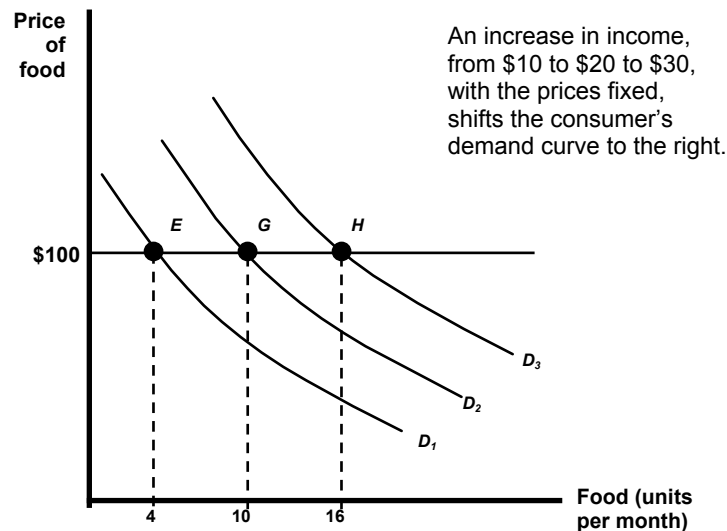
### INCOME CHANGES

Using the figures developed in the previous chapter, the impact of a change in the income can be illustrated using indifference curves.

### Effects of Income Changes



### Effects of Income Changes



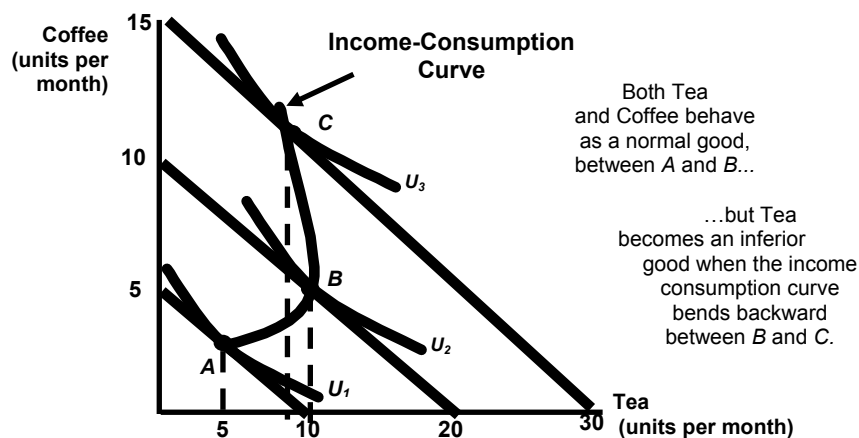
The income-consumption curve traces out the utility-maximizing combinations of food and clothing associated with every income level. An increase in income shifts the budget line to the right, increasing consumption along the income-consumption curve. Simultaneously, the increase in income shifts the demand curve to the right.

### NORMAL GOOD VERSUS INFERIOR GOOD

When the income-consumption curve has a positive slope, the quantity demanded increases with income. The income elasticity of demand is positive. *The good is a normal good.*

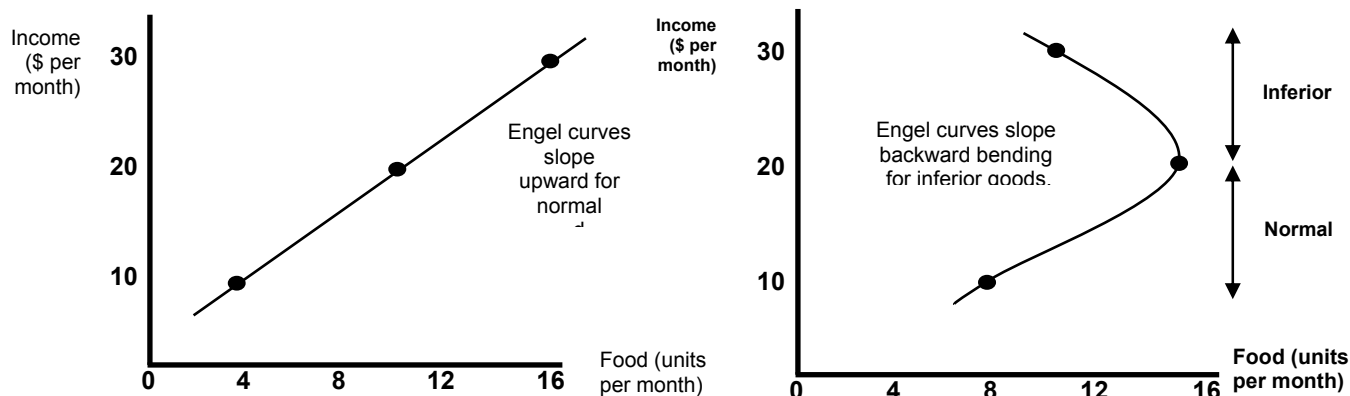
When the income-consumption curve has a negative slope, the quantity demanded decreases with income. The income elasticity of demand is negative. *The good is an inferior good.*

### An Inferior Good



### ENGEL CURVES

Engel curves relate the quantity of good consumed to income. If the good is a normal good, the Engel curve is upward sloping. If the good is an inferior good, the Engel curve is downward sloping.



### Consumer Expenditures in US

#### Income Group (1997 \$)

| Expenditure (\$ on) | Less than 1,000-<br>\$10,000 | 1,000-19,000 | 20,000-29,000 | 30,000-39,000 | 40,000-49,000 | 50,000-69,000 | 70,000-<br>and above |
|---------------------|------------------------------|--------------|---------------|---------------|---------------|---------------|----------------------|
| Entertainment       | 700                          | 947          | 1274          | 1514          | 2054          | 2654          | 4300                 |
| Owned Res.          | 1116                         | 1725         | 253           | 3243          | 4454          | 5793          | 9898                 |
| Rented Res.         | 1957                         | 2170         | 2371          | 2536          | 2137          | 1540          | 1266                 |
| Health Care         | 1031                         | 1697         | 1918          | 820           | 2052          | 2214          | 2642                 |
| Food                | 656                          | 3385         | 4109          | 4888          | 5429          | 6220          | 8279                 |
| Clothing            | 859                          | 978          | 1363          | 1772          | 1778          | 2614          | 3442                 |

**SUBSTITUTES AND COMPLEMENTS**

Two goods are considered substitutes if an increase (decrease) in the price of one leads to an increase (decrease) in the quantity demanded of the other. e.g. movie tickets and video rentals. If the price consumption curve is downward-sloping, the two goods are considered substitutes.

Two goods are considered complements if an increase (decrease) in the price of one leads to a decrease (increase) in the quantity demanded of the other. e.g. gasoline and motor oil. If the price consumption curve is upward-sloping, the two goods are considered complements. They could be both!

Two goods are independent when a change in the price of one good has no effect on the quantity demanded of the other.

## LESSON 11

## INDIVIDUAL AND MARKET DEMAND (Continued)

## INCOME &amp; SUBSTITUTION EFFECTS

A fall in the price of a good has two effects: **Substitution & Income**

Consumers will tend to buy more of the good that has become relatively cheaper, and less of the good that is now relatively more expensive.

Consumers experience an increase in real purchasing power when the price of one good falls.

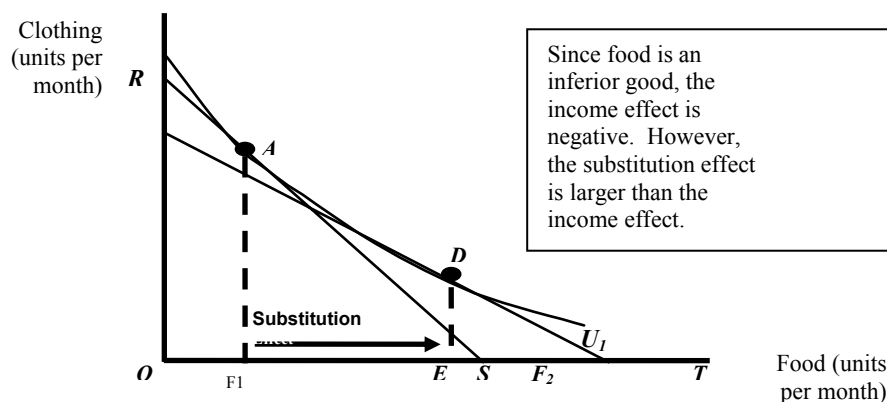
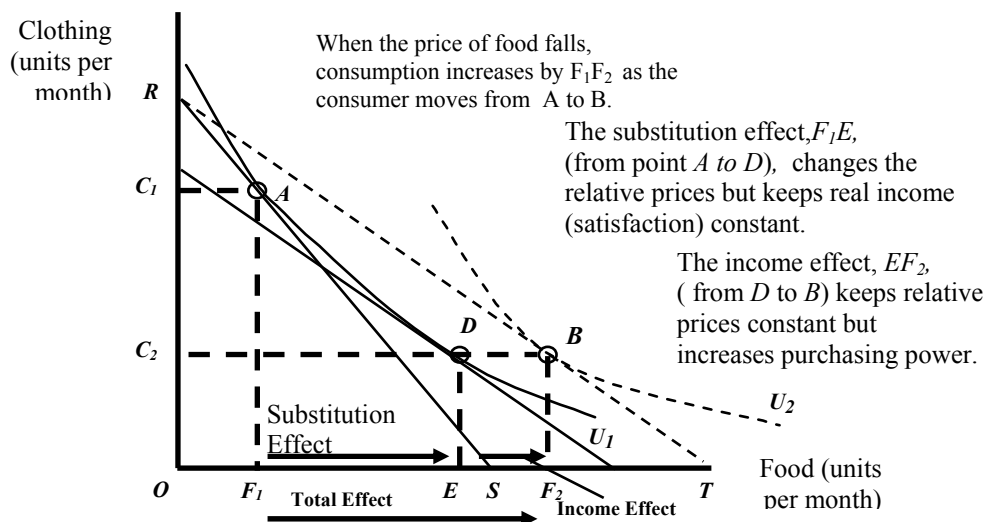
## SUBSTITUTION EFFECT

The **substitution effect** is the change in an item's consumption associated with a change in the price of the item, with the level of utility held constant. When the price of an item declines, the substitution effect always leads to an increase in the quantity of the item demanded.

## INCOME EFFECT

The **income effect** is the change in an item's consumption brought about by the increase in purchasing power, with the price of the item held constant. When a person's income increases, the quantity demanded for the product may increase or decrease. Even with inferior goods, the income effect is rarely large enough to outweigh the substitution effect.

## INCOME &amp; SUBSTITUTION EFFECTS: NORMAL GOOD



### A SPECIAL CASE--THE GIFFEN GOOD

The income effect may theoretically be large enough to cause the demand curve for a good to slope upward. This rarely occurs and is of little practical interest.

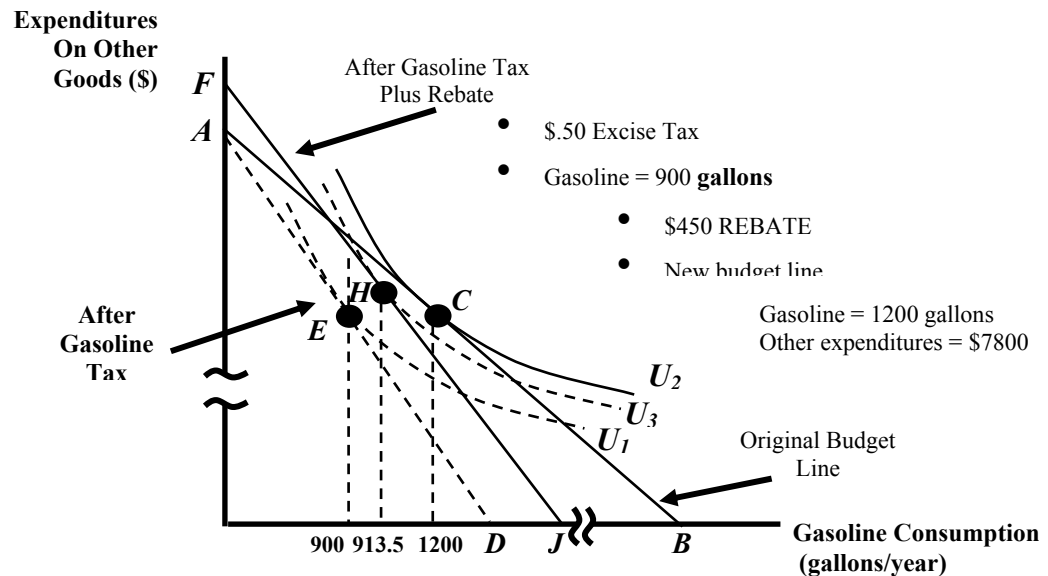
### EFFECT OF A GASOLINE TAX WITH A REBATE

Assume

$$P_e^d = -0.5$$

Income = \$9,000

Price of gasoline = \$1



### MARKET DEMAND

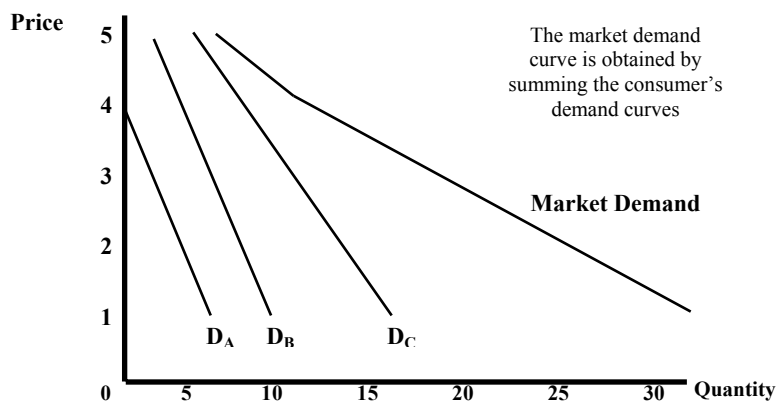
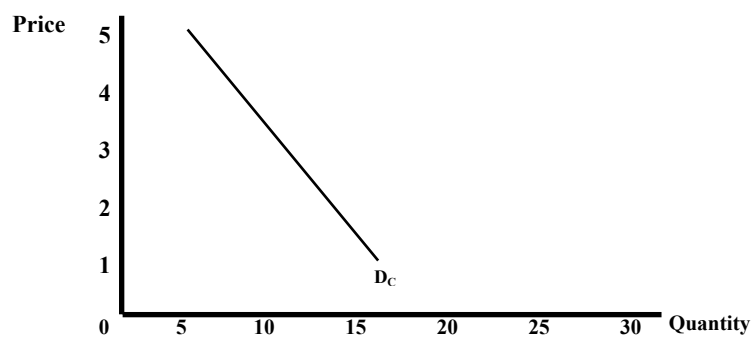
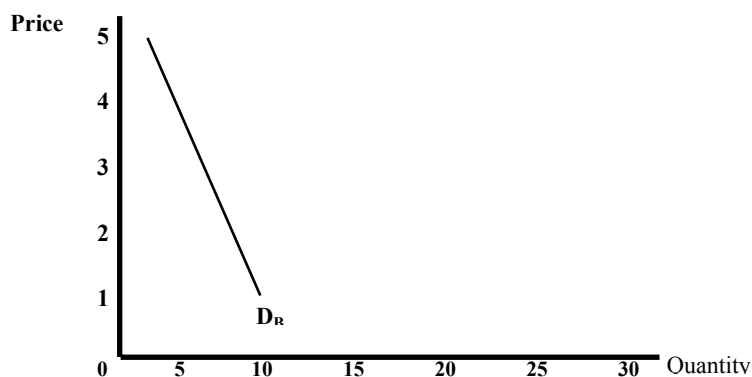
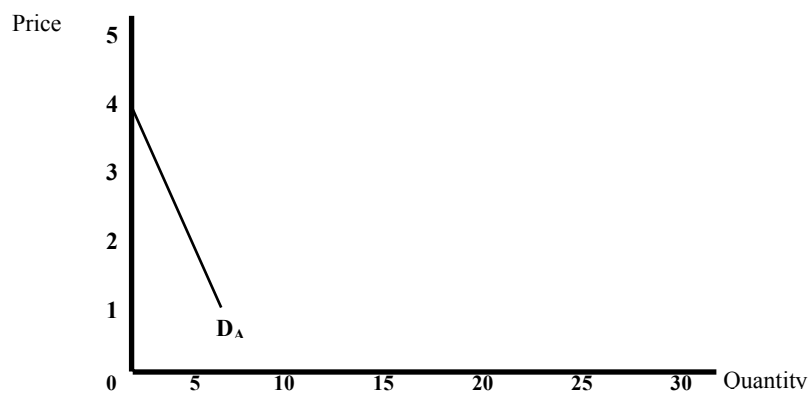
#### MARKET DEMAND CURVES

A curve that relates the quantity of a good that all consumers in a market buy to the price of that good is called market demand curve.

#### DETERMINING THE MARKET DEMAND CURVE

| Price | Individual A | Individual B | Individual C | Market  |
|-------|--------------|--------------|--------------|---------|
| (\$)  | (units)      | (units)      | (units)      | (units) |
| 1     | 6            | 10           | 16           | 32      |
| 2     | 4            | 8            | 13           | 25      |
| 3     | 2            | 6            | 10           | 18      |
| 4     | 0            | 4            | 7            | 11      |
| 5     | 0            | 2            | 4            | 6       |



**SUMMING TO OBTAIN A MARKET DEMAND CURVE**

**TWO IMPORTANT POINTS**

The market demand will shift to the right as more consumers enter the market.

Factors that influence the demands of many consumers will also affect the market demand.

**ELASTICITY OF DEMAND**

Recall: Price elasticity of demand measures the percentage change in the quantity demanded resulting from a 1-percent change in price.

$$E_P = \frac{\Delta Q/Q}{\Delta P/P} = \frac{\Delta Q / \Delta P}{Q / P}$$

**Price Elasticity and Consumer Expenditure**

| Demand                     | If Price Increases,<br>Expenditures: | If Price Decreases,<br>Expenditures: |
|----------------------------|--------------------------------------|--------------------------------------|
| Inelastic ( $E_p < 1$ )    | Increase                             | Decrease                             |
| Unit Elastic ( $E_p = 1$ ) | Are unchanged                        | Are unchanged                        |
| Elastic ( $E_p > 1$ )      | Decrease                             | Increase                             |

**POINT ELASTICITY OF DEMAND**

For large price changes (e.g. 20%), the value of elasticity will depend upon where the price and quantity lie on the demand curve. Point elasticity measures elasticity at a point on the demand curve.

Its formula is:

$$E_P = (P/Q)(1/\text{slope})$$

**PROBLEMS USING POINT ELASTICITY**

We may need to calculate price elasticity over portion of the demand curve rather than at a single point. The price and quantity used as the base will alter the price elasticity of demand.

Point Elasticity of Demand: An Example

Assume

- Price increases from 8\$ to \$10 quantity demanded falls from 6 to 4
- Percent change in price equals:  $\$2/\$8 = 25\%$  or  $\$2/\$10 = 20\%$
- Percent change in quantity equals:  $-2/6 = -33.33\%$  or  $-2/4 = -50\%$

Elasticity equals:

$$-33.33/.25 = -1.33 \text{ or } -.50/.20 = -2.54$$

Which one is correct?

**ARC ELASTICITY OF DEMAND**

Arc elasticity calculates elasticity over a range of prices

Its formula is:

$$E_P = (\Delta Q / \Delta P) (\bar{P} / \bar{Q})$$

$\bar{P}$  = the average price

$\bar{Q}$  = the average quantity

Arc Elasticity of Demand: An Example

$$E_P = (\Delta Q / \Delta P) (\bar{P} / \bar{Q})$$

$$P_1 = 8, P_2 = 10, Q_1 = 6, Q_2 = 4$$

$$\bar{P} = 18 / 2 = 9 \text{ \& } \bar{Q} = 10 / 2 = 5$$

$$E_p = (-2 / \$2) (\$9 / 5) = -1.8$$

## LESSON 12

## INDIVIDUAL AND MARKET DEMAND (Continued)

## THE AGGREGATE DEMAND FOR WHEAT

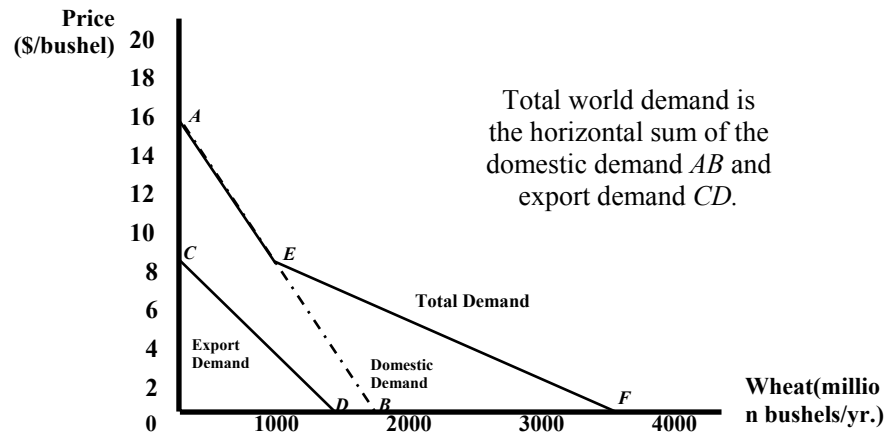
The demand for U.S. wheat is comprised of domestic demand and export demand. The domestic demand for wheat is given by the equation:

$$Q_{DD} = 1700 - 107P$$

The export demand for wheat is given by the equation:

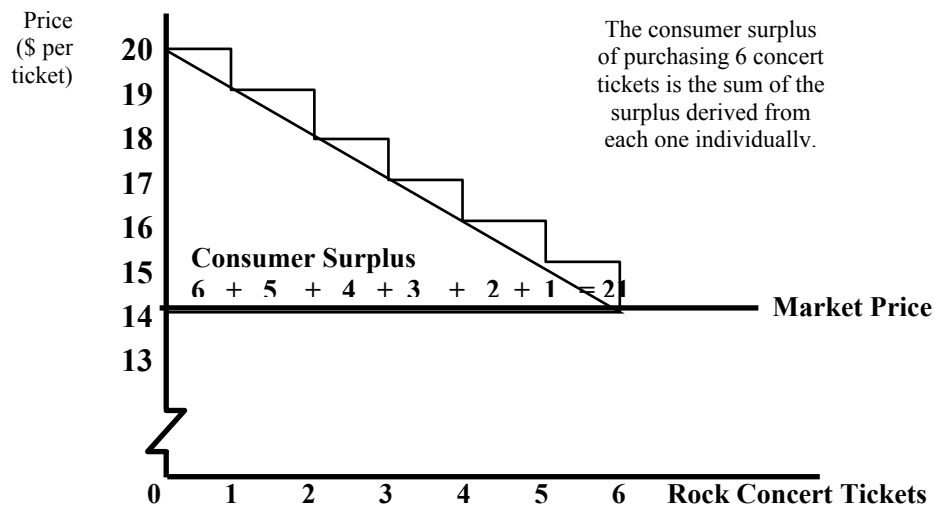
$$Q_{DE} = 1544 - 176P$$

Domestic demand is relatively price inelastic (-0.2), while export demand is more price elastic (-0.4).

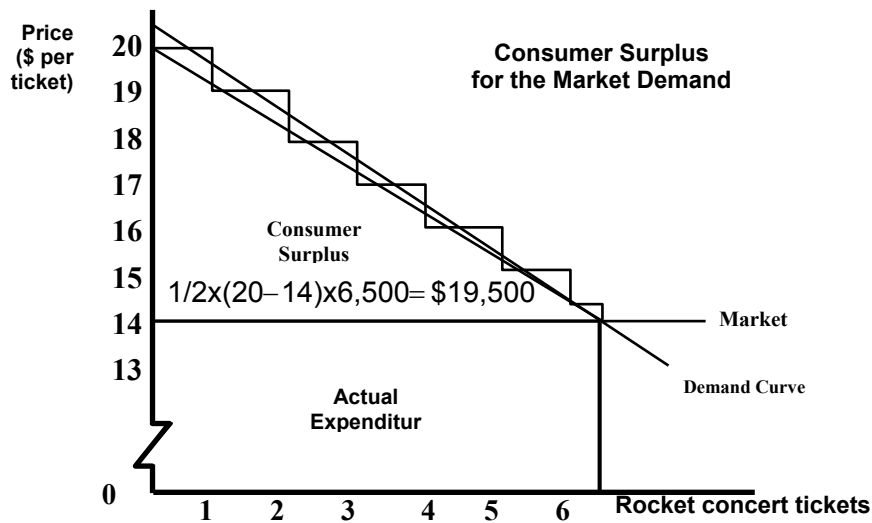


## CONSUMER SURPLUS

Consumer Surplus is the difference between the maximum amount a consumer is willing to pay for a good and the amount actually paid.



The stepladder demand curve can be converted into a straight-line demand curve by making the units of the good smaller.

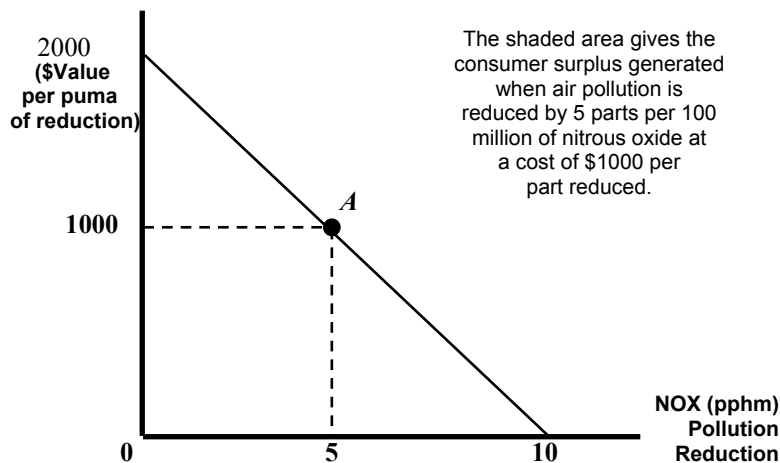


Combining consumer surplus with the aggregate profits that producers obtain we can evaluate:

- 1) Costs and benefits of different market structures
- 2) Public policies that alter the behavior of consumers and firms

### AN EXAMPLE: THE VALUE OF CLEAN AIR

Air is free in the sense that we don't pay to breathe it. Question: Are the benefits of cleaning up the air worth the costs? People pay more to buy houses where the air is clean. Data for house prices among neighborhoods of Lahore and Rawalpindi were compared with the various air pollutants.



### NETWORK EXTERNALITIES

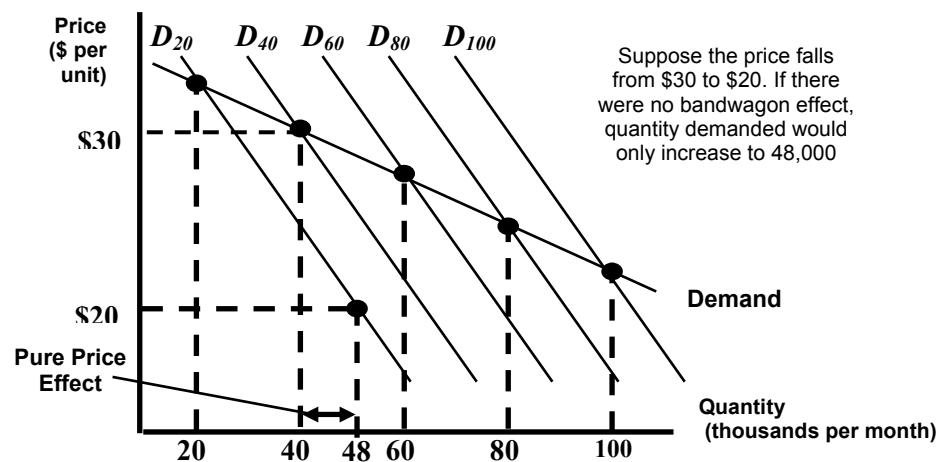
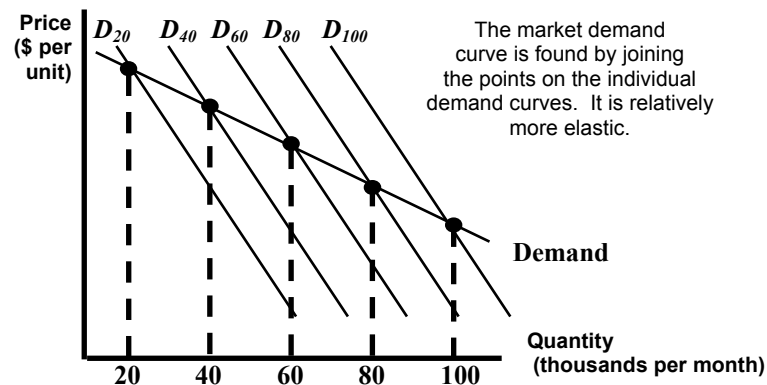
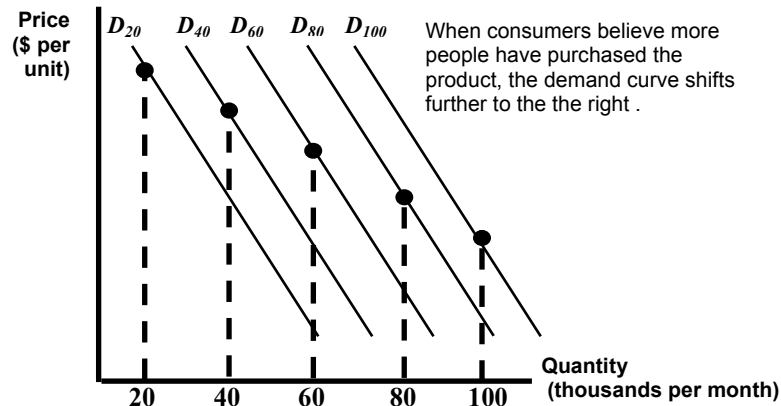
Up to this point we have assumed that people's demands for a good are independent of one another. In fact, a person's demand may be affected by the number of other people who have purchased the good. If this is the case, a network externality exists. Network externalities can be positive or negative.

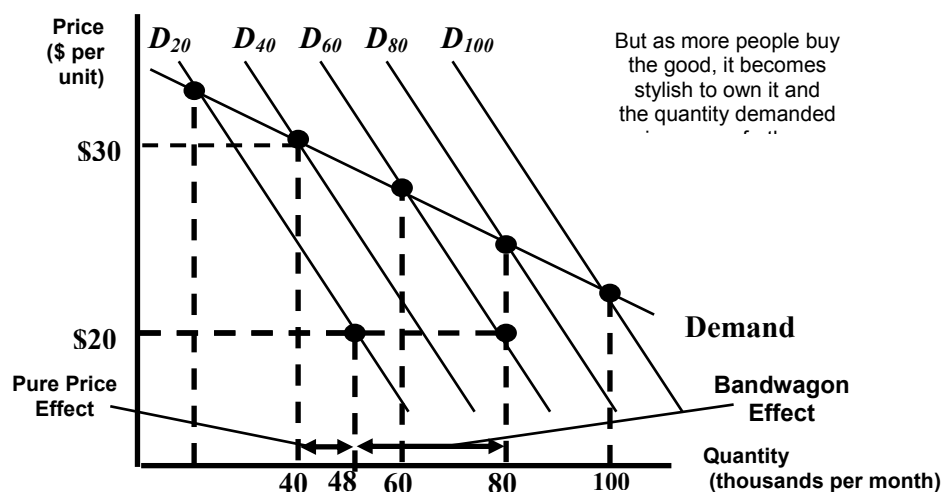
A positive network externality exists if the quantity of a good demanded by a consumer increases in response to an increase in purchases by other consumers. Negative network externalities are just the opposite.

## THE BANDWAGON EFFECT

This is the desire to be in style, to have a good because almost everyone else has it, or to indulge in a fad. This is the major objective of marketing and advertising campaigns (e.g. toys, clothing).

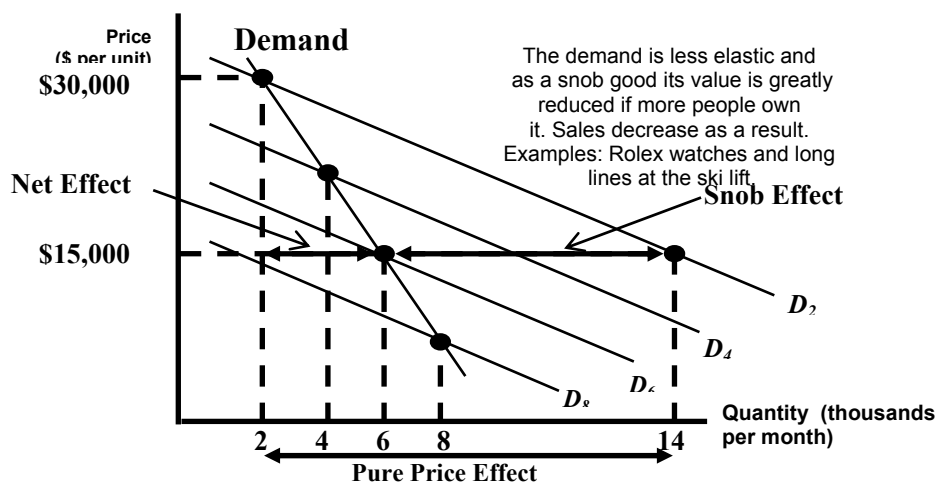
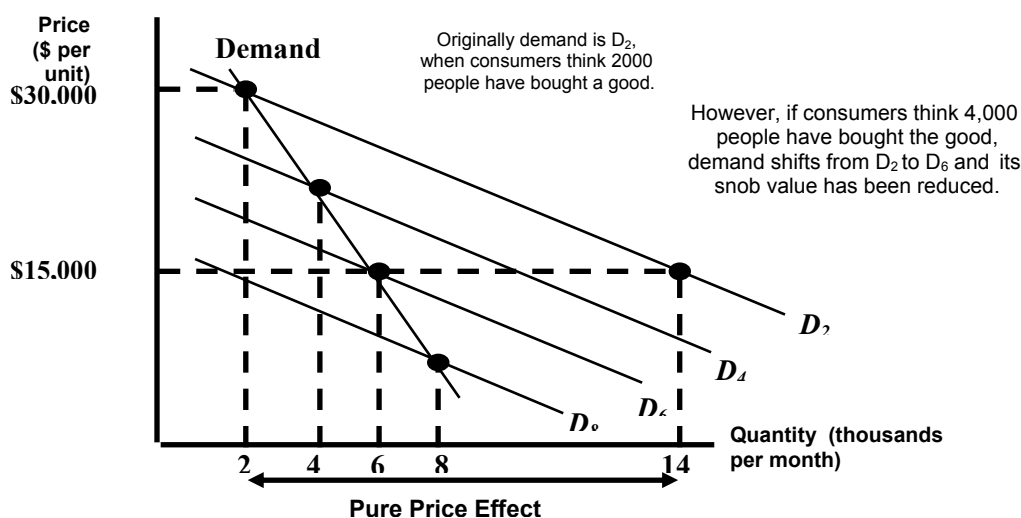
### POSITIVE NETWORK EXTERNALITY: BANDWAGON EFFECT





### THE SNOB EFFECT

If the network externality is negative, a snob effect exists. The snob effect refers to the desire to own exclusive or unique goods. The quantity demanded of a “snob” good is higher the fewer the people who own it.



**NETWORK EXTERNALITIES AND THE DEMANDS FOR COMPUTERS AND FAX MACHINES**

Examples of Positive Feedback Externalities

- Mainframe computers: 1954 - 1965
- Microsoft Windows PC operating system
- Fax-machines and e-mail



**LESSON 13****CHOICE UNDER UNCERTANTY****INTRODUCTION**

Choice with certainty is reasonably straightforward. How do we choose when certain variables such as income and prices are uncertain (i.e. making choices with risk)?

**DESCRIBING RISK**

To measure risk we must know:

- 1) All of the possible outcomes.
- 2) The likelihood that each outcome will occur (its probability).

**INTERPRETING PROBABILITY**

Probability is the likelihood that a given outcome will occur. Objective Interpretation of probability is based on the observed frequency of past events whereas subjective interpretation is based on perception or experience with or without an observed frequency. Different information or different abilities to process the same information can influence the subjective probability.

**EXPECTED VALUE**

The expected value is the weighted average of the payoffs or values resulting from all possible outcomes. The probabilities of each outcome are used as weights. Expected value measures the central tendency; the payoff or value expected on average

**AN EXAMPLE****INVESTMENT IN DRILLING EXPLORATION:**

Two outcomes are possible

- Success -- the stock price increase from \$30 to \$40/share
- Failure -- the stock price falls from \$30 to \$20/share

Objective Probability

100 explorations, 25 successes and 75 failures

Probability (Pr) of success = 1/4 and the probability of failure = 3/4

$$EV = \text{Pr}(\text{success})(\$40/\text{share}) + \text{Pr}(\text{failure})(\$20/\text{share})$$

$$EV = 1/4 (\$40/\text{share}) + 3/4 (\$20/\text{share})$$

$$EV = \$25/\text{share}$$

Given two possible outcomes having payoffs  $X_1$  and  $X_2$ , probabilities of each outcome is given by  $\text{Pr}_1$  &  $\text{Pr}_2$ . Generally, expected value is written as:

$$E(X) = \text{Pr}_1 X_1 + \text{Pr}_2 X_2 + \dots + \text{Pr}_n X_n$$

**VARIABILITY**

Variability is the extent to which possible outcomes of an uncertain event may differ.

**VARIABILITY: A SCENARIO**

Suppose you are choosing between two part-time sales jobs that have the same expected income (\$1,500). The first job is based entirely on commission. The second is a salaried position. There are two equally likely outcomes in the first job--\$2,000 for a good sales job and \$1,000 for a modestly successful one. The second pays \$1,510 most of the time (.99 probability), but you will earn \$510 if the company goes out of business (.01 probability).

| Income from Sales Jobs |             |            |             |            |                 |
|------------------------|-------------|------------|-------------|------------|-----------------|
| Outcome 1              |             |            | Outcome 2   |            |                 |
|                        | Probability | Income(\$) | probability | Income(\$) | Expected income |
| Job 1: Commission      | .5          | 2000       | .5          | 1000       | 1500            |
| Job 2: Fixed salary    | .99         | 1510       | .01         | 510        | 1500            |

Job 1 Expected Income

$$E(X_1) = .5(\$2000) + .5(\$1000) = \$1500$$

Job 2 Expected Income

$$E(X_2) = .99(\$1510) + .01(\$510) = \$1500$$

While the expected values are the same, the variability is not. Greater variability from expected values signals greater risk.

### DEVIATION

Deviation is the difference between expected payoff and actual payoff.

| Deviations from Expected Income (\$) |           |           |           |           |
|--------------------------------------|-----------|-----------|-----------|-----------|
|                                      | Outcome 1 | Deviation | Outcome 2 | Deviation |
| Job 1                                | \$2,000   | \$500     | \$1,000   | -\$500    |
| Job 2                                | 1,510     | 10        | 510       | -900      |

### ADJUSTING FOR NEGATIVE NUMBERS

The standard deviation measures the square root of the average of the squares of the deviations of the payoffs associated with each outcome from their expected value.

The standard deviation is written:

$$\sigma = \sqrt{\text{Pr}_1[X_1 - E(X)]^2 + \text{Pr}_2[X_2 - E(X)]^2}$$

### CALCULATING VARIANCE (\$)

|       | Outcome 1 | Deviation Squared | Outcome 2 | Deviation Squared | Deviation Squared | Standard Deviation |
|-------|-----------|-------------------|-----------|-------------------|-------------------|--------------------|
| Job 1 | \$2,000   | \$250,000         | \$1,000   | \$250,000         | \$250,000         | \$500.00           |
| Job 2 | 1,510     | 100               | 510       | 98,010            | 9,900             | 99.50              |

The standard deviations of the two jobs are:

$$\sigma_1 = \sqrt{.5(\$250,000) + .5(\$250,000)}$$

$$\sigma_1 = \sqrt{\$250,000}$$

$$\sigma_1 = 500 \quad \text{*Greater Risk}$$

$$\sigma_2 = \sqrt{.99(\$100) + .01(\$980,100)}$$

$$\sigma_2 = \sqrt{\$9,900}$$

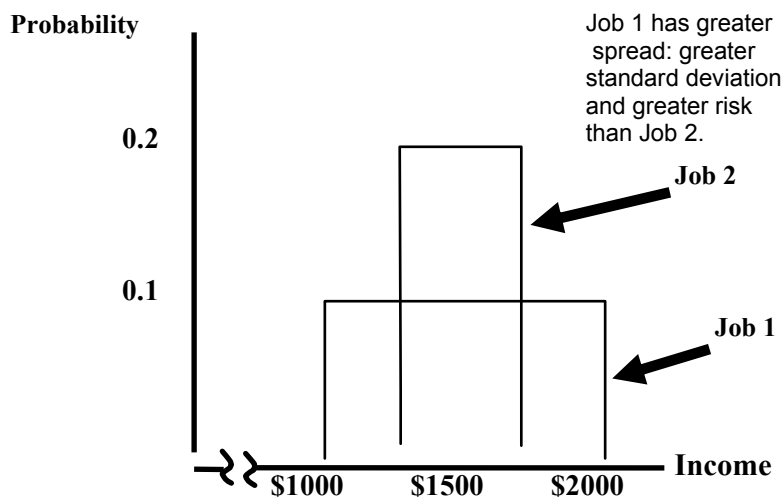
$$\sigma_2 = 99.50$$

The standard deviation can be used when there are many outcomes instead of only two.

### AN EXAMPLE

Job 1 is a job in which the income ranges from \$1000 to \$2000 in increments of \$100 that are all equally likely.

Job 2 is a job in which the income ranges from \$1300 to \$1700 in increments of \$100 that, also, are all equally likely.



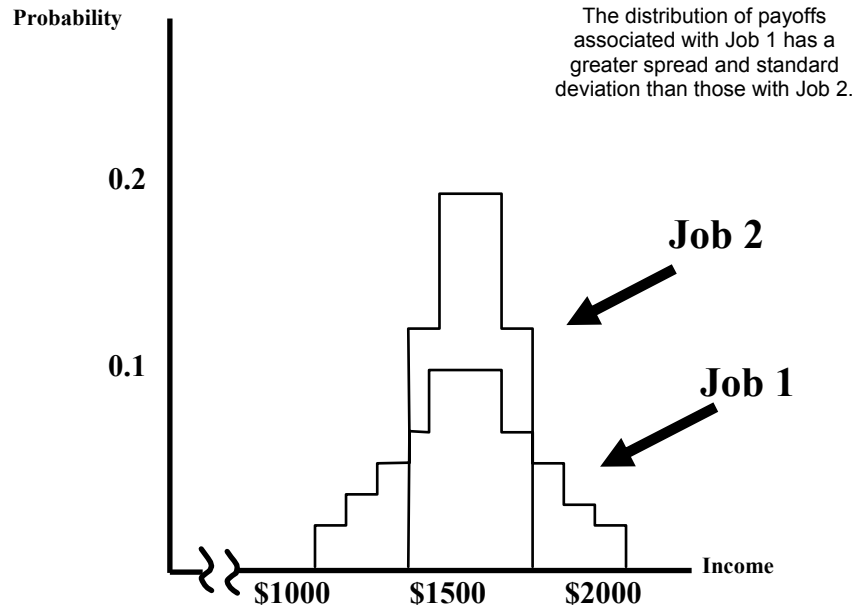
### OUTCOME PROBABILITIES OF TWO JOBS (UNEQUAL PROBABILITY OF OUTCOMES)

- Job 1: greater spread & standard deviation
- Peaked distribution: extreme payoffs are less likely

### DECISION MAKING

A risk avoider would choose Job 2: same expected income as Job 1 with less risk. Suppose we add \$100 to each payoff in Job 1 which makes the expected payoff = \$1600.

### Unequal Probability Outcomes



#### INCOME FROM SALES JOBS--MODIFIED (\$)

|       | Outcome 1 | Deviation Squared | Outcome 2 | Deviation Squared | Deviation Squared | Standard Deviation |
|-------|-----------|-------------------|-----------|-------------------|-------------------|--------------------|
| Job 1 | \$2,100   | \$250,000         | \$1,100   | \$250,000         | \$1, 600          | \$500              |
| Job 2 | 1510      | 100               | 510       | 980,100           | 1, 500            | 99.50              |

**Recall: The standard deviation is the square root of the deviation squared.**

Decision Making

- Job 1: expected income \$1,600 and a standard deviation of \$500.
- Job 2: expected income of \$1,500 and a standard deviation of \$99.50

**Which job? Greater value or less risk?**

Example

Suppose a city wants to deter people from wrong parking. The alternatives .....

Assumptions:

- 1) Wrong parking saves a person \$5 in terms of time spent searching for a parking space.
- 2) The driver is risk neutral.
- 3) Cost of apprehension is zero.

A fine of \$5.01 would deter the driver from double parking. Benefit of wrong parking (\$5) is less than the cost (\$5.01) equals a net benefit that is less than 0.

Increasing the fine can reduce enforcement cost:

- A \$50 fine with a .1 probability of being caught results in an expected penalty of \$5.
- A \$500 fine with a .01 probability of being caught results in an expected penalty of \$5.

The more risk averse drivers are, the lower the fine needs to be in order to be effective.

**LESSON 14****CHOICE UNDER UNCERTAINTY (Continued)****CHOOSING AMONG RISKY ALTERNATIVES**

Assume

- Consumption of a single commodity
- The consumer knows all probabilities
- Payoffs measured in terms of utility
- Utility function given

**EXAMPLE**

A person is earning \$15,000 and receiving 13 units of utility from the job. She is considering a new, but risky job. She has a .50 chance of increasing her income to \$30,000 and a .50 chance of decreasing her income to \$10,000. She will evaluate the position by calculating the expected value (utility) of the resulting income. The expected utility of the new position is the sum of the utilities associated with all her possible incomes weighted by the probability that each income will occur.

The expected utility can be written:

$$\begin{aligned} E(u) &= (1/2)u(\$10,000) + (1/2)u(\$30,000) \\ &= 0.5(10) + 0.5(18) \\ &= 14 \end{aligned}$$

E(u) of new job is 14 which is greater than the current utility of 13 and therefore preferred.

**DIFFERENT PREFERENCES TOWARD RISK**

People can be

- Risk averse
- Risk neutral or
- Risk loving

**RISK AVERSE:**

A person who prefers a certain given income to a risky income with the same expected value. A person is considered risk averse if they have a diminishing marginal utility of income. The use of insurance demonstrates risk averse behavior.

**RISK AVERSE: A SCENARIO**

A person can have a \$20,000 job with 100% probability and receive a utility level of 16. The person could have a job with a .5 chance of earning \$30,000 and a .5 chance of earning \$10,000.

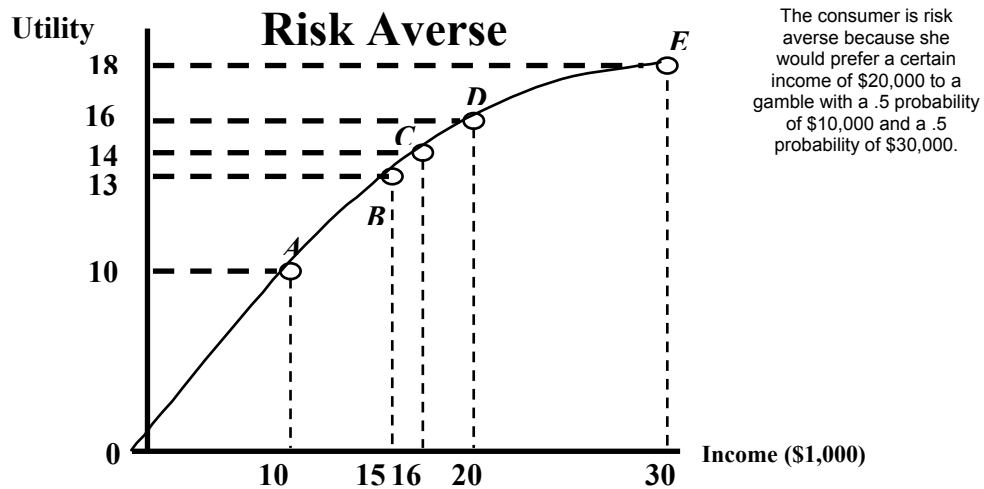
$$\text{Expected Income} = (0.5)(\$30,000) + (0.5)(\$10,000) = \$20,000$$

Expected income from both jobs is the same -- risk averse may choose current job.

The expected utility from the new job is found:

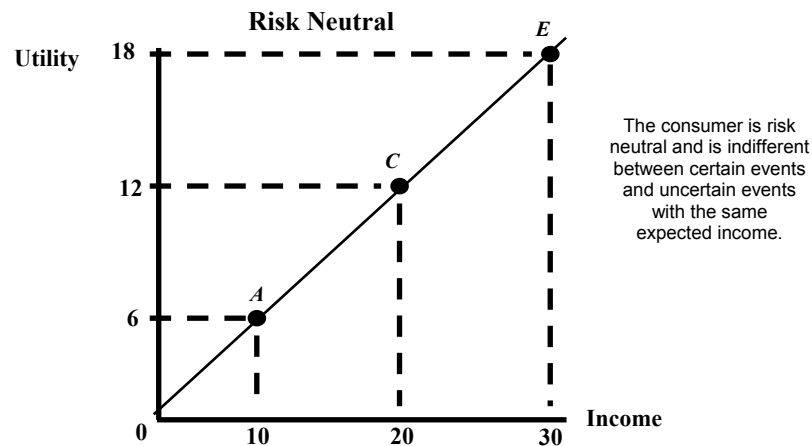
$$\begin{aligned} E(u) &= (1/2)u(\$10,000) + (1/2)u(\$30,000) \\ E(u) &= (0.5)(10) + (0.5)(18) = 14 \end{aligned}$$

E(u) of Job 1 is 16 which is greater than the E(u) of Job 2 which is 14. This individual would keep their present job since it provides them with more utility than the risky job. They are said to be risk averse.



### RISK NEUTRAL

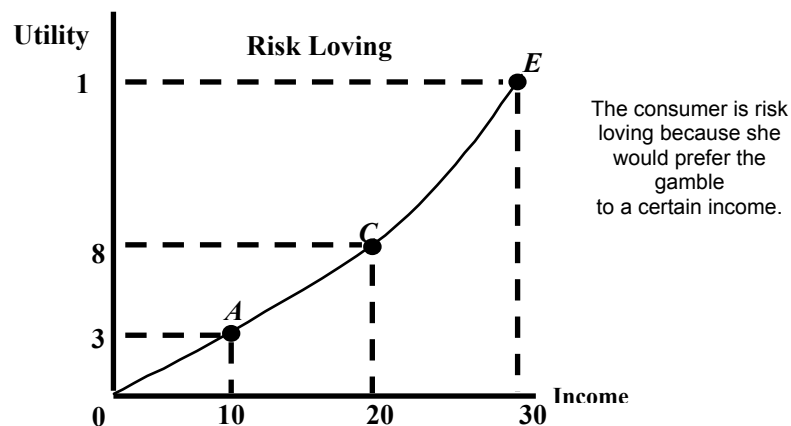
A person is said to be **risk neutral** if they show no preference between a certain income, and an uncertain one with the same expected value.



### RISK LOVING

A person is said to be **risk loving** if they show a preference toward an uncertain income over a certain income with the same expected value.

Examples: Gambling, some criminal activity



## RISK PREMIUM

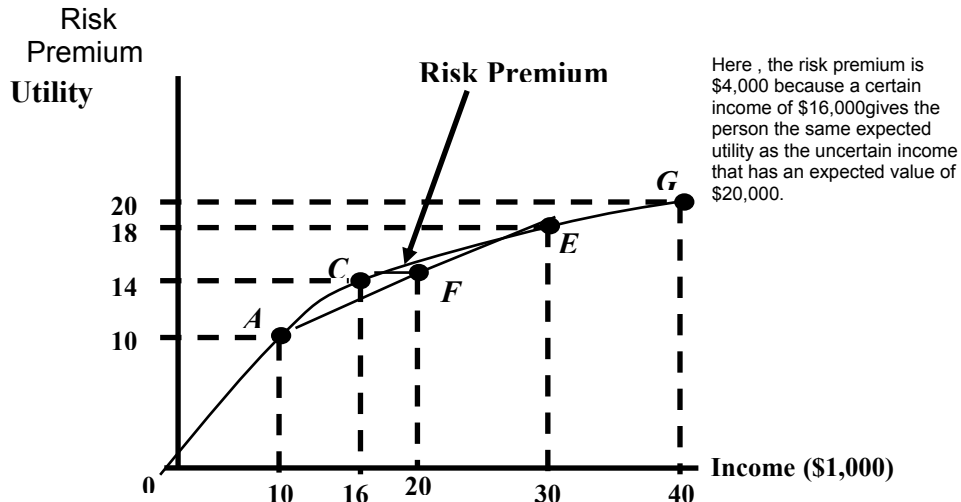
The risk premium is the amount of money that a risk-averse person would pay to avoid taking a risk.

### RISK PREMIUM: A SCENARIO

The person has a .5 probability of earning \$30,000 and a .5 probability of earning \$10,000 (expected income = \$20,000). The expected utility of these two outcomes can be found:

$$E(u) = .5(18) + .5(10) = 14$$

**Question: How much would the person pay to avoid risk?**



## RISK AVERSION AND INCOME

Variability in potential payoffs increases the risk premium.

### EXAMPLE:

A job has a .5 probability of paying \$40,000 (utility of 20) and a .5 chance of paying 0 (utility of 0). The expected income is still \$20,000, but the expected utility falls to 10.

$$\begin{aligned}\text{Expected utility} &= .5u(\$) + .5u(\$40,000) \\ &= 0 + .5(20) = 10\end{aligned}$$

The certain income of \$20,000 has a utility of 16. If the person is required to take the new position, their utility will fall by 6.

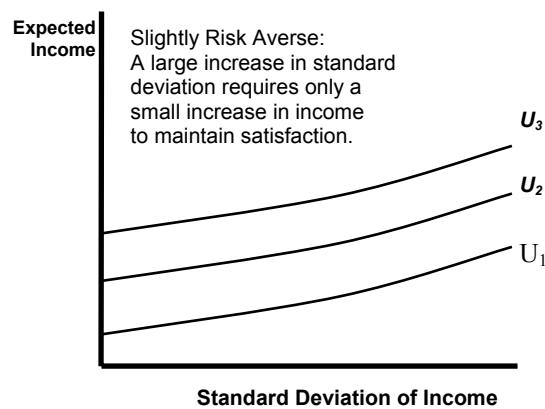
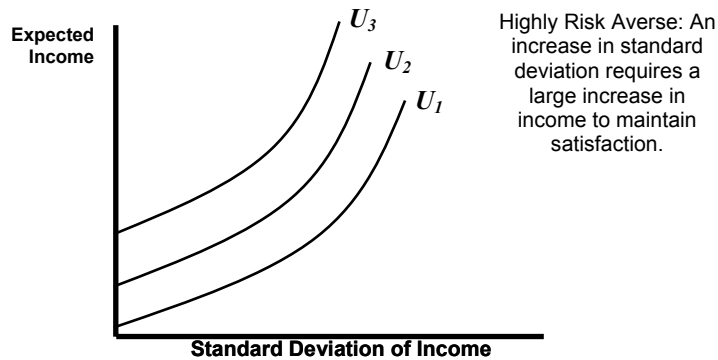
The risk premium is \$10,000 (i.e. they would be willing to give up \$10,000 of the \$20,000 and have the same  $E(u)$  as the risky job.

Therefore, it can be said that the greater the variability, the greater the risk premium.

## INDIFFERENCE CURVE

Indifference curves are combinations of expected income & standard deviation of income that yield the same utility.

## RISK AVERSION AND INDIFFERENCE CURVES



## BUSINESS EXECUTIVES AND THE CHOICE OF RISK

### Example

Study of 464 executives found that:

- 20% were risk neutral
- 40% were risk takers
- 20% were risk averse
- 20% did not respond

Those who liked risky situations did so when losses were involved. When risks involved gains the same, executives opted for less risky situations. The executives made substantial efforts to reduce or eliminate risk by delaying decisions and collecting more information.



## LESSON 15

## CHOICE UNDER UNCERTANTY (Continued)

## REDUCING RISK

Three ways consumers attempt to reduce risk are:

- 1) Diversification
- 2) Insurance
- 3) Obtaining more information

## DIVERSIFICATION

Suppose a firm has a choice of selling air conditioners, heaters, or both. The probability of it being hot or cold is 0.5. The firm would probably be better off by diversification.

|                       | Income from Sales of Appliances |              |
|-----------------------|---------------------------------|--------------|
|                       | Hot Weather                     | Cold Weather |
| Air conditioner sales | \$30,000                        | \$12,000     |
| Heater sales          | 12,000                          | 30,000       |

\* 0.5 probability of hot or cold weather

If the firm sells only heaters or air conditioners their income will be either \$12,000 or \$30,000. Their expected income would be:

$$1/2(\$12,000) + 1/2(\$30,000) = \$21,000$$

If the firm divides their time evenly between appliances their air conditioning and heating sales would be half their original values. If it were hot, their expected income would be \$15,000 from air conditioners and \$6,000 from heaters, or \$21,000. If it were cold, their expected income would be \$6,000 from air conditioners and \$15,000 from heaters, or \$21,000. With diversification, expected income is \$21,000 with no risk. Firms can reduce risk by diversifying among a variety of activities that are not closely related.

## STOCK MARKET

How can diversification reduce the risk of investing in the stock market? Can diversification eliminate the risk of investing in the stock market?

## INSURANCE

Risk averse are willing to pay to avoid risk. If the cost of insurance equals the expected loss, risk averse people will buy enough insurance to recover fully from a potential financial loss.

## The Decision to Insure

| Insurance | Burglary<br>(Pr = .1) | No Burglary<br>(Pr = .9) | Expected<br>Wealth | Standard<br>Deviation |
|-----------|-----------------------|--------------------------|--------------------|-----------------------|
| No        | \$40,000              | \$50,000                 | \$49,000           | \$9,055               |
| Yes       | 49,000                | 49,000                   | 49,000             | 0                     |

While the expected wealth is the same, the expected utility with insurance is greater because the marginal utility in the event of the loss is greater than if no loss occurs. Purchases of insurance transfers wealth and increases expected utility.

**THE LAW OF LARGE NUMBERS**

Although single events are random and largely unpredictable, the average outcome of many similar events can be predicted.

Examples

A single coin toss vs. large number of coins

Whom will have a car wreck vs. the number of wrecks for a large group of drivers?

Assume:

10% chance of a \$10,000 loss from a home burglary

Expected loss =  $.10 \times \$10,000 = \$1,000$  with a high risk (10% chance of a \$10,000 loss)

100 people face the same risk

Then:

\$1,000 premium generates a \$100,000 fund to cover losses

Actuarial Fairness

When the insurance premium = expected payout

**THE VALUE OF TITLE INSURANCE WHEN BUYING A HOUSE**

A Scenario:

Price of a house is \$200,000. There is 5% chance that the seller does not own the house

Risk neutral buyer would pay:

$$(.95 [200,000] + .05 [0]) = 190,000$$

Risk averse buyer would pay much less. By reducing risk, title insurance increases the value of the house by an amount far greater than the premium.

**Value Of Complete Information is the difference between the expected value of a choice with complete information and the expected value when information is incomplete.**

Suppose a store manager must determine how many fall suits to order: 100 suits cost \$180/suit. 50 suits cost \$200/suit. The price of the suits is \$300

Suppose a store manager must determine how many fall suits to order:

- Unsold suits can be returned for half cost.
- The probability of selling each quantity is .50.

**THE DECISION TO INSURE**

|                  | Sale of 50 | Sale of 100 | Expected Profit |
|------------------|------------|-------------|-----------------|
| 1. Buy 50 suits  | \$5,000    | \$5,000     | \$5,000         |
| 2. Buy 100 suits | 1,500      | 12,000      | 6,750           |

**With incomplete information:**

Risk Neutral: Buy 100 suits

Risk Averse: Buy 50 suits

The expected value with complete information is \$8,500.

$$8,500 = .5(5,000) + .5(12,000)$$

The expected value with uncertainty (buy 100 suits) is \$6,750.

The value of complete information is \$1,750, or the difference between the two (the amount the store owner would be willing to pay for a marketing study).

### AN EXAMPLE

Per capita packed milk consumption has fallen over the years. The milk producers engaged in market research to develop new sales strategies to encourage the consumption of packed milk.

### FINDINGS

Packed milk demand is seasonal with the greatest demand in the summer.  $E_p$  is negative and small and  $E_l$  is positive and large. Milk advertising increases sales most in the summer. Allocating advertising based on this information in Karachi increased sales by Rs. 400,000 and profits by 9%. The cost of the information was relatively low, while the value was substantial.

### THE DEMAND FOR RISKY ASSETS

**Assets** are something that provides a flow of money or services to its owner. The flow of money or services can be explicit (dividends) or implicit (capital gain).

**Capital Gain** is an increase in the value of an asset, while a decrease is a capital loss.

### RISKY & RISKLESS ASSETS

Risky Asset provides an uncertain flow of money or services to its owner. For example apartment rent, capital gains, corporate bonds, and stock prices. Whereas Riskless Asset provides a flow of money or services that is known with certainty. For example short-term government bonds, short-term certificates of deposit.

### ASSET RETURNS

Return on an Asset is the total monetary flow of an asset as a fraction of its price. Real Return of an Asset is the simple (or nominal) return less the rate of inflation.

$$\text{Asset Return} = \frac{\text{Monetary Flow}}{\text{Purchase Price}}$$

$$\text{Asset Return} = \frac{\text{Flow}}{\text{Bond Price}} = \frac{\$100/\text{yr.}}{\$1,000} = 10\%$$

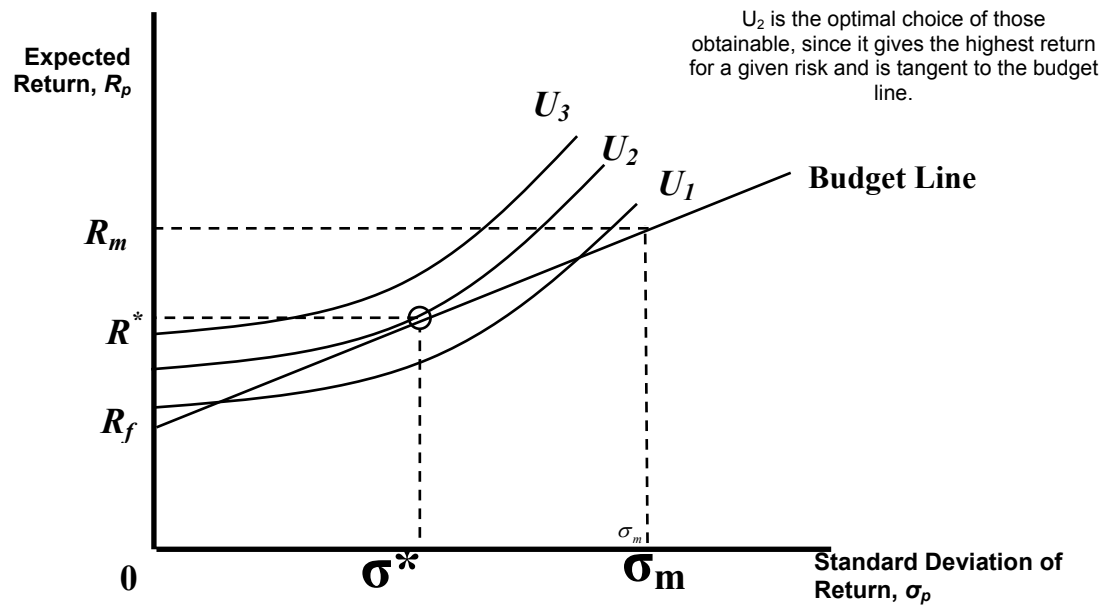
### EXPECTED VS. ACTUAL RETURNS

Expected Return is return that an asset should earn on average whereas actual Return is the Return that an asset earns. Higher returns are associated with greater risk. The risk-averse investor must balance risk relative to return.

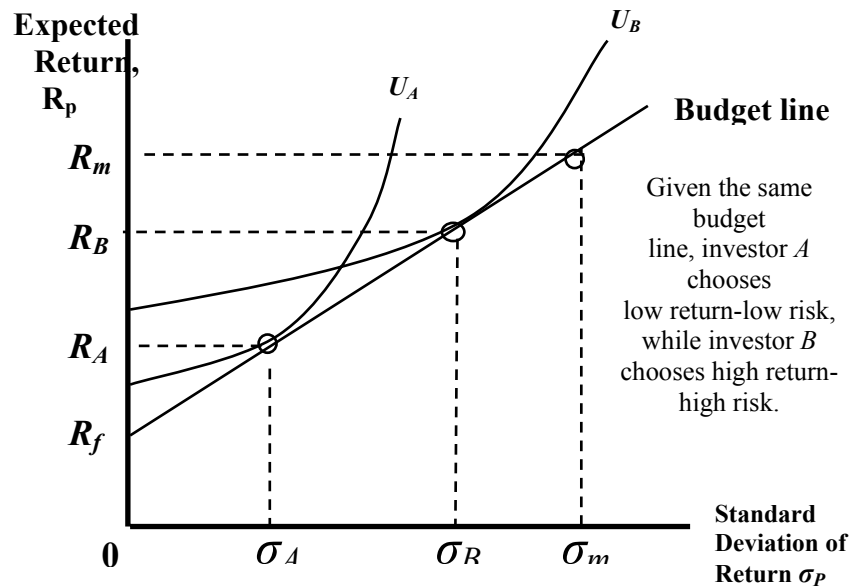
### RISK AND BUDGET LINE

Expected return,  $R_P$ , increases as risk increases. The slope is the price of risk or the risk-return trade-off.

## CHOOSING BETWEEN RISK AND RETURN



## THE CHOICES OF TWO DIFFERENT INVESTORS



## LESSON 16

## PRODUCTION

## INTRODUCTION

Our focus is the supply side. The theory of the firm will address:

- How a firm makes cost-minimizing production decisions
- How cost varies with output
- Characteristics of market supply

## THE TECHNOLOGY OF PRODUCTION

The Production Process is the Combining inputs or factors of production to achieve an output.

## CATEGORIES OF INPUTS (FACTORS OF PRODUCTION)

- Labor
- Materials
- Capital

## PRODUCTION FUNCTION

Production Function indicates the highest output that a firm can produce for every specified combination of inputs given the state of technology. It shows what is technically feasible when the firm operates efficiently.

The production functions for two inputs:

$$Q = F(K, L)$$

Where Q = Output, K = Capital, L = Labor

For a given technology

## ISOQUANTS

Assumptions

Food producer has two inputs: Labor (L) & Capital (K)

Observations:

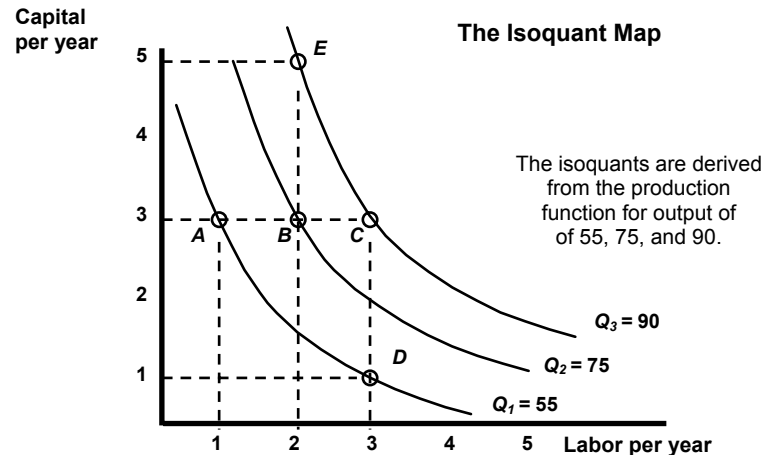
- 1) For any level of K, output increases with more L.
- 2) For any level of L, output increases with more K.
- 3) Various combinations of inputs produce the same output.

Isoquants are curves showing all possible combinations of inputs that yield the same output

Production Function for Food

| Capitan Input | Labor Input |    |     |     |     |
|---------------|-------------|----|-----|-----|-----|
|               | 1           | 2  | 3   | 4   | 5   |
| 1             | 20          | 40 | 55  | 65  | 75  |
| 2             | 40          | 60 | 75  | 85  | 90  |
| 3             | 55          | 75 | 90  | 100 | 105 |
| 4             | 65          | 85 | 100 | 110 | 115 |
| 5             | 75          | 90 | 105 | 115 | 120 |

## PRODUCTION WITH TWO VARIABLE INPUTS (L,K)



### INPUT FLEXIBILITY

The isoquants emphasize how different input combinations can be used to produce the same output. This information allows the producer to respond efficiently to changes in the markets for inputs.

### THE SHORT RUN VS. LONG RUN

Short-run is the period of time in which quantities of one or more production factors cannot be changed. These inputs are called fixed inputs. Long-run is the amount of time needed to make all production inputs variable.

### PRODUCTION WITH ONE VARIABLE INPUT (LABOR)

| Amount of Labor (L) | Amount of Capital (K) | Total Output (Q) | Average Product | Marginal Product |
|---------------------|-----------------------|------------------|-----------------|------------------|
| 0                   | 10                    | 0                | ---             | ---              |
| 1                   | 10                    | 10               | 10              | 10               |
| 2                   | 10                    | 30               | 15              | 20               |
| 3                   | 10                    | 60               | 20              | 30               |
| 4                   | 10                    | 80               | 20              | 20               |
| 5                   | 10                    | 95               | 19              | 15               |
| 6                   | 10                    | 108              | 18              | 13               |
| 7                   | 10                    | 112              | 16              | 4                |
| 8                   | 10                    | 112              | 14              | 0                |
| 9                   | 10                    | 108              | 12              | -4               |
| 10                  | 10                    | 100              | 10              | -8               |

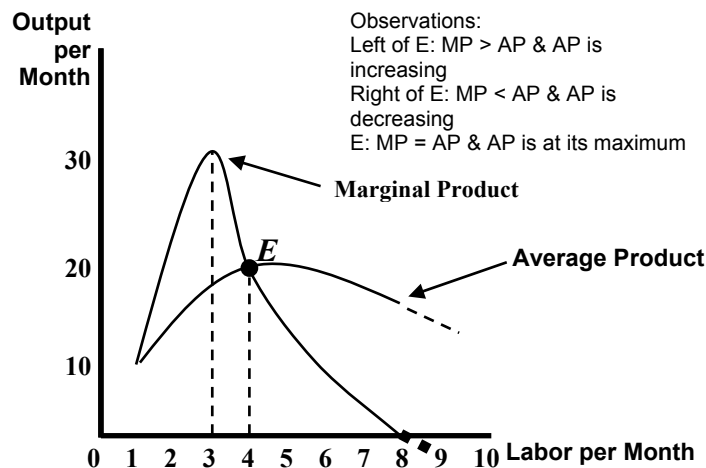
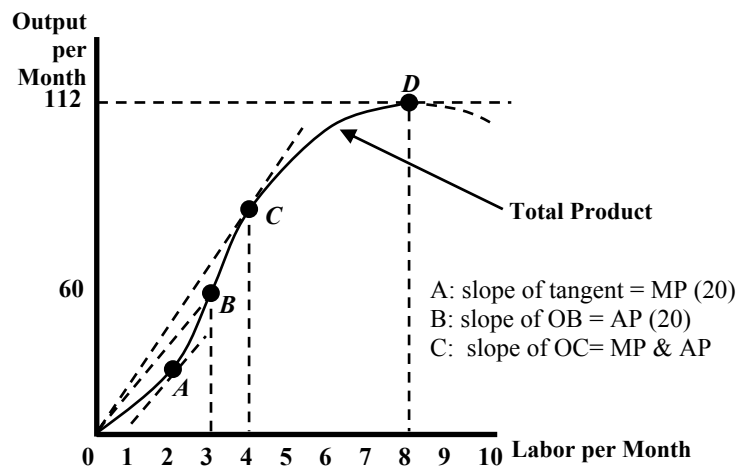
**OBSERVATIONS:**

- 1) With additional workers, output (Q) increases, reaches a maximum, and then decreases.
- 2) The average product of labor (AP), or output per worker, increases and then decreases.

$$AP = \frac{\text{Output}}{\text{Labor Input}} = \frac{Q}{L}$$

- 3) The marginal product of labor (MP), or output of the additional worker, increases rapidly initially and then decreases and becomes negative..

$$MP_L = \frac{\Delta \text{Output}}{\Delta \text{Labor Input}} = \frac{\Delta Q}{\Delta L}$$

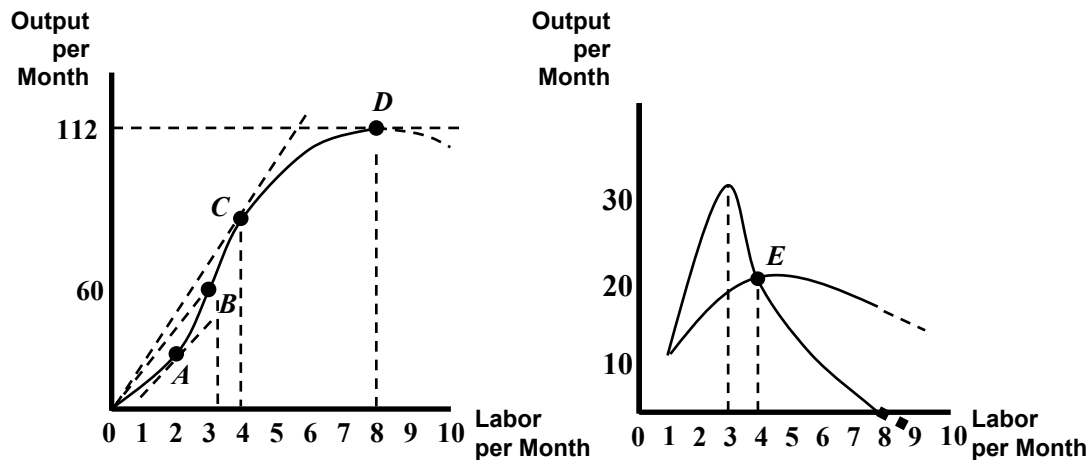
**Observations:**

- When MP = 0, TP is at its maximum
- When MP > AP, AP is increasing
- When MP < AP, AP is decreasing
- When MP = AP, AP is at its maximum

**AP** = slope of line from origin to a point on **TP**, lines **b**, & **c**.

**MP** = slope of a tangent to any point on the **TP** line, lines **a** & **c**.

### THE LAW OF DIMINISHING MARGINAL RETURNS

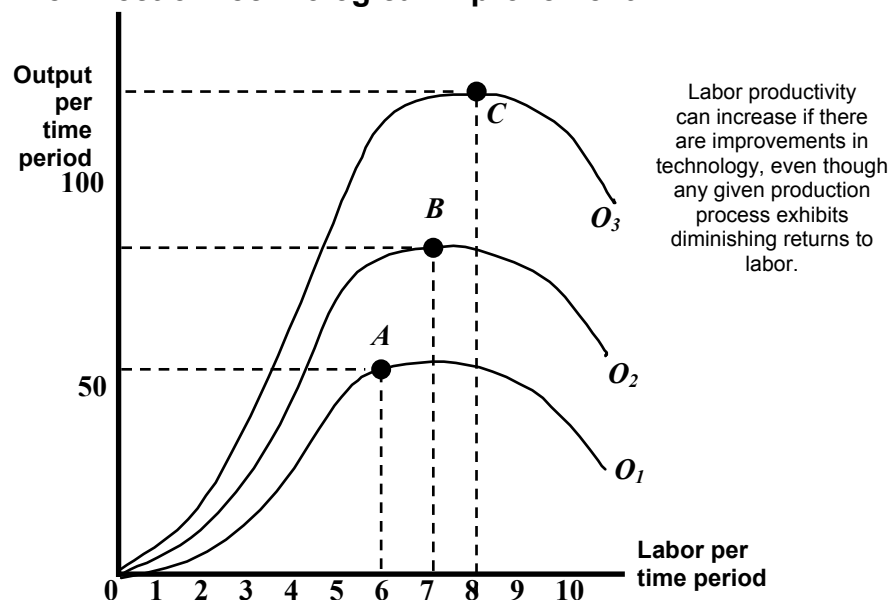


As the use of an input increases in equal increments, a point will be reached at which the resulting additions to output decreases (i.e. MP declines). When the labor input is small, MP increases due to specialization. When the labor input is large, MP decreases due to inefficiencies.

#### The Law of Diminishing Marginal Returns

- Can be used for long-run decisions to evaluate the trade-offs of different plant configurations
- Assumes the quality of the variable input is constant
- Explains a declining *MP*, not necessarily a negative one
- Assumes a constant technology

#### The Effect of Technological Improvement





**LABOR PRODUCTIVITY**

$$\text{Average Productivity} = \frac{\text{Total Output}}{\text{Total Labor Input}}$$

**LABOR PRODUCTIVITY AND THE STANDARD OF LIVING**

Consumption can increase only if productivity increases.

**DETERMINANTS OF PRODUCTIVITY**

- Stock of capital
- Technological change

**Labor Productivity in Developed Countries**

|                                                        | France   | Germany  | Japan    | United Kingdom | United States |
|--------------------------------------------------------|----------|----------|----------|----------------|---------------|
| <b>Output per Employed Person (1997)</b>               |          |          |          |                |               |
|                                                        | \$54,507 | \$55,644 | \$46,048 | \$42,630       | \$60,915      |
| <b>Annual Rate of Growth of Labor Productivity (%)</b> |          |          |          |                |               |
| 1960-1973                                              | 4.75     | 4.04     | 8.30     | 2.89           | 2.36          |
| 1974-1986                                              | 2.10     | 1.85     | 2.50     | 1.69           | 0.71          |
| 1987-1997                                              | 1.48     | 2.00     | 1.94     | 1.02           | 1.09          |

**TRENDS IN PRODUCTIVITY**

- 1) U.S. productivity is growing at a slower rate than other countries.
- 2) Productivity growth in developed countries has been decreasing.

**EXPLANATIONS FOR PRODUCTIVITY GROWTH SLOWDOWN**

- 1) Growth in the stock of capital is the primary determinant of the growth in productivity.
- 2) Rate of capital accumulation in the U.S. was slower than other developed countries because the others were rebuilding after WWII.
- 3) Depletion of natural resources
- 4) Environment regulations

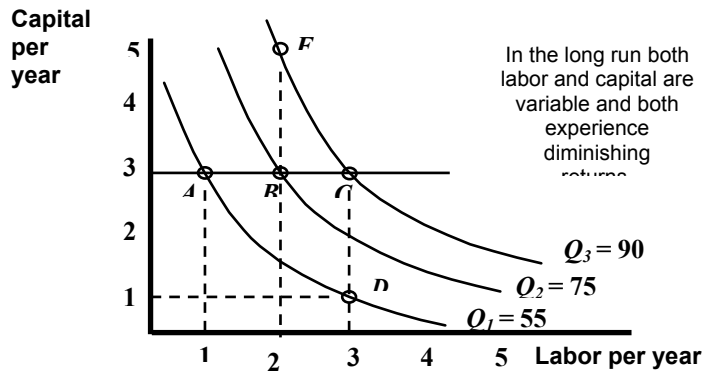
## LESSON 17

## PRODUCTION (Continued)

## PRODUCTION WITH TWO VARIABLE INPUTS

There is a relationship between production and productivity. Long-run production K & L are variable. Isoquants analyze and compare the different combinations of K & L and output.

## The Shape of Isoquants



## DIMINISHING MARGINAL RATE OF SUBSTITUTION

Reading the Isoquant Model

- 1) Assume capital is 3 and labor increases from 0 to 1 to 2 to 3. Notice output increases at a decreasing rate (55, 20, 15) illustrating diminishing returns from labor in the short-run and long-run.
- 2) Assume labor is 3 and capital increases from 0 to 1 to 2 to 3. Output also increases at a decreasing rate (55, 20, 15) due to diminishing returns from capital.

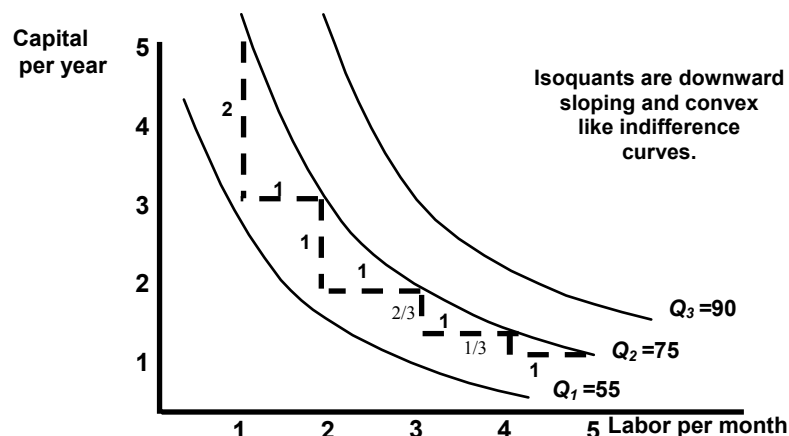
## SUBSTITUTING AMONG INPUTS

Managers want to determine what combination of inputs to use. They must deal with the trade-off between inputs. The slope of each isoquant gives the trade-off between two inputs while keeping output constant. The marginal rate of technical substitution equals:

$$MRTS = - \text{Change in capital} / \text{Change in labor input}$$

$$MRTS = - \Delta K / \Delta L \text{ (for a fixed level of } Q \text{)}$$

## MARGINAL RATE OF TECHNICAL SUBSTITUTION



**OBSERVATIONS:**

- 1) Increasing labor in one unit increments from 1 to 5 results in a decreasing MRTS from 1 to 1/2.
- 2) Diminishing MRTS occurs because of diminishing returns and implies isoquants are convex.

**MRTS AND MARGINAL PRODUCTIVITY**

The change in output from a change in labor equals:

$$(MP_L)(\Delta L)$$

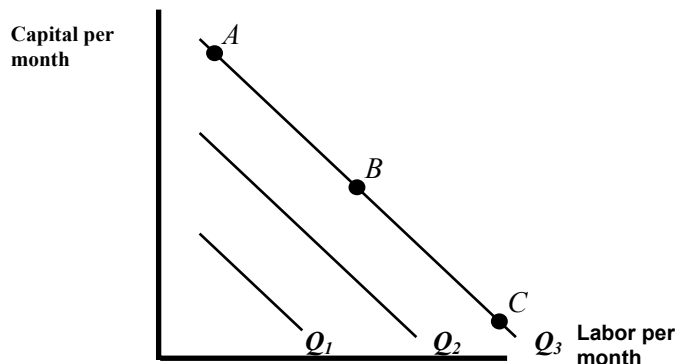
The change in output from a change in capital equals:

$$(MP_K)(\Delta K)$$

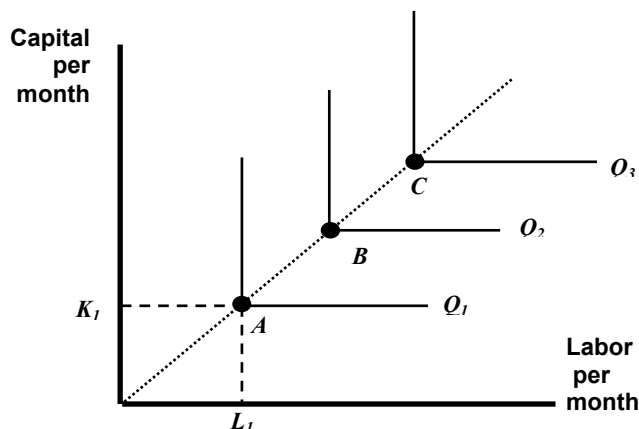
If output is constant and labor is increased, then:

$$(MP_L)(\Delta L) + (MP_K)(\Delta K) = 0$$

$$(MP_L)(MP_K) = -(\Delta K/\Delta L) = MRTS$$

**Isoquants When Inputs are perfectly substitutable****OBSERVATIONS WHEN INPUTS ARE PERFECTLY SUBSTITUTABLE:**

- 1) The MRTS is constant at all points on the isoquant.
- 2) For a given output, any combination of inputs can be chosen (A, B, or C) to generate the same level of output (e.g. toll booths & musical instruments).

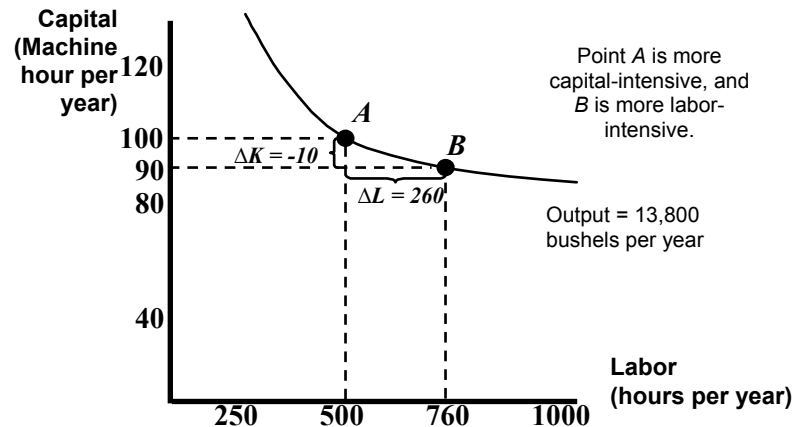
**Fixed-Proportions Production Function**

**OBSERVATIONS WHEN INPUTS MUST BE IN A FIXED-PROPORTION:**

- 1) No substitution is possible. Each output requires a specific amount of each input (e.g. labor and jackhammers).
- 2) To increase output requires more labor and capital (i.e. moving from A to B to C which is technically efficient).

**A PRODUCTION FUNCTION FOR WHEAT**

Farmers must choose between a capital intensive or labor intensive technique of production.

**ISOQUANT DESCRIBING THE PRODUCTION OF WHEAT****OBSERVATIONS:**

- 1) Operating at A:

$L = 500$  hours and  $K = 100$  machine hours.

- 2) Operating at B

Increase  $L$  to 760 and decrease  $K$  to 90 the  $MRTS < 1$ :

$$MRTS = -\frac{\Delta K}{\Delta L} = -(10 / 260) = 0.04$$

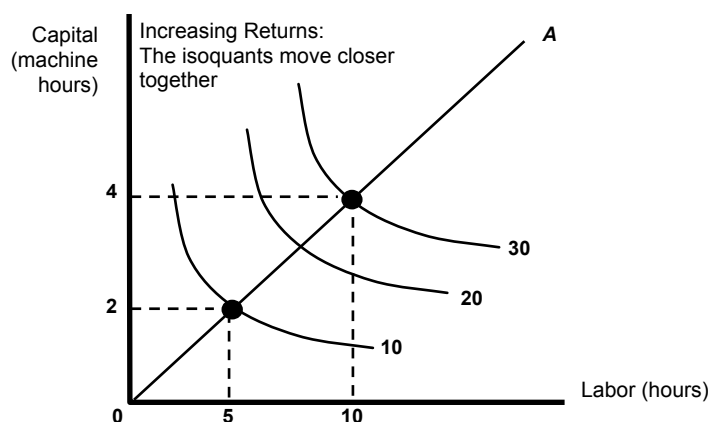
- 3)  $MRTS < 1$ , therefore the cost of labor must be less than capital in order for the farmer substitute labor for capital.
- 4) If labor is expensive, the farmer would use more capital (e.g. U.S.).
- 5) If labor is inexpensive, the farmer would use more labor (e.g. India).

**RETURNS TO SCALE**

Measuring the relationship between the scale (size) of a firm and output

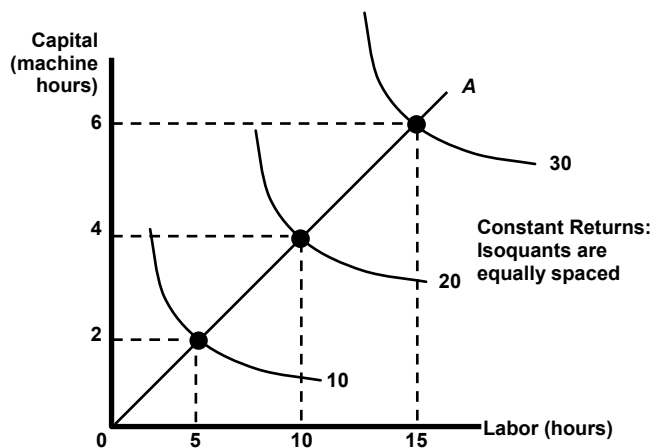
**INCREASING RETURNS TO SCALE:** output more than doubles when all inputs are doubled

- Larger output associated with lower cost (autos)
- One firm is more efficient than many (utilities)
- The isoquants get closer together



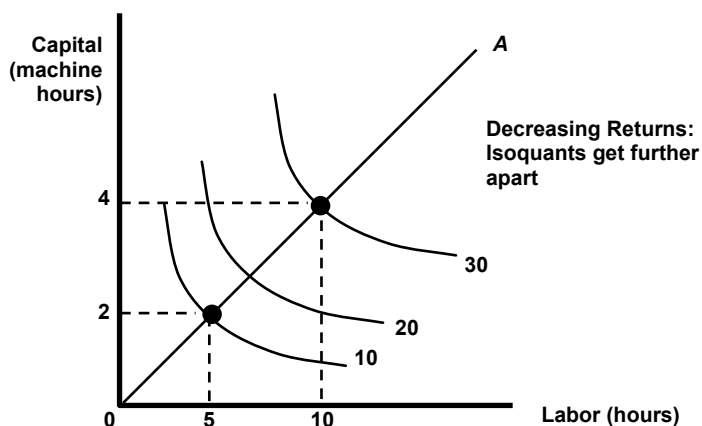
**CONSTANT RETURNS TO SCALE:** output doubles when all inputs are doubled.

- Size does not affect productivity
- May have a large number of producers
- Isoquants are equidistant apart



**DECREASING RETURNS TO SCALE:** output less than doubles when all inputs are doubled

- Decreasing efficiency with large size
- Reduction of entrepreneurial abilities
- Isoquants become farther apart



**RETURNS TO SCALE IN THE CARPET INDUSTRY**

The carpet industry has grown from a small industry to a large industry with some very large firms.

Question:

**Can the growth be explained by the presence of economies to scale?**

**The U.S. Carpet Industry  
Carpet Shipments, 1996  
(Millions of Dollars per Year)**

|                        |         |                          |       |
|------------------------|---------|--------------------------|-------|
| 1. Shaw Industries     | \$3,202 | 6. World Carpets         | \$475 |
| 2. Mohawk Industries   | 1,795   | 7. Burlington Industries | 450   |
| 3. Beaulieu of America | 1,006   | 8. Collins & Aikman      | 418   |
| 4. Interface Flooring  | 820     | 9. Masland Industries    | 380   |
| 5. Queen Carpet        | 775     | 10. Dixied Yarns         | 280   |

Are there economies of scale?

Costs (percent of cost)

- Capital -- 77%
- Labor -- 23%

Large Manufacturers

- Increased in machinery & labor
- Doubling inputs has more than doubled output
- Economies of scale exist for large producers

Small Manufacturers

- Small increases in scale have little or no impact on output
- Proportional increases in inputs increase output proportionally
- Constant returns to scale for small producers

**LESSON 18****THEORY OF COSTS****INTRODUCTION**

The production technology measures the relationship between input and output. Given the production technology, managers must choose how to produce. To determine the optimal level of output and the input combinations, we must convert from the unit measurements of the production technology to dollar measurements or costs.

**MEASURING COST: WHICH COSTS MATTER?****ACCOUNTING COST VS. ECONOMIC COST**

Accounting Cost is the actual expenses plus depreciation charges for capital equipment whereas economic cost is the cost to a firm of utilizing economic resources in production, including opportunity cost.

**OPPORTUNITY COST**

Cost associated with opportunities that are foregone when a firm's resources are not put to their highest-value use.

**AN EXAMPLE**

A firm owns its own building and pays no rent for office space. Does this mean the cost of office space is zero?

**SUNK COST**

Sunk cost is expenditure that has been made and cannot be recovered. It should not influence a firm's decisions.

**AN EXAMPLE**

A firm pays \$500,000 for an option to buy a building. The cost of the building is \$5 million or a total of \$5.5 million. The firm finds another building for \$5.25 million. Which building should the firm buy?

**FIXED AND VARIABLE COSTS**

Total output is a function of variable inputs and fixed inputs. Therefore, the total cost of production equals the fixed cost (the cost of the fixed inputs) plus the variable cost (the cost of the variable inputs), or...

$$TC = FC + VC$$

Fixed Cost does not vary with the level of output whereas Variable Cost is the cost that varies as output varies.

Fixed Cost is the Cost paid by a firm that is in business regardless of the level of output and sunk cost is the Cost that have been incurred and cannot be recovered.

Personal Computers: most costs are variable

- Components, labor

Software: most costs are sunk

- Cost of developing the software

Pizza

- Largest cost component is fixed

### **COST IN THE SHORT RUN**

Marginal Cost (MC) is the cost of expanding output by one unit. Since fixed costs have no impact on marginal cost, it can be written as:

$$MC = \frac{\Delta VC}{\Delta Q} = \frac{\Delta TC}{\Delta Q}$$

Average Total Cost (ATC) is the cost per unit of output, or average fixed cost (AFC) plus average variable cost (AVC). This can be written:

$$ATC = \frac{TFC}{Q} + \frac{TVC}{Q}$$

Average Total Cost (ATC) is the cost per unit of output, or average fixed cost (AFC) plus average variable cost (AVC). This can be written:

$$ATC = AFC + AVC \quad \text{or} \quad \frac{TC}{Q}$$

### **THE DETERMINANTS OF SHORT-RUN COST**

The relationship between the production function and cost can be exemplified by either increasing returns and cost or decreasing returns and cost. With increasing returns, output is increasing relative to input and variable cost and total cost will fall relative to output. With decreasing returns, output is decreasing relative to input and variable cost and total cost will rise relative to output.

For Example: Assume the wage rate ( $w$ ) is fixed relative to the number of workers hired. Then:

$$MC = \frac{\Delta VC}{\Delta Q}$$

$$VC = wL$$

$$MC = \frac{w\Delta L}{\Delta Q}$$

$$\Delta MP_L = \frac{\Delta Q}{\Delta L}$$

$$\Delta L \text{ for a 1 unit } \Delta Q = \frac{\Delta L}{\Delta Q} = \frac{1}{\Delta MP_L}$$

♦In conclusion:

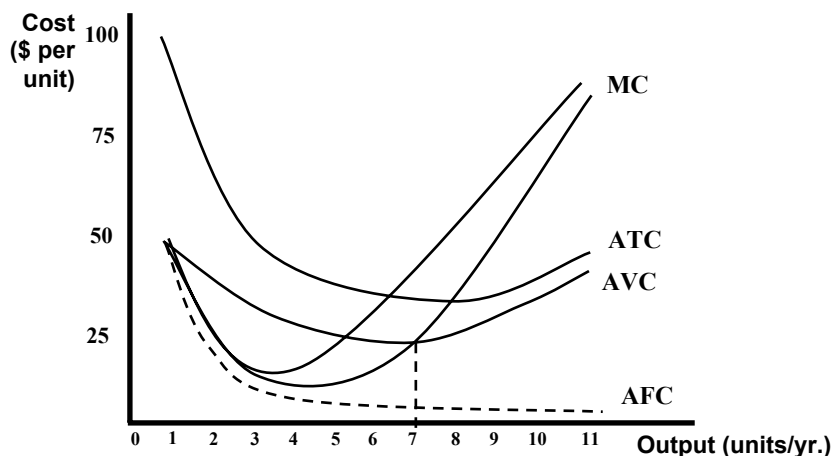
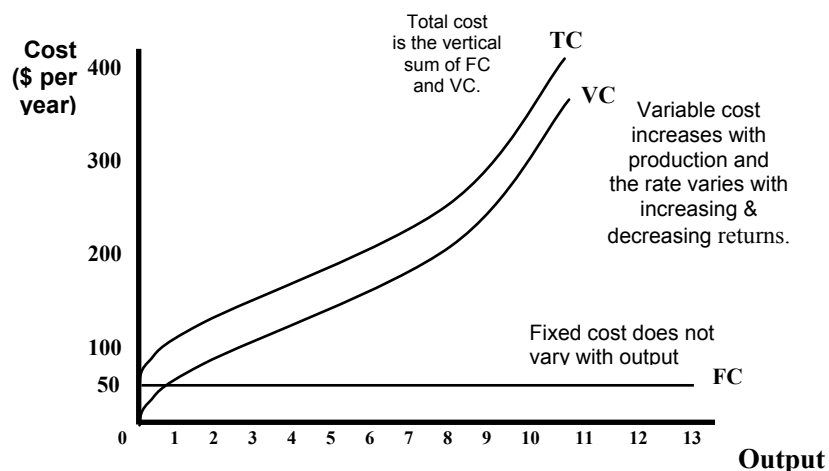
$$MC = \frac{w}{MP_L}$$



**A FIRM'S SHORT-RUN COSTS (\$)**

| Rate of output | Fixed cost (FC) | Variable cost (VC) | Total cost (TC) | Marginal cost (MC) | Average fixed cost (AFC) | Average variable cost (AVC) | Average total cost (ATC) |
|----------------|-----------------|--------------------|-----------------|--------------------|--------------------------|-----------------------------|--------------------------|
| 0              | 50              | 0                  | 50              | ----               | ----                     | ----                        | ----                     |
| 1              | 50              | 50                 | 100             | 50                 | 50                       | 50                          | 100                      |
| 2              | 50              | 78                 | 128             | 28                 | 25                       | 39                          | 64                       |
| 3              | 50              | 98                 | 148             | 20                 | 16.5                     | 32.7                        | 49.3                     |
| 4              | 50              | 112                | 162             | 14                 | 12.5                     | 28                          | 40.5                     |
| 5              | 50              | 130                | 180             | 18                 | 10                       | 26                          | 36                       |
| 6              | 50              | 150                | 200             | 20                 | 8.3                      | 25                          | 33.3                     |
| 7              | 50              | 175                | 225             | 25                 | 7.1                      | 25                          | 32.1                     |
| 8              | 50              | 204                | 254             | 29                 | 6.3                      | 25.5                        | 31.8                     |
| 9              | 50              | 242                | 292             | 38                 | 5.6                      | 26.9                        | 32.4                     |
| 10             | 50              | 300                | 350             | 58                 | 5                        | 30                          | 35                       |
| 11             | 50              | 385                | 435             | 85                 | 4.5                      | 35                          | 39.5                     |

Consequently (from the table), MC decreases initially with increasing returns i.e; 0 through 4 units of output and then MC increases with decreasing returns i.e; 5 through 11 units of output.



The line drawn from the origin to the tangent of the variable cost curve:

- Its slope equals AVC
- The slope of a point on VC equals MC
- Therefore,  $MC = AVC$  at 7 units of output (point A)

#### Unit Costs

- AFC falls continuously
- When  $MC < AVC$  or  $MC < ATC$ , AVC & ATC decrease
- When  $MC > AVC$  or  $MC > ATC$ , AVC & ATC increase
- $MC = AVC$  and  $ATC$  at minimum AVC and ATC
- Minimum AVC occurs at a lower output than minimum ATC due to FC

## LESSON 19

## THEORY OF COSTS (Continued)

## THE USER COST OF CAPITAL

User Cost of Capital = Economic Depreciation + (Interest Rate)(Value of Capital)

## EXAMPLE

An Airline buys a Boeing 737 for \$150 million with an expected life of 30 years. Its Annual economic depreciation = \$150 million/30 = \$5 million and Interest rate = 10%

User Cost of Capital = \$5 million + (.10) (\$150 million – depreciation)

Year 1 = \$5 million + (.10)(\$150 million) = \$20 million

Year 10 = \$5 million + (.10) (\$100 million) = \$15 million

Rate per dollar of capital

$$r = \text{Depreciation Rate} + \text{Interest Rate}$$

## AIRLINE EXAMPLE

If Depreciation Rate =  $1/30 = 3.33/\text{yr}$  and rate of Return = 10%/yr

User Cost of Capital

$$r = 3.33 + 10 = 13.33\%/\text{yr}$$

## THE COST MINIMIZING INPUT CHOICE

Assumptions

- Two Inputs: Labor (L) & capital (K)
- Price of labor: wage rate (w)
- The price of capital
- $R = \text{depreciation rate} + \text{interest rate}$

Question

- If capital was rented, would it change the value of  $r$ ?

## THE ISOCOST LINE

$$C = wL + rK$$

Isocost is a line showing all combinations of L & K that can be purchased for the same cost

Rewriting C as linear:

$$K = C/r - (w/r)L$$

Slope of the Isocost is the ratio of the wage rate to rental cost of capital.

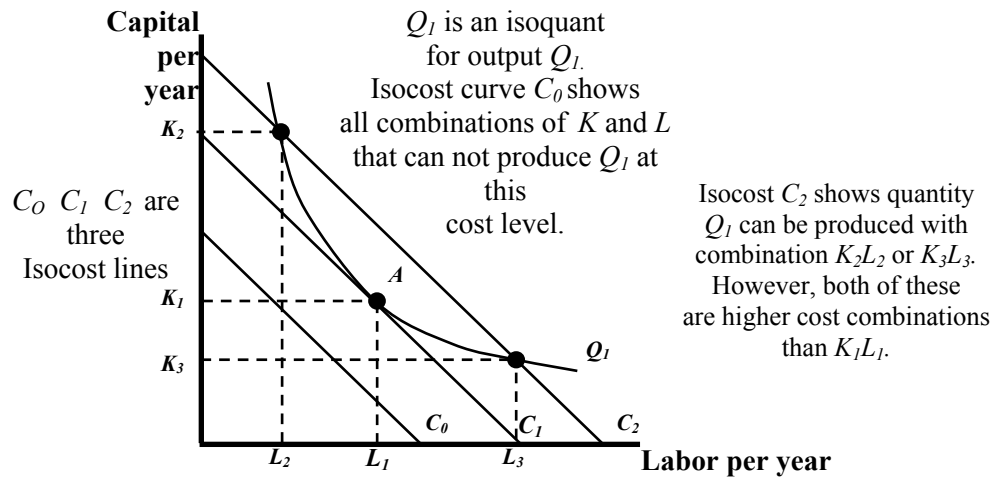
$$\Delta K / \Delta L = -\left(\frac{w}{r}\right)$$

This shows the rate at which capital can be substituted for labor with no change in cost.

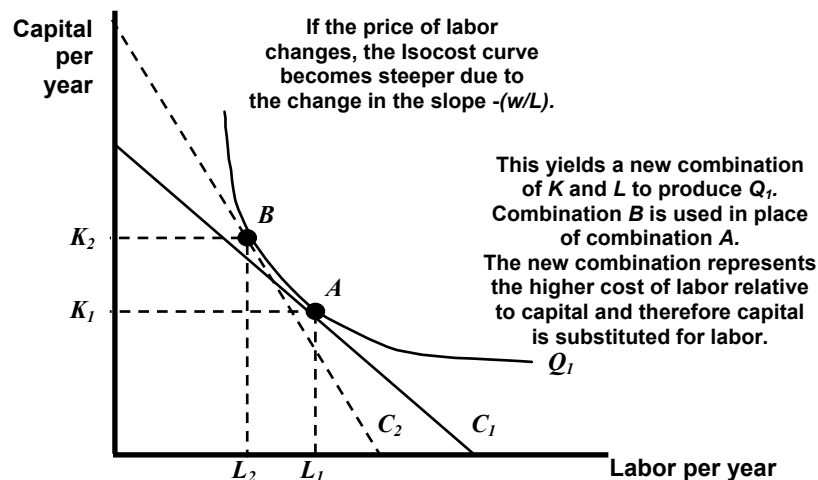
## CHOOSING INPUTS

We will address how to minimize cost for a given level of output. We will do so by combining Isocosts with Isoquants.

### Producing a Given Output at Minimum Cost



### Input Substitution When an Input Price Change



### ISOQUANTS AND ISOCOSTS AND THE PRODUCTION FUNCTION

$$MRTS = -\frac{\Delta K}{\Delta L} = \frac{MP_L}{MP_K}$$

$$\text{Slope of isocost line} = \frac{\Delta K}{\Delta L} = -\frac{w}{r}$$

$$\text{and } = \frac{MP_L}{MP_K} = \frac{w}{r}$$

The minimum cost combination can then be written as:

$$\frac{MP_L}{w} = \frac{MP_K}{r}$$

Minimum cost for a given output will occur when each dollar of input added to the production process will add an equivalent amount of output.

## Question

If  $w = \$10$ ,  $r = \$2$ , and  $MP_L = MP_K$ , which input would the producer use more of? Why?

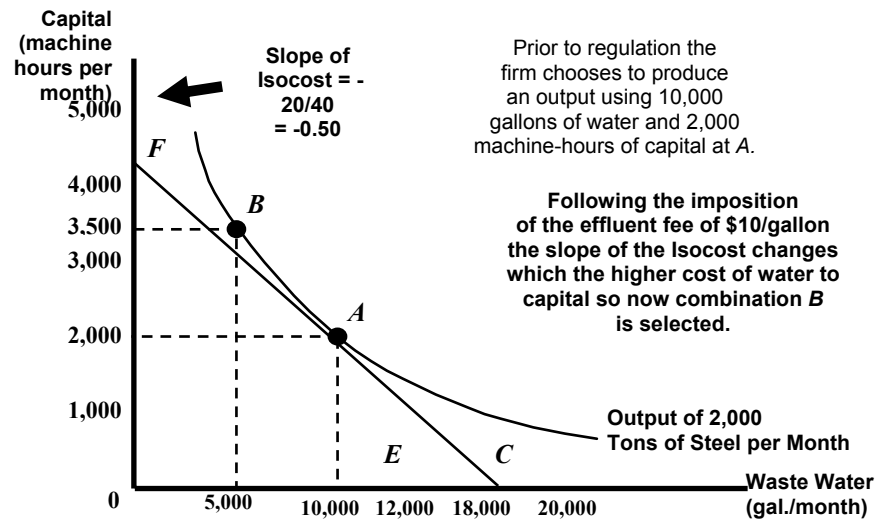
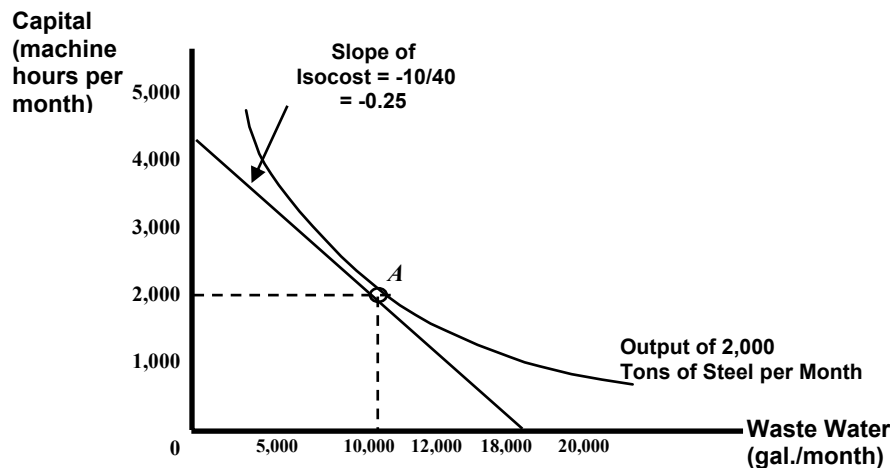
### THE EFFECT OF EFFLUENT FEES ON FIRMS' INPUT CHOICES

Firms that have a by-product to production produce an effluent. An effluent fee is a per-unit fee that firms must pay for the effluent that they emit. How would a producer respond to an effluent fee on production?

The Scenario: Steel Producer

- 1) Located on a river: Low cost transportation and emission disposal (effluent).
- 2) EPA imposes a per unit effluent fee to reduce the environmentally harmful effluent.
- 3) How should the firm respond?

### THE COST-MINIMIZING RESPONSE TO AN EFFLUENT FEE



Observations:

- The more easily factors can be substituted; the more effective the fee is in reducing the effluent.

The greater the degree of substitutes, the less the firm will have to pay (e.g.: \$50,000 with combination B instead of \$100,000 with combination A).

## LESSON 20

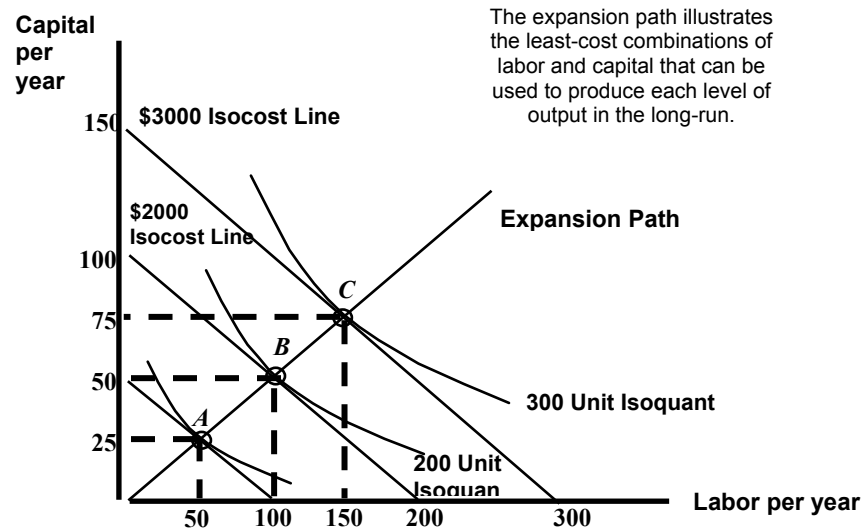
## THEORY OF COSTS (Continued)

## COST IN THE LONG RUN

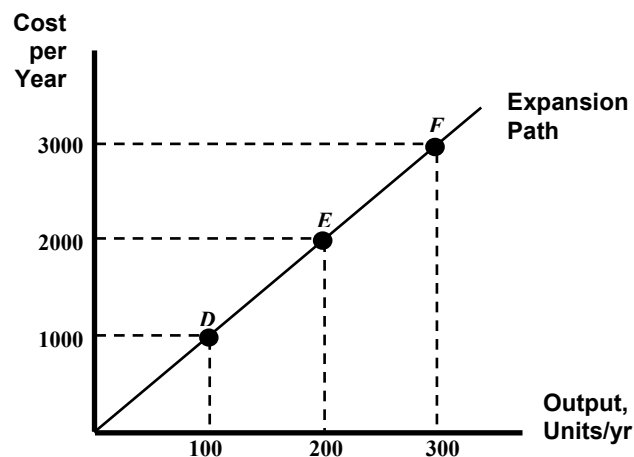
Cost minimization with Varying Output Levels

A firm's expansion path shows the minimum cost combinations of labor and capital at each level of output.

## A FIRM'S EXPANSION PATH



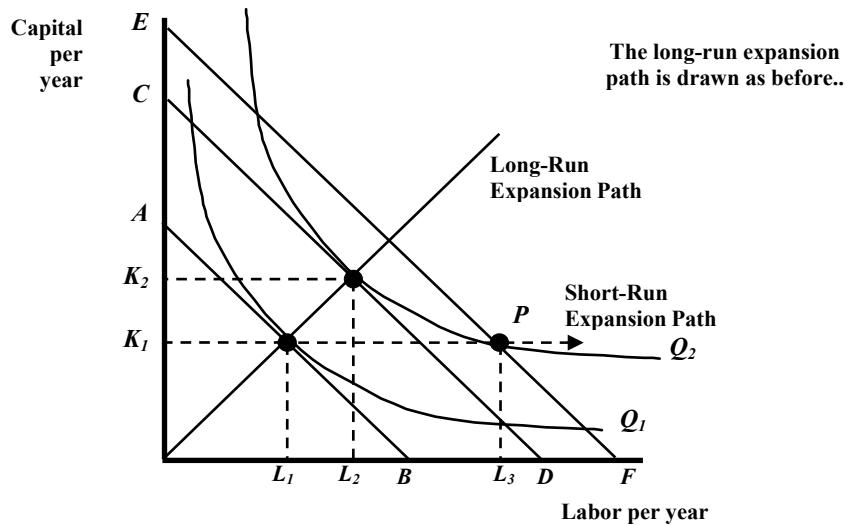
## A FIRM'S LONG RUN TOTAL COST CURVE



## LONG-RUN VERSUS SHORT-RUN COST CURVES

What happens to average costs when both inputs are variable (long run) versus only having one input that is variable (short run)?

### THE INFLEXIBILITY OF SHORT-RUN PRODUCTION



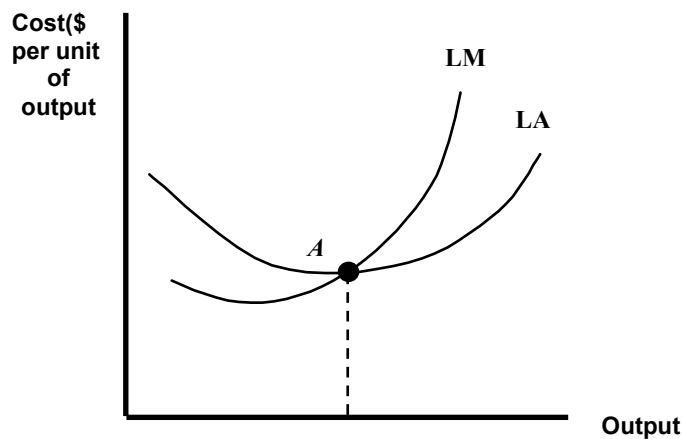
### LONG-RUN AVERAGE COST (LAC)

If input is doubled, output will double and average cost is constant at all levels of output. If input is doubled, output will more than double and average cost decreases at all levels of output. If input is doubled, the increase in output is less than twice as large and average cost increases with output. In the long-run, Firms experience increasing and decreasing returns to scale and therefore long-run average cost is “U” shaped.

Long-run marginal cost leads long-run average cost:

- If  $LMC < LAC$ , LAC will fall
- If  $LMC > LAC$ , LAC will rise
- Therefore,  $LMC = LAC$  at the minimum of LAC

### LONG-RUN AVERAGE AND MARGINAL COST



Question

What is the relationship between long-run average cost and long-run marginal cost when long-run average cost is constant?

### ECONOMIES AND DISECONOMIES OF SCALE

In the case of economies of scale, increase in output is greater than the increase in inputs. Whereas, in case of diseconomies of scale, increase in output is less than the increase in inputs.

**MEASURING ECONOMIES OF SCALE**

$$E_c = \text{Cost} - \text{output elasticity}$$

$$= \% \Delta \text{ in cost from a 1\% increase}$$

$$\text{in output}$$

Therefore, the following is true:

$$E_c < 1: MC < AC$$

Average cost indicate decreasing economies of scale

$$E_c = 1: MC = AC$$

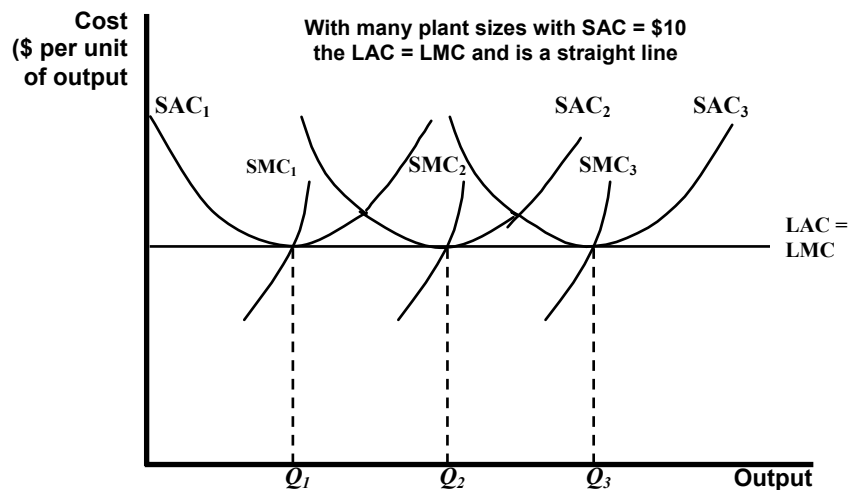
Average cost indicate constant economies of scale

$$E_c > 1: MC > AC$$

Average cost indicate increasing economies of scale

**THE RELATIONSHIP BETWEEN SHORT-RUN AND LONG-RUN COST**

We will use short and long-run cost to determine the optimal plant size.

**LONG-RUN COST WITH CONSTANT RETURNS TO SCALE****Observation**

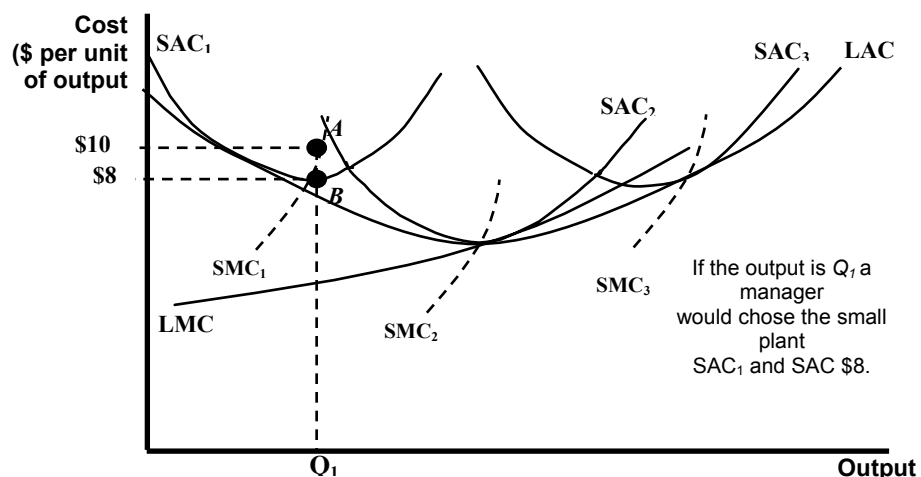
The optimal plant size will depend on the anticipated output (e.g. Q<sub>1</sub> choose SAC<sub>1</sub>, etc). The long-run average cost curve is the envelope of the firm's short-run average cost curves.

**Question**

What would happen to average cost if an output level other than that shown is chosen?



### Long-Run Cost with Economies & Diseconomies of Scale



What is the firms' long-run cost curve?

Firms can change scale to change output in the long-run. The long-run cost curve is the dark blue portion of the SAC curve which represents the minimum cost for any level of output.

Observations

The LAC does not include the minimum points of small and large size plants? Why not?

LMC is not the envelope of the short-run marginal cost. Why not?

## LESSON 21

## THEORY OF COSTS (Continued)

**PRODUCTION WITH TWO OUTPUTS--ECONOMIES OF SCOPE**

Economies of scope exist when the joint output of a single firm is greater than the output that could be achieved by two different firms each producing a single output.

Examples:

- Chicken farm--poultry and eggs
- Automobile company--cars and trucks
- University--Teaching and research

What are the advantages of joint production?

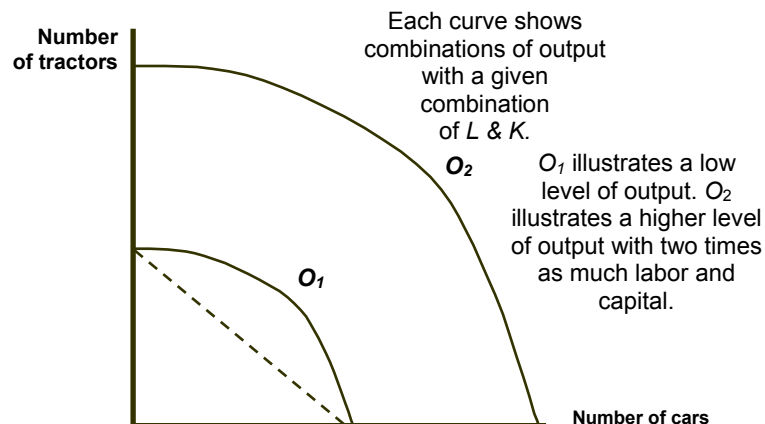
Consider an automobile company producing cars and tractors

Advantages

- Both use capital and labor.
- The firms share management resources.
- Both use the same labor skills and type of machinery.

Production:

Firms must choose how much of each to produce. The alternative quantities can be illustrated using product transformation curves.

**PRODUCT TRANSFORMATION CURVE**

Observations

- Product transformation curves are negatively sloped
- Constant returns exist in this example
- Since the production transformation curve is concave is joint production desirable?
- There is no direct relationship between economies of scope and economies of scale.
- May experience economies of scope and diseconomies of scale
- May have economies of scale and not have economies of scope

**The degree of economies of scope** measures the savings in cost and can be written:

$$SC = \frac{C(Q_1) + C(Q_2) - C(Q_1, Q_2)}{C(Q_1, Q_2)}$$

$C(Q_1)$  is the cost of producing  $Q_1$

$C(Q_2)$  is the cost of producing  $Q_2$

$C(Q_1Q_2)$  is the joint cost of producing both products

Interpretation:

- If  $SC > 0$  -- Economies of scope
- If  $SC < 0$  -- Diseconomies of scope

Issues

- Truckload versus less than truck load
- Direct versus indirect routing
- Length of haul

## ECONOMIES OF SCOPE IN THE TRUCKING INDUSTRY

Questions:

Are large-scale, direct hauls cheaper and more profitable than individual hauls by small trucks? Are there cost advantages from operating both direct and indirect hauls?

Empirical Findings

An analysis of 105 trucking firms examined four distinct outputs.

- 1) Short hauls with partial loads
- 2) Intermediate hauls with partial loads
- 3) Long hauls with partial loads
- 4) Hauls with total loads

Results

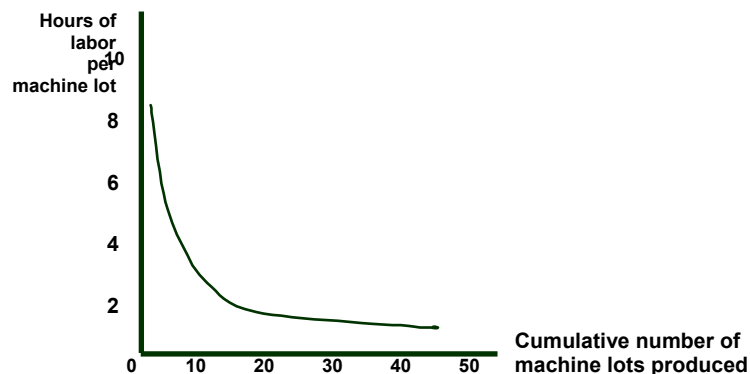
- $SC = 1.576$  for reasonably large firm
- $SC = 0.104$  for very large firms

Interpretation

Combining partial loads at an intermediate location lowers cost management difficulties with very large firms.

## DYNAMIC CHANGES IN COSTS--THE LEARNING CURVE

The **learning curve** measures the impact of worker's experience on the costs of production. It describes the relationship between a firm's cumulative output and amount of inputs needed to produce a unit of output.

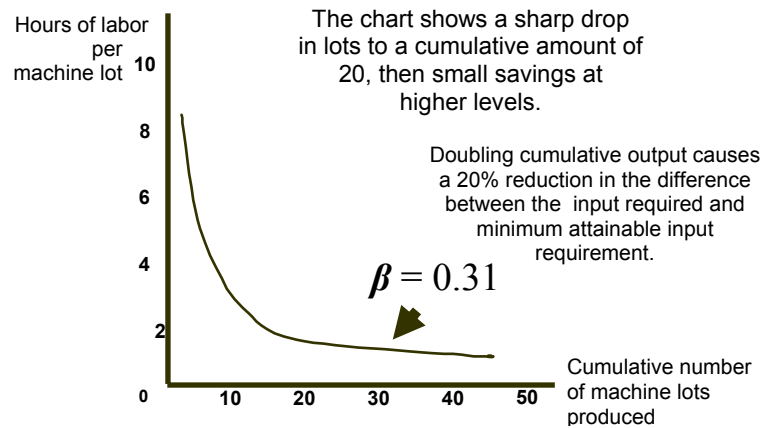


The horizontal axis measures the cumulative number of hours of machine tools the firm has produced. The vertical axis measures the number of hours of labor needed to produce each lot.

The learning curve in the figure is based on the relationship:

$$L = A + BN^{\beta}$$

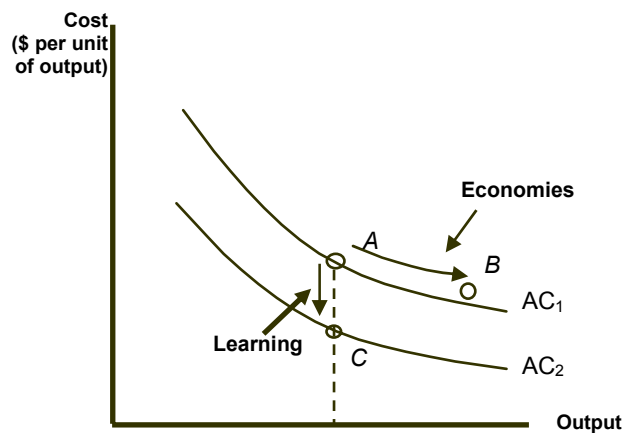
If  $N=1$ ,  $L$  equals  $A + B$  and this measures labor input to produce the first unit of output. If  $\beta = 0$ , Labor input remains constant as the cumulative level of output increases, so there is no learning. If  $\beta > 0$  and  $N$  increases,  $L$  approaches  $A$ , and  $A$  represent minimum labor input/unit of output after all learning has taken place. The larger  $\beta$ , The more important the learning effect.



### Observations

- 1) New firms may experience a learning curve, not economies of scale.
- 2) Older firms have relatively small gains from learning.

### ECONOMIES OF SCALE VERSUS LEARNING



**PREDICTING THE LABOR REQUIREMENTS OF PRODUCING A GIVEN OUTPUT**

| <b>Cumulative Output<br/>(N)</b> | <b>Per-Unit Labor Requirement<br/>for each 10 units of Output (L)</b> | <b>Total Labor<br/>Requirement</b> |
|----------------------------------|-----------------------------------------------------------------------|------------------------------------|
| 10                               | 1.00                                                                  | 10.0                               |
| 20                               | .80                                                                   | 18.0 (10.0 + 8.0)                  |
| 30                               | .70                                                                   | 25.0 (18.0 + 7.0)                  |
| 40                               | .64                                                                   | 31.4 (25.0 + 6.4)                  |
| 50                               | .60                                                                   | 37.4 (31.4 + 6.0)                  |
| 60                               | .56                                                                   | 43.0 (37.4 + 5.6)                  |
| 70                               | .53                                                                   | 48.3 (43.0 + 5.3)                  |
| 80 and over                      | .51                                                                   | 53.4 (48.3 + 5.1)                  |

The learning curve implies:

- 1) The labor requirement falls per unit.
- 2) Costs will be high at first and then will fall with learning.
- 3) After 8 years the labor requirement will be 0.51 and per unit cost will be half what it was in the first year of production?

**LEARNING CURVE IN PRACTICE**

Scenario

A new firm enters the chemical processing industry. *Do they:*

- 1) Produce a low level of output and sell at a high price?
- 2) Produce a high level of output and sell at a low price?

How would the learning curve influence your decision?

The Empirical Findings

- Study of 37 chemical products
- Average cost fell 5.5% per year
- For each doubling of plant size, average production costs fall by 11%
- For each doubling of cumulative output, the average cost of production falls by 27%

Which is more important, the economies of scale or learning effects?

Other Empirical Findings

- In the semi-conductor industry a study of seven generations of DRAM semiconductors from 1974-1992 found learning rates averaged 20%.
- In the aircraft industry the learning rates are as high as 40%.

Applying Learning Curves

- 1) To determine if it is profitable to enter an industry.
- 2) To determine when profits will occur based on plant size and cumulative output.

**LESSON 22****PERFECTLY COMPETITIVE MARKETS****CHARACTERISTICS OF PERFECTLY COMPETITIVE MARKETS**

- 1) Price taking
- 2) Product homogeneity
- 3) Free entry and exit

**PRICE TAKING**

The individual firm sells a very small share of the total market output and, therefore, cannot influence market price. The individual consumer buys too small a share of industry output to have any impact on market price.

**PRODUCT HOMOGENEITY**

The products of all firms are perfect substitutes. Examples: Agricultural products, oil, copper, iron, lumber

**FREE ENTRY AND EXIT**

Buyers can easily switch from one supplier to another. Suppliers can easily enter or exit a market.

**Discussion Questions**

What are some barriers to entry and exit?  
Are all markets competitive?  
When is a market highly competitive?  
Do firms maximize profits?

**Possibility of other objectives**

- Revenue maximization
- Dividend maximization
- Short-run profit maximization

**Implications of non-profit objective**

- Over the long-run investors would not support the company
- Without profits, survival unlikely

Long-run profit maximization is valid and does not exclude the possibility of altruistic behavior.

**MARGINAL REVENUE, MARGINAL COST & PROFIT MAXIMIZATION**

Determining the profit maximizing level of output

$$\text{Profit } (\pi) = \text{Total Revenue} - \text{Total Cost}$$

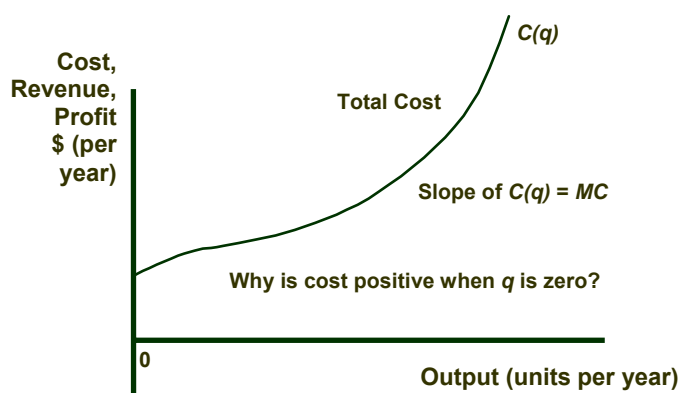
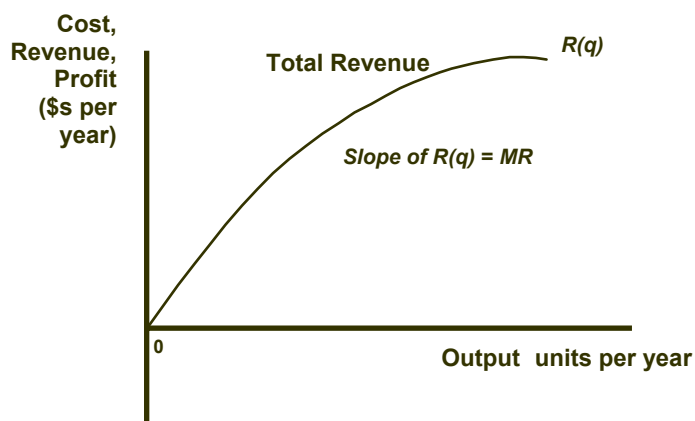
$$\text{Total Revenue } (R) = Pq$$

$$\text{Total Cost } (C) = Cq$$

Therefore:

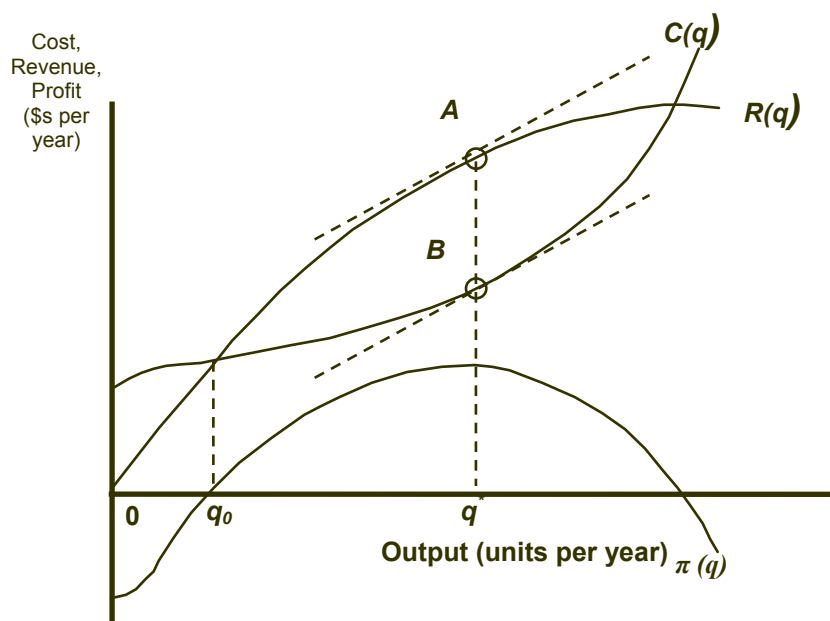
$$\pi(q) = R(q) - C(q)$$

## PROFIT MAXIMIZATION IN THE SHORT RUN



## MARGINAL REVENUE, MARGINAL COST & PROFIT MAXIMIZATION

**Marginal revenue** is the additional revenue from producing one more unit of output. **Marginal cost** is the additional cost from producing one more unit of output.



Comparing  $R(q)$  and  $C(q)$

Output levels: 0-  $q_0$ :  
 $C(q) > R(q)$

### NEGATIVE PROFIT

$$FC + VC > R(q)$$

$$MR > MC$$

Indicates higher profit at higher output

Question: Why is profit negative when output is zero?

Output levels:  $q_0 - q^*$

- $R(q) > C(q)$
- $MR > MC$

Indicates higher profit at higher output while Profit is increasing

Output level:  $q^*$

$$R(q) = C(q)$$

$$MR = MC$$

Profit is maximized

Question

Why is profit reduced when producing more or less than  $q^*$ ?

Output levels beyond  $q^*$ :

$$R(q) > C(q)$$

$$MC > MR$$

Profit is decreasing

Therefore, it can be said:

Profits are maximized when  $MC = MR$ .

$$\pi = R - C \quad MR = \frac{\Delta R}{\Delta q} \quad MC = \frac{\Delta C}{\Delta q}$$

Profits are maximized when :

$$\frac{\Delta \pi}{\Delta q} = \frac{\Delta R}{\Delta q} - \frac{\Delta C}{\Delta q} = 0 \text{ or } MR - MC = 0 \text{ so that}$$

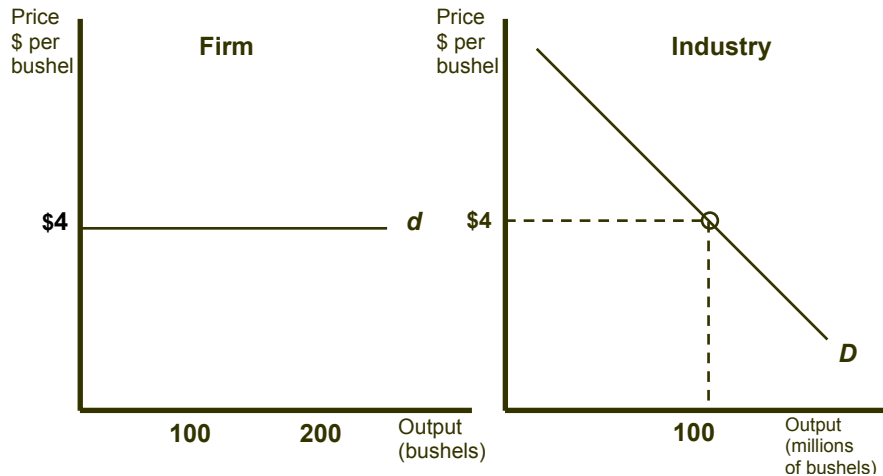
$$MR(q) = MC(q)$$

The Competitive Firm

- Price taker
- Market output (Q) and firm output (q)
- Market demand (D) and firm demand (d)
- $R(q)$  is a straight line



### DEMAND & MARGINAL REVENUE FACED BY A COMPETITIVE FIRM



Individual producer sells all units for \$4 regardless of the producer's level of output. If the producer tries to raise price, sales are zero. If the producers tries to lower price he cannot increase sales

$$P = D = MR = AR$$

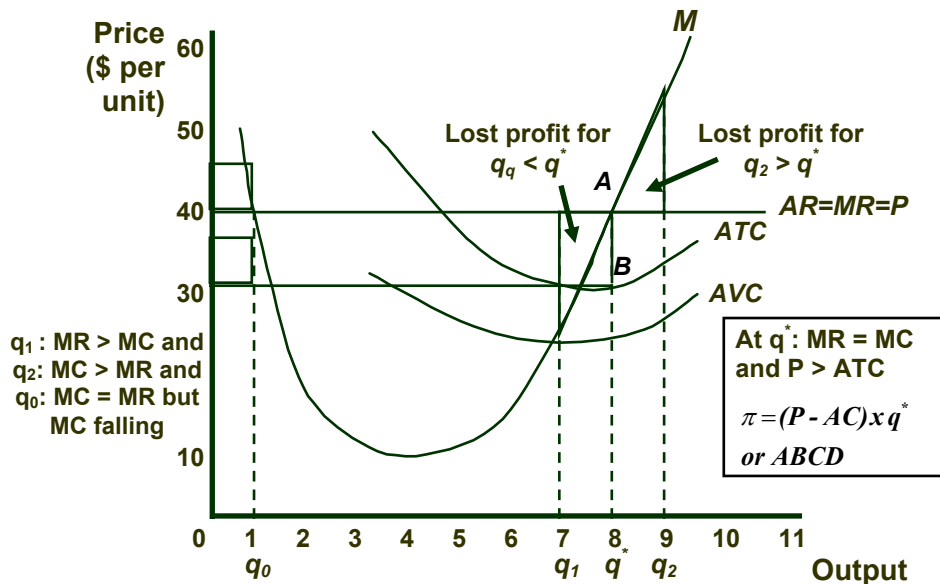
Profit Maximization point

$$MC(q) = MR = P$$

### CHOOSING OUTPUT IN SHORT RUN

We will combine production and cost analysis with demand to determine output and profitability.

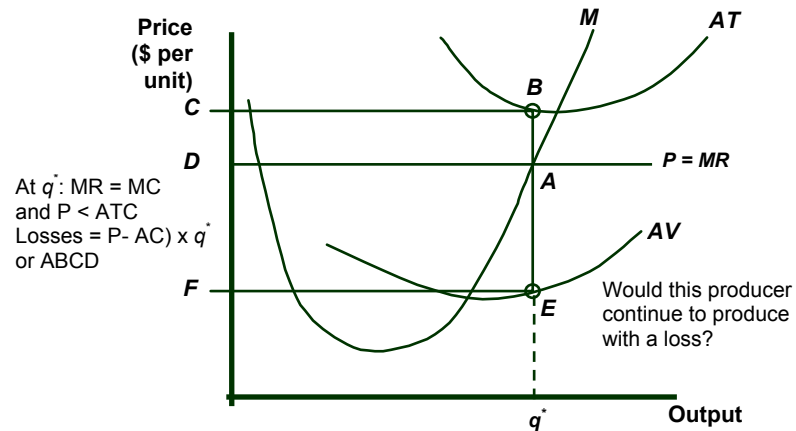
### A COMPETITIVE FIRM MAKING A POSITIVE PROFIT



## LESSON 23

## PERFECTLY COMPETITIVE MARKETS (Continued)

## A COMPETITIVE FIRM INCURRING LOSSES

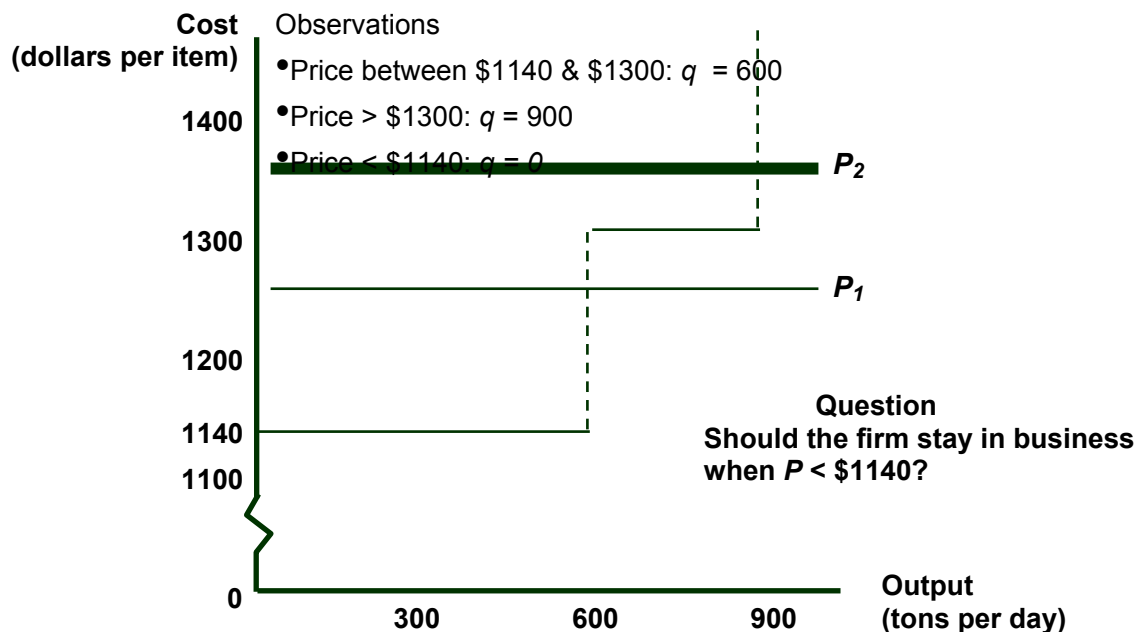


## CHOOSING OUTPUT IN SHORT RUN

## Summary of Production Decisions

- Profit is maximized when  $MC = MR$
- If  $P > ATC$  the firm is making profits.
- If  $AVC < P < ATC$  the firm should produce at a loss.
- If  $P < AVC < ATC$  the firm should shut-down.

## THE SHORT-RUN OUTPUT OF AN ALUMINUM SMELTING PLANT

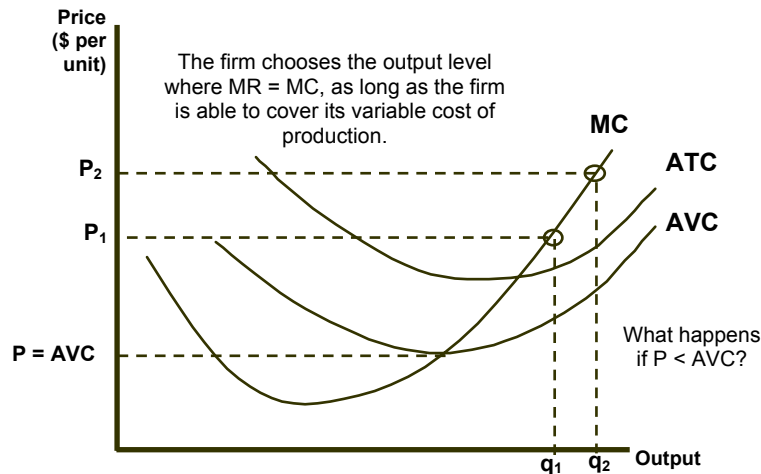


## SOME COST CONSIDERATIONS FOR MANAGERS

Three guidelines for estimating marginal cost:

- 1) Average variable cost should not be used as a substitute for marginal cost.
- 2) A single item on a firm's accounting ledger may have two components, only one of which involves marginal cost.
- 3) All opportunity cost should be included in determining marginal cost.

### A COMPETITIVE FIRM'S SHORT-RUN SUPPLY CURVE

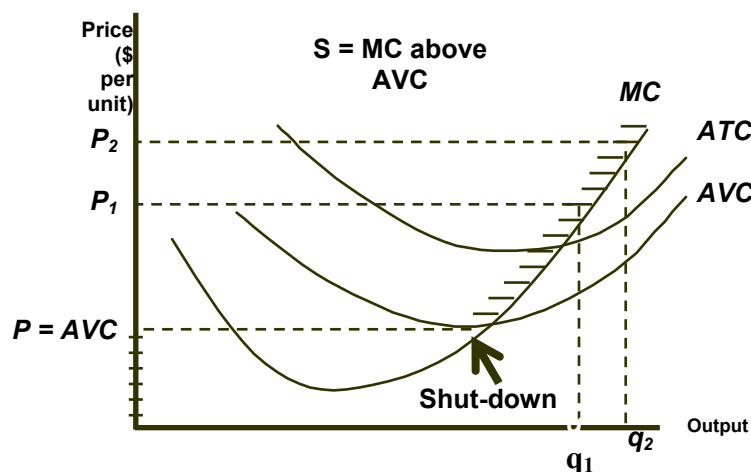


Observations:

- $P = MR$
- $MR = MC$
- $P = MC$

Supply is the amount of output for every possible price. Therefore:

- If  $P = P_1$ , then  $q = q_1$
- If  $P = P_2$ , then  $q = q_2$

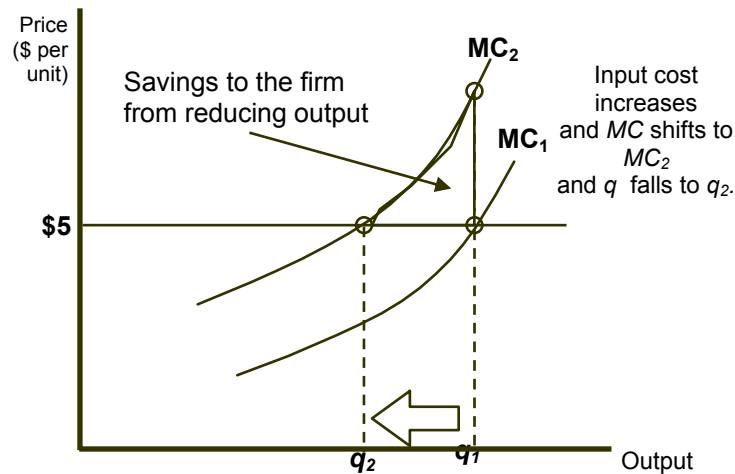


Observations:

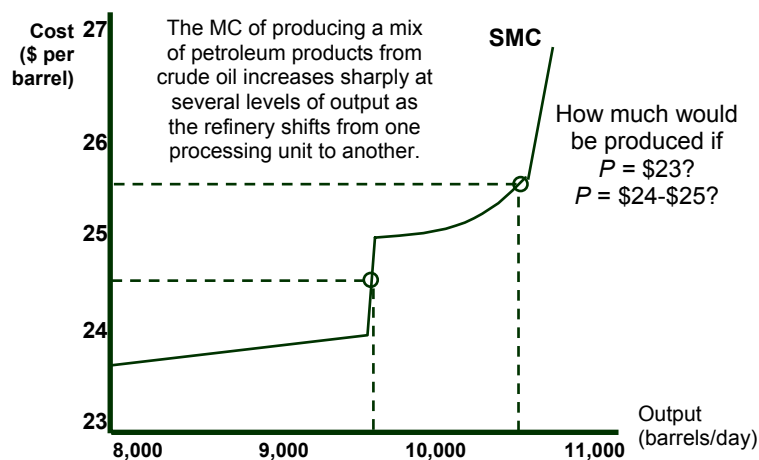
Supply is upward sloping due to diminishing returns. Higher price compensates the firm for higher cost of additional output and increases total profit because it applies to all units.

### FIRM'S RESPONSE TO AN INPUT PRICE CHANGE

When the price of a firm's product changes, the firm changes its output level, so that the marginal cost of production remains equal to the price.



### THE SHORT-RUN PRODUCTION OF PETROLEUM PRODUCTS



Stepped SMC indicates a different production (cost) process at various capacity levels.

Observation:

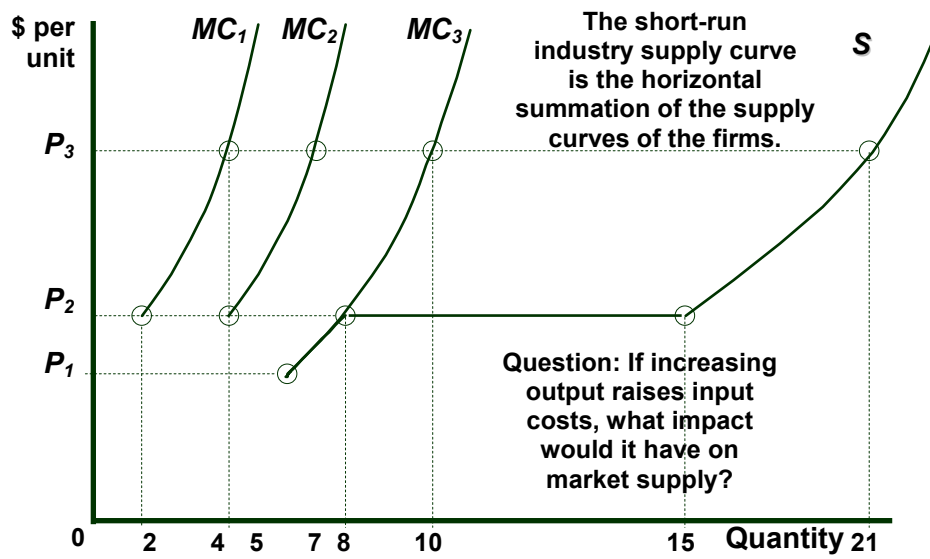
With a stepped MC function, small changes in price may not trigger a change in output.

### THE SHORT-RUN MARKET SUPPLY CURVE

The short-run market supply curve shows the amount of output that the industry will produce in the short-run for every possible price.

Consider, for simplicity, a competitive market with three firms:

## Industry Supply in Short Run



## LESSON 24

## EQUILIBRIUM IN PERFECTLY COMPETITIVE MARKETS

## ELASTICITY OF MARKET SUPPLY

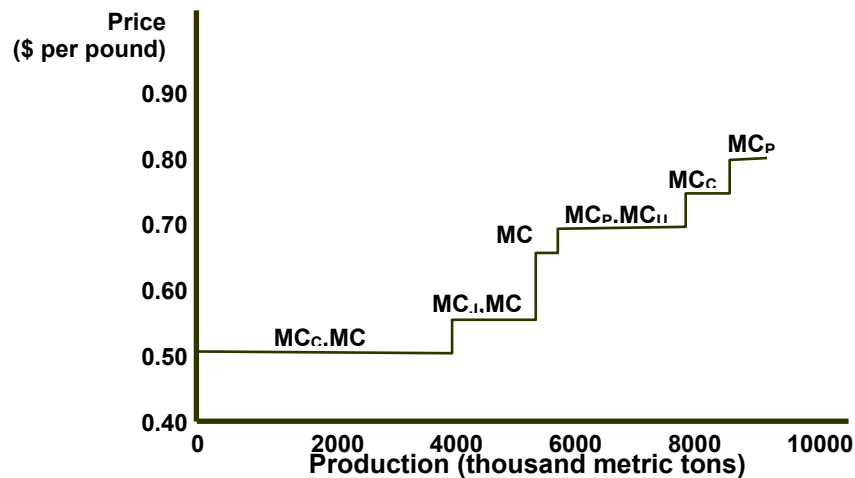
$$E_s = (\Delta Q / Q) / (\Delta P / P)$$

Perfectly inelastic short-run supply arises when the industry's plant and equipment are so fully utilized that new plants must be built to achieve greater output. Perfectly elastic short-run supply arises when marginal costs are constant.

## THE WORLD COPPER INDUSTRY (1999)

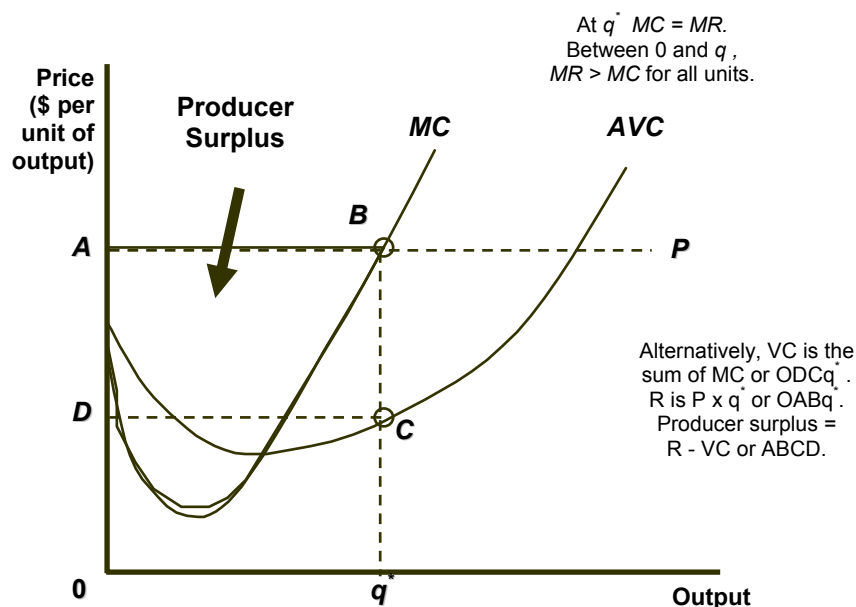
| Country       | Annual Production<br>(thousand metric tons) | Marginal Cost<br>(dollars/pound) |
|---------------|---------------------------------------------|----------------------------------|
| Australia     | 600                                         | 0.65                             |
| Canada        | 710                                         | 0.75                             |
| Chile         | 3660                                        | 0.50                             |
| Indonesia     | 750                                         | 0.55                             |
| Peru          | 450                                         | 0.70                             |
| Poland        | 420                                         | 0.80                             |
| Russia        | 450                                         | 0.50                             |
| United States | 1850                                        | 0.55                             |

## THE SHORT-RUN WORLD SUPPLY OF COPPER



## PRODUCER SURPLUS IN THE SHORT RUN

Firms earn a surplus on all but the last unit of output. The **producer surplus** is the sum over all units produced of the difference between the market price of the good and the marginal cost of production.



Producer Surplus in the Short-Run

$$\text{Producer Surplus} = PS = R - VC$$

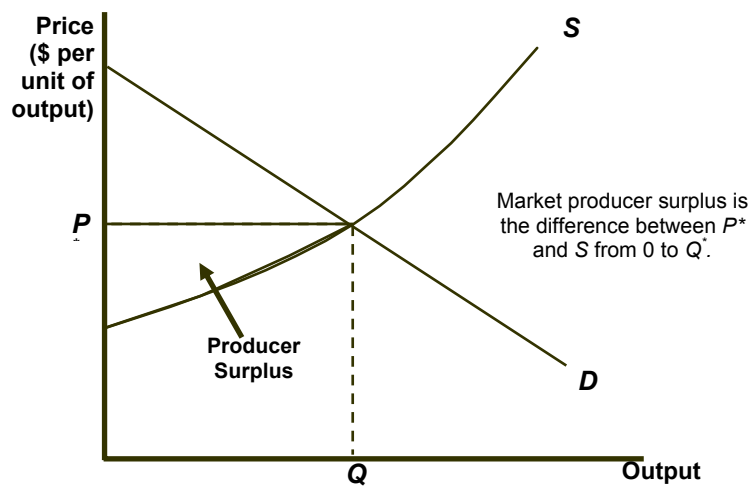
$$\text{Profit} = \pi = R - VC - FC$$

Observation

Short-run with positive fixed cost

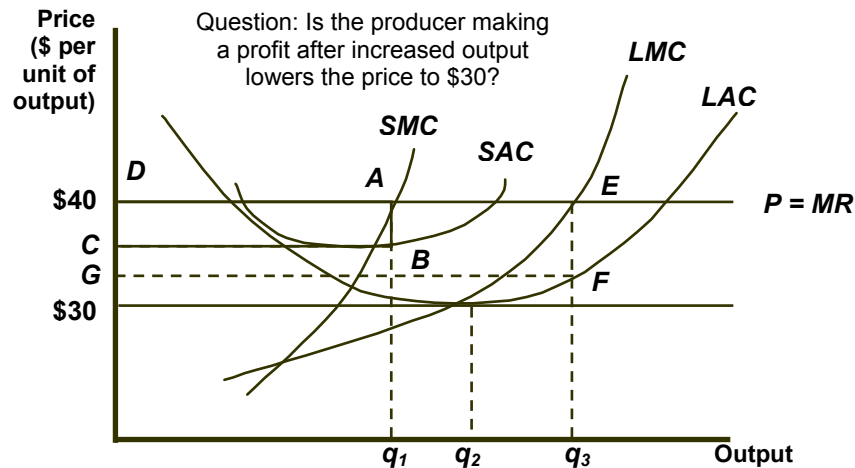
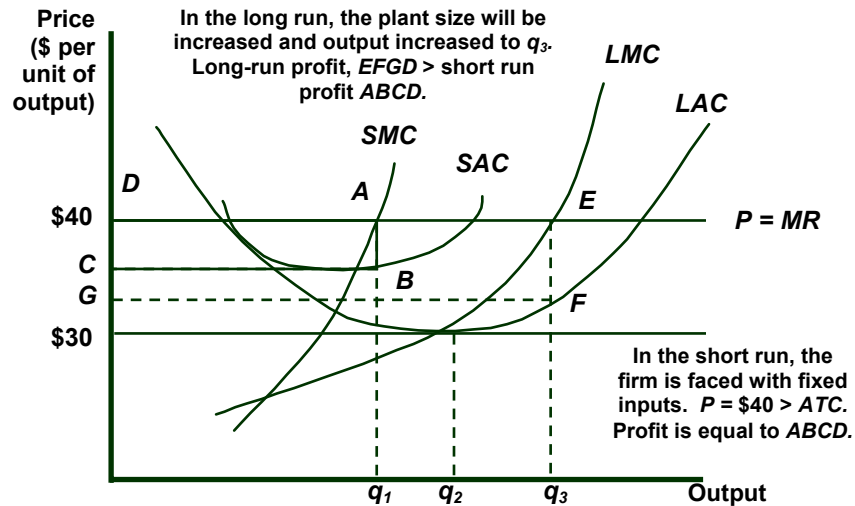
$$PS > \pi$$

### PRODUCER SURPLUS FOR A MARKET



### CHOOSING OUTPUT IN LONG RUN

In the long run, a firm can alter all its inputs, including the size of the plant. We assume free entry and free exit.



### ACCOUNTING PROFIT & ECONOMIC PROFIT

$$\text{Accounting profit } (\pi) = R - wL$$

$$\text{Economic profit } (\pi) = R - wL - rK$$

Where

$wL$  = labor cost

$rK$  = opportunity cost of capital

### ZERO-PROFIT

- If  $R > wL + rK$ , economic profits are positive
- If  $R = wL + rK$ , zero economic profits, but the firm is earning a normal rate of return; indicating the industry is competitive
- If  $R < wL + rK$ , consider going out of business

### ENTRY AND EXIT

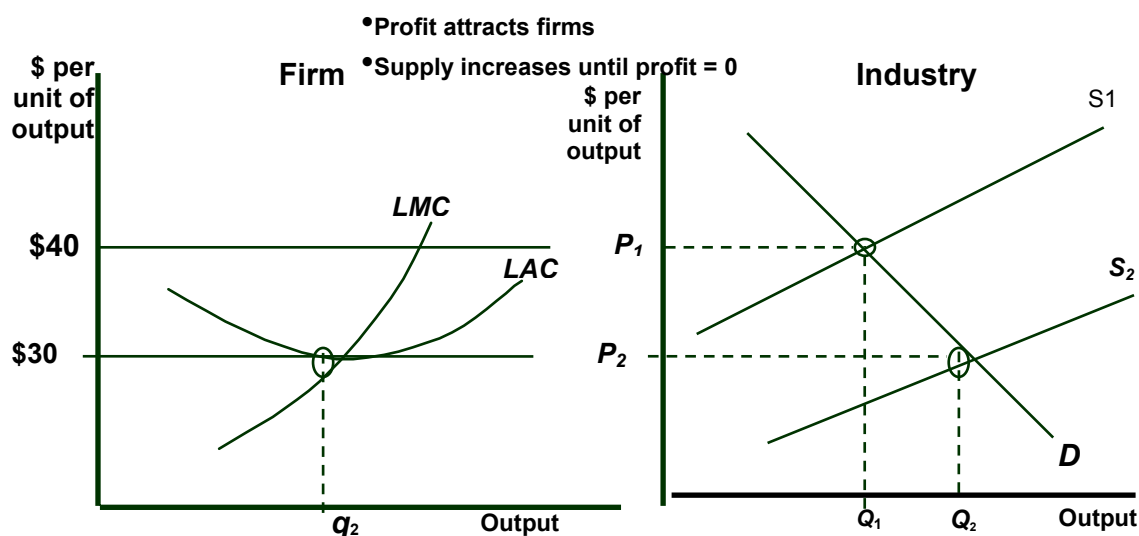
The long-run response to short-run profits is to increase output and profits. Profits will attract other producers. More producers increase industry supply which lowers the market price.



## LESSON 25

## EQUILIBRIUM IN PERFECTLY COMPETITIVE MARKETS (CONTINUED)

## LONG-RUN COMPETITIVE EQUILIBRIUM



## LONG-RUN COMPETITIVE EQUILIBRIUM

- 1)  $MC = MR$
- 2)  $P = LAC$ 
  - No incentive to leave or enter
  - Profit = 0

## 3) Equilibrium Market Price

## Questions

- 1) Explain the market adjustment when  $P < LAC$  and firms have identical costs.
- 2) Explain the market adjustment when firms have different costs.
- 3) What is the opportunity cost of land?

## ECONOMIC RENT

Economic rent is the difference between what firms are willing to pay for an input less the minimum amount necessary to obtain it.

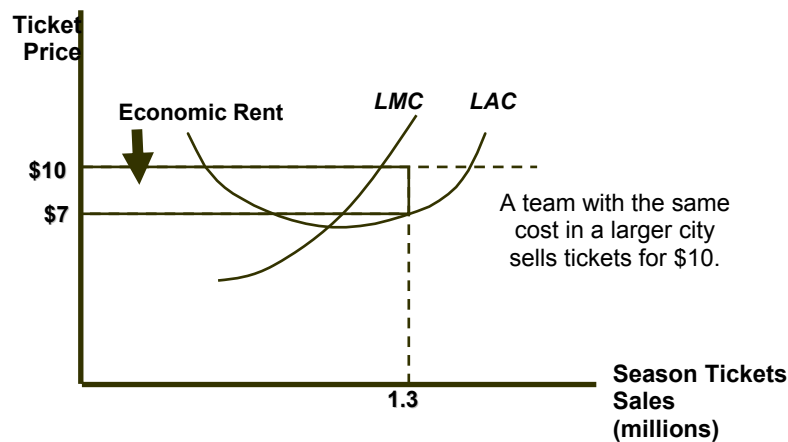
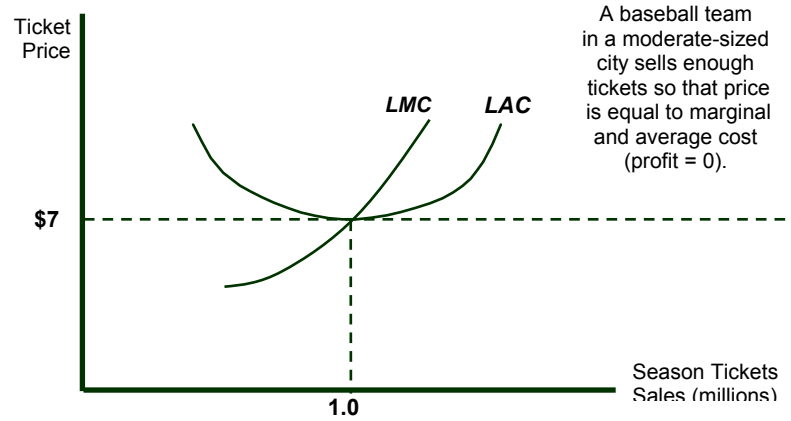
## An Example

Suppose there are two firms A & B. Both own their land. A is located on a river which lowers A's shipping cost by \$10,000 compared to B. The demand for A's river location will increase the price of A's land to \$10,000

- Economic rent = \$10,000
- \$10,000 - zero cost for the land

Economic rent increases. Economic profit of A = 0

## FIRMS EARN ZERO PROFIT IN LONG-RUN EQUILIBRIUM



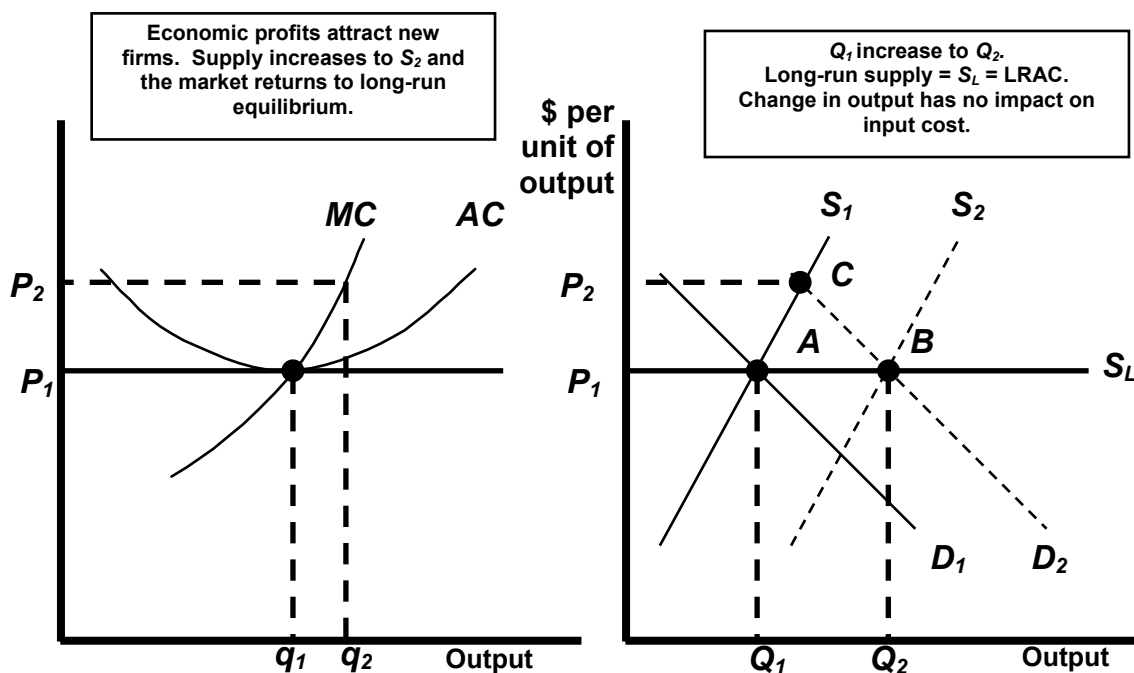
With a fixed input such as a unique location, the difference between the cost of production ( $LAC = 7$ ) and price (\$10) is the value or opportunity cost of the input (location) and represents the economic rent from the input. If the opportunity cost of the input (rent) is not taken into consideration it may appear that economic profits exist in the long-run.

### The Industry's Long-Run Supply Curve

The Shape of the long-run supply curve depends on the extent to which changes in industry output affect the prices the firms must pay for inputs. To determine long-run supply, we assume:

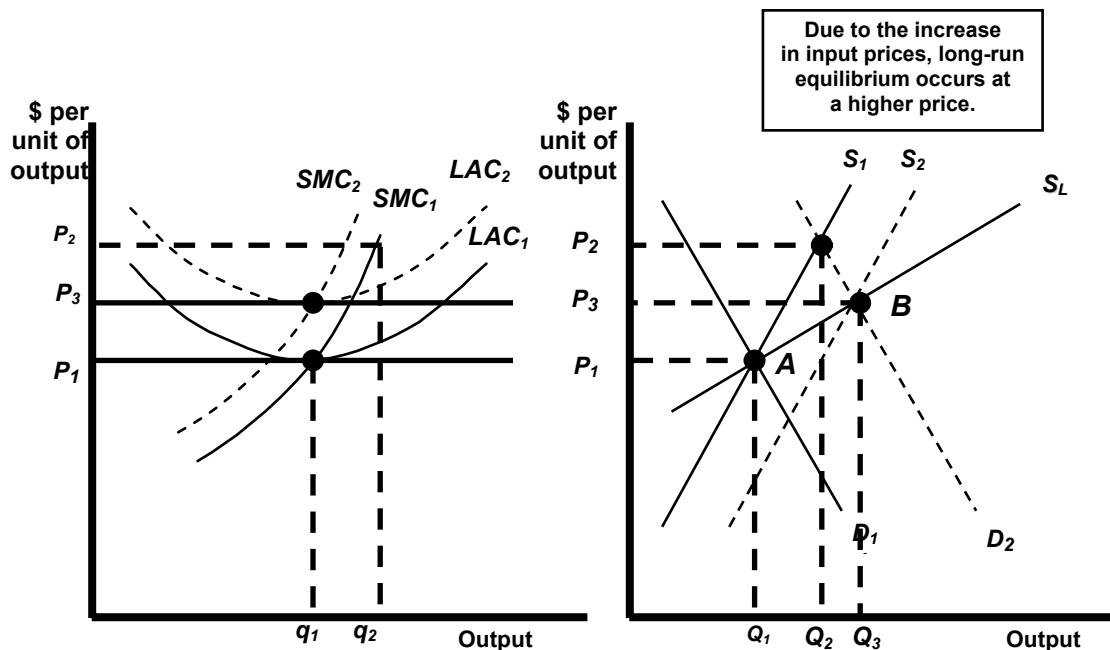
- All firms have access to the available production technology.
- Output is increasing by using more inputs, not by invention.
- The market for inputs does not change with expansions and contractions of the industry.

### Long-Run Supply in a Constant-Cost Industry



In a constant-cost industry, long-run supply is a horizontal line at a price that is equal to the minimum average cost of production.

### Long-Run Supply in an Increasing-Cost Industry



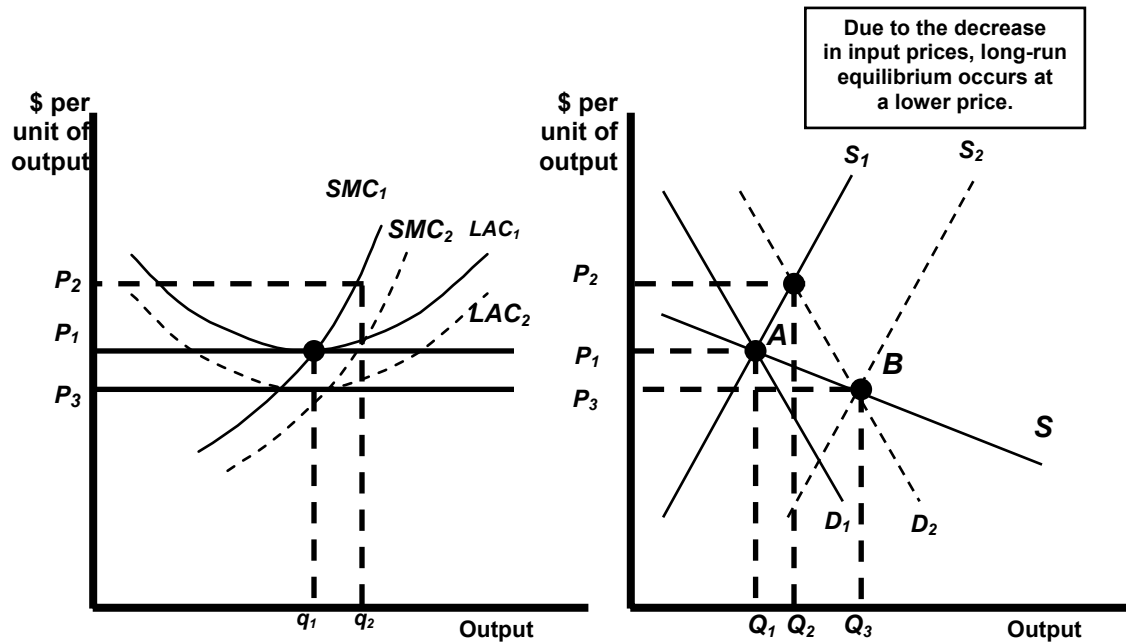
In an increasing-cost industry, long-run supply curve is upward sloping.

### The Industry's Long-Run Supply Curve

Questions:

- 1) Explain how decreasing-cost is possible.
- 2) Illustrate a decreasing cost industry.
- 3) What is the slope of the SL in a decreasing-cost industry?

#### Long-Run Supply in a Decreasing-Cost Industry



In a decreasing-cost industry, long-run supply curve is downward sloping.

## LESSON 26

## PROFIT MAXIMIZATION AND COMPETITIVE SUPPLY

## THE INDUSTRY'S LONG-RUN SUPPLY CURVE

In a constant-cost industry, long-run supply is horizontal. Small increase in price will induce an extremely large output increase. Long-run supply elasticity is infinitely large. Inputs would be readily available.

In an increasing-cost industry, long-run supply is upward-sloping and elasticity is positive. The slope (elasticity) will depend on the rate of increase in input cost. Long-run elasticity will generally be greater than short-run elasticity of supply.

Question:

Describe the long-run elasticity of supply in a decreasing-cost industry.

## THE LONG-RUN SUPPLY OF HOUSING

Scenario 1: Owner-occupied housing

- Suburban or rural areas
- National market for inputs

Questions

Is this an increasing or a constant-cost industry? What would you predict about the elasticity of supply?

Scenario 2: Rental property

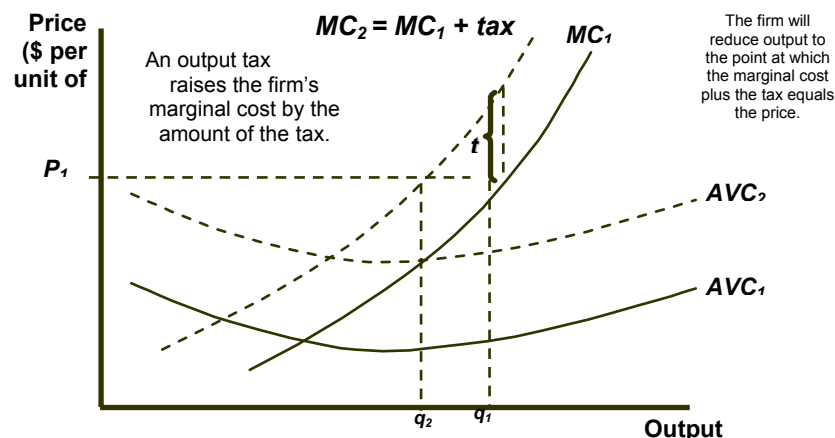
- Urban location
- High-rise construction cost

Questions

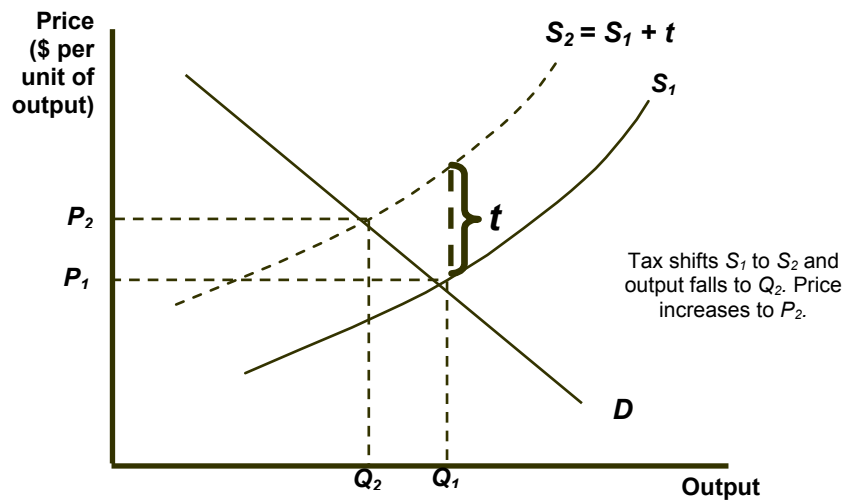
Is this an increasing or a constant-cost industry? What would you predict about the elasticity of supply?

## EFFECT OF AN OUTPUT TAX ON A COMPETITIVE FIRM'S OUTPUT

In an earlier chapter we studied how firms respond to taxes on an input. Now, we will consider how a firm responds to a tax on its output.



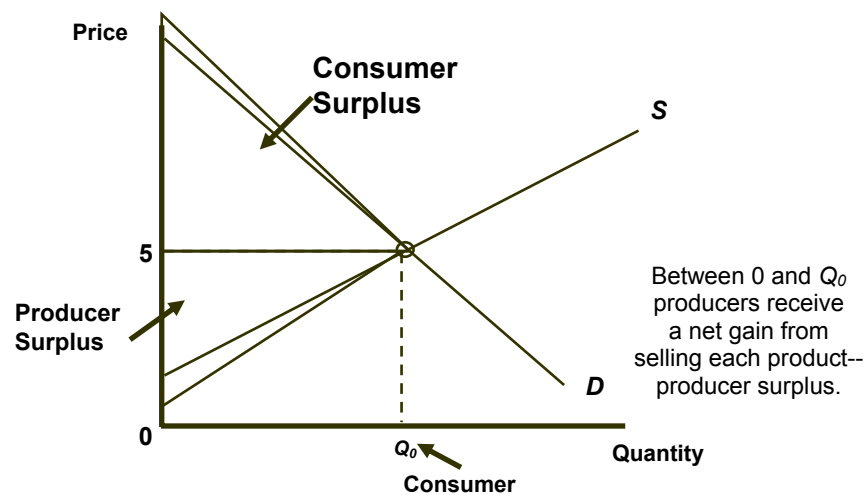
### EFFECT OF AN OUTPUT TAX ON INDUSTRY OUTPUT



### EVALUATING THE GAINS & LOSSES FROM GOVERNMENT POLICIES:

#### CONSUMER & PRODUCER SURPLUS

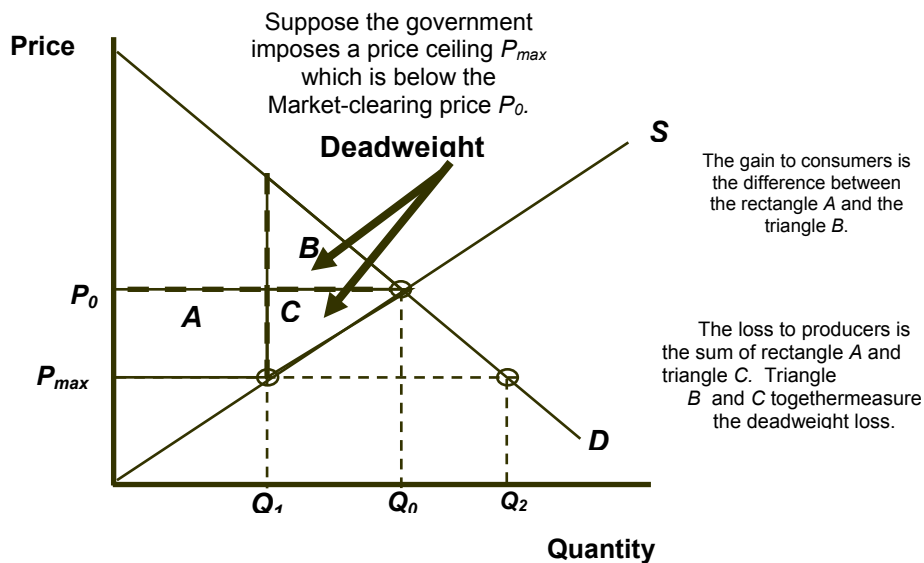
Consumer surplus is the total benefit or value that consumers receive beyond what they pay for the good. Producer surplus is the total benefit or revenue that producers receive beyond what it cost to produce a good.



#### WELFARE EFFECTS

To determine the welfare effect of a governmental policy we can measure the gain or loss in consumer and producer surplus.

## GAINS AND LOSSES CAUSED BY GOVERNMENT INTERVENTION IN THE MARKET



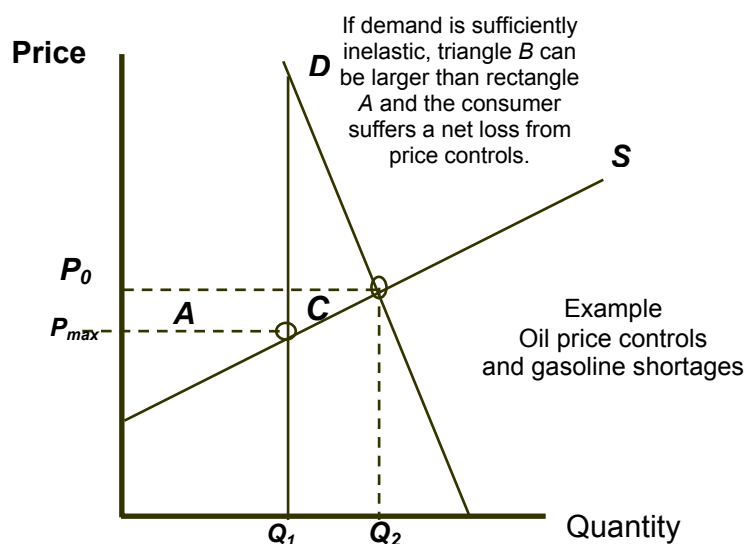
### CHANGE IN CONSUMER & PRODUCER SURPLUS FROM PRICE CONTROLS

Observations:

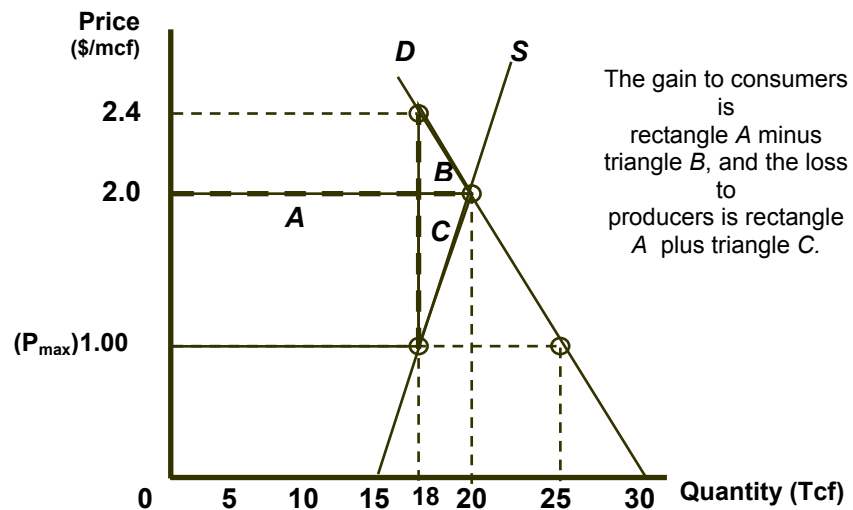
The total loss is equal to area B + C. The total change in surplus =  $(A - B) + (-A - C) = -B - C$ .

The deadweight loss is the inefficiency of the price controls or the loss of the producer surplus exceeds the gain from consumer surplus. Consumers can experience a net loss in consumer surplus when the demand is sufficiently inelastic.

### EFFECT OF PRICE CONTROLS WHEN DEMAND IS INELASTIC



### PRICE CONTROLS AND NATURAL GAS SHORTAGES



### THE EFFICIENCY OF A COMPETITIVE MARKET

When do competitive markets generate an inefficient allocation of resources or market failure?

Externalities: Costs or benefits that do not show up as part of the market price (e.g. pollution)

Lack of Information: Imperfect information prevents consumers from making utility-maximizing decisions.

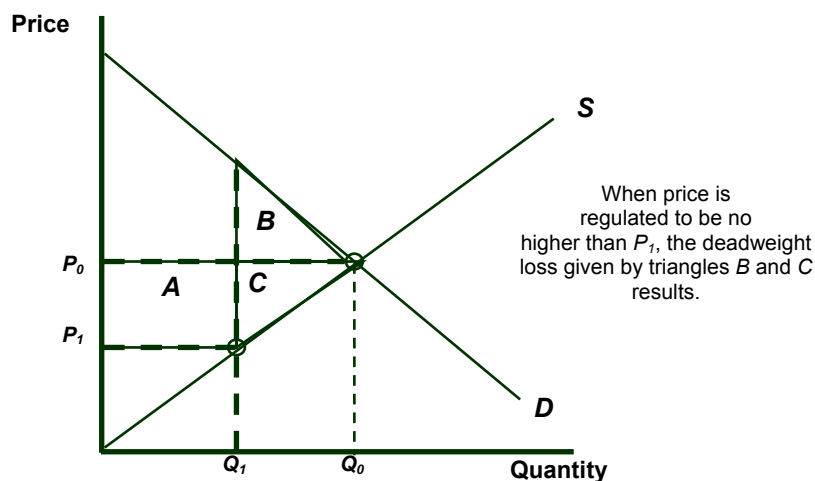
Government intervention in these markets can increase efficiency. Government intervention without a market failure creates inefficiency or deadweight loss.



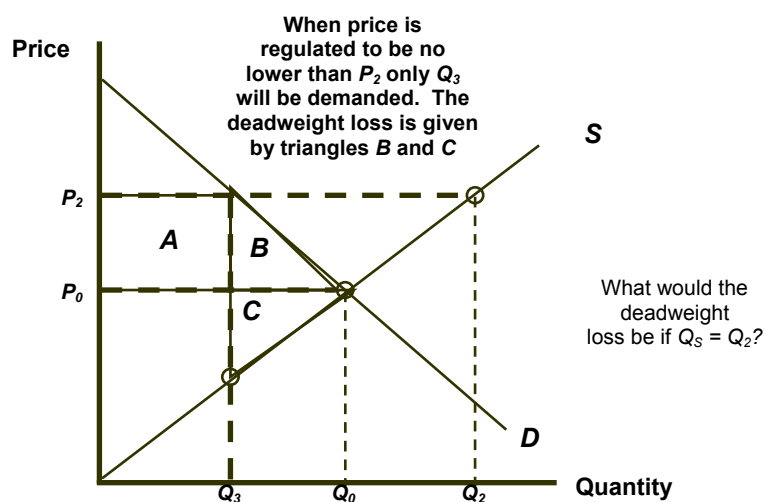
## LESSON 27

## THE ANALYSIS OF COMPETITIVE MARKETS

## WELFARE LOSS IF PRICE IS HELD BELOW MARKET-CLEARING LEVEL

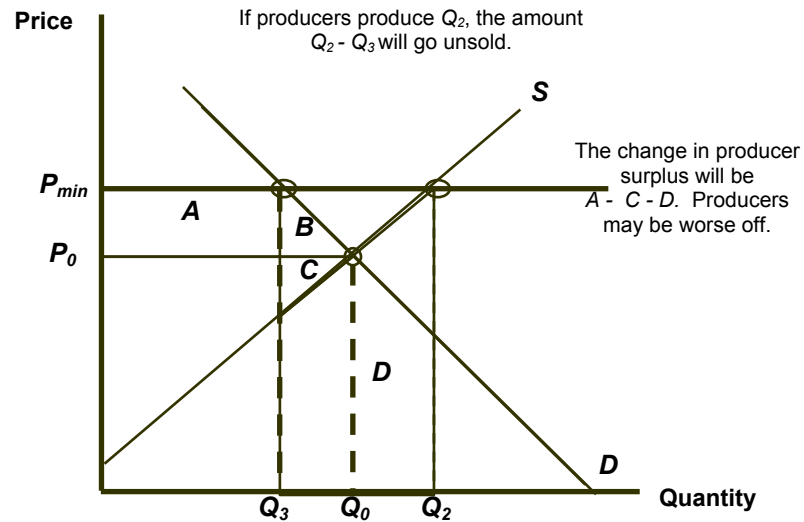
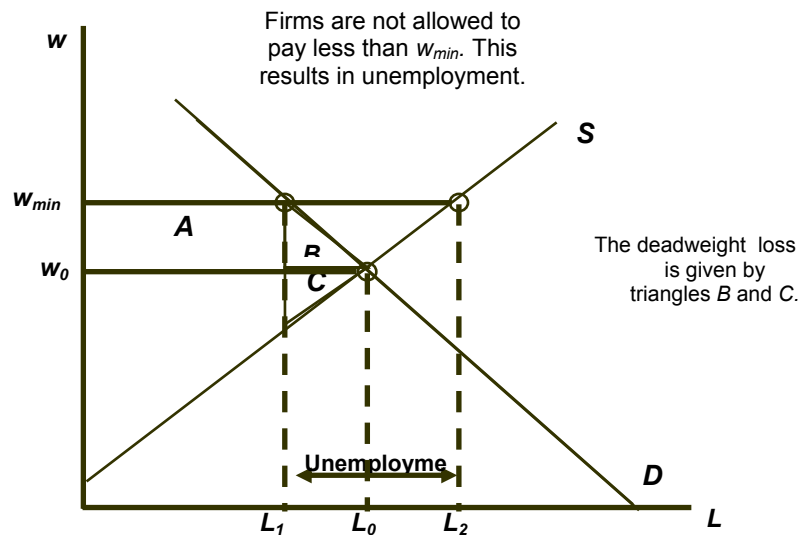


## WELFARE LOSS IF PRICE IS HELD ABOVE MARKET-CLEARING LEVEL

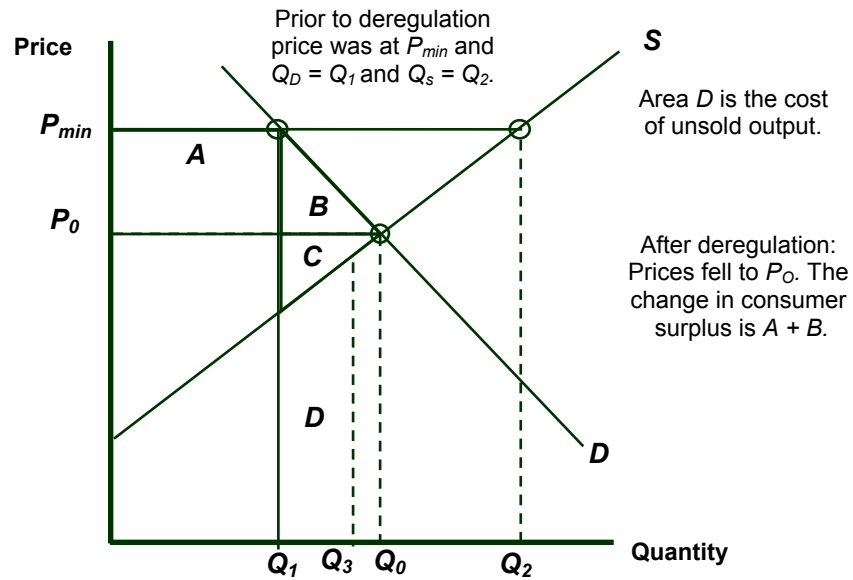


## MINIMUM PRICES

Periodically government policy seeks to raise prices above market-clearing levels. We will investigate this by looking at a price floor and the minimum wage.

**PRICE MINIMUM****THE MINIMUM WAGE****AIRLINE REGULATION**

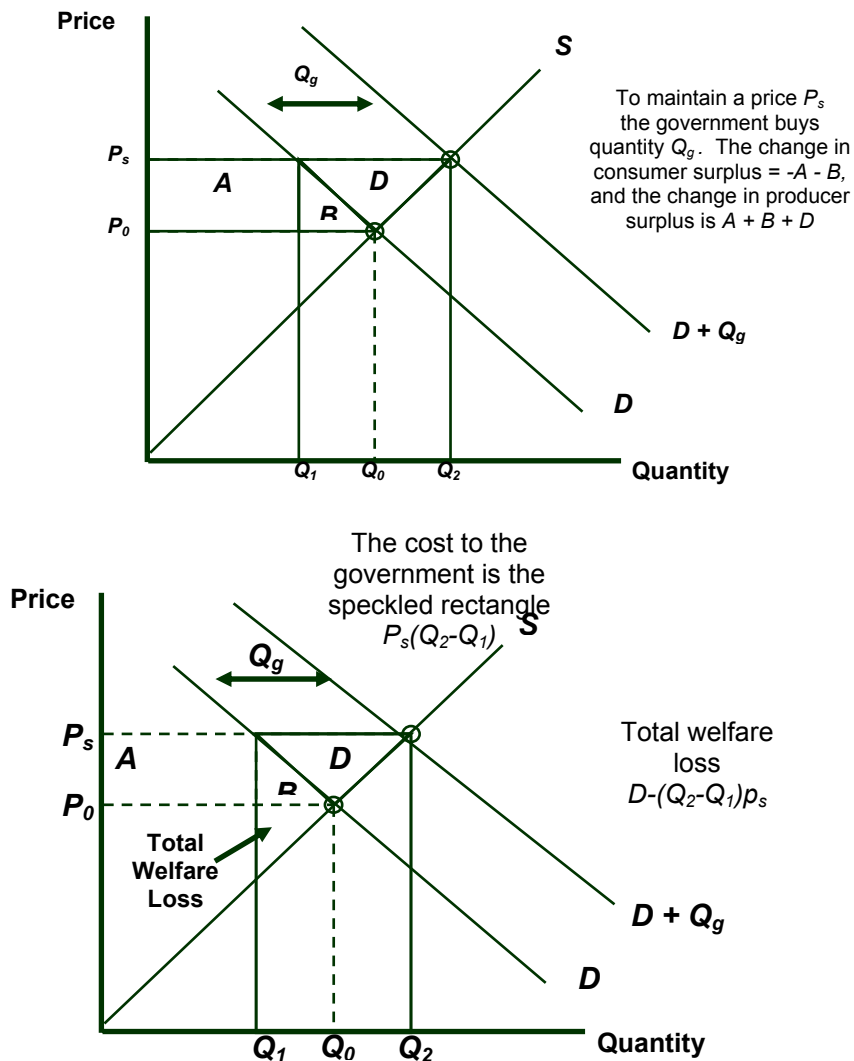
- During 1976-1981 the airline industry in the U.S. changed dramatically.
- Deregulation lead to major changes in the industry.
- Some airlines merged or went out of business as new airlines entered the industry.

**EFFECT OF AIRLINE REGULATION BY THE CIVIL AERONAUTICS BOARD**

## LESSON 28

## THE ANALYSIS OF COMPETITIVE MARKETS (Continued)

## PRICE SUPPORTS



Question:

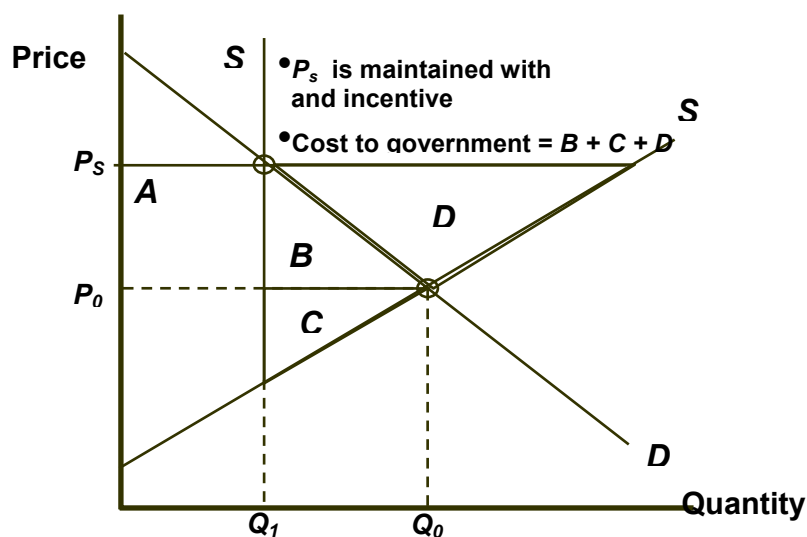
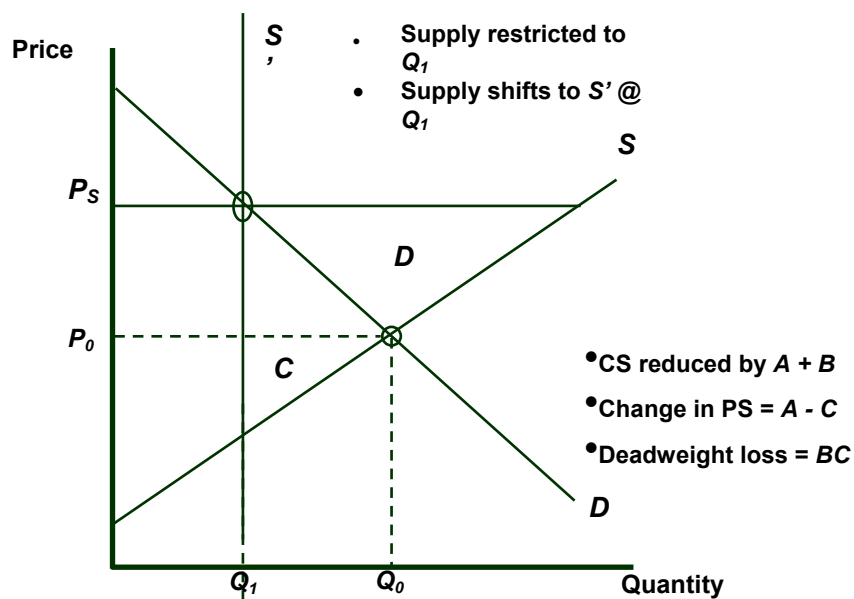
Is there a more efficient way to increase farmer's income by  $A + B + D$ ?

### PRICE SUPPORTS AND PRODUCTION QUOTAS

Production Quotas: The government can also cause the price of a good to rise by reducing supply.

What is the impact of controlling entry into the taxicab market?

## SUPPLY RESTRICTIONS



$$\begin{aligned}\Delta PS &= A - C + B + C + D \\ &= A + B + D.\end{aligned}$$

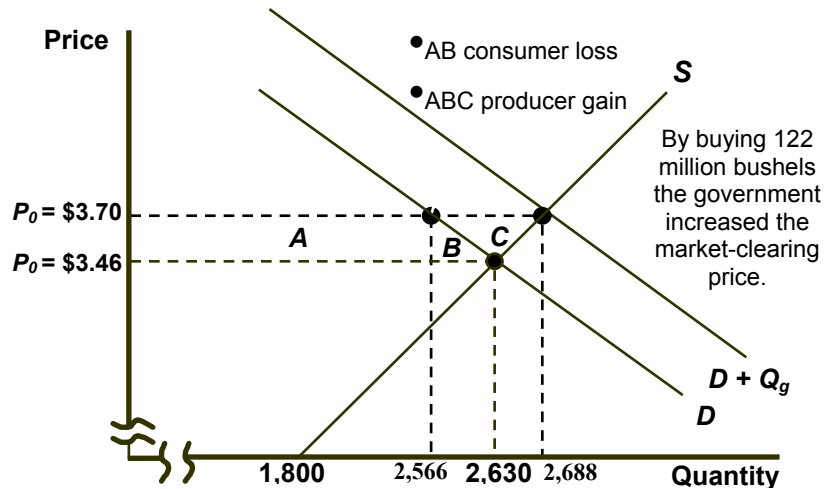
The change in consumer and producer surplus is the same as with price supports.

$$\Delta \text{welfare} = -A - B + A + B + D - B - C - D = -B - C.$$

Questions:

- How could the government reduce the cost and still subsidize the farmer?
- Which is more costly: supports or acreage limitations?

### THE WHEAT MARKET IN 1981



### SUPPORTING THE PRICE OF WHEAT

In 1981

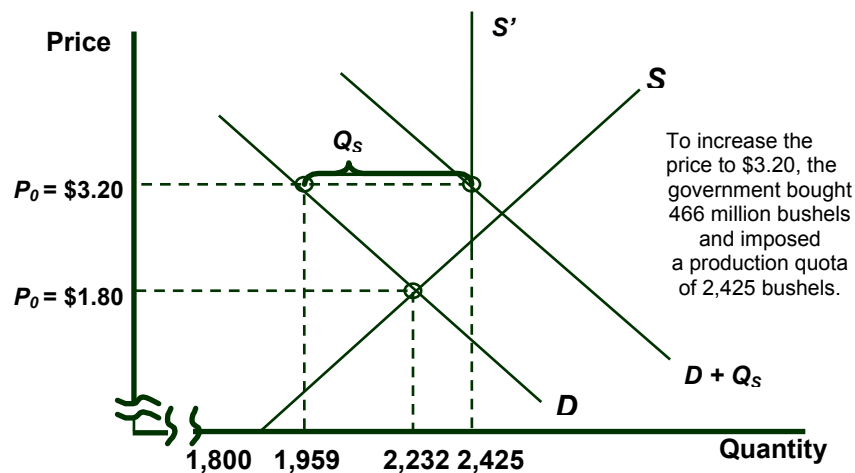
$$\begin{aligned} \text{Change in consumer surplus} &= (-A - B) \\ A &= (3.70 - 3.46)(2,566) = \$616 \text{ million} \\ B &= (1/2)(3.70 - 3.46)(2,630 - 2,566) \\ &= \$8 \text{ million} \end{aligned}$$

Change in consumer surplus: -\$624 million.

Cost to the government:

$$\begin{aligned} \$3.70 \times 122 \text{ million bushels} &= \$452 \text{ million} \\ \text{Total cost} &= \$624 + 452 = \$1,076 \text{ million} \\ \text{Total gain} &= A + B + C = \$638 \text{ million} \\ \text{Government also paid 30 cents/bushel} &= \$806 \text{ million} \end{aligned}$$

### THE WHEAT MARKET IN 1985



**In 1985, Government Purchase:**

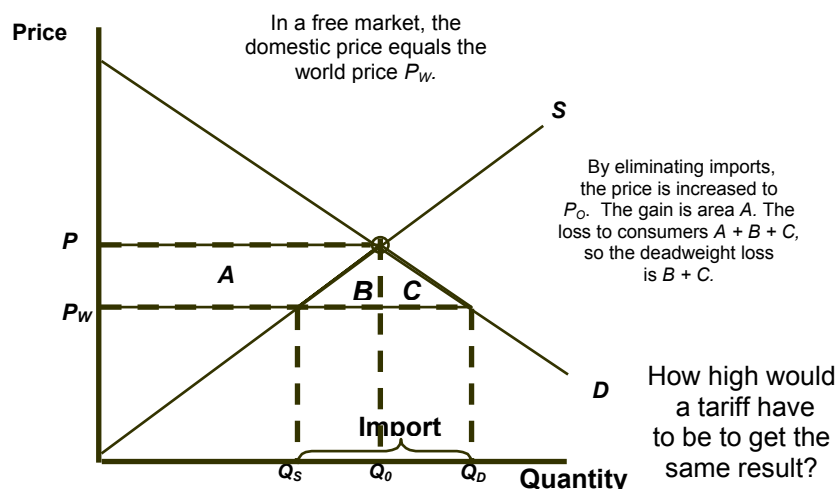
Government cost =  $\$3.20 \times 466 = \$1,491$  million

80 cent subsidy =  $.80 \times 2,425 = \$1,940$  million

Total cost = \$3.5 billion

**Import Quotas and Tariffs**

Many countries use import quotas and tariffs to keep the domestic price of a product above world levels

**IMPORT TARIFF OR QUOTA THAT ELIMINATES IMPORTS**

The increase in price can be achieved by a quota or a tariff. Area A is again the gain to domestic producers. The loss to consumers is  $A + B + C + D$ . If a tariff is used the government gains D, so the net domestic product loss is  $B + C$ . If a quota is used instead, rectangle D becomes part of the profits of foreign producers, and the net domestic loss is  $B + C + D$ .

Question:

Would a country be better off or worse off with a quota instead of a tariff?

## LESSON 29

## THE ANALYSIS OF COMPETITIVE MARKETS (Continued)

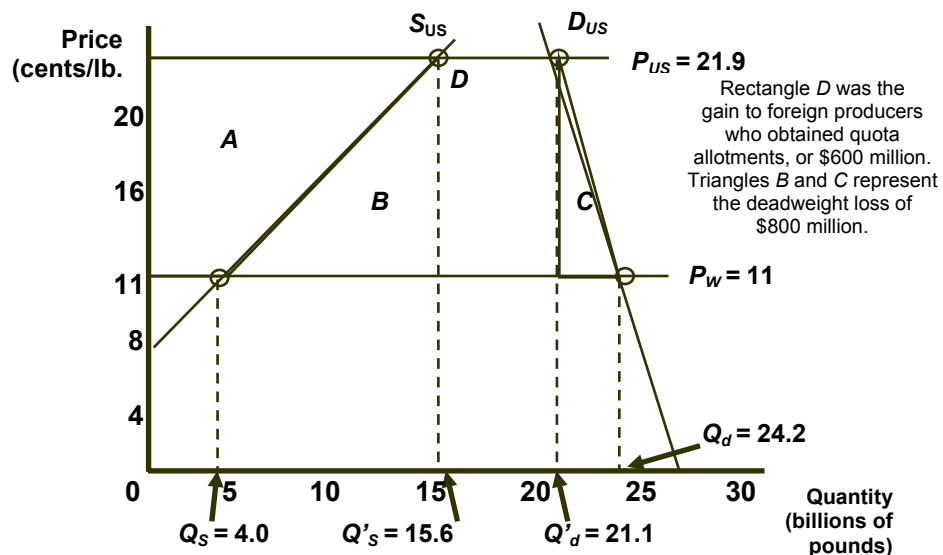
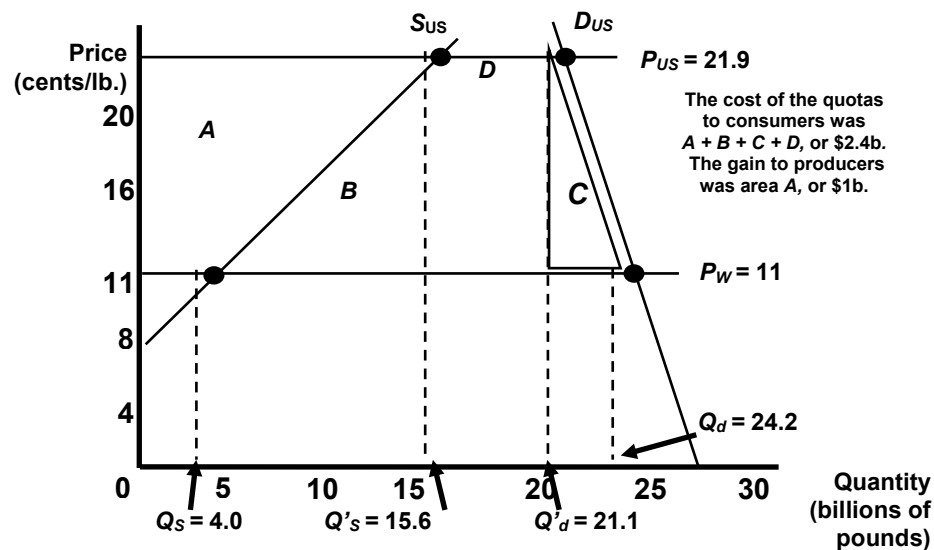
## THE SUGAR QUOTA

The world price of sugar has been as low as 4 cents per pound, while in the U.S. the price has been 20-25 cents per pound.

The Impact of a Restricted Market (1997)

- U.S. production = 15.6 billion pounds
- U.S. consumption = 21.1 billion pounds
- U.S. price = 22 cents/pound
- World price = 11 cents/pound

## SUGAR QUOTA IN 1997

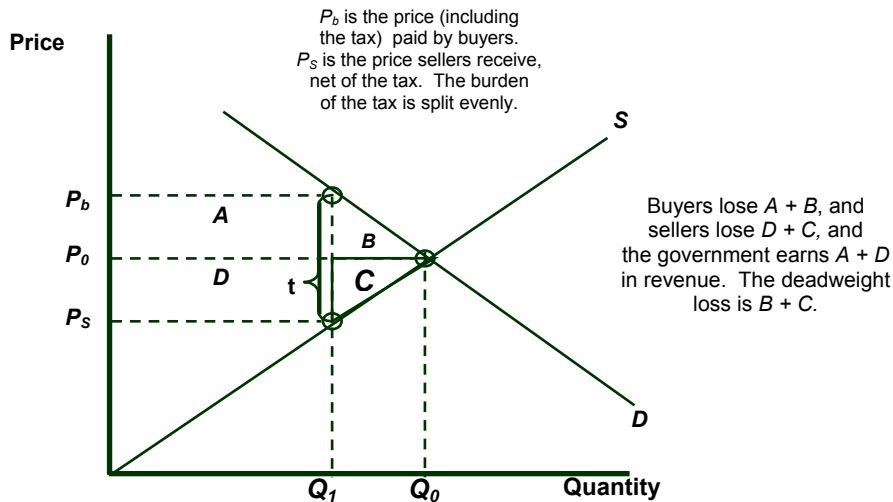




### THE IMPACT OF A TAX OR SUBSIDY

The burden of a tax (or the benefit of a subsidy) falls partly on the consumer and partly on the producer. We will consider a specific tax which is a tax of a certain amount of money per unit sold.

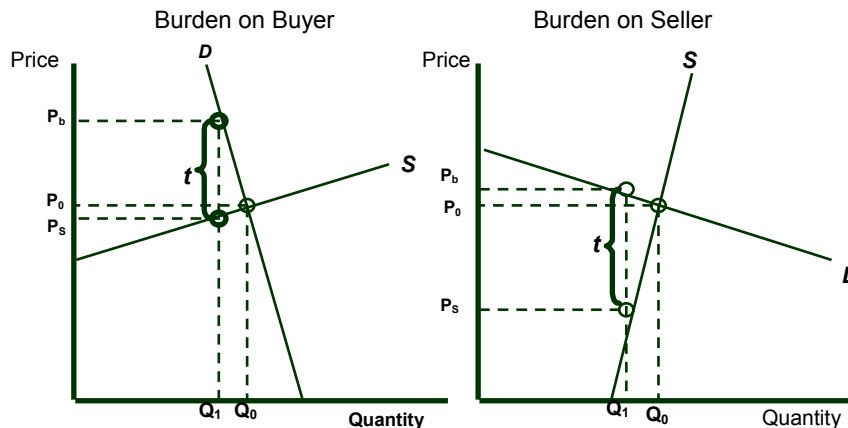
#### INCIDENCE OF A SPECIFIC TAX



Four conditions that must be satisfied after the tax is in place:

- 1) Quantity sold and  $P_b$  must be on the demand line:  $Q^D = Q^D(P_b)$
- 2) Quantity sold and  $P_s$  must be on the supply line:  $Q^S = Q^S(P_s)$
- 3)  $Q^D = Q^S$
- 4)  $P_b - P_s = \text{tax}$

#### IMPACT OF TAX DEPENDS ON ELASTICITIES OF SUPPLY & DEMAND



### THE IMPACT OF A TAX OR SUBSIDY

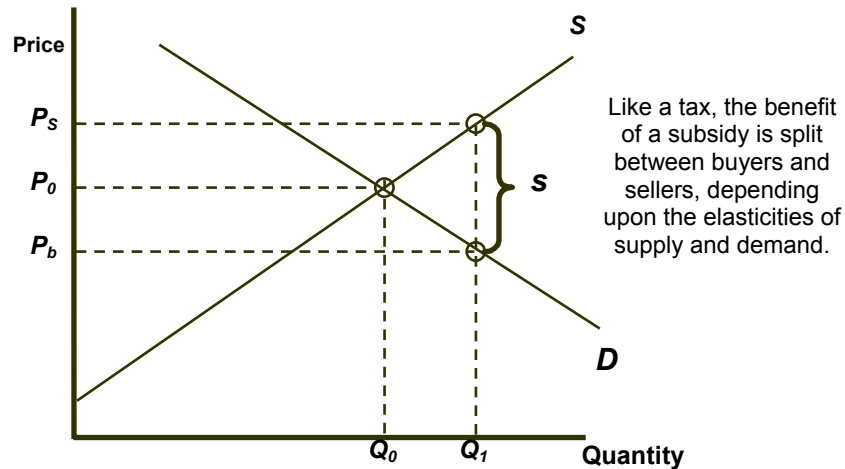
#### PASS-THROUGH FRACTION

$$E_s / (E_s - E_d)$$

For example, when demand is perfectly inelastic ( $E_d = 0$ ), the pass-through fraction is 1, and all the tax is borne by the consumer.

A subsidy can be analyzed in much the same way as a tax. It can be treated as a negative tax. The seller's price exceeds the buyer's price.

## SUBSIDY

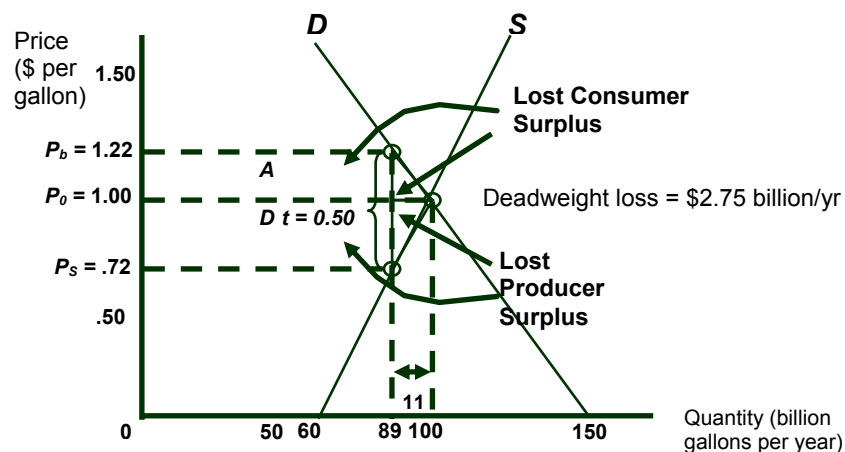
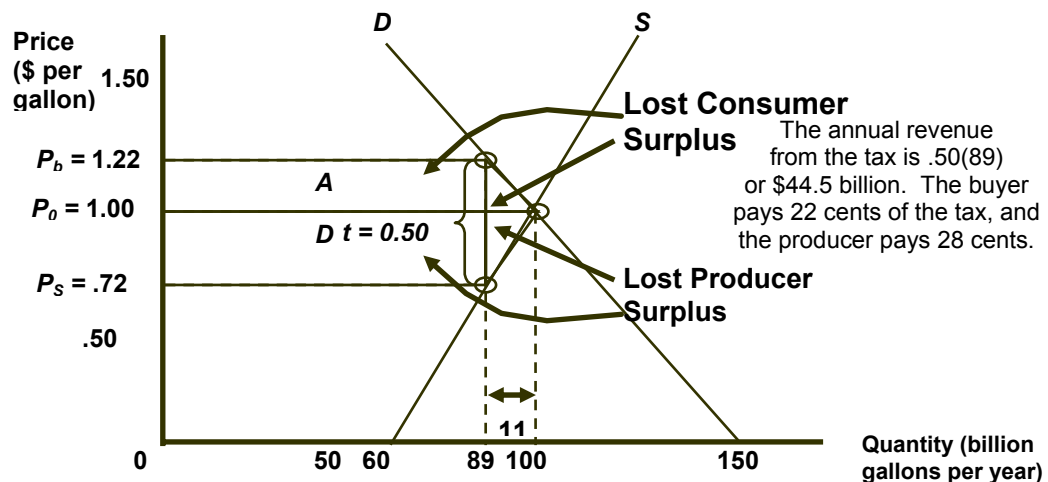


With a subsidy ( $s$ ), the selling price  $P_b$  is below the subsidized price  $P_s$  so that:

$$s = P_s - P_b$$

The benefit of the subsidy depends upon  $E_d / E_s$ . If the ratio is small, most of the benefit accrues to the consumer. If the ratio is large, the producer benefits most.

## IMPACT OF A \$0.50 GASOLINE TAX

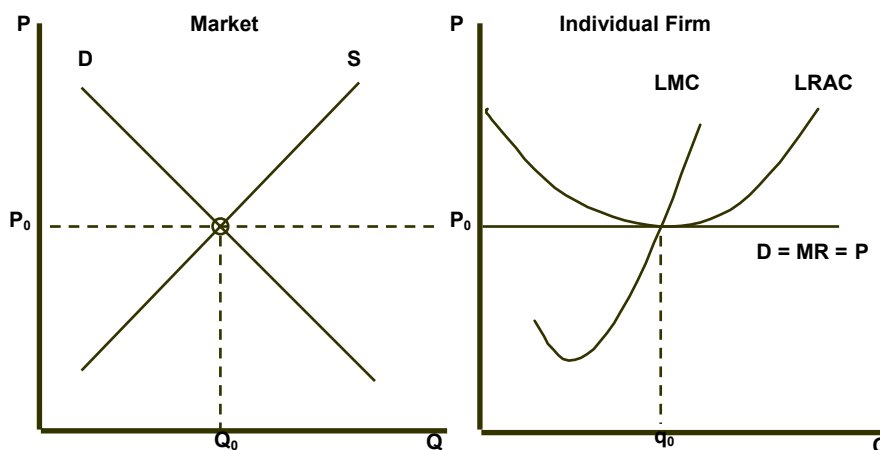


## LESSON 30

## MARKET STRUCTURE AND COMPETITIVE STRATEGY

## REVIEW OF PERFECT COMPETITION

- $P = LMC = LRAC$
- Normal profits or zero economic profits in the long run
- Large number of buyers and sellers
- Homogenous product
- Perfect information
- Firm is a price taker



## MONOPOLY

- 1) One seller - many buyers
- 2) One product (no good substitutes)
- 3) Barriers to entry

The monopolist is the supply-side of the market and has complete control over the amount offered for sale. Profits will be maximized at the level of output where marginal revenue equals marginal cost.

## FINDING MARGINAL REVENUE

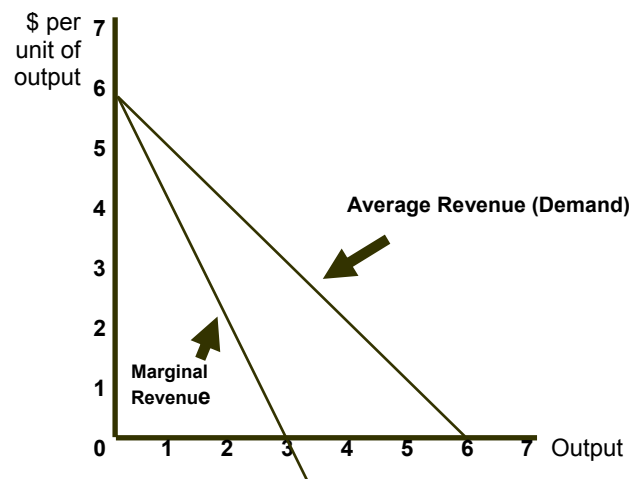
As the sole producer, the monopolist works with the market demand to determine output and price.

Assume a firm with demand:

$$P = 6 - Q$$

**TOTAL, MARGINAL, AND AVERAGE REVENUE**

| Price<br>P | Quantity<br>Q | Total<br>Revenue<br>R | Marginal<br>Revenue<br>MR | Average<br>Revenue<br>AR |
|------------|---------------|-----------------------|---------------------------|--------------------------|
| \$6        | 0             | \$0                   | ---                       | ---                      |
| 5          | 1             | 5                     | \$5                       | \$5                      |
| 4          | 2             | 8                     | 3                         | 4                        |
| 3          | 3             | 9                     | 1                         | 3                        |
| 2          | 4             | 8                     | -1                        | 2                        |
| 1          | 5             | 5                     | -3                        | 1                        |

**OBSERVATIONS**

- 1) To increase sales the price must fall
- 2)  $MR < P$
- 3) Compared to perfect competition
  - No change in price to change sales
  - $MR = P$

**MONOPOLIST'S OUTPUT DECISION**

Profits maximized at the output level where  $MR = MC$  and cost functions are the same.

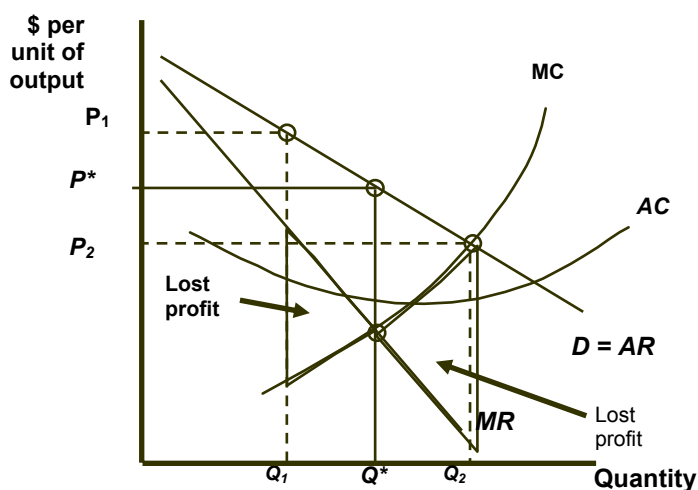
$$\pi(Q) = R(Q) - C(Q)$$

$$\Delta \pi / \Delta Q = \Delta R / \Delta Q - \Delta C / \Delta Q = 0 = MC - MR$$

$$\text{or } MC = MR$$

**MAXIMIZING PROFIT WHEN MARGINAL REVENUE EQUALS MARGINAL COST****THE MONOPOLIST'S OUTPUT DECISION**

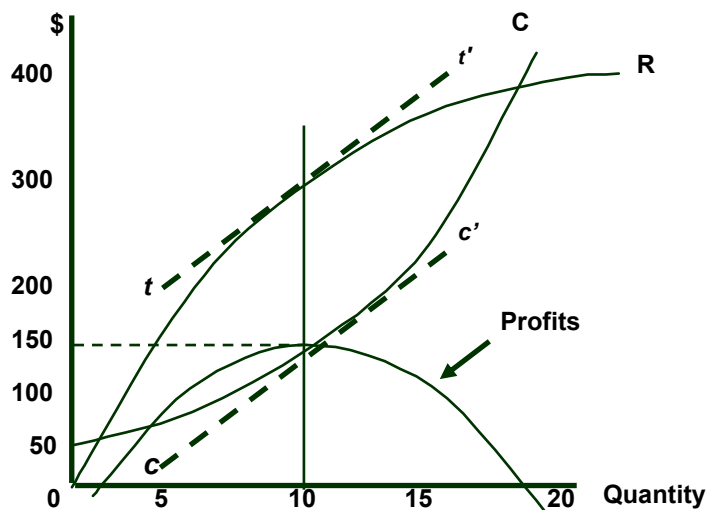
At output levels below  $MR = MC$  the decrease in revenue is greater than the decrease in cost ( $MR > MC$ ) and at output levels above  $MR = MC$  the increase in cost is greater than the decrease in revenue ( $MR < MC$ ).



### THE MONOPOLIST'S OUTPUT DECISION: AN EXAMPLE

By setting marginal revenue equal to marginal cost, it can be verified that profit is maximized at  $P = \$30$  and  $Q = 10$ . This can be seen graphically:

### EXAMPLE OF PROFIT MAXIMIZATION



#### Observations

Slope of  $rr'$  = slope  $cc'$  and they are parallel at 10 units

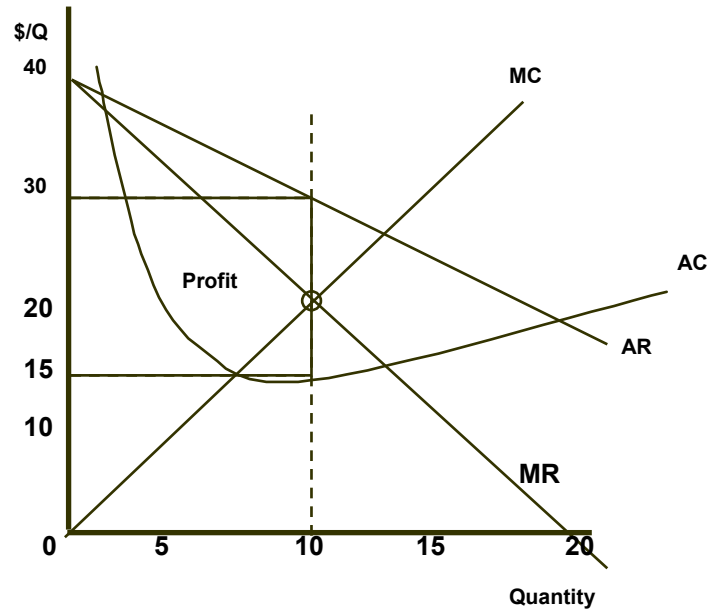
Profits are maximized at 10 units

$$P = \$30, Q = 10, \quad TR = P \times Q = \$300$$

$$AC = \$15, Q = 10, \quad TC = AC \times Q = \$150$$

$$\text{Profit} = TR - TC$$

$$\$150 = \$300 - \$150$$



### Observations

$$AC = \$15, Q = 10, \quad TC = AC \times Q = 150$$

$$\text{Profit} = TR - TC = \$300 - \$150 = \$150$$

or

$$\text{Profit} = (P - AC) \times Q = (\$30 - \$15)(10) = \$150$$

## LESSON 31

## MARKET STRUCTURE AND COMPETITIVE STRATEGY (Continued)

**A RULE OF THUMB FOR PRICING**

We want to translate the condition that marginal revenue should equal marginal cost into a rule of thumb that can be more easily applied in practice. This can be demonstrated using the following steps:

$$1. MR = \frac{\Delta R}{\Delta Q} = \frac{\Delta(PQ)}{\Delta Q}$$

$$2. MR = P + Q \frac{\Delta P}{\Delta Q} = P + P \left( \frac{Q}{P} \right) \left( \frac{\Delta P}{\Delta Q} \right)$$

$$3. E_d = \left( \frac{P}{Q} \right) \left( \frac{\Delta Q}{\Delta P} \right)$$

$$4. \left( \frac{Q}{P} \right) \left( \frac{\Delta P}{\Delta Q} \right) = \frac{1}{E_d}$$

$$5. MR = P + P \left( \frac{1}{E_d} \right)$$

$$6. \pi \text{ is maximized @ } MR = MC$$

$$P + P \left[ \frac{1}{E_d} \right] = - \frac{1}{E_d}$$

$$P = \frac{MC}{1 + (1/E_d)}$$

$$7. - \frac{1}{E_d} = \text{the markup over MC as a percentage of price } (P-MC)/P$$

8. The markup should equal the inverse of the elasticity of demand.

$$9. P = \frac{MC}{1 + \left( \frac{1}{E_d} \right)}$$

Assume

$$E_d = -4 \quad MC = 9$$

$$P = \frac{9}{1 + \left( \frac{1}{-4} \right)} = \frac{9}{.75} = \$12$$

**MONOPOLY PRICING COMPARED TO PERFECT COMPETITION PRICING:**

- In monopoly,  $P > MC$  whereas in perfect competition,  $P = MC$ .
- The more elastic the demand, the closer price is to marginal cost.
- If  $E_d$  is a large negative number, price is close to marginal cost and vice versa.

## A MONOPOLIST'S PRICING

Suppose:

- Price of Medicine A = \$3.50/daily dose
- Price of Medicine B and Medicine C = \$1.50 - \$2.25/daily dose
- MC of Medicine A = 30 - 40 cents/daily dose

The Monopolist's Output Decision

$$P = \frac{MC}{1 + \left[ \frac{1}{E_D} \right]} = \frac{.35}{1 + \left[ \frac{1}{-1.1} \right]} =$$

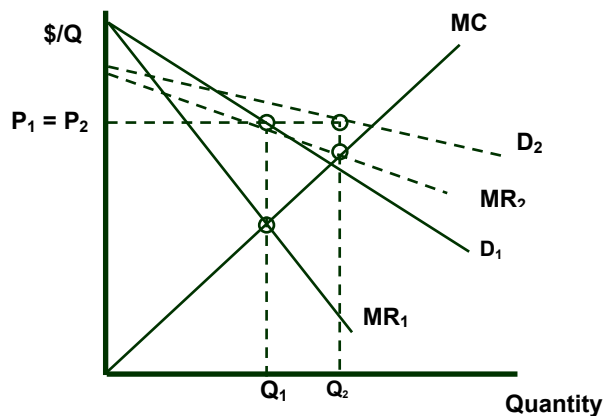
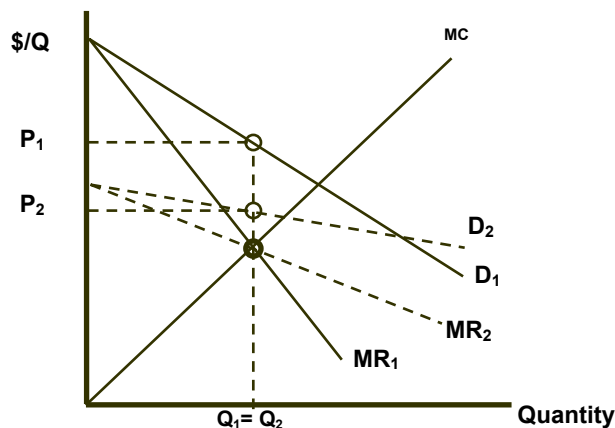
$$\frac{MC}{1 + (-.91)} = \frac{.35}{.09} = \$3.89$$

Price of \$3.50 is consistent with “the rule of thumb pricing”.

## SHIFTS IN DEMAND

In perfect competition, the market supply curve is determined by marginal cost. For a monopoly, output is determined by marginal cost and the shape of the demand curve.

### SHIFT IN DEMAND LEADS TO CHANGE IN PRICE BUT SAME OUTPUT





### Observations

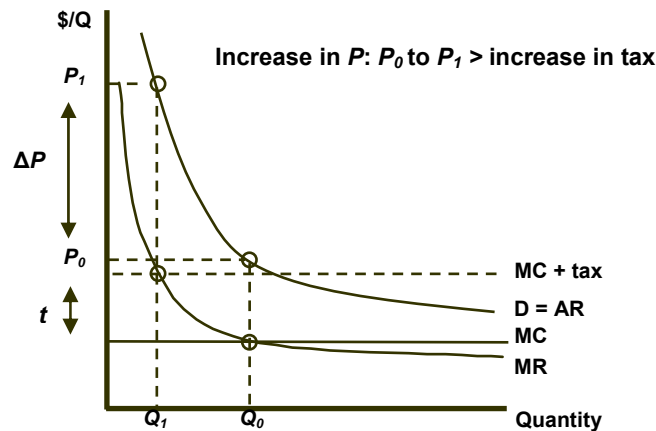
- Shifts in demand usually cause a change in both price and quantity.
- A monopolistic market has no supply curve.
- Monopolist may supply many different quantities at the same price.
- Monopolist may supply the same quantity at different prices.

### THE EFFECT OF A TAX

Under monopoly price can sometimes rise by more than the amount of the tax. To determine the impact of a tax:

- $t$  = specific tax
- $MC = MC + t$
- $MR = MC + t$  : optimal production decision

### EFFECT OF EXCISE TAX ON MONOPOLIST



### Question

Suppose:  $E_d = -2$ , how much would the price change?

### Answer

$$P = \frac{MC}{1 + \left( \frac{1}{E_d} \right)}$$

$$\text{If } E_d = -2 \rightarrow P = 2MC$$

If  $MC$  increases to  $MC + t$

$$\Delta P = 2(MC + t) = 2MC + 2t$$

Price increases by twice the tax.

What would happen to profits?

## LESSON 32

## MARKET STRUCTURE AND COMPETITIVE STRATEGY (Continued)

## THE MULTIPLANT FIRM

For many firms, production takes place in two or more different plants whose operating cost can differ. Choosing total output and the output for each plant:

- The marginal cost in each plant should be equal.
- The marginal cost should equal the marginal revenue for each plant.

Algebraically:

$Q_1$  &  $C_1 \Rightarrow$  Output & Cost for Plant 1

$Q_2$  &  $C_2 \Rightarrow$  Output & Cost for Plant 2

Total Output =  $Q_T = Q_1 + Q_2$

Algebraically:

$$\pi = PQ_T - C_1(Q_1) - C_2(Q_2)$$

$$\frac{\Delta \pi}{\Delta Q_1} = \frac{\Delta (PQ_T)}{\Delta Q_1} - \frac{\Delta C_1}{\Delta Q_1} = 0$$

$$(MR) \frac{\Delta (PQ_T)}{\Delta Q_1} - (MC) \frac{\Delta C_1}{\Delta Q_1} = 0$$

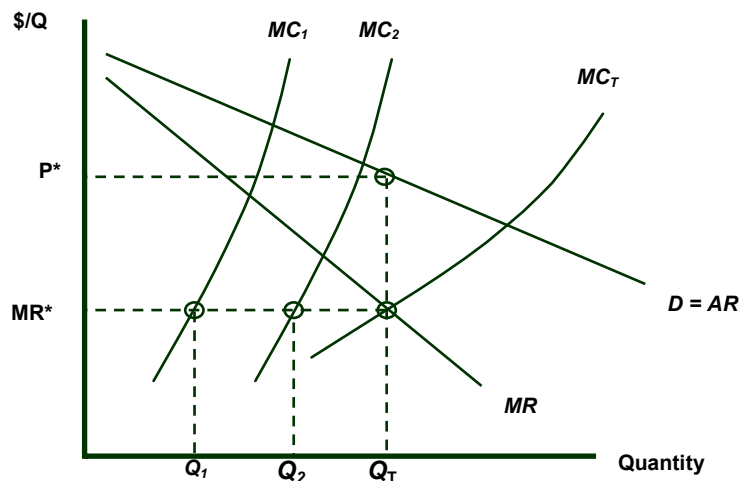
$$MR = MC_1$$

$$MR = MC_1$$

$$MR = MC_2$$

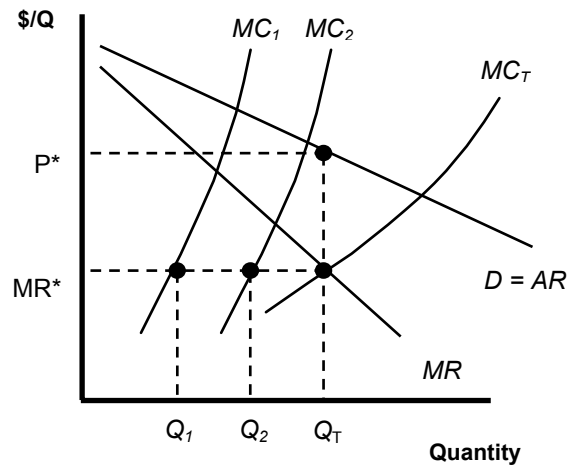
$$MR = MC_1 = MC_2$$

## PRODUCTION WITH TWO PLANTS



Observations:

- 1)  $MC_T = MC_1 + MC_2$
- 2) Profit maximizing output:
  - $MC_T = MR$  at  $Q_T$  and  $P^*$
  - $MR = MR^*$
  - $MR^* = MC_1$  at  $Q_1$ ,  $MR^* = MC_2$  at  $Q_2$
  - $MC_1 + MC_2 = MC_T$ ,  $Q_1 + Q_2 = Q_T$ , and  $MR = MC_1 + MC_2$

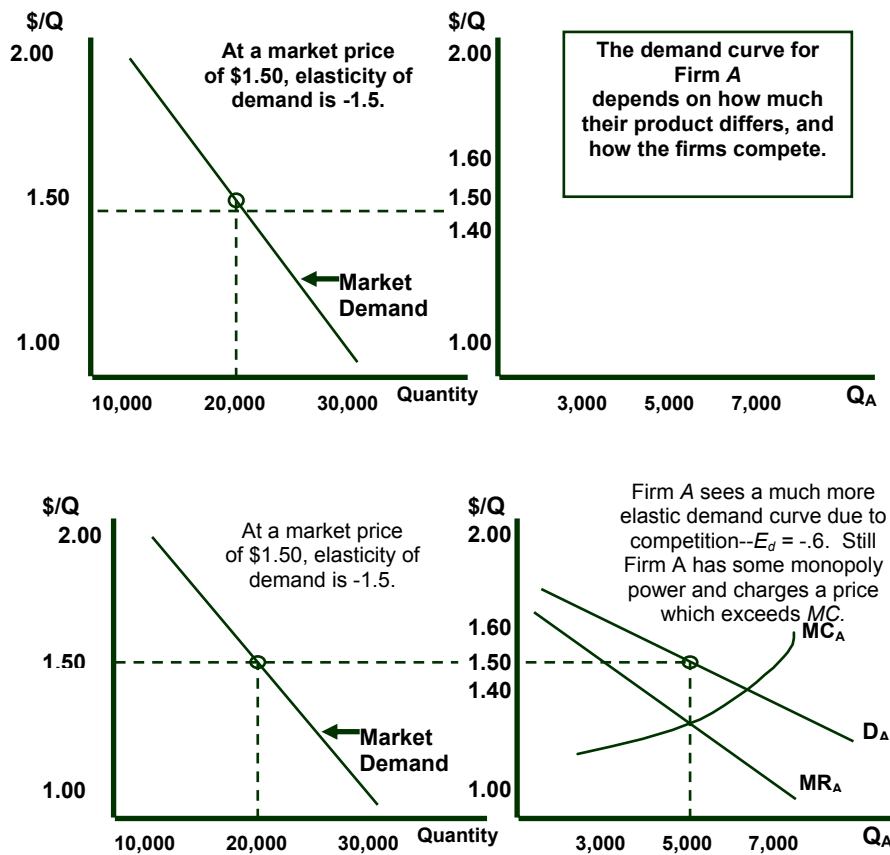


### MONOPOLY POWER

Monopoly is rare. However, a market with several firms, each facing a downward sloping demand curve will produce so that price exceeds marginal cost.

Scenario: Four firms with equal share (5,000) of a market for 20,000 toothbrushes at a price of \$1.50.

### THE DEMAND FOR TOOTHBRUSHES



**MEASURING MONOPOLY POWER**

- In perfect competition:  $P = MR = MC$
- Monopoly power:  $P > MC$

**LERNER'S INDEX OF MONOPOLY POWER**

$$L = (P - MC)/P$$

The larger the value of  $L$  (between 0 and 1) the greater the monopoly power.  $L$  is expressed in terms of  $E_d$

- $L = (P - MC)/P = -1/E_d$
- $E_d$  is elasticity of demand for a firm, not the market

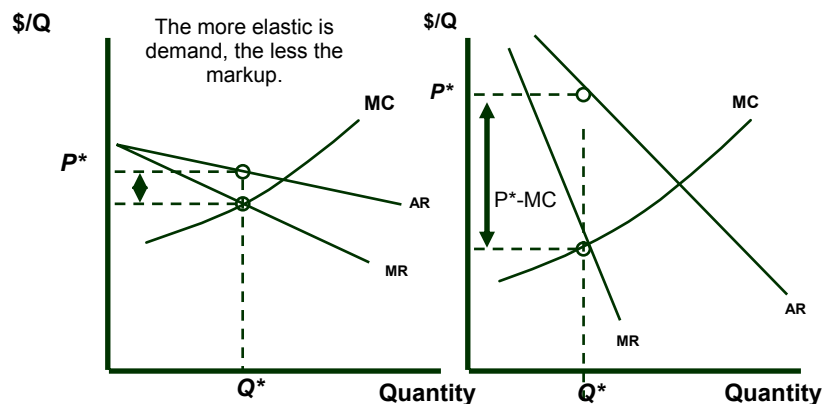
Monopoly power does not guarantee profits. Profit depends on average cost relative to price.

**THE RULE OF THUMB FOR PRICING**

$$P = \frac{MC}{1 + (1 / E_d)}$$

Pricing for any firm with monopoly power

- If  $E_d$  is large, markup is small
- If  $E_d$  is small, markup is large

**ELASTICITY OF DEMAND AND PRICE MARKUP****MARKUP PRICING: SUPERMARKETS TO DESIGNER JEANS**

Supermarkets

- Several firms
- Similar product
- $E_d = -10$  for individual stores

$$P = \frac{MC}{1 + (1/-1)} = \frac{MC}{0.9} = 1.11(MC)$$

Prices set about 10 – 11% above MC.

Convenience Stores

- Higher prices than supermarkets
- Convenience differentiates them
- $E_d = -5$

$$P = \frac{MC}{1 + (1/-5)} = \frac{MC}{0.8} = 1.25(MC)$$

Prices set about 25% above MC. Convenience stores have more monopoly power.

**Question: Do convenience stores have higher profits than supermarkets?**

Designer jeans

- $E_d = -3$  to  $-4$
- Price 33 - 50% > MC
- MC = \$12 - \$18/pair
- Wholesale price = \$18 - \$27

**SOURCES OF MONOPOLY POWER**

Why do some firm's have considerable monopoly power, and others have little or none?

A firm's monopoly power is determined by the firm's elasticity of demand. The firm's elasticity of demand is determined by:

- 1) Elasticity of market demand
- 2) Number of firms
- 3) The interaction among firms

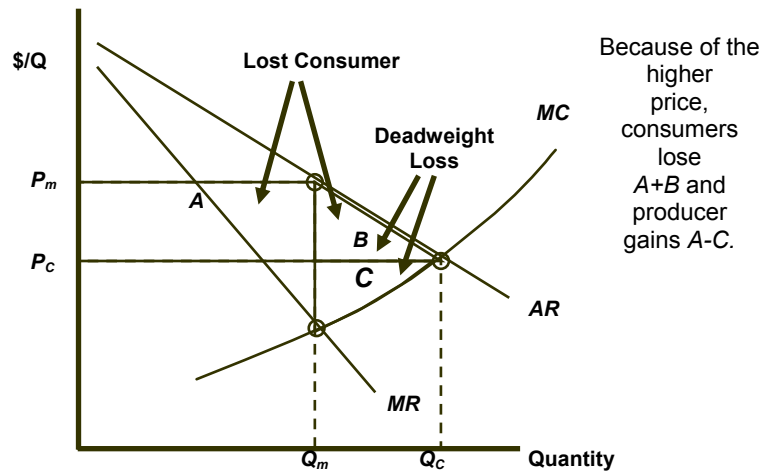
## LESSON 33

## MARKET STRUCTURE AND COMPETITIVE STRATEGY (Continued)

## THE SOCIAL COSTS OF MONOPOLY POWER

Monopoly power results in higher prices and lower quantities. However, does monopoly power make consumers and producers in the aggregate better or worse off?

## DEADWEIGHT LOSS FROM MONOPOLY POWER



## RENT SEEKING

Firms may spend to gain monopoly power

- Lobbying
- Advertising
- Building excess capacity

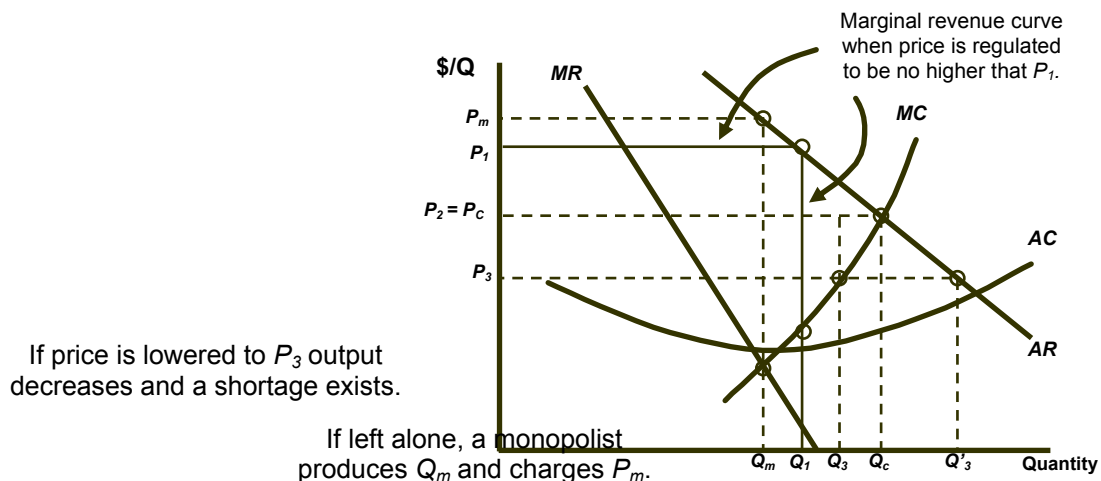
The incentive to engage in monopoly practices is determined by the profit to be gained. The larger the transfer from consumers to the firm, the larger the social cost of monopoly.

## PRICE REGULATION

Recall that in competitive markets, price regulation created a deadweight loss.

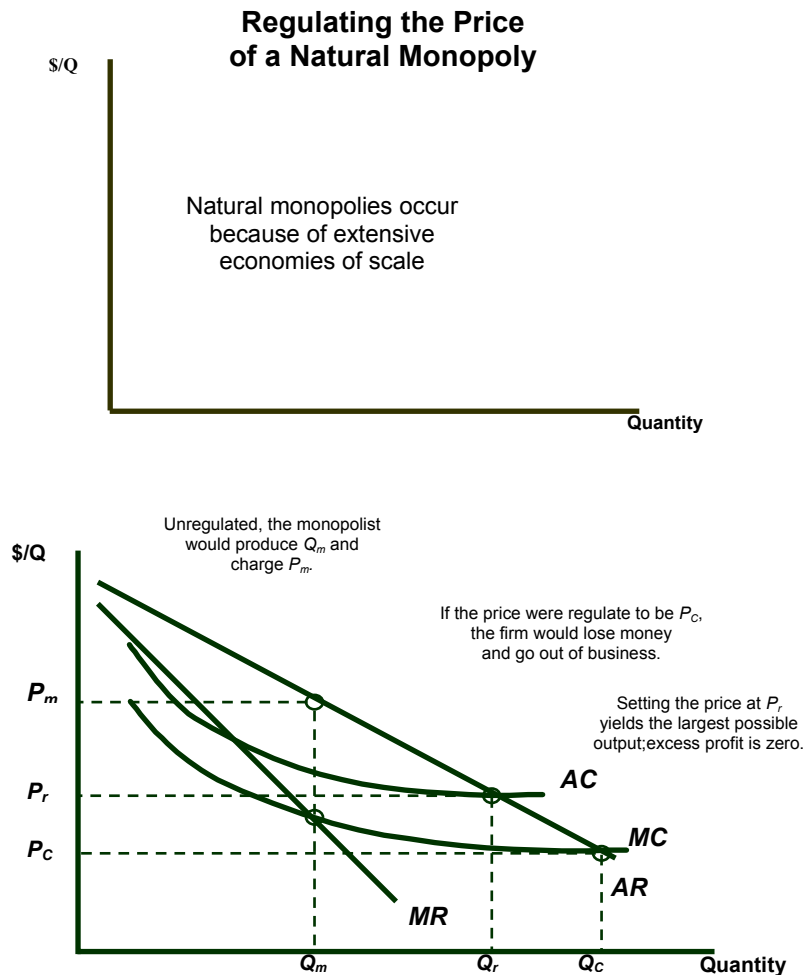
Question: What about a monopoly?

## PRICE REGULATION



## NATURAL MONOPOLY

A firm that can produce the entire output of an industry at a cost lower than what it would be if there were several firms.



## REGULATION IN PRACTICE

It is very difficult to estimate the firm's cost and demand functions because they change with evolving market conditions. An alternative pricing technique---rate-of-return regulation allows the firms to set a maximum price based on the expected rate or return that the firm will earn.

$$P = AVC + (D + T + sK)/Q, \text{ where}$$

$P$  = price,  $AVC$  = average variable cost

$D$  = depreciation,  $T$  = taxes

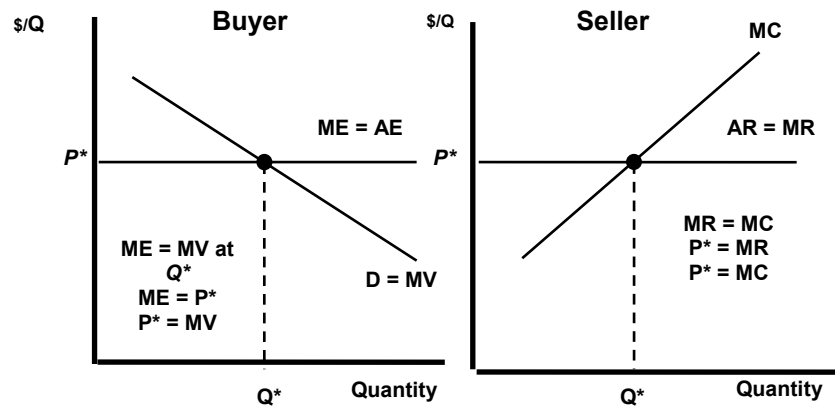
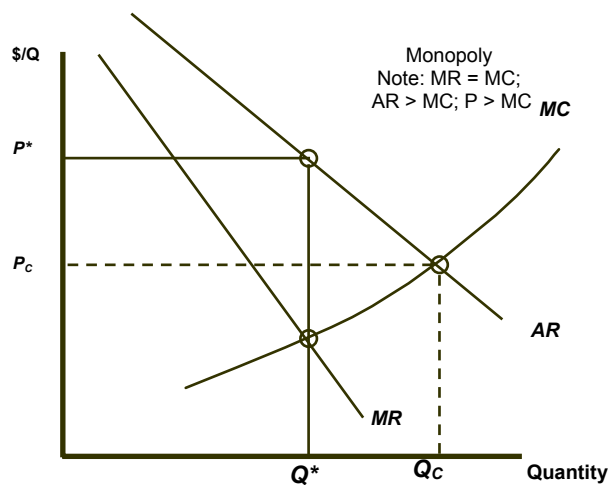
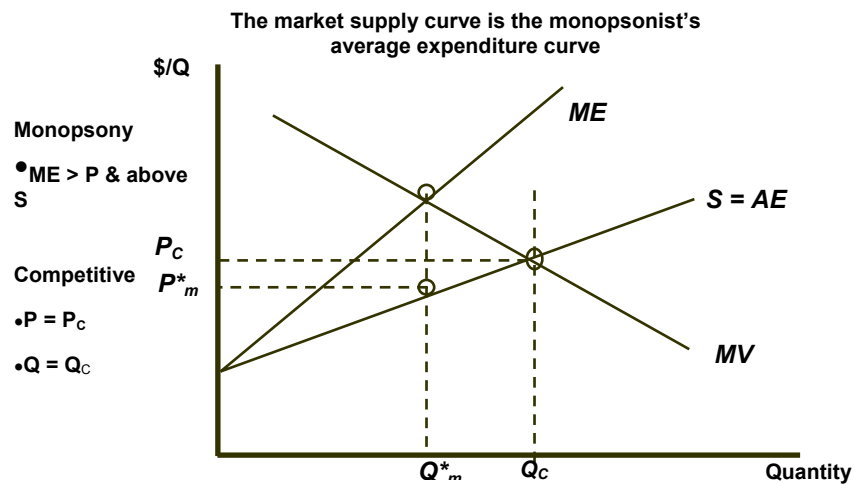
$s$  = allowed rate of return,  $K$  = firm's capital stock

## MONOPSONY

A monopsony is a market in which there is a single buyer. Monopsony power is the ability of the buyer to affect the price of the good and pay less than the price that would exist in a competitive market.

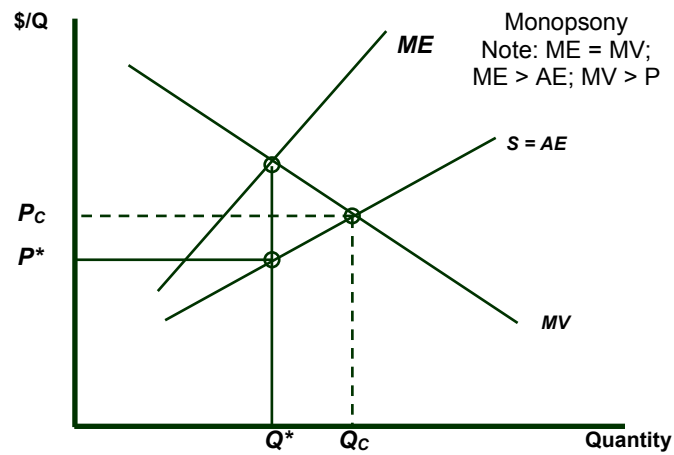
Competitive Buyer

- Price taker
- $P$  = Marginal expenditure = Average expenditure
- $D$  = Marginal value

**COMPETITIVE BUYER: COMPARED TO COMPETITIVE SELLER****MONOPSONIST BUYER**



# MONOPOLY AND MONOPSONY



## Monopoly

- $MR < P$
- $P > MC$
- $Q_m < Q_c$
- $P_m > P_c$

## Monopsony

- $ME > P$
- $P < MV$
- $Q_m < Q_c$
- $P_m < P_c$

## LESSON 34

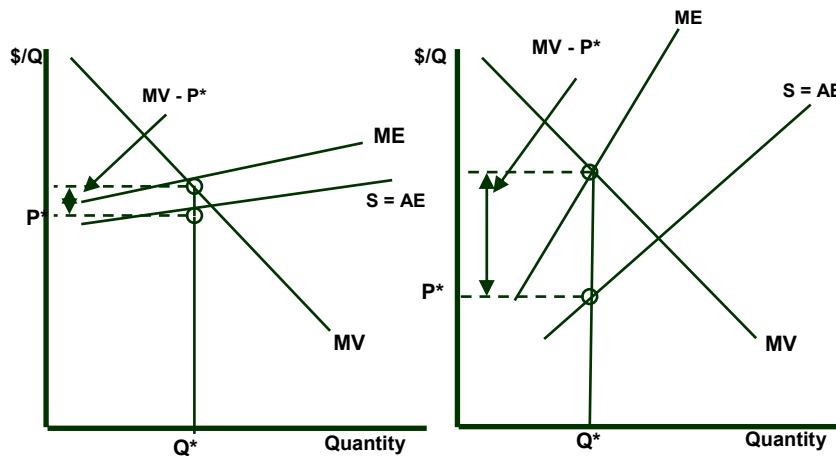
## PRICING WITH MARKET POWER

## MONOPSONY POWER

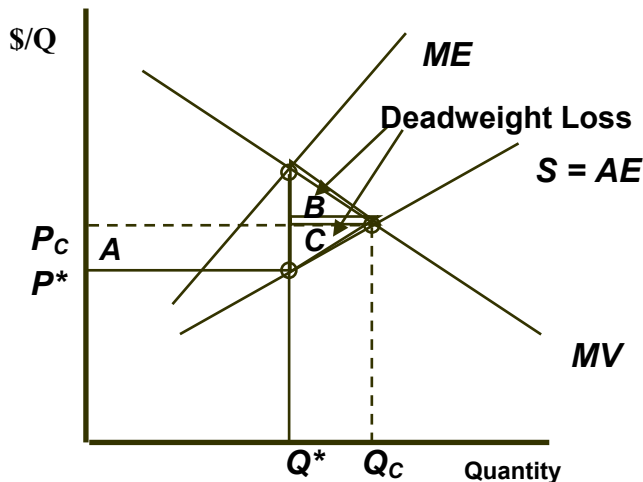
A few buyers can influence price (e.g. automobile industry). Monopsony power gives them the ability to pay a price that is less than marginal value. The degree of monopsony power depends on three similar factors.

- 1) Elasticity of market supply: The less elastic the market supply, the greater the monopsony power.
- 2) Number of buyers: The fewer the number of buyers, the less elastic the supply and the greater the monopsony power.
- 3) Interaction Among Buyers: The less the buyers compete, the greater the monopsony power.

**MONOPSONY POWER:  
IF THE ELASTIC VERSUS INELASTIC SUPPLY**



## DEADWEIGHT LOSS FROM MONOPSONY POWER



## DETERMINING THE DEADWEIGHT LOSS IN MONOPSONY

- Change in seller's surplus =  $-A - C$
- Change in buyer's surplus =  $A - B$
- Change in welfare =  $-A - C + A - B = -C - B$
- Inefficiency occurs because less is purchased

## THE SOCIAL COST OF MONOPSONY POWER

### BILATERAL MONOPOLY

Bilateral monopoly is rare, however, markets with a small number of sellers with monopoly power selling to a market with few buyers with monopsony power is more common.

**Question: In this case, what is likely to happen to price?**

### LIMITING MARKET POWER: THE ANTITRUST LAWS

Antitrust Laws promote a competitive economy. Rules and regulations designed to promote a competitive economy by:

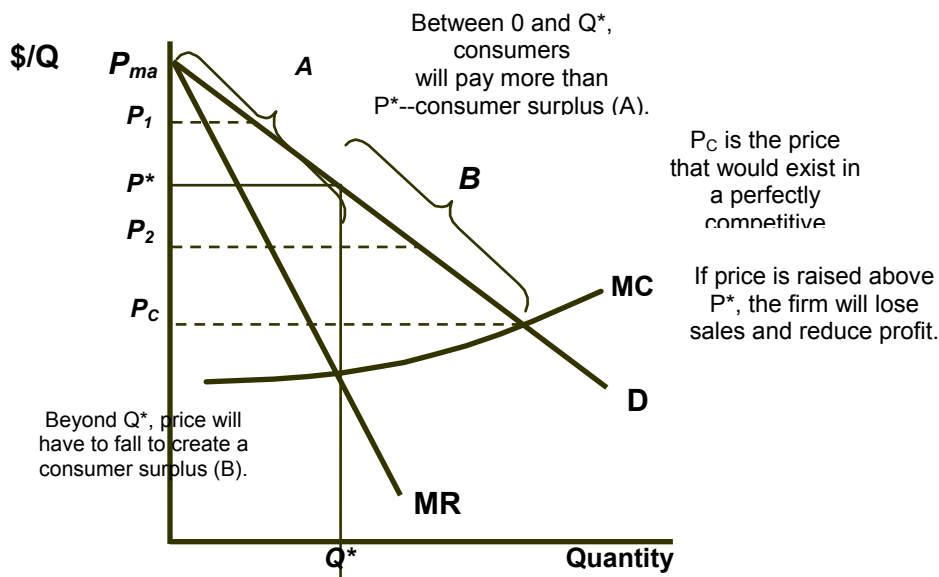
- Prohibiting actions that restrain or are likely to restrain competition
- Restricting the forms of market structures that are allowable

### PRICING WITH MARKET POWER

Pricing without market power (perfect competition) is determined by market supply and demand. The individual producer must be able to forecast the market and then concentrate on managing production (cost) to maximize profits.

Pricing with market power (imperfect competition) requires the individual producer to know much more about the characteristics of demand as well as manage production.

### CAPTURING CONSUMER SURPLUS



### CAPTURING CONSUMER SURPLUS

- $P^*Q^*$ : single  $P$  &  $Q$  @  $MC=MR$
- A: consumer surplus with  $P^*$
- B:  $P > MC$  & consumer would buy at a lower price
- $P_1$ : less sales and profits
- $P_2$ : increase sales & and reduce revenue and profits
- $P_C$ : competitive price

Question How can the firm capture the consumer surplus in A and sell profitably in B?

Answer Price discrimination Two-part tariffs Bundling

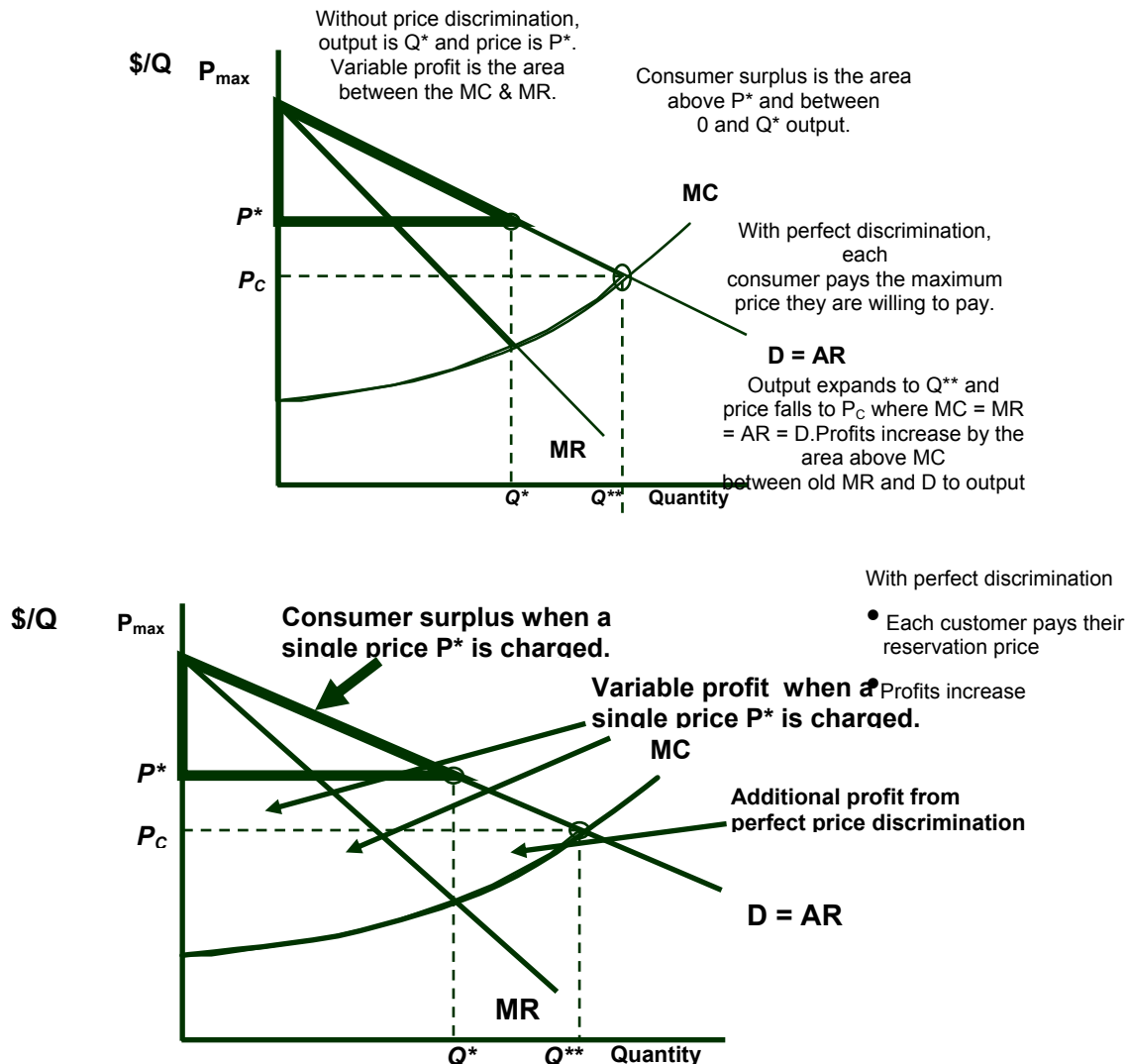
## PRICE DISCRIMINATION

Price discrimination is the charging of different prices to different consumers for similar goods.

### FIRST DEGREE PRICE DISCRIMINATION

**First Degree Price Discrimination** charge a separate price to each customer: the maximum or reservation price they are willing to pay.

#### ADDITIONAL PROFIT FROM PERFECT FIRST-DEGREE PRICE DISCRIMINATION



**Question: Why would a producer have difficulty in achieving first-degree price discrimination?**

Answer

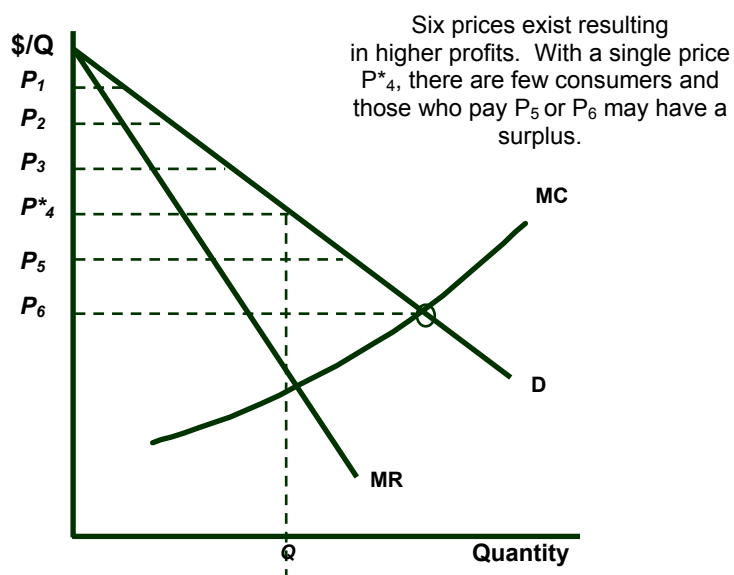
- 1) Too many customers (impractical)
- 2) Could not estimate the reservation price for each customer

The model does demonstrate the potential profit (incentive) of practicing price discrimination to some degree. Examples of imperfect price discrimination where the seller has the ability to segregate the market to some extent and charge different prices for the same product:

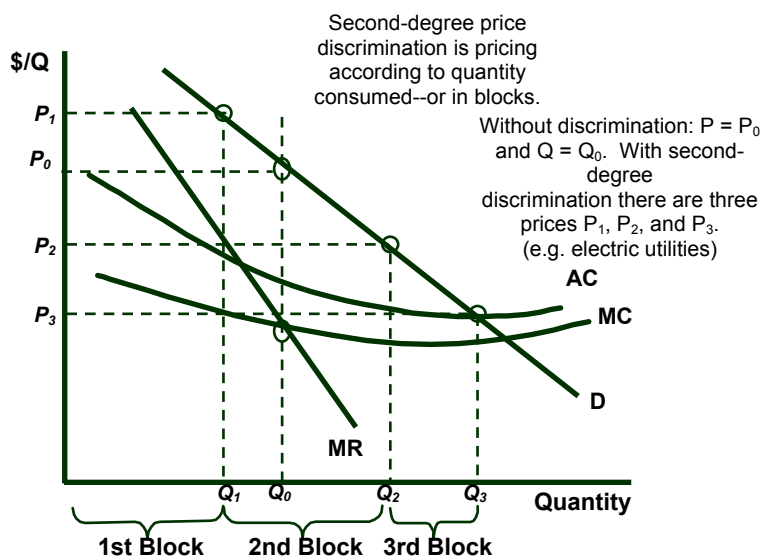
- Lawyers, doctors, accountants
- Car salesperson (15% profit margin)

- Colleges and universities

### FIRST-DEGREE PRICE DISCRIMINATION IN PRACTICE



### SECOND-DEGREE PRICE DISCRIMINATION



Economies of scale permit:

- Increase consumer welfare
- Higher profits

## LESSON 35

## PRICING WITH MARKET POWER (Continued)

## THIRD DEGREE PRICE DISCRIMINATION

Third degree price discrimination divides the market into two-groups. Each group has its own demand function. It is most common type of price discrimination. Examples: airlines, vegetables, discounts to students and senior citizens. Third-degree price discrimination is feasible when the seller can separate his/her market into groups who have different price elasticities of demand. (e.g. business air travelers versus vacation air travelers)

## OBJECTIVES

$$\begin{aligned} MR_1 &= MR_2 \\ MC_1 &= MR_1 \text{ and } MC_2 = MR_2 \\ MR_1 &= MR_2 = MC \\ P_1 &: \text{price first group} \\ P_2 &: \text{price second group} \\ C(Q_r) &= \text{total cost of } Q_T = Q_1 + Q_2 \\ \text{Profit } (\pi) &= P_1 Q_1 + P_2 Q_2 - C(Q_r) \end{aligned}$$

Set incremental  $\pi$  for sales to group 1=0

$$\frac{\Delta \pi}{\Delta Q_1} = \frac{\Delta(P_1 Q_1)}{\Delta Q_1} - \frac{\Delta C}{\Delta Q_1} = 0$$

$$\frac{\Delta(P_1 Q_1)}{\Delta Q_1} = MR_1 - \frac{\Delta C}{\Delta Q_1} = MC$$

Second group of customers:  $MR_2 = MC$

$$MR_1 = MR_2 = MC$$

Determining relative prices

$$\text{Recall: } MR = P (1 + 1/E_d)$$

$$\text{Then : } MR_1 = P_1 (1 + 1/E_1) = MR_2 = P_2 (1 + 1/E_2)$$

Determining relative prices

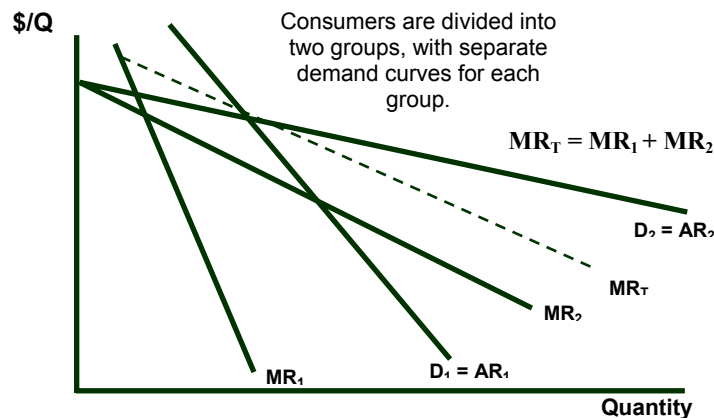
$$\text{And : } \frac{P_1}{P_2} = \frac{(1 + 1/E_2)}{(1 + 1/E_1)}$$

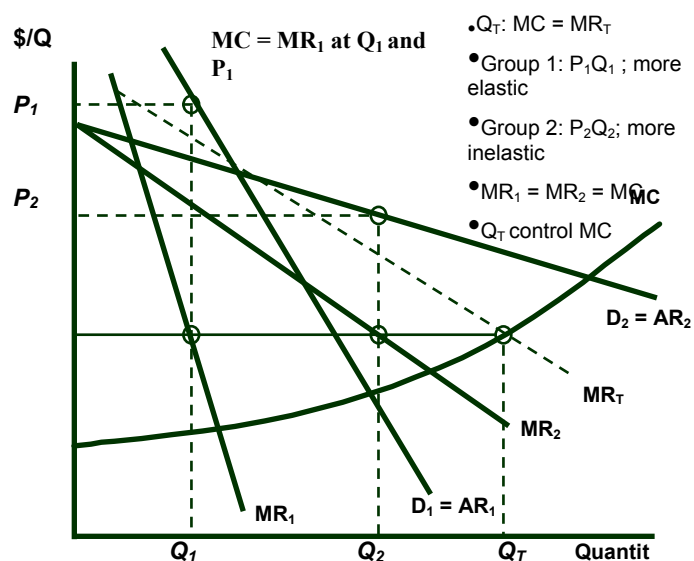
Pricing: Charge higher price to group with a low demand elasticity

Example:  $E_1 = -2$  &  $E_2 = -4$

$$\frac{P_1}{P_2} = \frac{(1 - 1/4)}{(1 - 1/2)} = 3/4 / 1/2 = 1.5$$

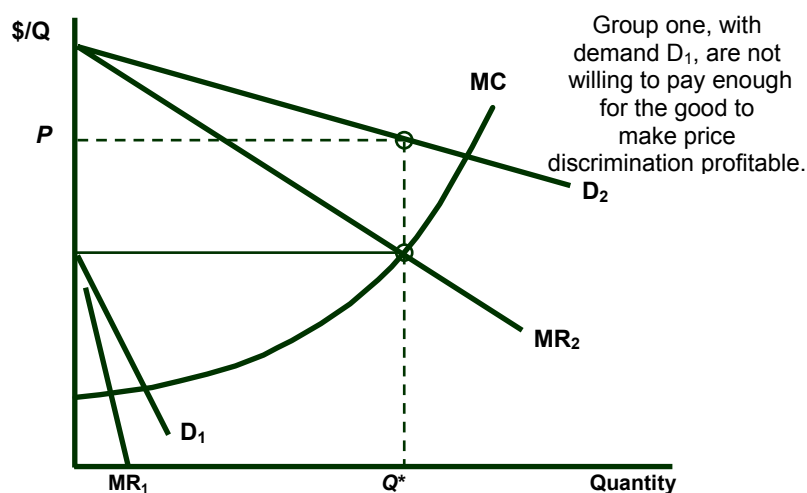
$P_1$  should be 1.5 times as high as  $P_2$





### NO SALES TO SMALLER MARKET

Even if third-degree price discrimination is feasible, it doesn't always pay to sell to both groups of consumers if marginal cost is rising.



### THE ECONOMICS OF COUPONS AND REBATES

#### PRICE DISCRIMINATION

Those consumers who are more price elastic will tend to use the coupon/rebate more often when they purchase the product than those consumers with a less elastic demand. Coupons and rebate programs allow firms to price discriminate.

**PRICE ELASTICITIES OF DEMAND FOR USERS VERSUS NONUSERS OF COUPONS**

| <u>Product</u>          | <u>Price Elasticity</u> |              |
|-------------------------|-------------------------|--------------|
|                         | <u>Nonusers</u>         | <u>Users</u> |
| Toilet tissue           | -0.60                   | -0.66        |
| Stuffing/dressing       | -0.71                   | -0.96        |
| Shampoo                 | -0.84                   | -1.04        |
| Cooking/salad oil       | -1.22                   | -1.32        |
| Dry mix dinner          | -0.88                   | -1.09        |
| Cake mix                | -0.21                   | -0.43        |
| Cat food                | -0.49                   | -1.13        |
| Frozen entrée           | -0.60                   | -0.95        |
| Gelatin                 | -0.97                   | -1.25        |
| Spaghetti sauce         | -1.65                   | -1.81        |
| Crème rinse/conditioner | -0.82                   | -1.12        |
| Soup                    | -1.05                   | -1.22        |
| Hot dogs                | -0.59                   | -0.77        |

**Cake Mix**

- Nonusers of coupons:  $P_E = -0.21$
- Users:  $P_E = -0.43$

**Cake Mix Brand A**

- $P_E$ : 8 to 10 times cake mix  $P_E$

**Example**

- $P_E$  Users: -4
- $P_E$  Nonusers: -2

Using:

$$\frac{P_1}{P_2} = \frac{(1 + 1/E_2)}{(1 + 1/E_1)}$$

Price of nonusers should be 1.5 times users or, if cake mix sells for \$1.50, coupons should be 50 cents.



## LESSON 36

## PRICING WITH MARKET POWER (Continued)

## AIRLINE FARES

Differences in elasticities imply that some customers will pay a higher fare than others. Business travelers have few choices and their demand is less elastic. Casual travelers have choices and are more price sensitive.

## Elasticities of Demand for Air Travel

| Elasticity | First-Class | Economy Plus | Economy |
|------------|-------------|--------------|---------|
| Price      | -0.3        | -0.4         | -0.9    |
| Income     | 1.2         | 1.2          | 1.8     |

The airlines separate the market by setting various restrictions on the tickets.

- Less expensive: notice, stay over the weekend, no refund
- Most expensive: no restrictions

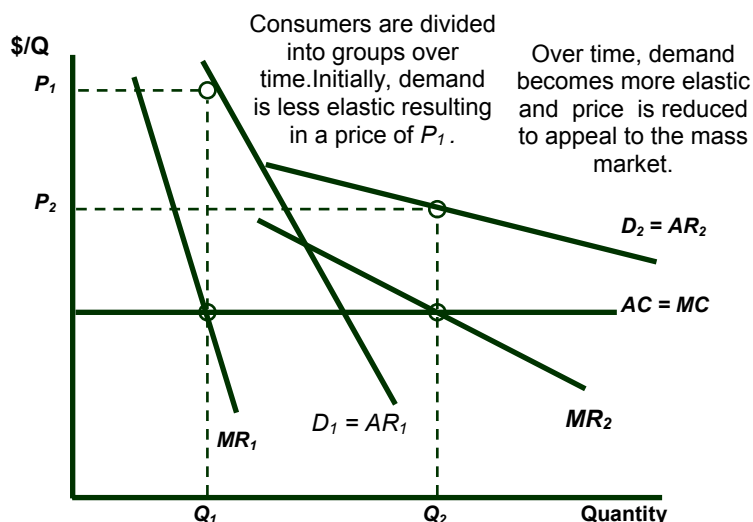
### INTERTEMPORAL PRICE DISCRIMINATION AND PEAK-LOAD PRICING SEPARATING THE MARKET WITH TIME

Initial release of a product, the demand is inelastic

- Book
- Movie
- Computer

Once this market has yielded a maximum profit, firms lower the price to appeal to a general market with a more elastic demand

- Paper back books
- Dollar Movies
- Discount computers

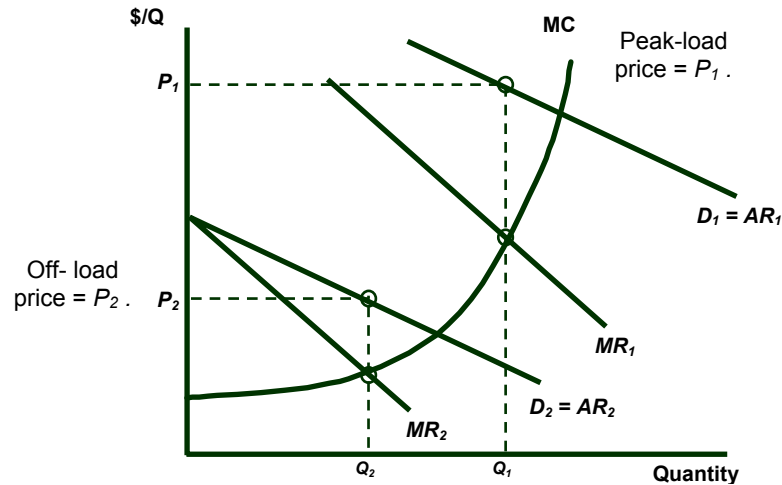


### PEAK-LOAD PRICING

Demand for some products may peak at particular times.

- Rush hour traffic
- Electricity - summer season
- Restaurants on weekends

Capacity restraints will also increase MC. Increased MR and MC would indicate a higher price. MR is not equal for each market because one market does not impact the other market.



### HOW TO PRICE A BEST SELLING NOVEL

What Do You Think?

- 1) How would you arrive at the price for the initial release of the hardbound edition of a book?
- 2) How long do you wait to release the paperback edition? Could the popularity of the book impact your decision?
- 3) How do you determine the price for the paperback edition?

### THE TWO-PART TARIFF

The purchase of some products and services can be separated into two decisions, and therefore, two prices.

Examples

Amusement Park

- Pay to enter
- Pay for rides and food within the park

Tennis Club

- Pay to join
- Pay to play

Safety Razor

- Pay for razor
- Pay for blades

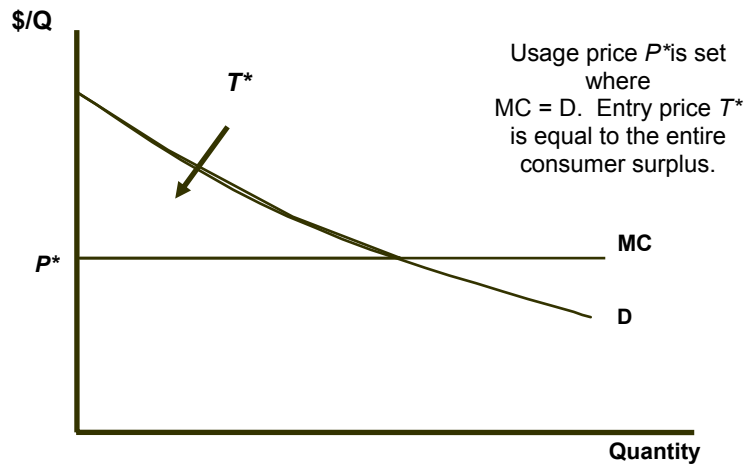
Polaroid Film

- Pay for the camera

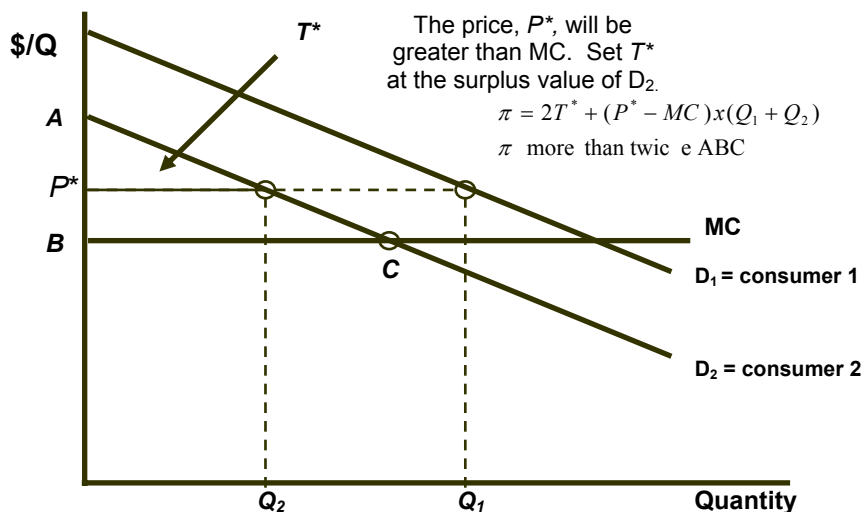
- Pay for the film

Pricing decision is setting the entry fee ( $T$ ) and the usage fee ( $P$ ). Choosing the trade-off between free-entry and high use prices or high-entry and zero use prices.

### TWO-PART TARIFF WITH A SINGLE CONSUMER



### TWO-PART TARIFF WITH TWO CONSUMERS

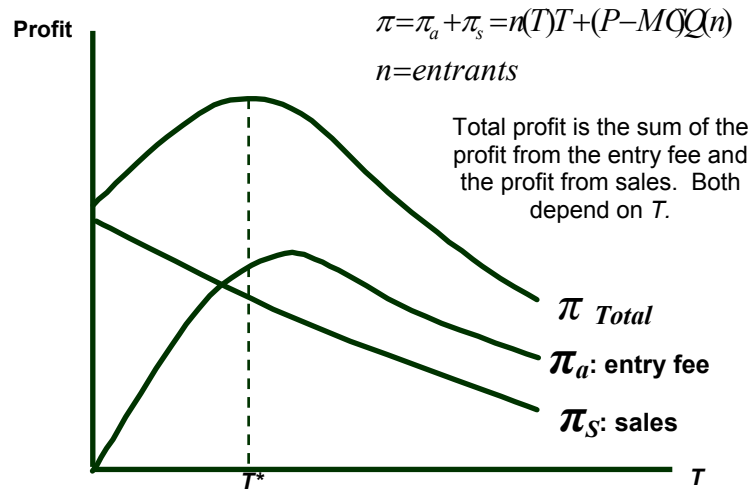


### THE TWO-PART TARIFF WITH MANY DIFFERENT CONSUMERS

No exact way to determine  $P^*$  and  $T^*$ . Must consider the trade-off between the entry fee  $T^*$  and the use fee  $P^*$ .

Low entry fee: High sales and falling profit with lower price and more entrants.

To find optimum combination, choose several combinations of  $P, T$ . Choose the combination that maximizes profit.

**RULE OF THUMB**

- Similar demand: Choose  $P$  close to  $MC$  and high  $T$
- Dissimilar demand: Choose high  $P$  and low  $T$ .

**TWO-PART TARIFF WITH A TWIST**

Entry price ( $T$ ) entitles the buyer to a certain number of free units

- Razors with several blades
- Amusement parks with some tokens
- On-line with free time

## LESSON 37

## PRICING WITH MARKET POWER (Continued)

**BUNDLING**

Bundling is packaging two or more products to gain a pricing advantage.

**CONDITIONS NECESSARY FOR BUNDLING**

- Heterogeneous customers
- Price discrimination is not possible
- Demands must be negatively correlated

**AN EXAMPLE: LEASING MOVIE X & MOVIE Y**

The reservation prices for each theater and movie are:

|           | Movie X  | Movie Y |
|-----------|----------|---------|
| Theater A | \$12,000 | \$3,000 |
| Theater B | \$10,000 | \$4,000 |

Renting the movies separately would result in each theater paying the lowest reservation price for each movie:

Maximum price X = \$10,000  
 Maximum price Y = \$3,000  
 Total Revenue = \$26,000

If the movies are bundled:

- Theater A will pay \$15,000 for both
- Theater B will pay \$14,000 for both

If each were charged the lower of the two prices, total revenue will be \$28,000.

**Relative Valuations**

Negative Correlated: Profitable to Bundle

- A pays more for X (\$12,000) than B (\$10,000).
- B pays more for Y (\$4,000) than A (\$3,000).

If the demands were positively correlated (Theater A would pay more for both films as shown) bundling would not result in an increase in revenue.

|           | Movie X  | Movie Y |
|-----------|----------|---------|
| Theater A | \$12,000 | \$4,000 |
| Theater B | \$10,000 | \$3,000 |

If the movies are bundled:

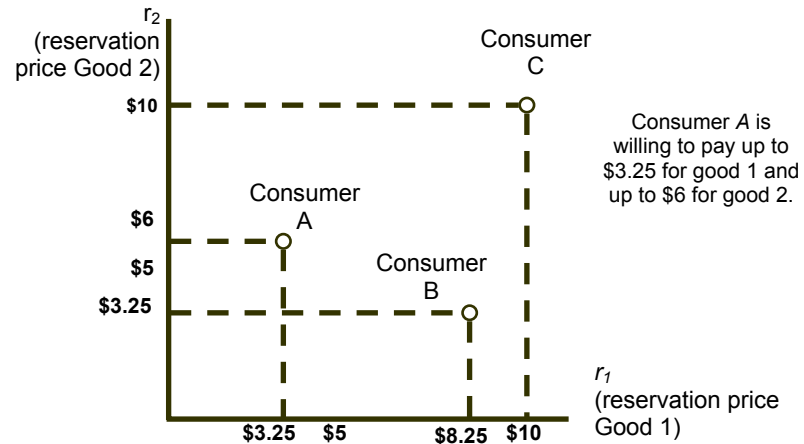
- Theater A will pay \$16,000 for both
- Theater B will pay \$13,000 for both

If each were charged the lower of the two prices, total revenue will be \$26,000, the same as by selling the films separately.

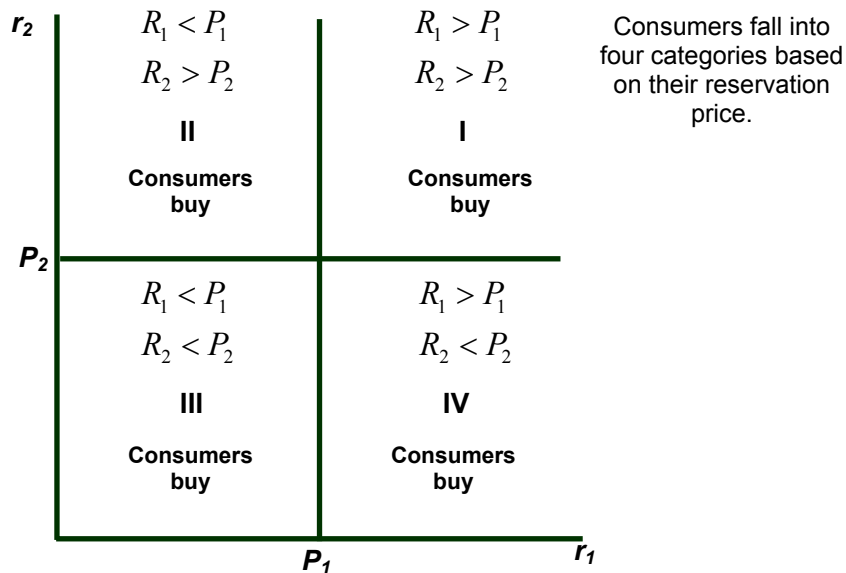
**BUNDLING SCENARIO: TWO DIFFERENT GOODS AND MANY CONSUMERS**

Many consumers with different reservation price combinations for two goods

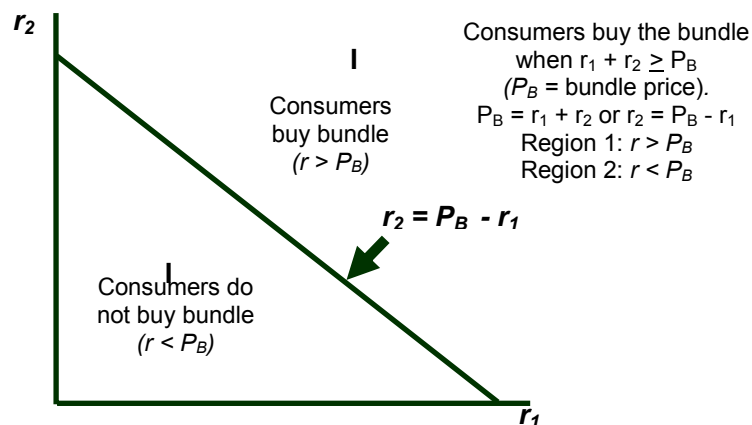
## RESERVATION PRICES



## CONSUMPTION DECISIONS WHEN PRODUCTS ARE SOLD SEPARATELY

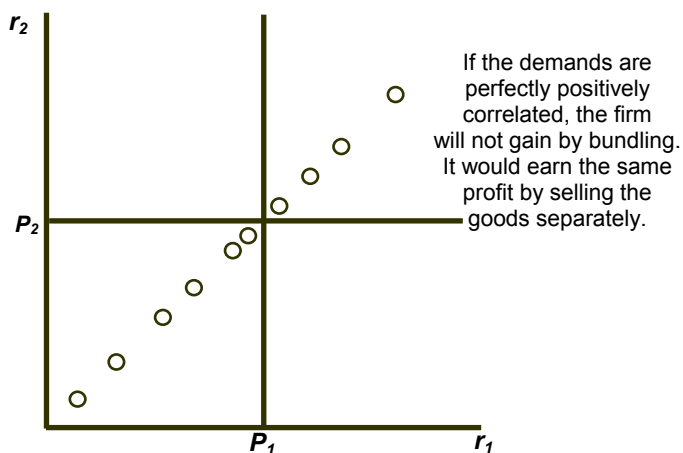


## CONSUMPTION DECISIONS WHEN PRODUCTS ARE BUNDLED

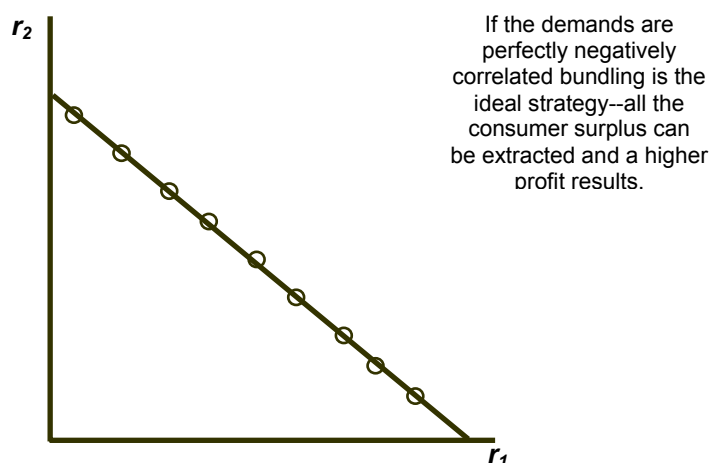


The effectiveness of bundling depends upon the degree of negative correlation between the two demands.

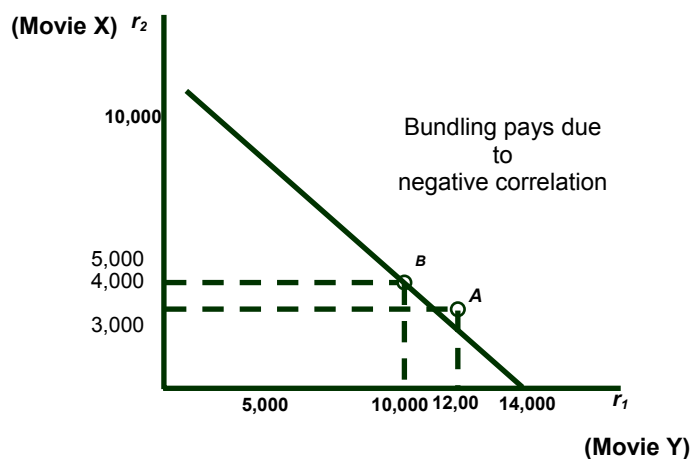
### RESERVATION PRICES



### RESERVATION PRICES



### MOVIE EXAMPLE



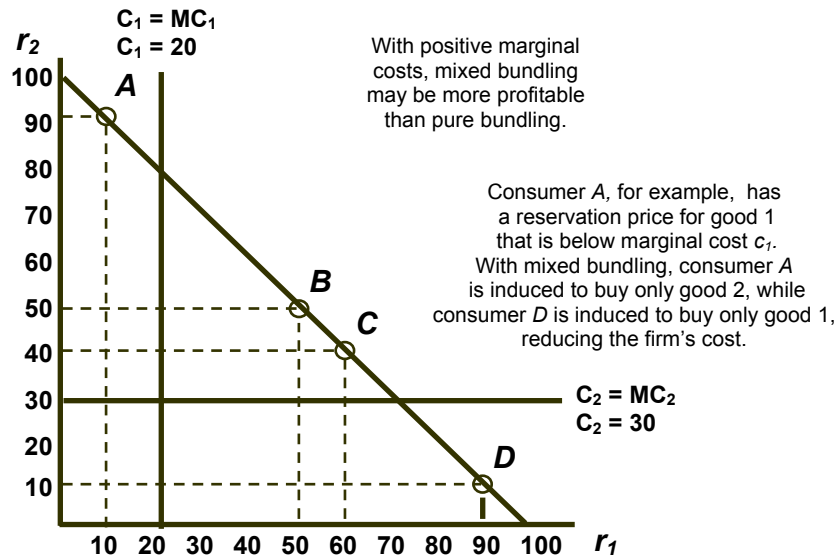
## LESSON 38

## PRICING WITH MARKET POWER (Continued)

## TYPES OF BUNDLING

- Mixed Bundling: Selling both as a bundle and separately
- Pure Bundling: Selling only a package

## MIXED VERSUS PURE BUNDLING



## MIXED VS. PURE BUNDLING: SCENARIO

- Perfect negative correlation
- Significant marginal cost

## Observations

Reservation price is below MC for some consumers. Mixed bundling induces the consumers to buy only goods for which their reservation price is greater than MC.

## BUNDLING EXAMPLE

- Sell Separately  
Consumers B, C, and D buy 1 and A buys 2
- Pure Bundling  
Consumers A, B, C, and D buy the bundle
- Mixed Bundling  
Consumer D buys 1, A buys 2, and B & C buy the bundle

|                 | P1      | P2      | PB    | Profit   |
|-----------------|---------|---------|-------|----------|
| Sell separately | \$50    | \$90    | ---   | \$150    |
| Pure bundling   | ---     | ---     | \$100 | \$200    |
| Mixed bundling  | \$89.95 | \$89.95 | \$100 | \$229.90 |



$$C_1 = \$20$$

$$C_2 = \$30$$

Sell Separately

$$3(\$50 - \$20) + 1(\$90 - \$30) = \$150$$

Pure Bundling

$$4(\$100 - \$20 - \$30) = \$200$$

Mixed Bundling

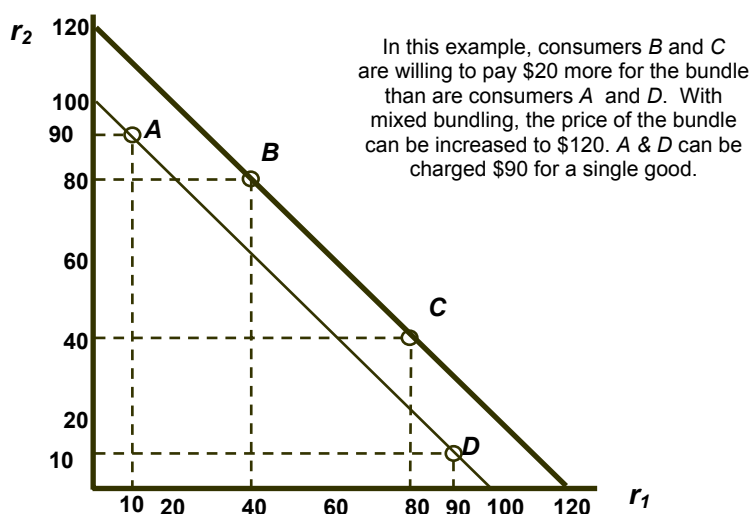
$$(\$89.95 - \$20) + (\$89.95 - \$30) - 2(\$100 - \$20 - \$30) = \$229.90$$

$$C_1 = \$20 \quad C_2 = \$30$$

Question

If  $MC = 0$ , would mixed bundling still be the most profitable strategy with perfect negative correlation?

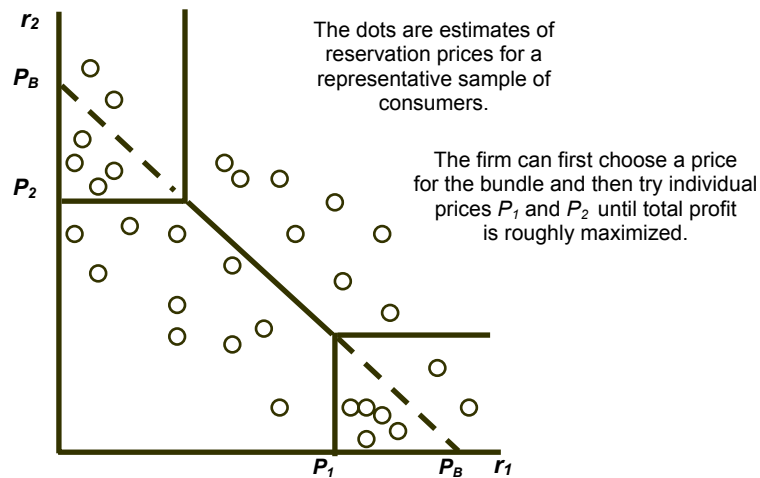
### MIXED BUNDLING WITH ZERO MARGINAL COSTS



|                 | P1   | P2   | PB    | Profit |
|-----------------|------|------|-------|--------|
| Sell separately | \$80 | \$80 | ----  | \$320  |
| Pure bundling   | ---- | ---- | \$100 | \$400  |
| Mixed bundling  | \$90 | \$90 | \$120 | \$420  |

### BUNDLING IN PRACTICE

- Automobile option packages
- Vacation travel
- Cable television
- ♦ Mixed Bundling in Practice
  - Use of market surveys to determine reservation prices
  - Design a pricing strategy from the survey results



### The Complete Dinner vs. a la Carte: A Restaurant's Pricing Problem

- ♦ Pricing to match consumer preferences for various selections
- ♦ Mixed bundling allows the customer to get maximum utility from a given expenditure by allowing a greater number of choices.

### Bundling

Tying is a practice of requiring a customer to purchase one good in order to purchase another.

#### Examples

- Xerox machines and the paper
- IBM mainframe and computer cards

Allows the seller to meter the customer and use a two-part tariff to discriminate against the heavy user

- McDonald's

Allows them to protect their brand name.

### ADVERTISING

#### Assumptions

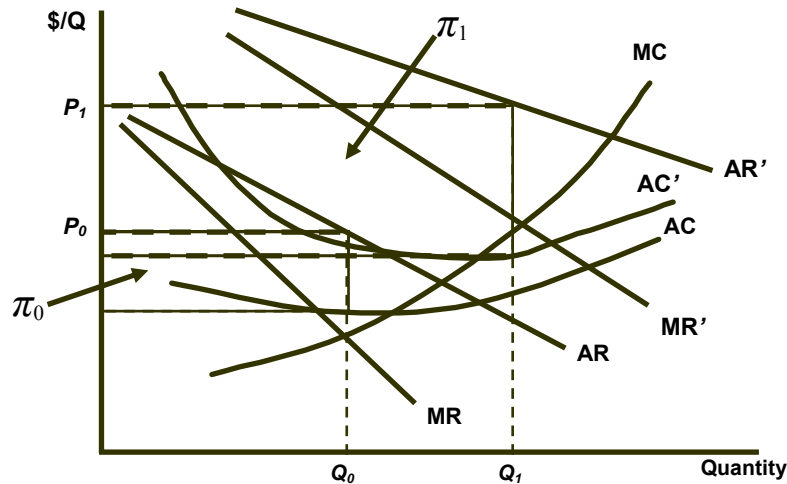
- Firm sets only one price
- Firm knows  $Q(P, A)$

How quantity demanded depends on price and advertising

## EFFECTS OF ADVERTISING

AR and MR are average and marginal revenue when the firm doesn't advertise.

If the firm advertises, its average and marginal revenue curves shift to the right -- average costs rise, but marginal cost does not.



## ADVERTISING

Choosing Price and Advertising Expenditure

$$\pi = PQ(P, A) - C(Q) - A$$

$$MR_{Ads} = P \frac{\Delta Q}{\Delta A} = 1 + MC \frac{\Delta Q}{\Delta A} = \text{full MC of adv.}$$

## A RULE OF THUMB FOR ADVERTISING

$$(A/Q)(\Delta Q/\Delta A) = E_A = \text{Adv. elasticity of demand}$$

$$(P - MC)/P = -1/E_p$$

$$A/PQ = -(E_A/E_p) = \text{Rule of Thumb}$$

To maximize profit, the firm's advertising-to-sales ratio should be equal to minus the ratio of the advertising and price elasticities of demand.

- $R(Q) = \$1 \text{ million/yr}$
- \$10,000 budget for A (advertising--1% of revenues)
- $E_A = .2$  (increase budget \$20,000, sales increase by 20%)
- $E_p = -4$  (markup price over MC is substantial)

Question

Should the firm increase advertising?

YES

- $A/PQ = -(2/-4) = 5\%$
- Increase budget to \$50,000

Questions

When  $E_A$  is large, do you advertise more or less?

When  $E_P$  is large, do you advertise more or less?

#### Advertising: In Practice

Estimate the level of advertising for each of the firms

- Supermarkets  $E_P = -10$ ;  $E_A = 0.1$  to  $0.3$
- Convenience stores  $E_P = -5$ ;  $E_A =$  very small
- Designer jeans  $E_P = -3$  to  $-4$ ;  $E_A = 0.3$  to  $1$
- Laundry detergents  $E_P = -3$  to  $-4$ ;  $E_A =$  very large

## LESSON 39

## MONOPOLISTIC COMPETITION

## CHARACTERISTICS

- Many firms
- Free entry and exit
- Differentiated product

The amount of monopoly power depends on the degree of differentiation.

Examples of this very common market structure include:

Toothpaste, Soap, Cold remedies

Toothpaste

Brand J and monopoly power

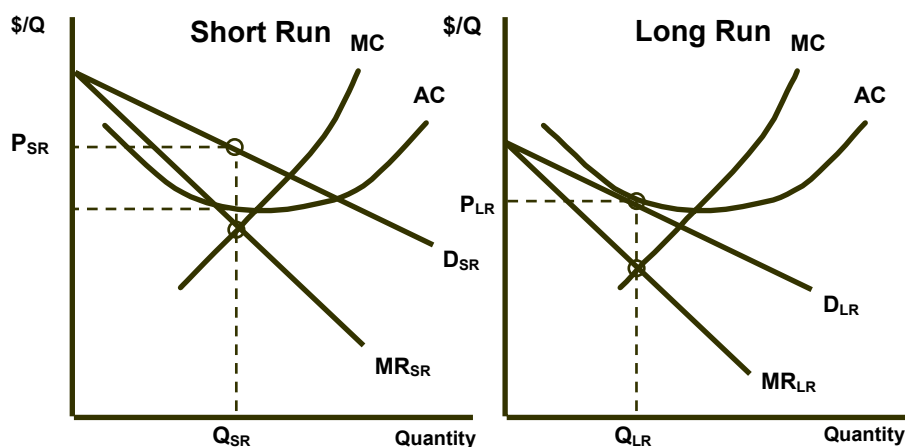
Suppose an MNC is the sole producer of Brand J. Consumers can have a preference for Brand J---taste, reputation, decay preventing efficacy. The greater the preference (differentiation) the higher the price.

## THE MAKINGS OF MONOPOLISTIC COMPETITION

Two important characteristics

- 1) Differentiated but highly substitutable products
- 2) Free entry and exit

## A MONOPOLISTICALLY COMPETITIVE FIRM IN THE SHORT AND LONG RUN



Observations (short-run)

- Downward sloping demand--differentiated product
- Demand is relatively elastic--good substitutes
- $MR < P$
- Profits are maximized when  $MR = MC$

This firm is making economic profits

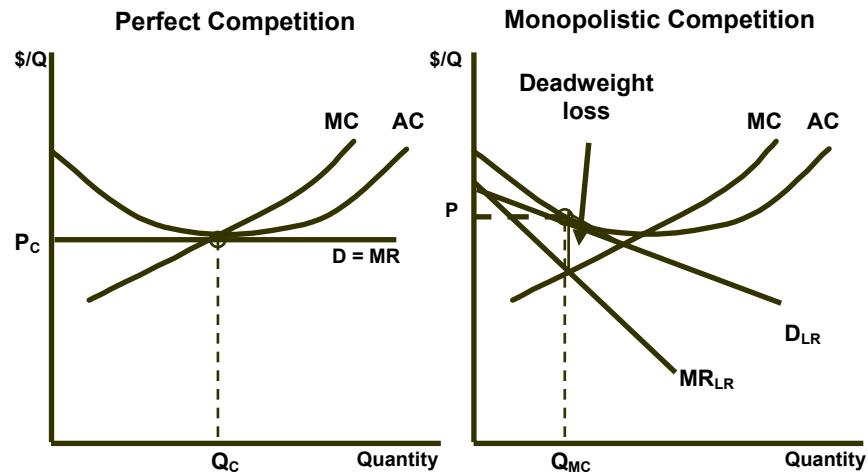
Observations (long-run)

- Profits will attract new firms to the industry (no barriers to entry)
- The old firm's demand will decrease to  $D_{LR}$
- Firm's output and price will fall
- Industry output will rise

No economic profit ( $P = AC$ )

$P > MC$  -- some monopoly power

### MONOPOLISTICALLY COMPETITIVE VS. PERFECTLY COMPETITIVE EQUILIBRIUM



### MONOPOLISTIC COMPETITION AND ECONOMIC EFFICIENCY

The monopoly power (differentiation) yields a higher price than perfect competition. If price was lowered to the point where  $MC = D$ , consumer surplus would increase by the shaded triangle. With no economic profits in the long run, the firm is still not producing at minimum AC and excess capacity exists.

#### Questions

If the market became competitive, what would happen to output and price? Should monopolistic competition be regulated?

### MONOPOLISTIC COMPETITION IN THE MARKET FOR COLAS AND COFFEE

The markets for soft drinks and coffee illustrate the characteristics of monopolistic competition.

#### ELASTICITIES OF DEMAND FOR BRANDS OF COLAS AND COFFEE

Colas: Brand X -2.4

Brand Y - 5.2 to -5.7

Ground Coffee: Hills Brothers -7.1

Maxwell House -8.9

Chase and Sanborn -5.6

#### Questions

- 1) Why is the demand for Brand X more price inelastic than for Brand Y?
- 2) Is there much monopoly power in these two markets?
- 3) Define the relationship between elasticity and monopoly power.

**LESSON 40****OLIGOPOLY****Characteristics**

- Small number of firms
- Product differentiation may or may not exist
- Barriers to entry

**Examples**

- Automobiles
- Steel
- Aluminum
- Petrochemicals
- Electrical equipment
- Computers

The barriers to entry are:

**Natural**

- Scale economies
- Patents
- Technology
- Name recognition

**Strategic action**

- Flooding the market
- Controlling an essential input

**Management Challenges**

- Strategic actions
- Rival behavior

**Question**

What are the possible rival responses to a 10% price cut by an automobile company?

**EQUILIBRIUM IN AN OLIGOPOLISTIC MARKET**

In perfect competition, monopoly, and monopolistic competition the producers did not have to consider a rival's response when choosing output and price. In oligopoly the producers must consider the response of competitors when choosing output and price.

Defining Equilibrium: Firms doing the best they can and have no incentive to change their output or price. All firms assume competitors are taking rival decisions into account.

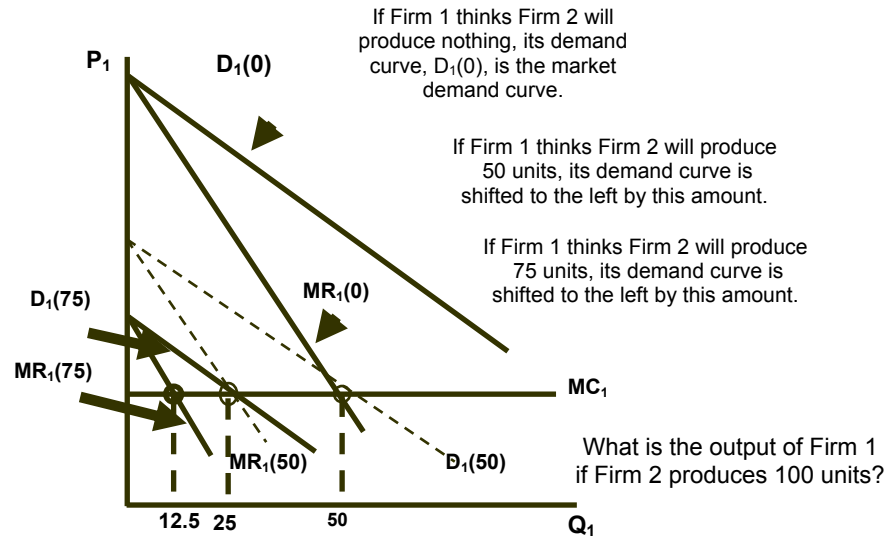
**NASH EQUILIBRIUM**

Each firm is doing the best it can given what its competitors are doing.

**THE COURNOT MODEL****DUOPOLY**

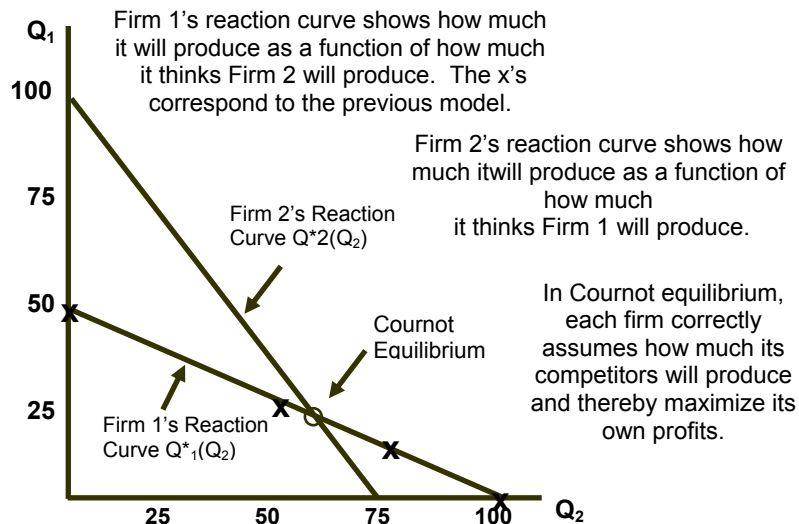
- Two firms competing with each other
- Homogenous good
- The output of the other firm is assumed to be fixed

### Firm 1's Output Decision



### THE REACTION CURVE

A firm's profit-maximizing output is a decreasing schedule of the expected output of Firm 2.



### Questions

If the firms are not producing at the Cournot equilibrium, will they adjust until the Cournot equilibrium is reached?

When is it rational to assume that its competitor's output is fixed?

### THE LINEAR DEMAND CURVE: AN EXAMPLE OF THE COURNOT EQUILIBRIUM

Duopoly

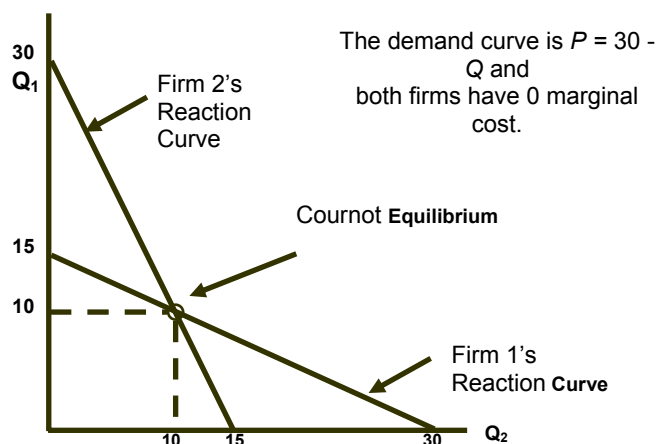
Market demand is  $P = 30 - Q$

where  $Q = Q_1 + Q_2$

$MC_1 = MC_2 = 0$



### Duopoly Example



### Profit Maximization with Collusion

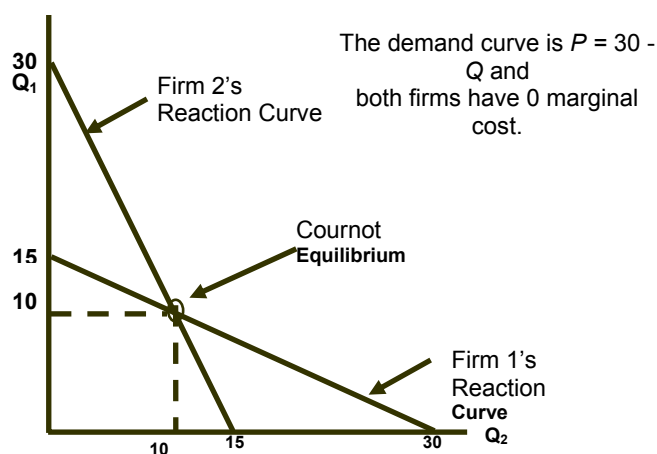
#### — Collusion Curve

- $Q_1 + Q_2 = 15$

—Shows all pairs of output  $Q_1$  and  $Q_2$  that maximizes total profits

- $Q_1 = Q_2 = 7.5$

—Less output and higher profits than the Cournot equilibrium



### FIRST MOVER ADVANTAGE- THE STACKELBERG MODEL

#### Assumptions

- One firm can set output first
- $MC = 0$
- Market demand is  $P = 30 - Q$  where  $Q = \text{total output}$
- Firm 1 sets output first and Firm 2 then makes an output decision

Firm 1 must consider the reaction of Firm 2. Firm 2 takes Firm 1's output as fixed and therefore determines output with the Cournot reaction curve.

Firm 1

Choose  $Q_1$  so that:

$$MR = MC, MC = 0 \text{ therefore } MR = 0$$

$$R_1 = PQ_1 = 30Q_1 - Q_1^2 - Q_2Q_1$$

Conclusion

Firm 1's output is twice as large as firm 2's. Firm 1's profit is twice as large as firm 2's

### PRICE COMPETITION

Competition in an oligopolistic industry may occur with price instead of output. The Bertrand Model is used to illustrate price competition in an oligopolistic industry with homogenous goods.

### BERTRAND MODEL

Assumptions

- Homogenous good
- Market demand is  $P = 30 - Q$  where  $Q = Q_1 + Q_2$
- $MC = \$3$  for both firms and  $MC_1 = MC_2 = \$3$

Assumptions

The Cournot equilibrium:

$$P = \$12$$

$$\pi \text{ for both firms} = \$81$$

Assume the firms compete with price, not quantity.

How will consumers respond to a price differential?

The Nash equilibrium:

$$\begin{aligned} P &= MC; P_1 = P_2 = \$3 \\ Q &= 27; Q_1 \text{ \& } Q_2 = 13.5 \\ \pi &= 0 \end{aligned}$$

### PRICE COMPETITION WITH DIFFERENTIATED PRODUCTS

Market shares are now determined not just by prices, but by differences in the design, performance, and durability of each firm's product.

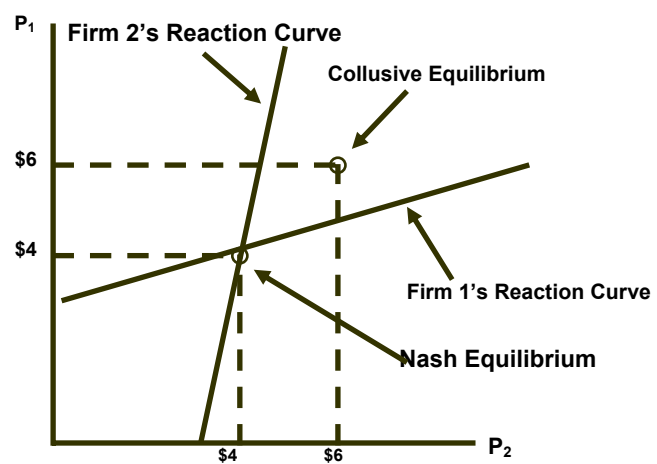
Differentiated Products

Assumptions

- Duopoly
- $FC = \$20$
- $VC = 0$
- Firm 1's demand is  $Q_1 = 12 - 2P_1 + P_2$
- Firm 2's demand is  $Q_2 = 12 - 2P_2 + P_1$

$P_1$  and  $P_2$  are prices firms 1 and 2 charge respectively.  $Q_1$  and  $Q_2$  are the resulting quantities they sell

### Nash Equilibrium in Prices



## LESSON 41

## COMPETITION VERSUS COLLUSION

## THE PRISONERS' DILEMMA

Why wouldn't each firm set the collusion price independently and earn the higher profits that occur with explicit collusion?

Assume:

$$FC = \$20 \text{ and } VC = \$0$$

$$\text{Firm 1's Demand : } Q = 12 - 2P_1 + P_2$$

$$\text{Firm 2's Demand : } Q = 12 - 2P_2 + P_1$$

$$\text{Nash Equilibrium : } P = \$4 \quad \pi = \$12$$

$$\text{Collusion : } P = \$6 \quad \pi = \$16$$

Possible Pricing Outcomes:

$$\text{Firm 1 : } P = \$6 \quad \text{Firm 2 : } P = \$6 \quad \pi = \$16$$

$$P = \$6 \quad P = \$4$$

$$\pi_2 = P_2 Q_2 - 20$$

$$= (4)[12 - (2)(4) + 6] - 20 = \$20$$

$$\pi_1 = P_1 Q_1 - 20$$

$$= (6)[12 - (2)(6) + 4] - 20 = \$4$$

## PAYOFF MATRIX FOR PRICING GAME

|        |            | Firm 2            |                   |
|--------|------------|-------------------|-------------------|
|        |            | Charge \$4        | Charge \$6        |
| Firm 1 | Charge \$4 | <b>\$12, \$12</b> | <b>\$20, \$4</b>  |
|        | Charge \$6 | <b>\$4, \$20</b>  | <b>\$16, \$16</b> |

These two firms are playing a non co-operative game. Each firm independently does the best it can taking its competitor into account.

## Question

Why will both firms both choose \$4 when \$6 will yield higher profits?

An example in game theory, called the Prisoners' Dilemma, illustrates the problem oligopolistic firms face.

**Scenario**

Two prisoners have been accused of collaborating in a crime. They are in separate jail cells and cannot communicate. Each has been asked to confess to the crime.

**PAYOFF MATRIX FOR PRISONERS' DILEMMA**

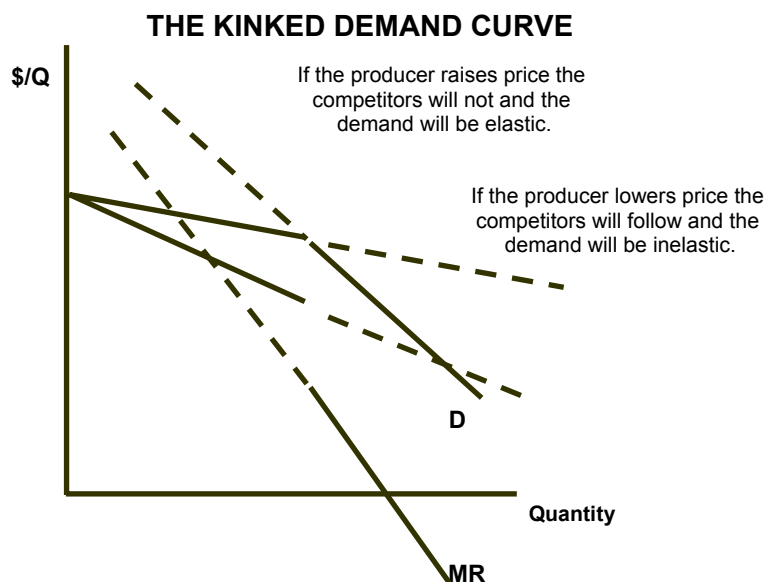
|            |               | Prisoner B |               |
|------------|---------------|------------|---------------|
|            |               | Confess    | Don't confess |
| Prisoner A | Confess       | -5,-5      | -1,-10        |
|            | Don't confess | -10,-1     | -1,-10        |

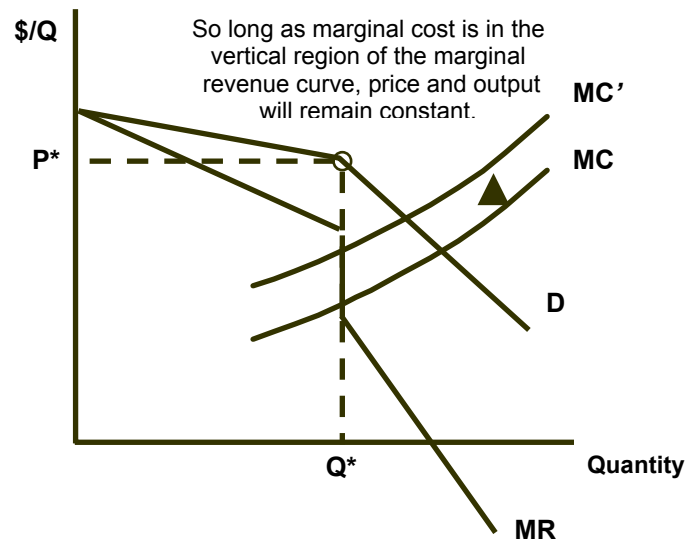
**CONCLUSIONS: OLIGIPOLISTIC MARKETS**

- 1) Collusion will lead to greater profits
- 2) Explicit and implicit collusion is possible
- 3) Once collusion exists, the profit motive to break and lower price is significant

**IMPLICATIONS OF THE PRISONERS' DILEMMA FOR OLIGIPOLISTIC PRICING****OBSERVATIONS OF OLIGOPOLY BEHAVIOR**

In some oligopoly markets, pricing behavior in time can create a predictable pricing environment and implied collusion may occur. In other oligopoly markets, the firms are very aggressive and collusion is not possible. Firms are reluctant to change price because of the likely response of their competitors. In this case prices tend to be relatively rigid.





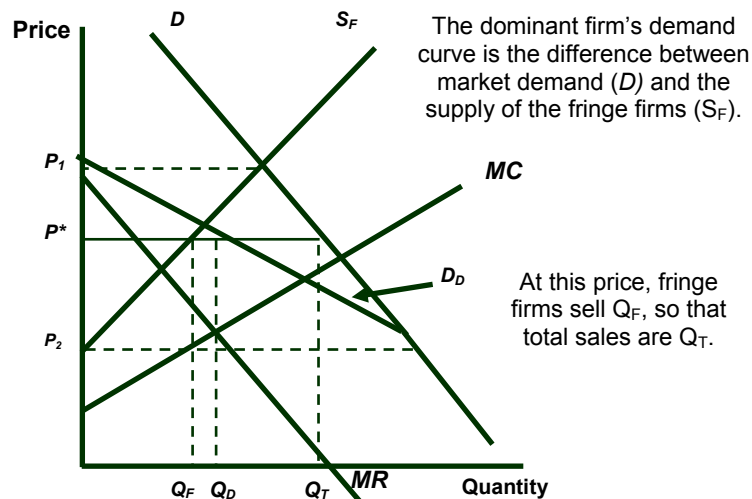
### PRICE SIGNALING & PRICE LEADERSHIP

Price signaling is an implicit collusion in which a firm announces a price increase in the hope that other firms will follow suit. Price leadership is a pattern of pricing in which one firm regularly announces price changes that other firms then match.

### THE DOMINANT FIRM MODEL

In some oligopolistic markets, one large firm has a major share of total sales, and a group of smaller firms supplies the remainder of the market. The large firm might then act as the dominant firm, setting a price that maximized its own profits.

### PRICE SETTING BY A DOMINANT FIRM



### CARTELS

Characteristics

- Explicit agreements to set output and price
- May not include all firms
- Most often international

#### Examples of successful cartels

- OPEC
- International Bauxite Association

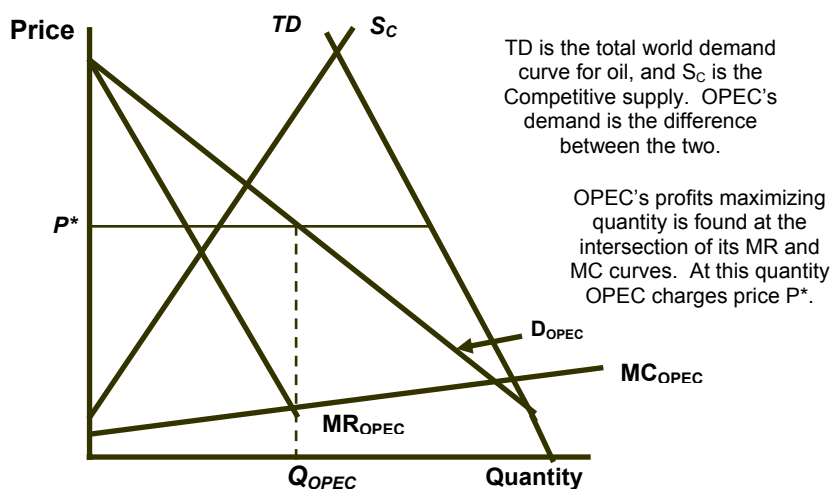
#### Examples of unsuccessful cartels

- Copper
- Tin
- Coffee
- Tea
- Cocoa

#### Conditions for success

- Competitive alternative sufficiently deters cheating
- Potential of monopoly power--inelastic demand

#### THE OPEC OIL CARTEL

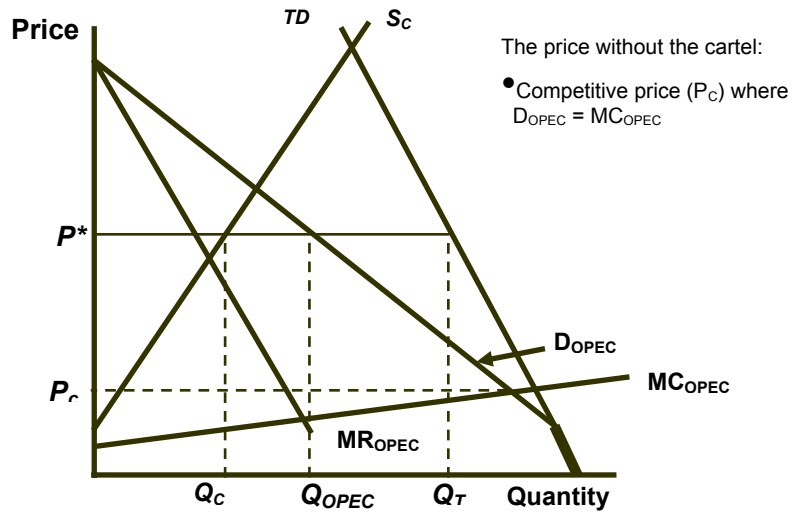


#### CARTELS

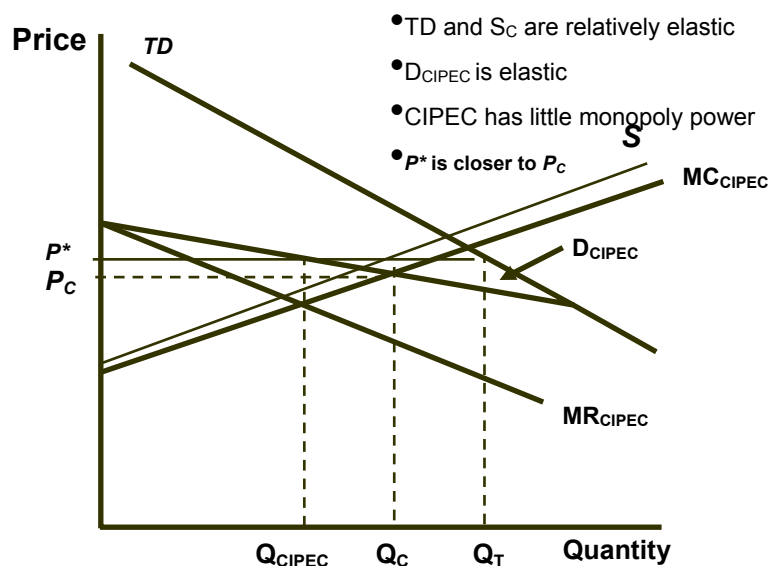
##### About OPEC

- Very low MC
- TD is inelastic
- Non-OPEC supply is inelastic
- $D_{OPEC}$  is relatively inelastic

### THE OPEC OIL CARTEL



### THE CIPEC COPPER CARTEL



### OBSERVATIONS

To be successful:

- Total demand must not be very price elastic
- Either the cartel must control nearly all of the world's supply or the supply of noncartel producers must not be price elastic.



## LESSON 42

## MARKETS FOR FACTOR INPUTS

## COMPETITIVE FACTOR MARKETS

## Characteristics

- 1) Large number of sellers of the factor of production
- 2) Large number of buyers of the factor of production
- 3) The buyers and sellers of the factor of production are price takers

## DEMAND FOR A FACTOR INPUT WHEN ONLY ONE INPUT IS VARIABLE

Demand for factor inputs is a derived demand from factor cost and output demand.

## Assume

- Two inputs: Capital ( $K$ ) and Labor ( $L$ )
- Cost of  $K$  is  $r$  and the cost of labor is  $w$
- $K$  is fixed and  $L$  is variable

Problem: How much labor to hire?

## MEASURING THE VALUE OF A WORKER'S OUTPUT

$$\text{Marginal Revenue Product of Labor (MRP}_L\text{)} = (\text{MP}_L)(\text{MR})$$

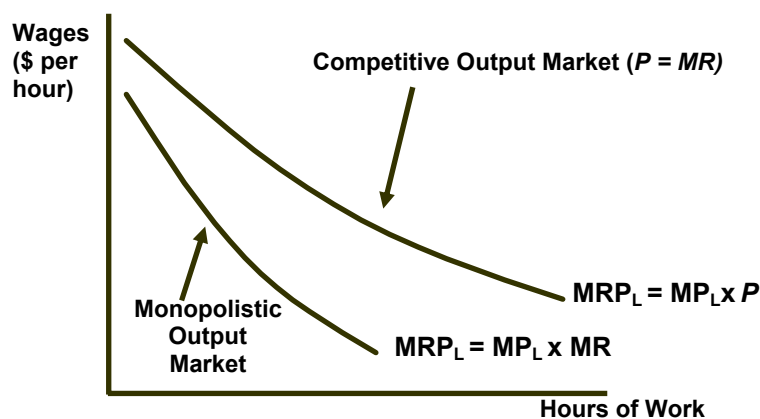
Assume perfect competition in the product market

Then  $\text{MR} = P$

## Question

What will happen to the value of  $\text{MRP}_L$  when more workers are hired?

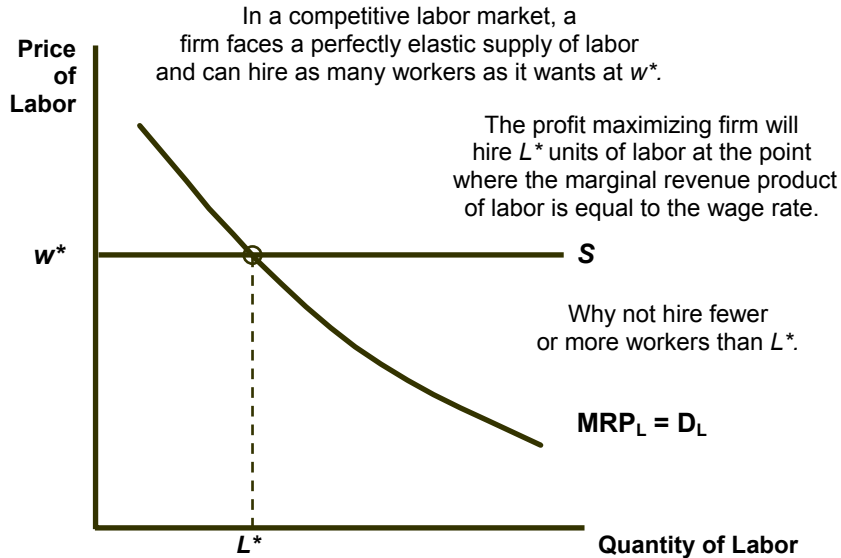
## MARGINAL REVENUE PRODUCT



Choosing the profit-maximizing amount of labor

- If  $\text{MRP}_L > w$  (the marginal cost of hiring a worker): hire the worker
- If  $\text{MRP}_L < w$ : hire less labor
- If  $\text{MRP}_L = w$ : profit maximizing amount of labor

### HIRING BY A FIRM IN THE LABOR MARKET (WITH CAPITAL FIXED)

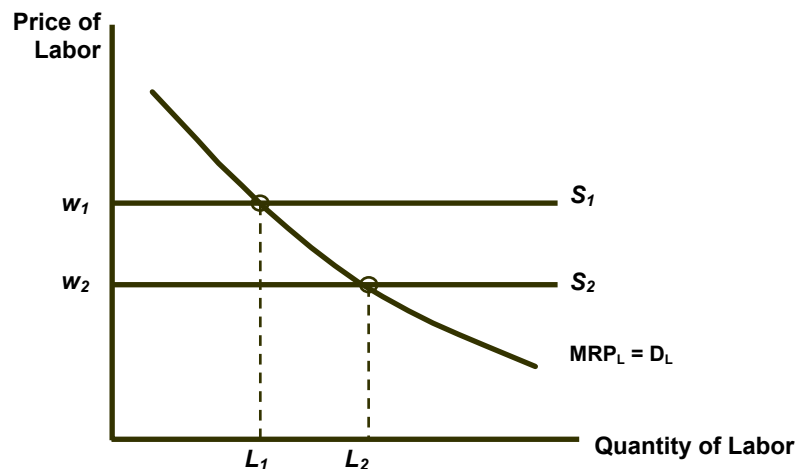


### DEMAND FOR A FACTOR INPUT WHEN ONLY ONE INPUT IS VARIABLE

If the market supply of labor increased relative to demand (baby boomers or female entry), a surplus of labor would exist and the wage rate would fall.

Question: How would this impact the quantity demanded for labor?

### A SHIFT IN THE SUPPLY OF LABOR



### COMPARING INPUT AND OUTPUT MARKETS

$$MRP_L = (MP_L)(MR)$$

and at profit maximizing

$$\text{number of workers } MRP_L = w$$

$$(MP_L)(MR) = w$$

$$MR = w/MP_L$$

$$w/MP_L = MC \text{ of production}$$

In both markets, input and output choices occur where  $MR = MC$

- MR from the sale of the output
- MC from the purchase of the input

### DEMAND FOR A FACTOR INPUT WHEN SEVERAL INPUTS ARE VARIABLE

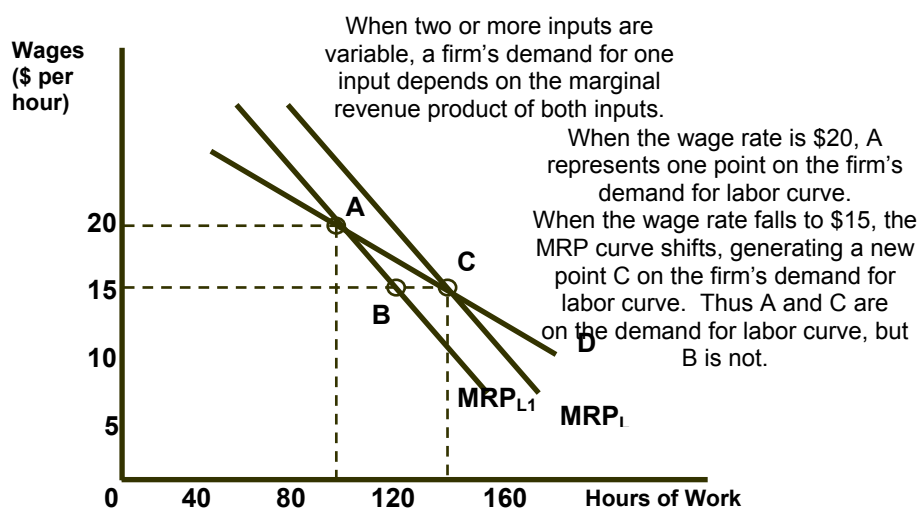
Scenario:

Producing farm equipment with two variable inputs: labor and assembly-line machinery.

Assume the wage rate falls.

Question: How will the decrease in the wage rate impact the demand for labor?

#### FIRM'S DEMAND CURVE FOR LABOR (WITH VARIABLE CAPITAL)



## LESSON 43

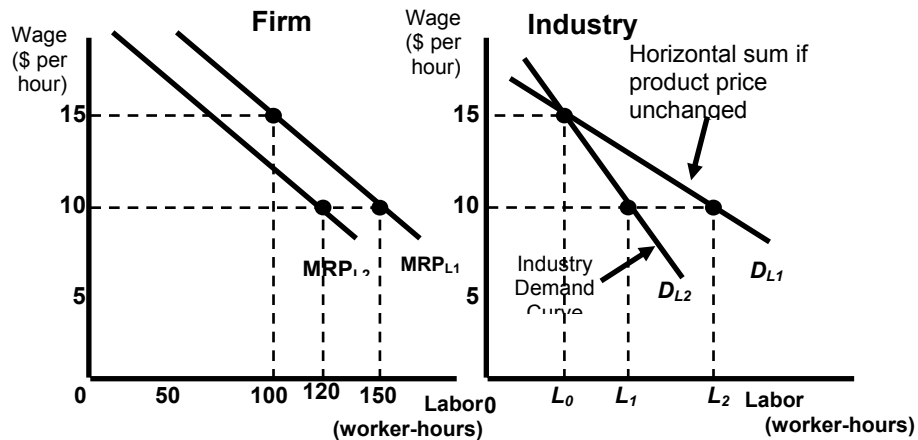
## MARKETS FOR FACTOR INPUTS (Continued)

## INDUSTRY DEMAND FOR LABOR

Assume that all firms respond to a lower wage

- All firms would hire more workers.
- Market supply would increase.
- The market price will fall.
- The quantity demanded for labor by the firm will be smaller.

## THE INDUSTRY DEMAND FOR LABOR



Question: How would a change to a non-competitive market impact the derivation of the market demand for labor?

## THE DEMAND FOR JET FUEL

Observations

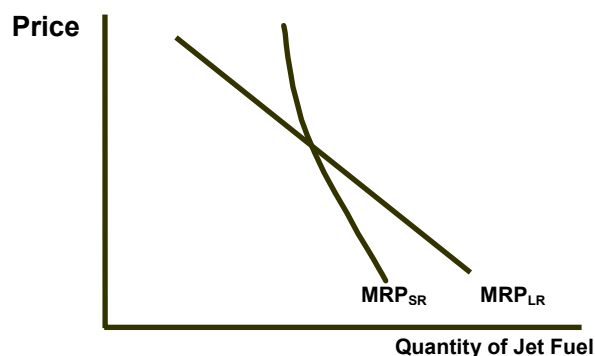
Jet fuel is a factor (input) cost. Cost of jet fuel

- 1971--Jet fuel cost equaled 12.4% of total operating cost
- 1980--Jet fuel cost equaled 30.0% of total operating cost
- 1990's--Jet fuel cost equaled 15.0% of total operating cost

The demand for jet fuel impacts the airlines and refineries alike. The short-run price elasticity of demand for jet-fuel is very inelastic

Question: How would the long-run price elasticity of demand compare to the short-run?

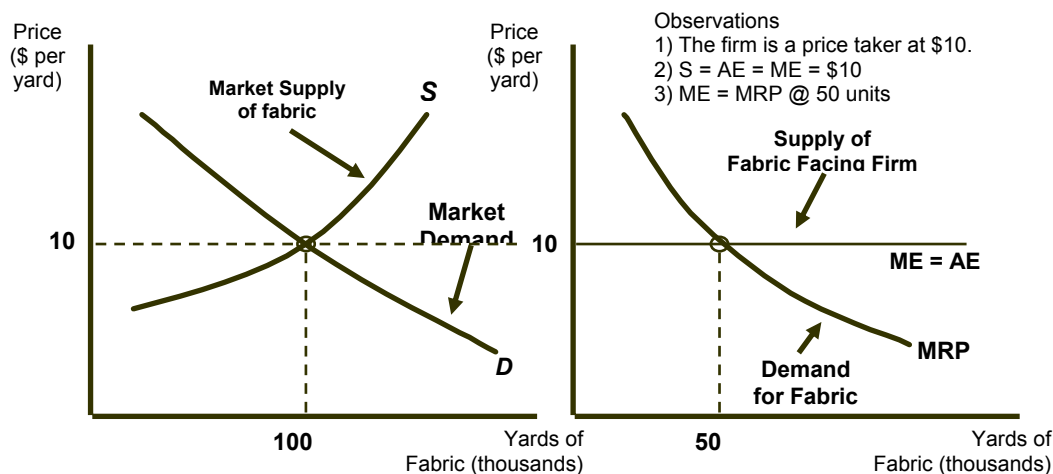
## THE SHORT- AND LONG-RUN



## THE SUPPLY OF INPUTS TO A FIRM

Determining how much of an input to purchase: Assume a perfectly competitive factor market

### A FIRM'S INPUT SUPPLY IN A COMPETITIVE FACTOR MARKET



## THE MARKET SUPPLY OF INPUTS

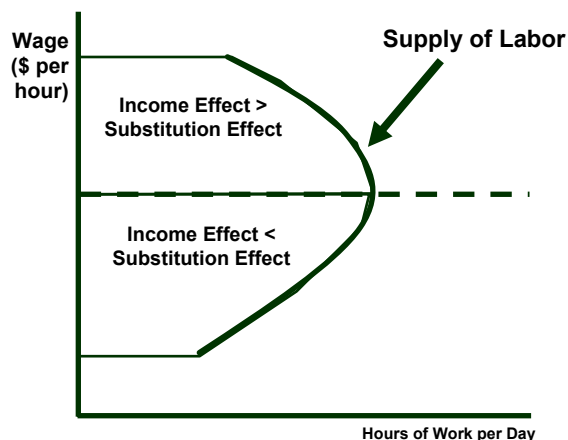
The market supply for physical inputs is upward sloping. Examples: jet fuel, fabric, steel. The market supply for labor may be upward sloping and backward bending

## THE SUPPLY OF LABOR

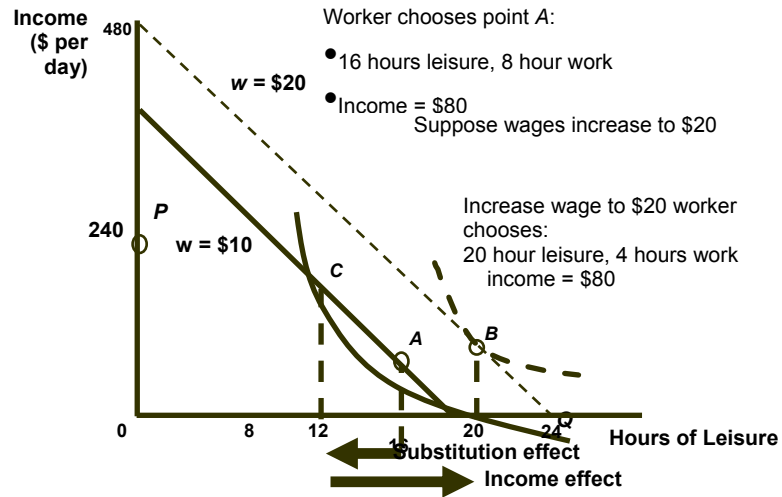
The choice to supply labor is based on utility maximization

- Leisure competes with labor for utility
- Wage rate measures the price of leisure
- Higher wage rate causes the price of leisure to increase
- Higher wages encourage workers to substitute work for leisure (i.e. the substitution effect)
- Higher wages allow the worker to purchase more goods, including leisure which reduces work hours (i.e. the income effect)
- If the income effect exceeds the substitution effect the supply curve is backward bending

### BACKWARD-BENDING SUPPLY OF LABOR



### SUBSTITUTION AND INCOME EFFECTS OF A WAGE INCREASE



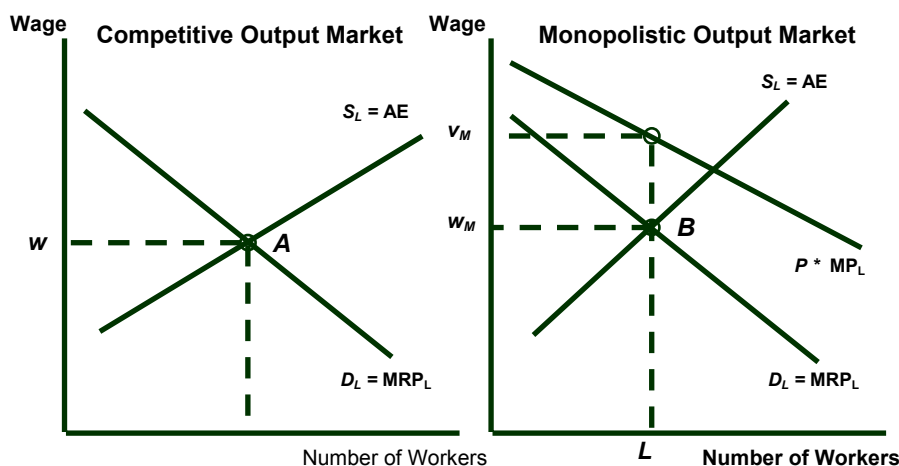
## LESSON 44

## MARKETS FOR FACTOR INPUTS (Continued)

## EQUILIBRIUM IN A COMPETITIVE FACTOR MARKET

A competitive factor market is in equilibrium when the price of the input equates the quantity demanded to the quantity supplied.

## LABOR MARKET EQUILIBRIUM



## EQUILIBRIUM IN A COMPETITIVE OUTPUT MARKET

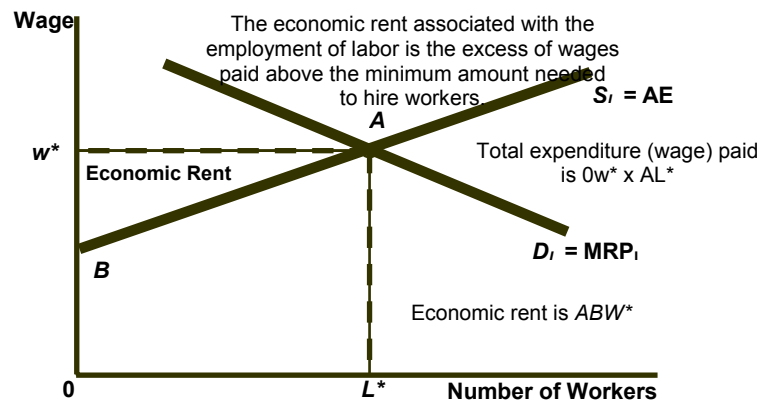
- $D_L(MRP_L) = S_L$
- $w_C = MRP_L$
- $MRP_L = (P)(MP_L)$
- Markets are efficient

## EQUILIBRIUM IN A MONOPOLISTIC OUTPUT MARKET

- $MR < P$
- $MRP = (MR)(MP_L)$
- Hire  $L_M$  at wage  $w_M$
- $v_M$  = marginal benefit to consumers
- $w_M$  = marginal cost to the firm
- Profits maximized
- Using less than efficient level of input

## ECONOMIC RENT

For a factor market, economic rent is the difference between the payments made to a factor of production and the minimum amount that must be spent to obtain the use of that factor.

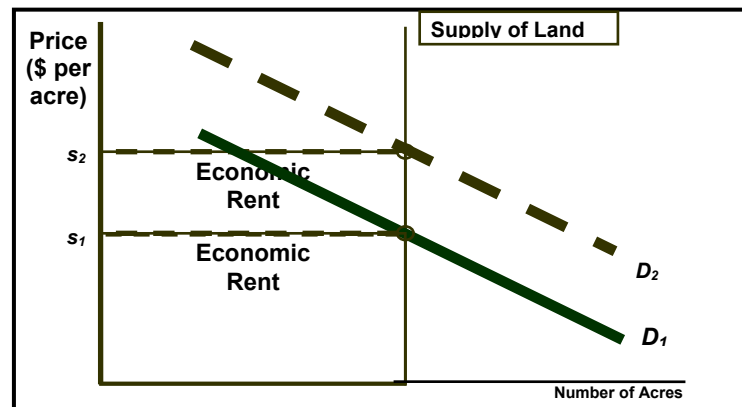


Question: What would be the economic rent if  $S_L$  is perfectly elastic or perfectly inelastic?

Land: A Perfectly Inelastic Supply

With land inelastically supplied, its price is determined entirely by demand, at least in the short run.

### LAND RENT



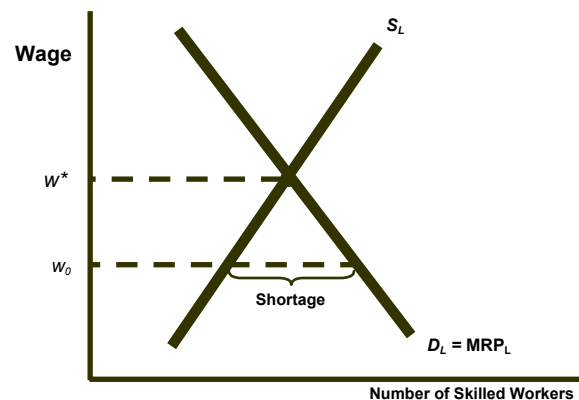
### PAY IN THE PUBLIC SECTOR

The percentage of personnel working in public sector has been declining.

Shortages of skilled personnel has occurred? Why?

If there is a shortage, the wage must be below the competitive wage rate

### THE SHORTAGE OF SKILLED PERSONNEL





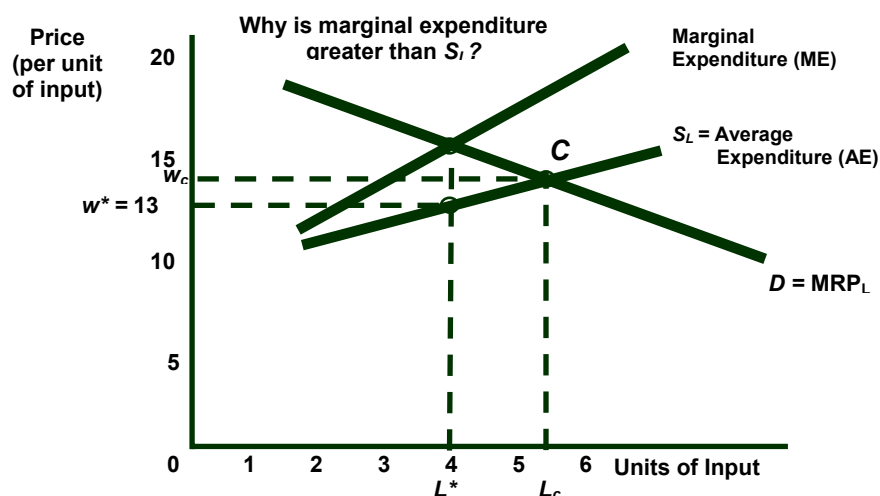
Public sector pay is based on years of service not MRP. MRP increases and the private sector pay is greater than public sector pay. Many leave the public sector.

### FACTOR MARKETS WITH MONOPSONY POWER

Assume

- The output market is perfectly competitive.
- Input market is pure monopsony.

#### MARGINAL AND AVERAGE EXPENDITURE



### FACTOR MARKETS WITH MONOPSONY POWER

Examples of Monopsony Power

Government

- Soldiers
- Missiles
- B2 Bombers

NASA

- Astronauts

Company town

### MONOPSONY POWER IN THE MARKET FOR BASEBALL PLAYERS

Baseball owners created a monopsonistic cartel

Reserve clause prevented competition for players

In 1969--Average salary was \$42,000

In 1997--Average salary was \$1,383,578

In 1975 salaries were 25% of team expenditures

In 1980 salaries were 40% of team expenditures

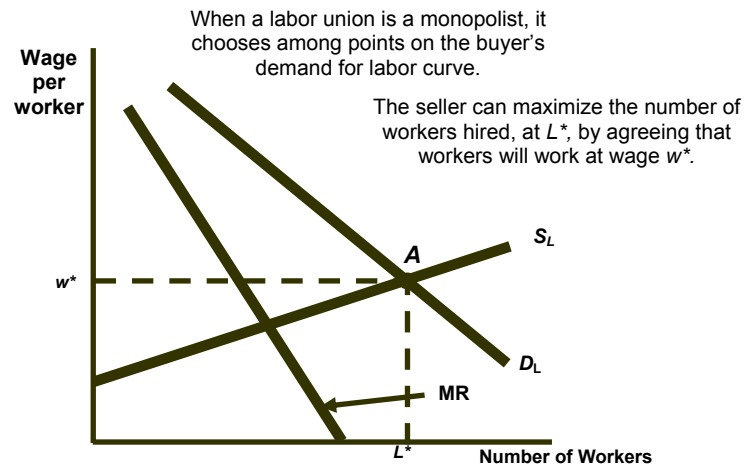
## LESSON 45

## MARKETS FOR FACTOR INPUTS (Continued)

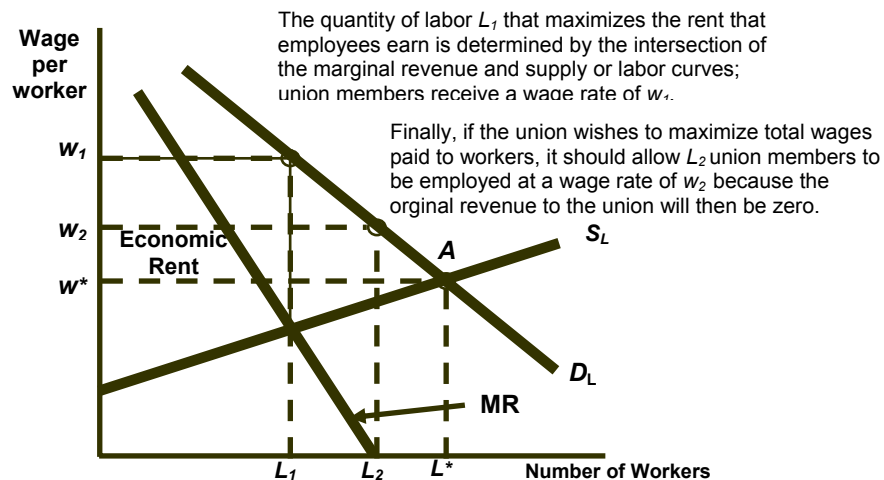
## FACTOR MARKETS WITH MONOPOLY POWER

Just as buyers of inputs can have monopsony power, sellers of inputs can have monopoly power. The most important example of monopoly power in factor markets involves labor unions.

## MONOPOLY POWER OF SELLERS OF LABOR



## MONOPOLY POWER OF SELLERS OF LABOR

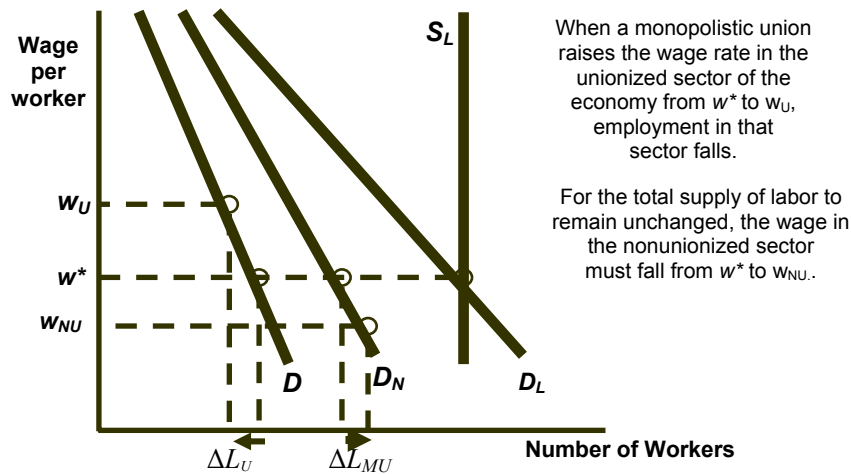


The primary determinant of controlling wage and economic rent is controlling the supply of labor

## A TWO-SECTOR MODEL OF LABOR EMPLOYMENT

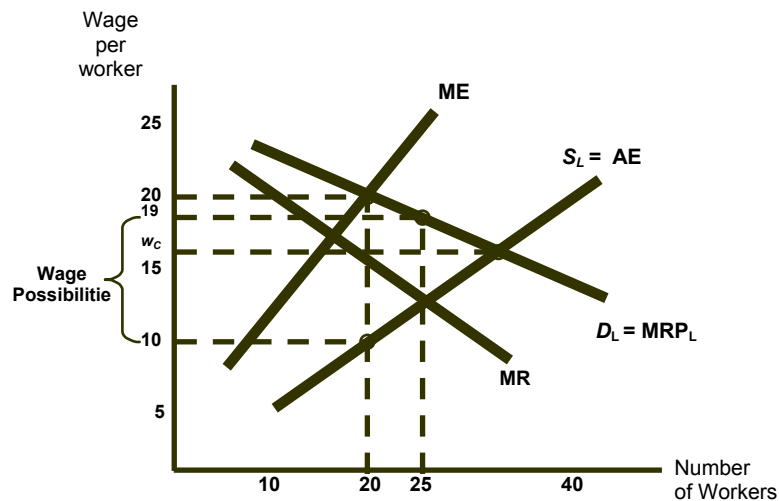
Union monopoly power impacts the nonunionized part of the economy.

### WAGE DETERMINATION IN UNIONIZED & NONUNIONIZED SECTORS



### BILATERAL MONOPOLY

Market in which a monopolist sells to a monopsonist.



#### Observations

Hiring without union monopoly power

$$MRP = ME \text{ at } 20 \text{ workers and } w = \$10/\text{hr}$$

Union's objective

$$MR = MC \text{ at } 25 \text{ workers and } w = \$19/\text{hr}$$

Who Will Win?

- The union will if its threat to strike is credible.
- The firm will if its threat to hire non-union workers is credible.
- If both make credible threats the wage will be at  $w_C$ .